095 to 077: consequential handoff

Illicit and black markets -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

095 to 078: consequential handoff

Illicit and black markets -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

095 to 079: consequential handoff

Illicit and black markets -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

095 to 080: consequential handoff

Illicit and black markets -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

095 to 081: consequential handoff

Illicit and black markets -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

095 to 082: consequential handoff

Illicit and black markets -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

095 to 083: consequential handoff

Illicit and black markets -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

095 to 084: consequential handoff

Illicit and black markets -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

095 to 085: consequential handoff

Illicit and black markets -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

095 to 086: consequential handoff

Illicit and black markets -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

095 to 087: consequential handoff

Illicit and black markets -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

095 to 088: consequential handoff

Illicit and black markets -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

095 to 089: consequential handoff

Illicit and black markets -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

095 to 090: consequential handoff

Illicit and black markets -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

095 to 091: consequential handoff

Illicit and black markets -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

095 to 092: consequential handoff

Illicit and black markets -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

095 to 093: consequential handoff

Illicit and black markets -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

095 to 094: consequential handoff

Illicit and black markets -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

095 to 095: consequential handoff

Illicit and black markets -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Illicit and black markets becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 718-724; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

095 to 096: consequential handoff

Illicit and black markets -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

095 to 097: consequential handoff

Illicit and black markets -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

095 to 098: consequential handoff

Illicit and black markets -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

095 to 099: consequential handoff

Illicit and black markets -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

095 to 100: consequential handoff

Illicit and black markets -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

095 to 101: consequential handoff

Illicit and black markets -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

095 to 102: consequential handoff

Illicit and black markets -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

095 to 103: consequential handoff

Illicit and black markets -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

095 to 104: consequential handoff

Illicit and black markets -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

095 to 105: consequential handoff

Illicit and black markets -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

095 to 106: consequential handoff

Illicit and black markets -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

095 to 107: consequential handoff

Illicit and black markets -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

095 to 108: consequential handoff

Illicit and black markets -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

095 to 109: consequential handoff

Illicit and black markets -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

095 to 110: consequential handoff

Illicit and black markets -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

095 to 111: consequential handoff

Illicit and black markets -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

095 to 112: consequential handoff

Illicit and black markets -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

095 to 113: consequential handoff

Illicit and black markets -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

095 to 114: consequential handoff

Illicit and black markets -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

095 to 115: consequential handoff

Illicit and black markets -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

095 to 116: consequential handoff

Illicit and black markets -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

095 to 117: consequential handoff

Illicit and black markets -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

095 to 118: consequential handoff

Illicit and black markets -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

095 to 119: consequential handoff

Illicit and black markets -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

095 to 120: consequential handoff

Illicit and black markets -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Illicit and black markets into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Official market data that excludes hidden exchange. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, communities, lawful firms, institutions. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Independent harm-reduction/public-safety evaluation' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 718-724 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

095: Illicit and black markets

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Illicit and black markets

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Illicit and black markets is described in the Atlas as: Hidden demand, supply, coercion, enforcement, prices and harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Profit and access externalize violence, exploitation or public harm. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, communities, lawful firms, institutions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Harm and illicit profitability fall while legitimate access/opportunity rises. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Independent harm-reduction/public-safety evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Evidence and source reconstruction

Illicit and black markets | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 095: Illicit and black markets; original author status: Safety-governed frontier. Native state: Hidden demand, supply, coercion, enforcement, prices and harm.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Official market data that excludes hidden exchange. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Profit and access externalize violence, exploitation or public harm. The relevant receiving carriers are: Victims, communities, lawful firms, institutions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Harm and illicit profitability fall while legitimate access/opportunity rises.

RETEST AND SCOPE CONTROL

Candidate decisive method: Independent harm-reduction/public-safety evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Boundary divergence

Illicit and black markets | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 095: Illicit and black markets; original author status: Safety-governed frontier. Native state: Hidden demand, supply, coercion, enforcement, prices and harm.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Official market data that excludes hidden exchange. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Profit and access externalize violence, exploitation or public harm. The relevant receiving carriers are: Victims, communities, lawful firms, institutions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Harm and illicit profitability fall while legitimate access/opportunity rises.

RETEST AND SCOPE CONTROL

Candidate decisive method: Independent harm-reduction/public-safety evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Causal diagram and identifiability

Illicit and black markets | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 095: Illicit and black markets; original author status: Safety-governed frontier. Native state: Hidden demand, supply, coercion, enforcement, prices and harm.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Official market data that excludes hidden exchange. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Profit and access externalize violence, exploitation or public harm. The relevant receiving carriers are: Victims, communities, lawful firms, institutions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Harm and illicit profitability fall while legitimate access/opportunity rises.

RETEST AND SCOPE CONTROL

Candidate decisive method: Independent harm-reduction/public-safety evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Receiving carrier identification

Illicit and black markets | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 095: Illicit and black markets; original author status: Safety-governed frontier. Native state: Hidden demand, supply, coercion, enforcement, prices and harm.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Official market data that excludes hidden exchange. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Profit and access externalize violence, exploitation or public harm. The relevant receiving carriers are: Victims, communities, lawful firms, institutions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Harm and illicit profitability fall while legitimate access/opportunity rises.

RETEST AND SCOPE CONTROL

Candidate decisive method: Independent harm-reduction/public-safety evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Hidden compensation accounting

Illicit and black markets | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 095: Illicit and black markets; original author status: Safety-governed frontier. Native state: Hidden demand, supply, coercion, enforcement, prices and harm.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Official market data that excludes hidden exchange. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Profit and access externalize violence, exploitation or public harm. The relevant receiving carriers are: Victims, communities, lawful firms, institutions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Harm and illicit profitability fall while legitimate access/opportunity rises.

RETEST AND SCOPE CONTROL

Candidate decisive method: Independent harm-reduction/public-safety evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Native units and dimensional integrity

Illicit and black markets | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 095: Illicit and black markets; original author status: Safety-governed frontier. Native state: Hidden demand, supply, coercion, enforcement, prices and harm.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Official market data that excludes hidden exchange. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Profit and access externalize violence, exploitation or public harm. The relevant receiving carriers are: Victims, communities, lawful firms, institutions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Harm and illicit profitability fall while legitimate access/opportunity rises.

RETEST AND SCOPE CONTROL

Candidate decisive method: Independent harm-reduction/public-safety evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Capacity specification

Illicit and black markets | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 095: Illicit and black markets; original author status: Safety-governed frontier. Native state: Hidden demand, supply, coercion, enforcement, prices and harm.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Official market data that excludes hidden exchange. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Profit and access externalize violence, exploitation or public harm. The relevant receiving carriers are: Victims, communities, lawful firms, institutions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Harm and illicit profitability fall while legitimate access/opportunity rises.

RETEST AND SCOPE CONTROL

Candidate decisive method: Independent harm-reduction/public-safety evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Intervention design

Illicit and black markets | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 095: Illicit and black markets; original author status: Safety-governed frontier. Native state: Hidden demand, supply, coercion, enforcement, prices and harm.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Official market data that excludes hidden exchange. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Profit and access externalize violence, exploitation or public harm. The relevant receiving carriers are: Victims, communities, lawful firms, institutions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Harm and illicit profitability fall while legitimate access/opportunity rises.

RETEST AND SCOPE CONTROL

Candidate decisive method: Independent harm-reduction/public-safety evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Incumbent parity contest

Illicit and black markets | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 095: Illicit and black markets; original author status: Safety-governed frontier. Native state: Hidden demand, supply, coercion, enforcement, prices and harm.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Official market data that excludes hidden exchange. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Profit and access externalize violence, exploitation or public harm. The relevant receiving carriers are: Victims, communities, lawful firms, institutions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Harm and illicit profitability fall while legitimate access/opportunity rises.

RETEST AND SCOPE CONTROL

Candidate decisive method: Independent harm-reduction/public-safety evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Time-window and recurrence

Illicit and black markets | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 095: Illicit and black markets; original author status: Safety-governed frontier. Native state: Hidden demand, supply, coercion, enforcement, prices and harm.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Official market data that excludes hidden exchange. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Profit and access externalize violence, exploitation or public harm. The relevant receiving carriers are: Victims, communities, lawful firms, institutions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Harm and illicit profitability fall while legitimate access/opportunity rises.

RETEST AND SCOPE CONTROL

Candidate decisive method: Independent harm-reduction/public-safety evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Ethical and legal safety

Illicit and black markets | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 095: Illicit and black markets; original author status: Safety-governed frontier. Native state: Hidden demand, supply, coercion, enforcement, prices and harm.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Official market data that excludes hidden exchange. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Profit and access externalize violence, exploitation or public harm. The relevant receiving carriers are: Victims, communities, lawful firms, institutions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Harm and illicit profitability fall while legitimate access/opportunity rises.

RETEST AND SCOPE CONTROL

Candidate decisive method: Independent harm-reduction/public-safety evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Sensitivity and negative controls

Illicit and black markets | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 095: Illicit and black markets; original author status: Safety-governed frontier. Native state: Hidden demand, supply, coercion, enforcement, prices and harm.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Official market data that excludes hidden exchange. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Profit and access externalize violence, exploitation or public harm. The relevant receiving carriers are: Victims, communities, lawful firms, institutions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Harm and illicit profitability fall while legitimate access/opportunity rises.

RETEST AND SCOPE CONTROL

Candidate decisive method: Independent harm-reduction/public-safety evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Replication and provenance

Illicit and black markets | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 095: Illicit and black markets; original author status: Safety-governed frontier. Native state: Hidden demand, supply, coercion, enforcement, prices and harm.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Official market data that excludes hidden exchange. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Profit and access externalize violence, exploitation or public harm. The relevant receiving carriers are: Victims, communities, lawful firms, institutions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Harm and illicit profitability fall while legitimate access/opportunity rises.

RETEST AND SCOPE CONTROL

Candidate decisive method: Independent harm-reduction/public-safety evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

095: Cross-domain invariant contest

Illicit and black markets | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 095: Illicit and black markets; original author status: Safety-governed frontier. Native state: Hidden demand, supply, coercion, enforcement, prices and harm.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Official market data that excludes hidden exchange. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Profit and access externalize violence, exploitation or public harm. The relevant receiving carriers are: Victims, communities, lawful firms, institutions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Harm and illicit profitability fall while legitimate access/opportunity rises.

RETEST AND SCOPE CONTROL

Candidate decisive method: Independent harm-reduction/public-safety evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 095, pp. 718-724 (Rufino 2026).

Adapter 096: Orbital and space economy

Frontier, illicit and cosmic | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Spacecraft, debris, spectrum, energy, logistics and operator incentives

RECORDED RECEIVING SCHEMA

Mission-local success metric

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Mission gain shifts debris/collision/spectrum burden. Candidate receivers: Other spacecraft, operators, orbital environment.

EXISTING CAPACITY DESIGN

Orbital consequential ledger + shared capacity

PROTECTED OUTCOME STATED BY THE AUTHOR

Mission value, safety, debris and resource efficiency improve

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Simulation + mission pilot. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096 | Orbital and space economy

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 096, Orbital and space economy. Native state: Spacecraft, debris, spectrum, energy, logistics and operator incentives. A human may be a researcher, operator, regulated party or downstream recipient of this Orbital and space economy system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Mission gain shifts debris/collision/spectrum burden.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Orbital consequential ledger + shared capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Mission value, safety, debris and resource efficiency improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

096 | Orbital and space economy

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 096, Orbital and space economy. Native state: Spacecraft, debris, spectrum, energy, logistics and operator incentives. A human may be a researcher, operator, regulated party or downstream recipient of this Orbital and space economy system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Mission gain shifts debris/collision/spectrum burden.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Orbital consequential ledger + shared capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Mission value, safety, debris and resource efficiency improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

096 | Orbital and space economy

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 096, Orbital and space economy. Native state: Spacecraft, debris, spectrum, energy, logistics and operator incentives. A human may be a researcher, operator, regulated party or downstream recipient of this Orbital and space economy system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Mission gain shifts debris/collision/spectrum burden.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Orbital consequential ledger + shared capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Mission value, safety, debris and resource efficiency improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

096 | Orbital and space economy

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 096, Orbital and space economy. Native state: Spacecraft, debris, spectrum, energy, logistics and operator incentives. A human may be a researcher, operator, regulated party or downstream recipient of this Orbital and space economy system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Mission gain shifts debris/collision/spectrum burden.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Orbital consequential ledger + shared capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Mission value, safety, debris and resource efficiency improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

096 | Orbital and space economy

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 096, Orbital and space economy. Native state: Spacecraft, debris, spectrum, energy, logistics and operator incentives. A human may be a researcher, operator, regulated party or downstream recipient of this Orbital and space economy system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Mission gain shifts debris/collision/spectrum burden.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Orbital consequential ledger + shared capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Mission value, safety, debris and resource efficiency improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

096 | Orbital and space economy

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 096, Orbital and space economy. Native state: Spacecraft, debris, spectrum, energy, logistics and operator incentives. A human may be a researcher, operator, regulated party or downstream recipient of this Orbital and space economy system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Mission gain shifts debris/collision/spectrum burden.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Orbital consequential ledger + shared capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Mission value, safety, debris and resource efficiency improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

096 to 001: consequential handoff

Orbital and space economy -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

096 to 002: consequential handoff

Orbital and space economy -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

096 to 003: consequential handoff

Orbital and space economy -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

096 to 004: consequential handoff

Orbital and space economy -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Clinical trials + patient-reported and partner-impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

096 to 005: consequential handoff

Orbital and space economy -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

096 to 006: consequential handoff

Orbital and space economy -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

096 to 007: consequential handoff

Orbital and space economy -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

096 to 008: consequential handoff

Orbital and space economy -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

096 to 009: consequential handoff

Orbital and space economy -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

096 to 010: consequential handoff

Orbital and space economy -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

096 to 011: consequential handoff

Orbital and space economy -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

096 to 012: consequential handoff

Orbital and space economy -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

096 to 013: consequential handoff

Orbital and space economy -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

096 to 014: consequential handoff

Orbital and space economy -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

096 to 015: consequential handoff

Orbital and space economy -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

096 to 016: consequential handoff

Orbital and space economy -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

096 to 017: consequential handoff

Orbital and space economy -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

096 to 018: consequential handoff

Orbital and space economy -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

096 to 019: consequential handoff

Orbital and space economy -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

096 to 020: consequential handoff

Orbital and space economy -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

096 to 021: consequential handoff

Orbital and space economy -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

096 to 022: consequential handoff

Orbital and space economy -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

096 to 023: consequential handoff

Orbital and space economy -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

096 to 024: consequential handoff

Orbital and space economy -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

096 to 025: consequential handoff

Orbital and space economy -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

096 to 026: consequential handoff

Orbital and space economy -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

096 to 027: consequential handoff

Orbital and space economy -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

096 to 028: consequential handoff

Orbital and space economy -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

096 to 029: consequential handoff

Orbital and space economy -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

096 to 030: consequential handoff

Orbital and space economy -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

096 to 031: consequential handoff

Orbital and space economy -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

096 to 032: consequential handoff

Orbital and space economy -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

096 to 033: consequential handoff

Orbital and space economy -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

096 to 034: consequential handoff

Orbital and space economy -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

096 to 035: consequential handoff

Orbital and space economy -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

096 to 036: consequential handoff

Orbital and space economy -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

096 to 037: consequential handoff

Orbital and space economy -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

096 to 038: consequential handoff

Orbital and space economy -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

096 to 039: consequential handoff

Orbital and space economy -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

096 to 040: consequential handoff

Orbital and space economy -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

096 to 041: consequential handoff

Orbital and space economy -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

096 to 042: consequential handoff

Orbital and space economy -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

096 to 043: consequential handoff

Orbital and space economy -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

096 to 044: consequential handoff

Orbital and space economy -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

096 to 045: consequential handoff

Orbital and space economy -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

096 to 046: consequential handoff

Orbital and space economy -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

096 to 047: consequential handoff

Orbital and space economy -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

096 to 048: consequential handoff

Orbital and space economy -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

096 to 049: consequential handoff

Orbital and space economy -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

096 to 050: consequential handoff

Orbital and space economy -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

096 to 051: consequential handoff

Orbital and space economy -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

096 to 052: consequential handoff

Orbital and space economy -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

096 to 053: consequential handoff

Orbital and space economy -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

096 to 054: consequential handoff

Orbital and space economy -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

096 to 055: consequential handoff

Orbital and space economy -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

096 to 056: consequential handoff

Orbital and space economy -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

096 to 057: consequential handoff

Orbital and space economy -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

096 to 058: consequential handoff

Orbital and space economy -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

096 to 059: consequential handoff

Orbital and space economy -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

096 to 060: consequential handoff

Orbital and space economy -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

096 to 061: consequential handoff

Orbital and space economy -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

096 to 062: consequential handoff

Orbital and space economy -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

096 to 063: consequential handoff

Orbital and space economy -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

096 to 064: consequential handoff

Orbital and space economy -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

096 to 065: consequential handoff

Orbital and space economy -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

096 to 066: consequential handoff

Orbital and space economy -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

096 to 067: consequential handoff

Orbital and space economy -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

096 to 068: consequential handoff

Orbital and space economy -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

096 to 069: consequential handoff

Orbital and space economy -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

096 to 070: consequential handoff

Orbital and space economy -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

096 to 071: consequential handoff

Orbital and space economy -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

096 to 072: consequential handoff

Orbital and space economy -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

096 to 073: consequential handoff

Orbital and space economy -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

096 to 074: consequential handoff

Orbital and space economy -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

096 to 075: consequential handoff

Orbital and space economy -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

096 to 076: consequential handoff

Orbital and space economy -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

096 to 077: consequential handoff

Orbital and space economy -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

096 to 078: consequential handoff

Orbital and space economy -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

096 to 079: consequential handoff

Orbital and space economy -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

096 to 080: consequential handoff

Orbital and space economy -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

096 to 081: consequential handoff

Orbital and space economy -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

096 to 082: consequential handoff

Orbital and space economy -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

096 to 083: consequential handoff

Orbital and space economy -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

096 to 084: consequential handoff

Orbital and space economy -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

096 to 085: consequential handoff

Orbital and space economy -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

096 to 086: consequential handoff

Orbital and space economy -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

096 to 087: consequential handoff

Orbital and space economy -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

096 to 088: consequential handoff

Orbital and space economy -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

096 to 089: consequential handoff

Orbital and space economy -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

096 to 090: consequential handoff

Orbital and space economy -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

096 to 091: consequential handoff

Orbital and space economy -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

096 to 092: consequential handoff

Orbital and space economy -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

096 to 093: consequential handoff

Orbital and space economy -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

096 to 094: consequential handoff

Orbital and space economy -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

096 to 095: consequential handoff

Orbital and space economy -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

096 to 096: consequential handoff

Orbital and space economy -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Orbital and space economy becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 725-731; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

096 to 097: consequential handoff

Orbital and space economy -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

096 to 098: consequential handoff

Orbital and space economy -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

096 to 099: consequential handoff

Orbital and space economy -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

096 to 100: consequential handoff

Orbital and space economy -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

096 to 101: consequential handoff

Orbital and space economy -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

096 to 102: consequential handoff

Orbital and space economy -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

096 to 103: consequential handoff

Orbital and space economy -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

096 to 104: consequential handoff

Orbital and space economy -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

096 to 105: consequential handoff

Orbital and space economy -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

096 to 106: consequential handoff

Orbital and space economy -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

096 to 107: consequential handoff

Orbital and space economy -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

096 to 108: consequential handoff

Orbital and space economy -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

096 to 109: consequential handoff

Orbital and space economy -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

096 to 110: consequential handoff

Orbital and space economy -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

096 to 111: consequential handoff

Orbital and space economy -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

096 to 112: consequential handoff

Orbital and space economy -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

096 to 113: consequential handoff

Orbital and space economy -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

096 to 114: consequential handoff

Orbital and space economy -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

096 to 115: consequential handoff

Orbital and space economy -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

096 to 116: consequential handoff

Orbital and space economy -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

096 to 117: consequential handoff

Orbital and space economy -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

096 to 118: consequential handoff

Orbital and space economy -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

096 to 119: consequential handoff

Orbital and space economy -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

096 to 120: consequential handoff

Orbital and space economy -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Orbital and space economy into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Mission-local success metric. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Other spacecraft, operators, orbital environment. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Orbital consequential ledger + shared capacity. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Simulation + mission pilot' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 725-731 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

096: Orbital and space economy

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Orbital and space economy

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Orbital and space economy is described in the Atlas as: Spacecraft, debris, spectrum, energy, logistics and operator incentives. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Mission gain shifts debris/collision/spectrum burden. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Other spacecraft, operators, orbital environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Orbital consequential ledger + shared capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Mission value, safety, debris and resource efficiency improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Simulation + mission pilot or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Evidence and source reconstruction

Orbital and space economy | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 096: Orbital and space economy; original author status: Hard engineering frontier. Native state: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Mission-local success metric. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Mission gain shifts debris/collision/spectrum burden. The relevant receiving carriers are: Other spacecraft, operators, orbital environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Orbital consequential ledger + shared capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Mission value, safety, debris and resource efficiency improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Simulation + mission pilot. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Boundary divergence

Orbital and space economy | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 096: Orbital and space economy; original author status: Hard engineering frontier. Native state: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Mission-local success metric. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Mission gain shifts debris/collision/spectrum burden. The relevant receiving carriers are: Other spacecraft, operators, orbital environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Orbital consequential ledger + shared capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Mission value, safety, debris and resource efficiency improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Simulation + mission pilot. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Causal diagram and identifiability

Orbital and space economy | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 096: Orbital and space economy; original author status: Hard engineering frontier. Native state: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Mission-local success metric. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Mission gain shifts debris/collision/spectrum burden. The relevant receiving carriers are: Other spacecraft, operators, orbital environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Orbital consequential ledger + shared capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Mission value, safety, debris and resource efficiency improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Simulation + mission pilot. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Receiving carrier identification

Orbital and space economy | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 096: Orbital and space economy; original author status: Hard engineering frontier. Native state: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Mission-local success metric. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Mission gain shifts debris/collision/spectrum burden. The relevant receiving carriers are: Other spacecraft, operators, orbital environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Orbital consequential ledger + shared capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Mission value, safety, debris and resource efficiency improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Simulation + mission pilot. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Hidden compensation accounting

Orbital and space economy | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 096: Orbital and space economy; original author status: Hard engineering frontier. Native state: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Mission-local success metric. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Mission gain shifts debris/collision/spectrum burden. The relevant receiving carriers are: Other spacecraft, operators, orbital environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Orbital consequential ledger + shared capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Mission value, safety, debris and resource efficiency improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Simulation + mission pilot. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Native units and dimensional integrity

Orbital and space economy | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 096: Orbital and space economy; original author status: Hard engineering frontier. Native state: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Mission-local success metric. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Mission gain shifts debris/collision/spectrum burden. The relevant receiving carriers are: Other spacecraft, operators, orbital environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Orbital consequential ledger + shared capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Mission value, safety, debris and resource efficiency improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Simulation + mission pilot. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Capacity specification

Orbital and space economy | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 096: Orbital and space economy; original author status: Hard engineering frontier. Native state: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Mission-local success metric. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Mission gain shifts debris/collision/spectrum burden. The relevant receiving carriers are: Other spacecraft, operators, orbital environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Orbital consequential ledger + shared capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Mission value, safety, debris and resource efficiency improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Simulation + mission pilot. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Intervention design

Orbital and space economy | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 096: Orbital and space economy; original author status: Hard engineering frontier. Native state: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Mission-local success metric. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Mission gain shifts debris/collision/spectrum burden. The relevant receiving carriers are: Other spacecraft, operators, orbital environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Orbital consequential ledger + shared capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Mission value, safety, debris and resource efficiency improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Simulation + mission pilot. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Incumbent parity contest

Orbital and space economy | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 096: Orbital and space economy; original author status: Hard engineering frontier. Native state: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Mission-local success metric. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Mission gain shifts debris/collision/spectrum burden. The relevant receiving carriers are: Other spacecraft, operators, orbital environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Orbital consequential ledger + shared capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Mission value, safety, debris and resource efficiency improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Simulation + mission pilot. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Time-window and recurrence

Orbital and space economy | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 096: Orbital and space economy; original author status: Hard engineering frontier. Native state: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Mission-local success metric. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Mission gain shifts debris/collision/spectrum burden. The relevant receiving carriers are: Other spacecraft, operators, orbital environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Orbital consequential ledger + shared capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Mission value, safety, debris and resource efficiency improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Simulation + mission pilot. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Ethical and legal safety

Orbital and space economy | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 096: Orbital and space economy; original author status: Hard engineering frontier. Native state: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Mission-local success metric. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Mission gain shifts debris/collision/spectrum burden. The relevant receiving carriers are: Other spacecraft, operators, orbital environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Orbital consequential ledger + shared capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Mission value, safety, debris and resource efficiency improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Simulation + mission pilot. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Sensitivity and negative controls

Orbital and space economy | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 096: Orbital and space economy; original author status: Hard engineering frontier. Native state: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Mission-local success metric. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Mission gain shifts debris/collision/spectrum burden. The relevant receiving carriers are: Other spacecraft, operators, orbital environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Orbital consequential ledger + shared capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Mission value, safety, debris and resource efficiency improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Simulation + mission pilot. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Replication and provenance

Orbital and space economy | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 096: Orbital and space economy; original author status: Hard engineering frontier. Native state: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Mission-local success metric. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Mission gain shifts debris/collision/spectrum burden. The relevant receiving carriers are: Other spacecraft, operators, orbital environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Orbital consequential ledger + shared capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Mission value, safety, debris and resource efficiency improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Simulation + mission pilot. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

096: Cross-domain invariant contest

Orbital and space economy | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 096: Orbital and space economy; original author status: Hard engineering frontier. Native state: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Mission-local success metric. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Mission gain shifts debris/collision/spectrum burden. The relevant receiving carriers are: Other spacecraft, operators, orbital environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Orbital consequential ledger + shared capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Mission value, safety, debris and resource efficiency improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Simulation + mission pilot. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 096, pp. 725-731 (Rufino 2026).

Adapter 097: Extraterrestrial settlement

Frontier, illicit and cosmic | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Life support, energy, materials, health, governance and closed ecology

RECORDED RECEIVING SCHEMA

Earth-derived subsystem silos

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Local subsystem success threatens another life-support loop. Candidate receivers: Crew, machines, habitat ecology.

EXISTING CAPACITY DESIGN

Closed-loop consequential state + autonomous capacity construction

PROTECTED OUTCOME STATED BY THE AUTHOR

Survival, health, resource closure and social stability improve

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Analog habitat + mission tests. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097 | Extraterrestrial settlement

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 097, Extraterrestrial settlement. Native state: Life support, energy, materials, health, governance and closed ecology. A human may be a researcher, operator, regulated party or downstream recipient of this Extraterrestrial settlement system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Local subsystem success threatens another life-support loop.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Closed-loop consequential state + autonomous capacity construction can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Survival, health, resource closure and social stability improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

097 | Extraterrestrial settlement

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 097, Extraterrestrial settlement. Native state: Life support, energy, materials, health, governance and closed ecology. A human may be a researcher, operator, regulated party or downstream recipient of this Extraterrestrial settlement system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Local subsystem success threatens another life-support loop.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Closed-loop consequential state + autonomous capacity construction can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Survival, health, resource closure and social stability improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

097 | Extraterrestrial settlement

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 097, Extraterrestrial settlement. Native state: Life support, energy, materials, health, governance and closed ecology. A human may be a researcher, operator, regulated party or downstream recipient of this Extraterrestrial settlement system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Local subsystem success threatens another life-support loop.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Closed-loop consequential state + autonomous capacity construction can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Survival, health, resource closure and social stability improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

097 | Extraterrestrial settlement

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 097, Extraterrestrial settlement. Native state: Life support, energy, materials, health, governance and closed ecology. A human may be a researcher, operator, regulated party or downstream recipient of this Extraterrestrial settlement system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Local subsystem success threatens another life-support loop.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Closed-loop consequential state + autonomous capacity construction can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Survival, health, resource closure and social stability improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

097 | Extraterrestrial settlement

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 097, Extraterrestrial settlement. Native state: Life support, energy, materials, health, governance and closed ecology. A human may be a researcher, operator, regulated party or downstream recipient of this Extraterrestrial settlement system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Local subsystem success threatens another life-support loop.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Closed-loop consequential state + autonomous capacity construction can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Survival, health, resource closure and social stability improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

097 | Extraterrestrial settlement

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 097, Extraterrestrial settlement. Native state: Life support, energy, materials, health, governance and closed ecology. A human may be a researcher, operator, regulated party or downstream recipient of this Extraterrestrial settlement system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Local subsystem success threatens another life-support loop.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Closed-loop consequential state + autonomous capacity construction can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Survival, health, resource closure and social stability improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

097 to 001: consequential handoff

Extraterrestrial settlement -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

097 to 002: consequential handoff

Extraterrestrial settlement -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

097 to 003: consequential handoff

Extraterrestrial settlement -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

097 to 004: consequential handoff

Extraterrestrial settlement -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

097 to 005: consequential handoff

Extraterrestrial settlement -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

097 to 006: consequential handoff

Extraterrestrial settlement -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

097 to 007: consequential handoff

Extraterrestrial settlement -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

097 to 008: consequential handoff

Extraterrestrial settlement -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

097 to 009: consequential handoff

Extraterrestrial settlement -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

097 to 010: consequential handoff

Extraterrestrial settlement -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

097 to 011: consequential handoff

Extraterrestrial settlement -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

097 to 012: consequential handoff

Extraterrestrial settlement -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

097 to 013: consequential handoff

Extraterrestrial settlement -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

097 to 014: consequential handoff

Extraterrestrial settlement -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

097 to 015: consequential handoff

Extraterrestrial settlement -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

097 to 016: consequential handoff

Extraterrestrial settlement -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

097 to 017: consequential handoff

Extraterrestrial settlement -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

097 to 018: consequential handoff

Extraterrestrial settlement -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

097 to 019: consequential handoff

Extraterrestrial settlement -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

097 to 020: consequential handoff

Extraterrestrial settlement -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

097 to 021: consequential handoff

Extraterrestrial settlement -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

097 to 022: consequential handoff

Extraterrestrial settlement -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

097 to 023: consequential handoff

Extraterrestrial settlement -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

097 to 024: consequential handoff

Extraterrestrial settlement -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

097 to 025: consequential handoff

Extraterrestrial settlement -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

097 to 026: consequential handoff

Extraterrestrial settlement -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

097 to 027: consequential handoff

Extraterrestrial settlement -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

097 to 028: consequential handoff

Extraterrestrial settlement -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

097 to 029: consequential handoff

Extraterrestrial settlement -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

097 to 030: consequential handoff

Extraterrestrial settlement -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

097 to 031: consequential handoff

Extraterrestrial settlement -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

097 to 032: consequential handoff

Extraterrestrial settlement -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

097 to 033: consequential handoff

Extraterrestrial settlement -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

097 to 034: consequential handoff

Extraterrestrial settlement -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

097 to 035: consequential handoff

Extraterrestrial settlement -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

097 to 036: consequential handoff

Extraterrestrial settlement -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

097 to 037: consequential handoff

Extraterrestrial settlement -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

097 to 038: consequential handoff

Extraterrestrial settlement -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

097 to 039: consequential handoff

Extraterrestrial settlement -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

097 to 040: consequential handoff

Extraterrestrial settlement -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

097 to 041: consequential handoff

Extraterrestrial settlement -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

097 to 042: consequential handoff

Extraterrestrial settlement -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

097 to 043: consequential handoff

Extraterrestrial settlement -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

097 to 044: consequential handoff

Extraterrestrial settlement -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

097 to 045: consequential handoff

Extraterrestrial settlement -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

097 to 046: consequential handoff

Extraterrestrial settlement -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

097 to 047: consequential handoff

Extraterrestrial settlement -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

097 to 048: consequential handoff

Extraterrestrial settlement -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

097 to 049: consequential handoff

Extraterrestrial settlement -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

097 to 050: consequential handoff

Extraterrestrial settlement -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

097 to 051: consequential handoff

Extraterrestrial settlement -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

097 to 052: consequential handoff

Extraterrestrial settlement -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

097 to 053: consequential handoff

Extraterrestrial settlement -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

097 to 054: consequential handoff

Extraterrestrial settlement -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

097 to 055: consequential handoff

Extraterrestrial settlement -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

097 to 056: consequential handoff

Extraterrestrial settlement -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

097 to 057: consequential handoff

Extraterrestrial settlement -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

097 to 058: consequential handoff

Extraterrestrial settlement -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

097 to 059: consequential handoff

Extraterrestrial settlement -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

097 to 060: consequential handoff

Extraterrestrial settlement -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

097 to 061: consequential handoff

Extraterrestrial settlement -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

097 to 062: consequential handoff

Extraterrestrial settlement -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

097 to 063: consequential handoff

Extraterrestrial settlement -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

097 to 064: consequential handoff

Extraterrestrial settlement -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

097 to 065: consequential handoff

Extraterrestrial settlement -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

097 to 066: consequential handoff

Extraterrestrial settlement -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

097 to 067: consequential handoff

Extraterrestrial settlement -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

097 to 068: consequential handoff

Extraterrestrial settlement -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

097 to 069: consequential handoff

Extraterrestrial settlement -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

097 to 070: consequential handoff

Extraterrestrial settlement -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

097 to 071: consequential handoff

Extraterrestrial settlement -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

097 to 072: consequential handoff

Extraterrestrial settlement -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

097 to 073: consequential handoff

Extraterrestrial settlement -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

097 to 074: consequential handoff

Extraterrestrial settlement -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

097 to 075: consequential handoff

Extraterrestrial settlement -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

097 to 076: consequential handoff

Extraterrestrial settlement -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

097 to 077: consequential handoff

Extraterrestrial settlement -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

097 to 078: consequential handoff

Extraterrestrial settlement -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

097 to 079: consequential handoff

Extraterrestrial settlement -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

097 to 080: consequential handoff

Extraterrestrial settlement -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

097 to 081: consequential handoff

Extraterrestrial settlement -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

097 to 082: consequential handoff

Extraterrestrial settlement -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

097 to 083: consequential handoff

Extraterrestrial settlement -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

097 to 084: consequential handoff

Extraterrestrial settlement -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

097 to 085: consequential handoff

Extraterrestrial settlement -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

097 to 086: consequential handoff

Extraterrestrial settlement -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

097 to 087: consequential handoff

Extraterrestrial settlement -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

097 to 088: consequential handoff

Extraterrestrial settlement -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

097 to 089: consequential handoff

Extraterrestrial settlement -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

097 to 090: consequential handoff

Extraterrestrial settlement -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

097 to 091: consequential handoff

Extraterrestrial settlement -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

097 to 092: consequential handoff

Extraterrestrial settlement -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

097 to 093: consequential handoff

Extraterrestrial settlement -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

097 to 094: consequential handoff

Extraterrestrial settlement -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

097 to 095: consequential handoff

Extraterrestrial settlement -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

097 to 096: consequential handoff

Extraterrestrial settlement -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

097 to 097: consequential handoff

Extraterrestrial settlement -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Extraterrestrial settlement becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 732-738; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

097 to 098: consequential handoff

Extraterrestrial settlement -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

097 to 099: consequential handoff

Extraterrestrial settlement -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

097 to 100: consequential handoff

Extraterrestrial settlement -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

097 to 101: consequential handoff

Extraterrestrial settlement -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

097 to 102: consequential handoff

Extraterrestrial settlement -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

097 to 103: consequential handoff

Extraterrestrial settlement -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

097 to 104: consequential handoff

Extraterrestrial settlement -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

097 to 105: consequential handoff

Extraterrestrial settlement -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

097 to 106: consequential handoff

Extraterrestrial settlement -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

097 to 107: consequential handoff

Extraterrestrial settlement -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

097 to 108: consequential handoff

Extraterrestrial settlement -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

097 to 109: consequential handoff

Extraterrestrial settlement -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

097 to 110: consequential handoff

Extraterrestrial settlement -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

097 to 111: consequential handoff

Extraterrestrial settlement -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

097 to 112: consequential handoff

Extraterrestrial settlement -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

097 to 113: consequential handoff

Extraterrestrial settlement -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

097 to 114: consequential handoff

Extraterrestrial settlement -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

097 to 115: consequential handoff

Extraterrestrial settlement -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

097 to 116: consequential handoff

Extraterrestrial settlement -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

097 to 117: consequential handoff

Extraterrestrial settlement -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

097 to 118: consequential handoff

Extraterrestrial settlement -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

097 to 119: consequential handoff

Extraterrestrial settlement -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

097 to 120: consequential handoff

Extraterrestrial settlement -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Extraterrestrial settlement into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-derived subsystem silos. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Crew, machines, habitat ecology. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop consequential state + autonomous capacity construction. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Analog habitat + mission tests' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 732-738 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

097: Extraterrestrial settlement

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Extraterrestrial settlement

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Extraterrestrial settlement is described in the Atlas as: Life support, energy, materials, health, governance and closed ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local subsystem success threatens another life-support loop. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Crew, machines, habitat ecology. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop consequential state + autonomous capacity construction. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Survival, health, resource closure and social stability improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Analog habitat + mission tests or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Evidence and source reconstruction

Extraterrestrial settlement | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 097: Extraterrestrial settlement; original author status: Hard engineering frontier. Native state: Life support, energy, materials, health, governance and closed ecology.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-derived subsystem silos. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local subsystem success threatens another life-support loop. The relevant receiving carriers are: Crew, machines, habitat ecology.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop consequential state + autonomous capacity construction. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Survival, health, resource closure and social stability improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Analog habitat + mission tests. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Boundary divergence

Extraterrestrial settlement | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 097: Extraterrestrial settlement; original author status: Hard engineering frontier. Native state: Life support, energy, materials, health, governance and closed ecology.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-derived subsystem silos. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local subsystem success threatens another life-support loop. The relevant receiving carriers are: Crew, machines, habitat ecology.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop consequential state + autonomous capacity construction. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Survival, health, resource closure and social stability improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Analog habitat + mission tests. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Causal diagram and identifiability

Extraterrestrial settlement | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 097: Extraterrestrial settlement; original author status: Hard engineering frontier. Native state: Life support, energy, materials, health, governance and closed ecology.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-derived subsystem silos. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local subsystem success threatens another life-support loop. The relevant receiving carriers are: Crew, machines, habitat ecology.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop consequential state + autonomous capacity construction. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Survival, health, resource closure and social stability improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Analog habitat + mission tests. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Receiving carrier identification

Extraterrestrial settlement | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 097: Extraterrestrial settlement; original author status: Hard engineering frontier. Native state: Life support, energy, materials, health, governance and closed ecology.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-derived subsystem silos. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local subsystem success threatens another life-support loop. The relevant receiving carriers are: Crew, machines, habitat ecology.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop consequential state + autonomous capacity construction. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Survival, health, resource closure and social stability improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Analog habitat + mission tests. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Hidden compensation accounting

Extraterrestrial settlement | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 097: Extraterrestrial settlement; original author status: Hard engineering frontier. Native state: Life support, energy, materials, health, governance and closed ecology.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-derived subsystem silos. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local subsystem success threatens another life-support loop. The relevant receiving carriers are: Crew, machines, habitat ecology.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop consequential state + autonomous capacity construction. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Survival, health, resource closure and social stability improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Analog habitat + mission tests. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Native units and dimensional integrity

Extraterrestrial settlement | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 097: Extraterrestrial settlement; original author status: Hard engineering frontier. Native state: Life support, energy, materials, health, governance and closed ecology.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-derived subsystem silos. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local subsystem success threatens another life-support loop. The relevant receiving carriers are: Crew, machines, habitat ecology.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop consequential state + autonomous capacity construction. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Survival, health, resource closure and social stability improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Analog habitat + mission tests. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Capacity specification

Extraterrestrial settlement | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 097: Extraterrestrial settlement; original author status: Hard engineering frontier. Native state: Life support, energy, materials, health, governance and closed ecology.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-derived subsystem silos. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local subsystem success threatens another life-support loop. The relevant receiving carriers are: Crew, machines, habitat ecology.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop consequential state + autonomous capacity construction. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Survival, health, resource closure and social stability improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Analog habitat + mission tests. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Intervention design

Extraterrestrial settlement | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 097: Extraterrestrial settlement; original author status: Hard engineering frontier. Native state: Life support, energy, materials, health, governance and closed ecology.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-derived subsystem silos. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local subsystem success threatens another life-support loop. The relevant receiving carriers are: Crew, machines, habitat ecology.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop consequential state + autonomous capacity construction. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Survival, health, resource closure and social stability improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Analog habitat + mission tests. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Incumbent parity contest

Extraterrestrial settlement | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 097: Extraterrestrial settlement; original author status: Hard engineering frontier. Native state: Life support, energy, materials, health, governance and closed ecology.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-derived subsystem silos. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local subsystem success threatens another life-support loop. The relevant receiving carriers are: Crew, machines, habitat ecology.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop consequential state + autonomous capacity construction. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Survival, health, resource closure and social stability improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Analog habitat + mission tests. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Time-window and recurrence

Extraterrestrial settlement | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 097: Extraterrestrial settlement; original author status: Hard engineering frontier. Native state: Life support, energy, materials, health, governance and closed ecology.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-derived subsystem silos. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local subsystem success threatens another life-support loop. The relevant receiving carriers are: Crew, machines, habitat ecology.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop consequential state + autonomous capacity construction. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Survival, health, resource closure and social stability improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Analog habitat + mission tests. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Ethical and legal safety

Extraterrestrial settlement | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 097: Extraterrestrial settlement; original author status: Hard engineering frontier. Native state: Life support, energy, materials, health, governance and closed ecology.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-derived subsystem silos. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local subsystem success threatens another life-support loop. The relevant receiving carriers are: Crew, machines, habitat ecology.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop consequential state + autonomous capacity construction. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Survival, health, resource closure and social stability improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Analog habitat + mission tests. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Sensitivity and negative controls

Extraterrestrial settlement | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 097: Extraterrestrial settlement; original author status: Hard engineering frontier. Native state: Life support, energy, materials, health, governance and closed ecology.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-derived subsystem silos. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local subsystem success threatens another life-support loop. The relevant receiving carriers are: Crew, machines, habitat ecology.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop consequential state + autonomous capacity construction. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Survival, health, resource closure and social stability improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Analog habitat + mission tests. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Replication and provenance

Extraterrestrial settlement | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 097: Extraterrestrial settlement; original author status: Hard engineering frontier. Native state: Life support, energy, materials, health, governance and closed ecology.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-derived subsystem silos. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local subsystem success threatens another life-support loop. The relevant receiving carriers are: Crew, machines, habitat ecology.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop consequential state + autonomous capacity construction. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Survival, health, resource closure and social stability improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Analog habitat + mission tests. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

097: Cross-domain invariant contest

Extraterrestrial settlement | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 097: Extraterrestrial settlement; original author status: Hard engineering frontier. Native state: Life support, energy, materials, health, governance and closed ecology.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-derived subsystem silos. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local subsystem success threatens another life-support loop. The relevant receiving carriers are: Crew, machines, habitat ecology.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop consequential state + autonomous capacity construction. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Survival, health, resource closure and social stability improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Analog habitat + mission tests. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 097, pp. 732-738 (Rufino 2026).

Adapter 098: Astrobiology and biosignatures

Frontier, illicit and cosmic | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Unknown life chemistries, environments, signals and contamination risks

RECORDED RECEIVING SCHEMA

Earth-centric life schema

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Narrow schema misses unfamiliar life or creates false positives. Candidate receivers: Scientific program, environment, future missions.

EXISTING CAPACITY DESIGN

Plural biosignature schema + provenance/contamination control

PROTECTED OUTCOME STATED BY THE AUTHOR

Detection sensitivity and specificity improve without contamination

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Blind benchmark + mission data. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098 | Astrobiology and biosignatures

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 098, Astrobiology and biosignatures. Native state: Unknown life chemistries, environments, signals and contamination risks. A human may be a researcher, operator, regulated party or downstream recipient of this Astrobiology and biosignatures system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Narrow schema misses unfamiliar life or creates false positives.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Plural biosignature schema + provenance/contamination control can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Detection sensitivity and specificity improve without contamination. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

098 | Astrobiology and biosignatures

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 098, Astrobiology and biosignatures. Native state: Unknown life chemistries, environments, signals and contamination risks. A human may be a researcher, operator, regulated party or downstream recipient of this Astrobiology and biosignatures system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Narrow schema misses unfamiliar life or creates false positives.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Plural biosignature schema + provenance/contamination control can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Detection sensitivity and specificity improve without contamination. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

098 | Astrobiology and biosignatures

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 098, Astrobiology and biosignatures. Native state: Unknown life chemistries, environments, signals and contamination risks. A human may be a researcher, operator, regulated party or downstream recipient of this Astrobiology and biosignatures system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Narrow schema misses unfamiliar life or creates false positives.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Plural biosignature schema + provenance/contamination control can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Detection sensitivity and specificity improve without contamination. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

098 | Astrobiology and biosignatures

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 098, Astrobiology and biosignatures. Native state: Unknown life chemistries, environments, signals and contamination risks. A human may be a researcher, operator, regulated party or downstream recipient of this Astrobiology and biosignatures system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Narrow schema misses unfamiliar life or creates false positives.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Plural biosignature schema + provenance/contamination control can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Detection sensitivity and specificity improve without contamination. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

098 | Astrobiology and biosignatures

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 098, Astrobiology and biosignatures. Native state: Unknown life chemistries, environments, signals and contamination risks. A human may be a researcher, operator, regulated party or downstream recipient of this Astrobiology and biosignatures system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Narrow schema misses unfamiliar life or creates false positives.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Plural biosignature schema + provenance/contamination control can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Detection sensitivity and specificity improve without contamination. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

098 | Astrobiology and biosignatures

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 098, Astrobiology and biosignatures. Native state: Unknown life chemistries, environments, signals and contamination risks. A human may be a researcher, operator, regulated party or downstream recipient of this Astrobiology and biosignatures system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Narrow schema misses unfamiliar life or creates false positives.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Plural biosignature schema + provenance/contamination control can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Detection sensitivity and specificity improve without contamination. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

098 to 001: consequential handoff

Astrobiology and biosignatures -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

098 to 002: consequential handoff

Astrobiology and biosignatures -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

098 to 003: consequential handoff

Astrobiology and biosignatures -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

098 to 004: consequential handoff

Astrobiology and biosignatures -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

098 to 005: consequential handoff

Astrobiology and biosignatures -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

098 to 006: consequential handoff

Astrobiology and biosignatures -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

098 to 007: consequential handoff

Astrobiology and biosignatures -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

098 to 008: consequential handoff

Astrobiology and biosignatures -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

098 to 009: consequential handoff

Astrobiology and biosignatures -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

098 to 010: consequential handoff

Astrobiology and biosignatures -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

098 to 011: consequential handoff

Astrobiology and biosignatures -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

098 to 012: consequential handoff

Astrobiology and biosignatures -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

098 to 013: consequential handoff

Astrobiology and biosignatures -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

098 to 014: consequential handoff

Astrobiology and biosignatures -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

098 to 015: consequential handoff

Astrobiology and biosignatures -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

098 to 016: consequential handoff

Astrobiology and biosignatures -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

098 to 017: consequential handoff

Astrobiology and biosignatures -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

098 to 018: consequential handoff

Astrobiology and biosignatures -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

098 to 019: consequential handoff

Astrobiology and biosignatures -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

098 to 020: consequential handoff

Astrobiology and biosignatures -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

098 to 021: consequential handoff

Astrobiology and biosignatures -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

098 to 022: consequential handoff

Astrobiology and biosignatures -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

098 to 023: consequential handoff

Astrobiology and biosignatures -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

098 to 024: consequential handoff

Astrobiology and biosignatures -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

098 to 025: consequential handoff

Astrobiology and biosignatures -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

098 to 026: consequential handoff

Astrobiology and biosignatures -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

098 to 027: consequential handoff

Astrobiology and biosignatures -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

098 to 028: consequential handoff

Astrobiology and biosignatures -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

098 to 029: consequential handoff

Astrobiology and biosignatures -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

098 to 030: consequential handoff

Astrobiology and biosignatures -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

098 to 031: consequential handoff

Astrobiology and biosignatures -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

098 to 032: consequential handoff

Astrobiology and biosignatures -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

098 to 033: consequential handoff

Astrobiology and biosignatures -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

098 to 034: consequential handoff

Astrobiology and biosignatures -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

098 to 035: consequential handoff

Astrobiology and biosignatures -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

098 to 036: consequential handoff

Astrobiology and biosignatures -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

098 to 037: consequential handoff

Astrobiology and biosignatures -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

098 to 038: consequential handoff

Astrobiology and biosignatures -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

098 to 039: consequential handoff

Astrobiology and biosignatures -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

098 to 040: consequential handoff

Astrobiology and biosignatures -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

098 to 041: consequential handoff

Astrobiology and biosignatures -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

098 to 042: consequential handoff

Astrobiology and biosignatures -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

098 to 043: consequential handoff

Astrobiology and biosignatures -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

098 to 044: consequential handoff

Astrobiology and biosignatures -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

098 to 045: consequential handoff

Astrobiology and biosignatures -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

098 to 046: consequential handoff

Astrobiology and biosignatures -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

098 to 047: consequential handoff

Astrobiology and biosignatures -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

098 to 048: consequential handoff

Astrobiology and biosignatures -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

098 to 049: consequential handoff

Astrobiology and biosignatures -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

098 to 050: consequential handoff

Astrobiology and biosignatures -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

098 to 051: consequential handoff

Astrobiology and biosignatures -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

098 to 052: consequential handoff

Astrobiology and biosignatures -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

098 to 053: consequential handoff

Astrobiology and biosignatures -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

098 to 054: consequential handoff

Astrobiology and biosignatures -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

098 to 055: consequential handoff

Astrobiology and biosignatures -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

098 to 056: consequential handoff

Astrobiology and biosignatures -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

098 to 057: consequential handoff

Astrobiology and biosignatures -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

098 to 058: consequential handoff

Astrobiology and biosignatures -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

098 to 059: consequential handoff

Astrobiology and biosignatures -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

098 to 060: consequential handoff

Astrobiology and biosignatures -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

098 to 061: consequential handoff

Astrobiology and biosignatures -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

098 to 062: consequential handoff

Astrobiology and biosignatures -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

098 to 063: consequential handoff

Astrobiology and biosignatures -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

098 to 064: consequential handoff

Astrobiology and biosignatures -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

098 to 065: consequential handoff

Astrobiology and biosignatures -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

098 to 066: consequential handoff

Astrobiology and biosignatures -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

098 to 067: consequential handoff

Astrobiology and biosignatures -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

098 to 068: consequential handoff

Astrobiology and biosignatures -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

098 to 069: consequential handoff

Astrobiology and biosignatures -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

098 to 070: consequential handoff

Astrobiology and biosignatures -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

098 to 071: consequential handoff

Astrobiology and biosignatures -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

098 to 072: consequential handoff

Astrobiology and biosignatures -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

098 to 073: consequential handoff

Astrobiology and biosignatures -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

098 to 074: consequential handoff

Astrobiology and biosignatures -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

098 to 075: consequential handoff

Astrobiology and biosignatures -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

098 to 076: consequential handoff

Astrobiology and biosignatures -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

098 to 077: consequential handoff

Astrobiology and biosignatures -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

098 to 078: consequential handoff

Astrobiology and biosignatures -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

098 to 079: consequential handoff

Astrobiology and biosignatures -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

098 to 080: consequential handoff

Astrobiology and biosignatures -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

098 to 081: consequential handoff

Astrobiology and biosignatures -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

098 to 082: consequential handoff

Astrobiology and biosignatures -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

098 to 083: consequential handoff

Astrobiology and biosignatures -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

098 to 084: consequential handoff

Astrobiology and biosignatures -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

098 to 085: consequential handoff

Astrobiology and biosignatures -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

098 to 086: consequential handoff

Astrobiology and biosignatures -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

098 to 087: consequential handoff

Astrobiology and biosignatures -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

098 to 088: consequential handoff

Astrobiology and biosignatures -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

098 to 089: consequential handoff

Astrobiology and biosignatures -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

098 to 090: consequential handoff

Astrobiology and biosignatures -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

098 to 091: consequential handoff

Astrobiology and biosignatures -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

098 to 092: consequential handoff

Astrobiology and biosignatures -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

098 to 093: consequential handoff

Astrobiology and biosignatures -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

098 to 094: consequential handoff

Astrobiology and biosignatures -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

098 to 095: consequential handoff

Astrobiology and biosignatures -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

098 to 096: consequential handoff

Astrobiology and biosignatures -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

098 to 097: consequential handoff

Astrobiology and biosignatures -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

098 to 098: consequential handoff

Astrobiology and biosignatures -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Astrobiology and biosignatures becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 739-745; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

098 to 099: consequential handoff

Astrobiology and biosignatures -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

098 to 100: consequential handoff

Astrobiology and biosignatures -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

098 to 101: consequential handoff

Astrobiology and biosignatures -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

098 to 102: consequential handoff

Astrobiology and biosignatures -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

098 to 103: consequential handoff

Astrobiology and biosignatures -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

098 to 104: consequential handoff

Astrobiology and biosignatures -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

098 to 105: consequential handoff

Astrobiology and biosignatures -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

098 to 106: consequential handoff

Astrobiology and biosignatures -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

098 to 107: consequential handoff

Astrobiology and biosignatures -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

098 to 108: consequential handoff

Astrobiology and biosignatures -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

098 to 109: consequential handoff

Astrobiology and biosignatures -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

098 to 110: consequential handoff

Astrobiology and biosignatures -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

098 to 111: consequential handoff

Astrobiology and biosignatures -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

098 to 112: consequential handoff

Astrobiology and biosignatures -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

098 to 113: consequential handoff

Astrobiology and biosignatures -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

098 to 114: consequential handoff

Astrobiology and biosignatures -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

098 to 115: consequential handoff

Astrobiology and biosignatures -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

098 to 116: consequential handoff

Astrobiology and biosignatures -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

098 to 117: consequential handoff

Astrobiology and biosignatures -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

098 to 118: consequential handoff

Astrobiology and biosignatures -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

098 to 119: consequential handoff

Astrobiology and biosignatures -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

098 to 120: consequential handoff

Astrobiology and biosignatures -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Astrobiology and biosignatures into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Earth-centric life schema. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Scientific program, environment, future missions. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Plural biosignature schema + provenance/contamination control. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blind benchmark + mission data' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 739-745 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

098: Astrobiology and biosignatures

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

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Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Astrobiology and biosignatures

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Astrobiology and biosignatures is described in the Atlas as: Unknown life chemistries, environments, signals and contamination risks. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Narrow schema misses unfamiliar life or creates false positives. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Scientific program, environment, future missions. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Plural biosignature schema + provenance/contamination control. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Detection sensitivity and specificity improve without contamination. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blind benchmark + mission data or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Evidence and source reconstruction

Astrobiology and biosignatures | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 098: Astrobiology and biosignatures; original author status: Hard scientific frontier. Native state: Unknown life chemistries, environments, signals and contamination risks.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-centric life schema. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Narrow schema misses unfamiliar life or creates false positives. The relevant receiving carriers are: Scientific program, environment, future missions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Plural biosignature schema + provenance/contamination control. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Detection sensitivity and specificity improve without contamination.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blind benchmark + mission data. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Boundary divergence

Astrobiology and biosignatures | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 098: Astrobiology and biosignatures; original author status: Hard scientific frontier. Native state: Unknown life chemistries, environments, signals and contamination risks.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-centric life schema. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Narrow schema misses unfamiliar life or creates false positives. The relevant receiving carriers are: Scientific program, environment, future missions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Plural biosignature schema + provenance/contamination control. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Detection sensitivity and specificity improve without contamination.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blind benchmark + mission data. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Causal diagram and identifiability

Astrobiology and biosignatures | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 098: Astrobiology and biosignatures; original author status: Hard scientific frontier. Native state: Unknown life chemistries, environments, signals and contamination risks.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-centric life schema. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Narrow schema misses unfamiliar life or creates false positives. The relevant receiving carriers are: Scientific program, environment, future missions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Plural biosignature schema + provenance/contamination control. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Detection sensitivity and specificity improve without contamination.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blind benchmark + mission data. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Receiving carrier identification

Astrobiology and biosignatures | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 098: Astrobiology and biosignatures; original author status: Hard scientific frontier. Native state: Unknown life chemistries, environments, signals and contamination risks.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-centric life schema. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Narrow schema misses unfamiliar life or creates false positives. The relevant receiving carriers are: Scientific program, environment, future missions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Plural biosignature schema + provenance/contamination control. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Detection sensitivity and specificity improve without contamination.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blind benchmark + mission data. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Hidden compensation accounting

Astrobiology and biosignatures | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 098: Astrobiology and biosignatures; original author status: Hard scientific frontier. Native state: Unknown life chemistries, environments, signals and contamination risks.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-centric life schema. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Narrow schema misses unfamiliar life or creates false positives. The relevant receiving carriers are: Scientific program, environment, future missions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Plural biosignature schema + provenance/contamination control. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Detection sensitivity and specificity improve without contamination.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blind benchmark + mission data. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Native units and dimensional integrity

Astrobiology and biosignatures | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 098: Astrobiology and biosignatures; original author status: Hard scientific frontier. Native state: Unknown life chemistries, environments, signals and contamination risks.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-centric life schema. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Narrow schema misses unfamiliar life or creates false positives. The relevant receiving carriers are: Scientific program, environment, future missions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Plural biosignature schema + provenance/contamination control. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Detection sensitivity and specificity improve without contamination.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blind benchmark + mission data. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Capacity specification

Astrobiology and biosignatures | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 098: Astrobiology and biosignatures; original author status: Hard scientific frontier. Native state: Unknown life chemistries, environments, signals and contamination risks.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-centric life schema. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Narrow schema misses unfamiliar life or creates false positives. The relevant receiving carriers are: Scientific program, environment, future missions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Plural biosignature schema + provenance/contamination control. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Detection sensitivity and specificity improve without contamination.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blind benchmark + mission data. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Intervention design

Astrobiology and biosignatures | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 098: Astrobiology and biosignatures; original author status: Hard scientific frontier. Native state: Unknown life chemistries, environments, signals and contamination risks.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-centric life schema. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Narrow schema misses unfamiliar life or creates false positives. The relevant receiving carriers are: Scientific program, environment, future missions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Plural biosignature schema + provenance/contamination control. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Detection sensitivity and specificity improve without contamination.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blind benchmark + mission data. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Incumbent parity contest

Astrobiology and biosignatures | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 098: Astrobiology and biosignatures; original author status: Hard scientific frontier. Native state: Unknown life chemistries, environments, signals and contamination risks.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-centric life schema. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Narrow schema misses unfamiliar life or creates false positives. The relevant receiving carriers are: Scientific program, environment, future missions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Plural biosignature schema + provenance/contamination control. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Detection sensitivity and specificity improve without contamination.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blind benchmark + mission data. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Time-window and recurrence

Astrobiology and biosignatures | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 098: Astrobiology and biosignatures; original author status: Hard scientific frontier. Native state: Unknown life chemistries, environments, signals and contamination risks.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-centric life schema. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Narrow schema misses unfamiliar life or creates false positives. The relevant receiving carriers are: Scientific program, environment, future missions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Plural biosignature schema + provenance/contamination control. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Detection sensitivity and specificity improve without contamination.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blind benchmark + mission data. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Ethical and legal safety

Astrobiology and biosignatures | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 098: Astrobiology and biosignatures; original author status: Hard scientific frontier. Native state: Unknown life chemistries, environments, signals and contamination risks.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-centric life schema. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Narrow schema misses unfamiliar life or creates false positives. The relevant receiving carriers are: Scientific program, environment, future missions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Plural biosignature schema + provenance/contamination control. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Detection sensitivity and specificity improve without contamination.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blind benchmark + mission data. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Sensitivity and negative controls

Astrobiology and biosignatures | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 098: Astrobiology and biosignatures; original author status: Hard scientific frontier. Native state: Unknown life chemistries, environments, signals and contamination risks.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-centric life schema. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Narrow schema misses unfamiliar life or creates false positives. The relevant receiving carriers are: Scientific program, environment, future missions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Plural biosignature schema + provenance/contamination control. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Detection sensitivity and specificity improve without contamination.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blind benchmark + mission data. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Replication and provenance

Astrobiology and biosignatures | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 098: Astrobiology and biosignatures; original author status: Hard scientific frontier. Native state: Unknown life chemistries, environments, signals and contamination risks.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-centric life schema. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Narrow schema misses unfamiliar life or creates false positives. The relevant receiving carriers are: Scientific program, environment, future missions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Plural biosignature schema + provenance/contamination control. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Detection sensitivity and specificity improve without contamination.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blind benchmark + mission data. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

098: Cross-domain invariant contest

Astrobiology and biosignatures | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 098: Astrobiology and biosignatures; original author status: Hard scientific frontier. Native state: Unknown life chemistries, environments, signals and contamination risks.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Earth-centric life schema. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Narrow schema misses unfamiliar life or creates false positives. The relevant receiving carriers are: Scientific program, environment, future missions.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Plural biosignature schema + provenance/contamination control. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Detection sensitivity and specificity improve without contamination.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blind benchmark + mission data. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 098, pp. 739-745 (Rufino 2026).

Adapter 099: Planetary systems and stewardship

Frontier, illicit and cosmic | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Atmosphere, hydrology, geology, biosphere, resources and settlements

RECORDED RECEIVING SCHEMA

Sector-local resource model

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Extraction/settlement shifts planetary burdens. Candidate receivers: Settlements, ecosystems, future generations.

EXISTING CAPACITY DESIGN

Closed-loop planetary resource and restoration architecture

PROTECTED OUTCOME STATED BY THE AUTHOR

Human value and planetary resilience improve

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Earth analog + simulation. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099 | Planetary systems and stewardship

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 099, Planetary systems and stewardship. Native state: Atmosphere, hydrology, geology, biosphere, resources and settlements. A human may be a researcher, operator, regulated party or downstream recipient of this Planetary systems and stewardship system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Extraction/settlement shifts planetary burdens.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Closed-loop planetary resource and restoration architecture can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Human value and planetary resilience improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

099 | Planetary systems and stewardship

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 099, Planetary systems and stewardship. Native state: Atmosphere, hydrology, geology, biosphere, resources and settlements. A human may be a researcher, operator, regulated party or downstream recipient of this Planetary systems and stewardship system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Extraction/settlement shifts planetary burdens.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Closed-loop planetary resource and restoration architecture can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Human value and planetary resilience improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

099 | Planetary systems and stewardship

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 099, Planetary systems and stewardship. Native state: Atmosphere, hydrology, geology, biosphere, resources and settlements. A human may be a researcher, operator, regulated party or downstream recipient of this Planetary systems and stewardship system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Extraction/settlement shifts planetary burdens.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Closed-loop planetary resource and restoration architecture can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Human value and planetary resilience improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

099 | Planetary systems and stewardship

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 099, Planetary systems and stewardship. Native state: Atmosphere, hydrology, geology, biosphere, resources and settlements. A human may be a researcher, operator, regulated party or downstream recipient of this Planetary systems and stewardship system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Extraction/settlement shifts planetary burdens.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Closed-loop planetary resource and restoration architecture can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Human value and planetary resilience improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

099 | Planetary systems and stewardship

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 099, Planetary systems and stewardship. Native state: Atmosphere, hydrology, geology, biosphere, resources and settlements. A human may be a researcher, operator, regulated party or downstream recipient of this Planetary systems and stewardship system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Extraction/settlement shifts planetary burdens.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Closed-loop planetary resource and restoration architecture can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Human value and planetary resilience improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

099 | Planetary systems and stewardship

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 099, Planetary systems and stewardship. Native state: Atmosphere, hydrology, geology, biosphere, resources and settlements. A human may be a researcher, operator, regulated party or downstream recipient of this Planetary systems and stewardship system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Extraction/settlement shifts planetary burdens.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Closed-loop planetary resource and restoration architecture can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Human value and planetary resilience improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

099 to 001: consequential handoff

Planetary systems and stewardship -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

099 to 002: consequential handoff

Planetary systems and stewardship -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

099 to 003: consequential handoff

Planetary systems and stewardship -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

099 to 004: consequential handoff

Planetary systems and stewardship -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Clinical trials + patient-reported and partner-impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

099 to 005: consequential handoff

Planetary systems and stewardship -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

099 to 006: consequential handoff

Planetary systems and stewardship -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

099 to 007: consequential handoff

Planetary systems and stewardship -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

099 to 008: consequential handoff

Planetary systems and stewardship -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

099 to 009: consequential handoff

Planetary systems and stewardship -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

099 to 010: consequential handoff

Planetary systems and stewardship -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

099 to 011: consequential handoff

Planetary systems and stewardship -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

099 to 012: consequential handoff

Planetary systems and stewardship -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

099 to 013: consequential handoff

Planetary systems and stewardship -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

099 to 014: consequential handoff

Planetary systems and stewardship -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

099 to 015: consequential handoff

Planetary systems and stewardship -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

099 to 016: consequential handoff

Planetary systems and stewardship -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

099 to 017: consequential handoff

Planetary systems and stewardship -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

099 to 018: consequential handoff

Planetary systems and stewardship -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

099 to 019: consequential handoff

Planetary systems and stewardship -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

099 to 020: consequential handoff

Planetary systems and stewardship -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

099 to 021: consequential handoff

Planetary systems and stewardship -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

099 to 022: consequential handoff

Planetary systems and stewardship -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

099 to 023: consequential handoff

Planetary systems and stewardship -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

099 to 024: consequential handoff

Planetary systems and stewardship -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

099 to 025: consequential handoff

Planetary systems and stewardship -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

099 to 026: consequential handoff

Planetary systems and stewardship -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

099 to 027: consequential handoff

Planetary systems and stewardship -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

099 to 028: consequential handoff

Planetary systems and stewardship -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

099 to 029: consequential handoff

Planetary systems and stewardship -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

099 to 030: consequential handoff

Planetary systems and stewardship -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

099 to 031: consequential handoff

Planetary systems and stewardship -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

099 to 032: consequential handoff

Planetary systems and stewardship -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

099 to 033: consequential handoff

Planetary systems and stewardship -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

099 to 034: consequential handoff

Planetary systems and stewardship -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

099 to 035: consequential handoff

Planetary systems and stewardship -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

099 to 036: consequential handoff

Planetary systems and stewardship -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

099 to 037: consequential handoff

Planetary systems and stewardship -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

099 to 038: consequential handoff

Planetary systems and stewardship -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

099 to 039: consequential handoff

Planetary systems and stewardship -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

099 to 040: consequential handoff

Planetary systems and stewardship -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

099 to 041: consequential handoff

Planetary systems and stewardship -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

099 to 042: consequential handoff

Planetary systems and stewardship -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

099 to 043: consequential handoff

Planetary systems and stewardship -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

099 to 044: consequential handoff

Planetary systems and stewardship -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

099 to 045: consequential handoff

Planetary systems and stewardship -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

099 to 046: consequential handoff

Planetary systems and stewardship -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

099 to 047: consequential handoff

Planetary systems and stewardship -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

099 to 048: consequential handoff

Planetary systems and stewardship -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

099 to 049: consequential handoff

Planetary systems and stewardship -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

099 to 050: consequential handoff

Planetary systems and stewardship -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

099 to 051: consequential handoff

Planetary systems and stewardship -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

099 to 052: consequential handoff

Planetary systems and stewardship -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

099 to 053: consequential handoff

Planetary systems and stewardship -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

099 to 054: consequential handoff

Planetary systems and stewardship -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

099 to 055: consequential handoff

Planetary systems and stewardship -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

099 to 056: consequential handoff

Planetary systems and stewardship -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

099 to 057: consequential handoff

Planetary systems and stewardship -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

099 to 058: consequential handoff

Planetary systems and stewardship -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

099 to 059: consequential handoff

Planetary systems and stewardship -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

099 to 060: consequential handoff

Planetary systems and stewardship -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

099 to 061: consequential handoff

Planetary systems and stewardship -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

099 to 062: consequential handoff

Planetary systems and stewardship -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

099 to 063: consequential handoff

Planetary systems and stewardship -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

099 to 064: consequential handoff

Planetary systems and stewardship -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

099 to 065: consequential handoff

Planetary systems and stewardship -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

099 to 066: consequential handoff

Planetary systems and stewardship -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

099 to 067: consequential handoff

Planetary systems and stewardship -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

099 to 068: consequential handoff

Planetary systems and stewardship -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

099 to 069: consequential handoff

Planetary systems and stewardship -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

099 to 070: consequential handoff

Planetary systems and stewardship -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

099 to 071: consequential handoff

Planetary systems and stewardship -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

099 to 072: consequential handoff

Planetary systems and stewardship -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

099 to 073: consequential handoff

Planetary systems and stewardship -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

099 to 074: consequential handoff

Planetary systems and stewardship -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

099 to 075: consequential handoff

Planetary systems and stewardship -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

099 to 076: consequential handoff

Planetary systems and stewardship -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

099 to 077: consequential handoff

Planetary systems and stewardship -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

099 to 078: consequential handoff

Planetary systems and stewardship -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

099 to 079: consequential handoff

Planetary systems and stewardship -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

099 to 080: consequential handoff

Planetary systems and stewardship -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

099 to 081: consequential handoff

Planetary systems and stewardship -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

099 to 082: consequential handoff

Planetary systems and stewardship -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

099 to 083: consequential handoff

Planetary systems and stewardship -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

099 to 084: consequential handoff

Planetary systems and stewardship -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

099 to 085: consequential handoff

Planetary systems and stewardship -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

099 to 086: consequential handoff

Planetary systems and stewardship -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

099 to 087: consequential handoff

Planetary systems and stewardship -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

099 to 088: consequential handoff

Planetary systems and stewardship -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

099 to 089: consequential handoff

Planetary systems and stewardship -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

099 to 090: consequential handoff

Planetary systems and stewardship -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

099 to 091: consequential handoff

Planetary systems and stewardship -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

099 to 092: consequential handoff

Planetary systems and stewardship -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

099 to 093: consequential handoff

Planetary systems and stewardship -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

099 to 094: consequential handoff

Planetary systems and stewardship -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

099 to 095: consequential handoff

Planetary systems and stewardship -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

099 to 096: consequential handoff

Planetary systems and stewardship -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

099 to 097: consequential handoff

Planetary systems and stewardship -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

099 to 098: consequential handoff

Planetary systems and stewardship -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

099 to 099: consequential handoff

Planetary systems and stewardship -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Planetary systems and stewardship becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 746-752; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

099 to 100: consequential handoff

Planetary systems and stewardship -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

099 to 101: consequential handoff

Planetary systems and stewardship -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

099 to 102: consequential handoff

Planetary systems and stewardship -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

099 to 103: consequential handoff

Planetary systems and stewardship -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

099 to 104: consequential handoff

Planetary systems and stewardship -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

099 to 105: consequential handoff

Planetary systems and stewardship -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

099 to 106: consequential handoff

Planetary systems and stewardship -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

099 to 107: consequential handoff

Planetary systems and stewardship -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

099 to 108: consequential handoff

Planetary systems and stewardship -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

099 to 109: consequential handoff

Planetary systems and stewardship -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

099 to 110: consequential handoff

Planetary systems and stewardship -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

099 to 111: consequential handoff

Planetary systems and stewardship -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

099 to 112: consequential handoff

Planetary systems and stewardship -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

099 to 113: consequential handoff

Planetary systems and stewardship -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

099 to 114: consequential handoff

Planetary systems and stewardship -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

099 to 115: consequential handoff

Planetary systems and stewardship -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

099 to 116: consequential handoff

Planetary systems and stewardship -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

099 to 117: consequential handoff

Planetary systems and stewardship -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

099 to 118: consequential handoff

Planetary systems and stewardship -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

099 to 119: consequential handoff

Planetary systems and stewardship -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

099 to 120: consequential handoff

Planetary systems and stewardship -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Planetary systems and stewardship into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Sector-local resource model. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Settlements, ecosystems, future generations. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Closed-loop planetary resource and restoration architecture. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Earth analog + simulation' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 746-752 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

099: Planetary systems and stewardship

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Planetary systems and stewardship

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Planetary systems and stewardship is described in the Atlas as: Atmosphere, hydrology, geology, biosphere, resources and settlements. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Extraction/settlement shifts planetary burdens. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Settlements, ecosystems, future generations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Closed-loop planetary resource and restoration architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Human value and planetary resilience improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Earth analog + simulation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Evidence and source reconstruction

Planetary systems and stewardship | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 099: Planetary systems and stewardship; original author status: Hard engineering frontier. Native state: Atmosphere, hydrology, geology, biosphere, resources and settlements.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Sector-local resource model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Extraction/settlement shifts planetary burdens. The relevant receiving carriers are: Settlements, ecosystems, future generations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop planetary resource and restoration architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Human value and planetary resilience improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Earth analog + simulation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Boundary divergence

Planetary systems and stewardship | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 099: Planetary systems and stewardship; original author status: Hard engineering frontier. Native state: Atmosphere, hydrology, geology, biosphere, resources and settlements.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Sector-local resource model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Extraction/settlement shifts planetary burdens. The relevant receiving carriers are: Settlements, ecosystems, future generations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop planetary resource and restoration architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Human value and planetary resilience improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Earth analog + simulation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Causal diagram and identifiability

Planetary systems and stewardship | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 099: Planetary systems and stewardship; original author status: Hard engineering frontier. Native state: Atmosphere, hydrology, geology, biosphere, resources and settlements.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Sector-local resource model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Extraction/settlement shifts planetary burdens. The relevant receiving carriers are: Settlements, ecosystems, future generations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop planetary resource and restoration architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Human value and planetary resilience improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Earth analog + simulation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Receiving carrier identification

Planetary systems and stewardship | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 099: Planetary systems and stewardship; original author status: Hard engineering frontier. Native state: Atmosphere, hydrology, geology, biosphere, resources and settlements.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Sector-local resource model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Extraction/settlement shifts planetary burdens. The relevant receiving carriers are: Settlements, ecosystems, future generations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop planetary resource and restoration architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Human value and planetary resilience improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Earth analog + simulation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Hidden compensation accounting

Planetary systems and stewardship | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 099: Planetary systems and stewardship; original author status: Hard engineering frontier. Native state: Atmosphere, hydrology, geology, biosphere, resources and settlements.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Sector-local resource model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Extraction/settlement shifts planetary burdens. The relevant receiving carriers are: Settlements, ecosystems, future generations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop planetary resource and restoration architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Human value and planetary resilience improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Earth analog + simulation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Native units and dimensional integrity

Planetary systems and stewardship | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 099: Planetary systems and stewardship; original author status: Hard engineering frontier. Native state: Atmosphere, hydrology, geology, biosphere, resources and settlements.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Sector-local resource model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Extraction/settlement shifts planetary burdens. The relevant receiving carriers are: Settlements, ecosystems, future generations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop planetary resource and restoration architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Human value and planetary resilience improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Earth analog + simulation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Capacity specification

Planetary systems and stewardship | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 099: Planetary systems and stewardship; original author status: Hard engineering frontier. Native state: Atmosphere, hydrology, geology, biosphere, resources and settlements.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Sector-local resource model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Extraction/settlement shifts planetary burdens. The relevant receiving carriers are: Settlements, ecosystems, future generations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop planetary resource and restoration architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Human value and planetary resilience improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Earth analog + simulation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Intervention design

Planetary systems and stewardship | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 099: Planetary systems and stewardship; original author status: Hard engineering frontier. Native state: Atmosphere, hydrology, geology, biosphere, resources and settlements.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Sector-local resource model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Extraction/settlement shifts planetary burdens. The relevant receiving carriers are: Settlements, ecosystems, future generations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop planetary resource and restoration architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Human value and planetary resilience improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Earth analog + simulation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Incumbent parity contest

Planetary systems and stewardship | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 099: Planetary systems and stewardship; original author status: Hard engineering frontier. Native state: Atmosphere, hydrology, geology, biosphere, resources and settlements.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Sector-local resource model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Extraction/settlement shifts planetary burdens. The relevant receiving carriers are: Settlements, ecosystems, future generations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop planetary resource and restoration architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Human value and planetary resilience improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Earth analog + simulation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Time-window and recurrence

Planetary systems and stewardship | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 099: Planetary systems and stewardship; original author status: Hard engineering frontier. Native state: Atmosphere, hydrology, geology, biosphere, resources and settlements.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Sector-local resource model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Extraction/settlement shifts planetary burdens. The relevant receiving carriers are: Settlements, ecosystems, future generations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop planetary resource and restoration architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Human value and planetary resilience improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Earth analog + simulation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Ethical and legal safety

Planetary systems and stewardship | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 099: Planetary systems and stewardship; original author status: Hard engineering frontier. Native state: Atmosphere, hydrology, geology, biosphere, resources and settlements.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Sector-local resource model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Extraction/settlement shifts planetary burdens. The relevant receiving carriers are: Settlements, ecosystems, future generations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop planetary resource and restoration architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Human value and planetary resilience improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Earth analog + simulation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Sensitivity and negative controls

Planetary systems and stewardship | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 099: Planetary systems and stewardship; original author status: Hard engineering frontier. Native state: Atmosphere, hydrology, geology, biosphere, resources and settlements.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Sector-local resource model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Extraction/settlement shifts planetary burdens. The relevant receiving carriers are: Settlements, ecosystems, future generations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop planetary resource and restoration architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Human value and planetary resilience improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Earth analog + simulation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Replication and provenance

Planetary systems and stewardship | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 099: Planetary systems and stewardship; original author status: Hard engineering frontier. Native state: Atmosphere, hydrology, geology, biosphere, resources and settlements.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Sector-local resource model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Extraction/settlement shifts planetary burdens. The relevant receiving carriers are: Settlements, ecosystems, future generations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop planetary resource and restoration architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Human value and planetary resilience improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Earth analog + simulation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

099: Cross-domain invariant contest

Planetary systems and stewardship | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 099: Planetary systems and stewardship; original author status: Hard engineering frontier. Native state: Atmosphere, hydrology, geology, biosphere, resources and settlements.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Sector-local resource model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Extraction/settlement shifts planetary burdens. The relevant receiving carriers are: Settlements, ecosystems, future generations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Closed-loop planetary resource and restoration architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Human value and planetary resilience improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Earth analog + simulation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 099, pp. 746-752 (Rufino 2026).

Adapter 100: Unknown and not-yet-named realities

Frontier, illicit and cosmic | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Any newly observed substrate, scale, agent, value dimension or exchange mode

RECORDED RECEIVING SCHEMA

No adequate schema exists yet

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Unrepresented state is invisible until consequences accumulate. Candidate receivers: Unknown until carrier tracing identifies it.

EXISTING CAPACITY DESIGN

Instantiate M_d, define schema, measure gap, trace carrier, construct capacity

PROTECTED OUTCOME STATED BY THE AUTHOR

All material carriers in the new domain cross positive consequential thresholds

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Freeze kernel before learning outcomes; then holdout/replicate. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100 | Unknown and not-yet-named realities

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 100, Unknown and not-yet-named realities. Native state: Any newly observed substrate, scale, agent, value dimension or exchange mode. A human may be a researcher, operator, regulated party or downstream recipient of this Unknown and not-yet-named realities system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Unrepresented state is invisible until consequences accumulate.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Instantiate M_d, define schema, measure gap, trace carrier, construct capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: All material carriers in the new domain cross positive consequential thresholds. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

100 | Unknown and not-yet-named realities

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 100, Unknown and not-yet-named realities. Native state: Any newly observed substrate, scale, agent, value dimension or exchange mode. A human may be a researcher, operator, regulated party or downstream recipient of this Unknown and not-yet-named realities system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Unrepresented state is invisible until consequences accumulate.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Instantiate M_d, define schema, measure gap, trace carrier, construct capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: All material carriers in the new domain cross positive consequential thresholds. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

100 | Unknown and not-yet-named realities

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 100, Unknown and not-yet-named realities. Native state: Any newly observed substrate, scale, agent, value dimension or exchange mode. A human may be a researcher, operator, regulated party or downstream recipient of this Unknown and not-yet-named realities system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Unrepresented state is invisible until consequences accumulate.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Instantiate M_d, define schema, measure gap, trace carrier, construct capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: All material carriers in the new domain cross positive consequential thresholds. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

100 | Unknown and not-yet-named realities

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 100, Unknown and not-yet-named realities. Native state: Any newly observed substrate, scale, agent, value dimension or exchange mode. A human may be a researcher, operator, regulated party or downstream recipient of this Unknown and not-yet-named realities system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Unrepresented state is invisible until consequences accumulate.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Instantiate M_d, define schema, measure gap, trace carrier, construct capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: All material carriers in the new domain cross positive consequential thresholds. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

100 | Unknown and not-yet-named realities

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 100, Unknown and not-yet-named realities. Native state: Any newly observed substrate, scale, agent, value dimension or exchange mode. A human may be a researcher, operator, regulated party or downstream recipient of this Unknown and not-yet-named realities system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Unrepresented state is invisible until consequences accumulate.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Instantiate M_d, define schema, measure gap, trace carrier, construct capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: All material carriers in the new domain cross positive consequential thresholds. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

100 | Unknown and not-yet-named realities

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 100, Unknown and not-yet-named realities. Native state: Any newly observed substrate, scale, agent, value dimension or exchange mode. A human may be a researcher, operator, regulated party or downstream recipient of this Unknown and not-yet-named realities system; a direct human causal effect must not be presumed.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Unrepresented state is invisible until consequences accumulate.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Instantiate M_d, define schema, measure gap, trace carrier, construct capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: All material carriers in the new domain cross positive consequential thresholds. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

100 to 001: consequential handoff

Unknown and not-yet-named realities -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

100 to 002: consequential handoff

Unknown and not-yet-named realities -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

100 to 003: consequential handoff

Unknown and not-yet-named realities -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

100 to 004: consequential handoff

Unknown and not-yet-named realities -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

100 to 005: consequential handoff

Unknown and not-yet-named realities -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

100 to 006: consequential handoff

Unknown and not-yet-named realities -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

100 to 007: consequential handoff

Unknown and not-yet-named realities -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

100 to 008: consequential handoff

Unknown and not-yet-named realities -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

100 to 009: consequential handoff

Unknown and not-yet-named realities -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

100 to 010: consequential handoff

Unknown and not-yet-named realities -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

100 to 011: consequential handoff

Unknown and not-yet-named realities -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

100 to 012: consequential handoff

Unknown and not-yet-named realities -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

100 to 013: consequential handoff

Unknown and not-yet-named realities -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

100 to 014: consequential handoff

Unknown and not-yet-named realities -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

100 to 015: consequential handoff

Unknown and not-yet-named realities -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

100 to 016: consequential handoff

Unknown and not-yet-named realities -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

100 to 017: consequential handoff

Unknown and not-yet-named realities -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

100 to 018: consequential handoff

Unknown and not-yet-named realities -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

100 to 019: consequential handoff

Unknown and not-yet-named realities -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

100 to 020: consequential handoff

Unknown and not-yet-named realities -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

100 to 021: consequential handoff

Unknown and not-yet-named realities -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

100 to 022: consequential handoff

Unknown and not-yet-named realities -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

100 to 023: consequential handoff

Unknown and not-yet-named realities -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

100 to 024: consequential handoff

Unknown and not-yet-named realities -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

100 to 025: consequential handoff

Unknown and not-yet-named realities -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

100 to 026: consequential handoff

Unknown and not-yet-named realities -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

100 to 027: consequential handoff

Unknown and not-yet-named realities -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

100 to 028: consequential handoff

Unknown and not-yet-named realities -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

100 to 029: consequential handoff

Unknown and not-yet-named realities -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

100 to 030: consequential handoff

Unknown and not-yet-named realities -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

100 to 031: consequential handoff

Unknown and not-yet-named realities -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

100 to 032: consequential handoff

Unknown and not-yet-named realities -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

100 to 033: consequential handoff

Unknown and not-yet-named realities -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

100 to 034: consequential handoff

Unknown and not-yet-named realities -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

100 to 035: consequential handoff

Unknown and not-yet-named realities -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

100 to 036: consequential handoff

Unknown and not-yet-named realities -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

100 to 037: consequential handoff

Unknown and not-yet-named realities -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

100 to 038: consequential handoff

Unknown and not-yet-named realities -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

100 to 039: consequential handoff

Unknown and not-yet-named realities -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

100 to 040: consequential handoff

Unknown and not-yet-named realities -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

100 to 041: consequential handoff

Unknown and not-yet-named realities -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

100 to 042: consequential handoff

Unknown and not-yet-named realities -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

100 to 043: consequential handoff

Unknown and not-yet-named realities -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

100 to 044: consequential handoff

Unknown and not-yet-named realities -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

100 to 045: consequential handoff

Unknown and not-yet-named realities -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

100 to 046: consequential handoff

Unknown and not-yet-named realities -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

100 to 047: consequential handoff

Unknown and not-yet-named realities -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

100 to 048: consequential handoff

Unknown and not-yet-named realities -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

100 to 049: consequential handoff

Unknown and not-yet-named realities -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

100 to 050: consequential handoff

Unknown and not-yet-named realities -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

100 to 051: consequential handoff

Unknown and not-yet-named realities -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

100 to 052: consequential handoff

Unknown and not-yet-named realities -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

100 to 053: consequential handoff

Unknown and not-yet-named realities -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

100 to 054: consequential handoff

Unknown and not-yet-named realities -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

100 to 055: consequential handoff

Unknown and not-yet-named realities -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

100 to 056: consequential handoff

Unknown and not-yet-named realities -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

100 to 057: consequential handoff

Unknown and not-yet-named realities -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

100 to 058: consequential handoff

Unknown and not-yet-named realities -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

100 to 059: consequential handoff

Unknown and not-yet-named realities -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

100 to 060: consequential handoff

Unknown and not-yet-named realities -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

100 to 061: consequential handoff

Unknown and not-yet-named realities -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

100 to 062: consequential handoff

Unknown and not-yet-named realities -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

100 to 063: consequential handoff

Unknown and not-yet-named realities -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

100 to 064: consequential handoff

Unknown and not-yet-named realities -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

100 to 065: consequential handoff

Unknown and not-yet-named realities -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

100 to 066: consequential handoff

Unknown and not-yet-named realities -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

100 to 067: consequential handoff

Unknown and not-yet-named realities -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

100 to 068: consequential handoff

Unknown and not-yet-named realities -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

100 to 069: consequential handoff

Unknown and not-yet-named realities -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

100 to 070: consequential handoff

Unknown and not-yet-named realities -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

100 to 071: consequential handoff

Unknown and not-yet-named realities -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

100 to 072: consequential handoff

Unknown and not-yet-named realities -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

100 to 073: consequential handoff

Unknown and not-yet-named realities -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

100 to 074: consequential handoff

Unknown and not-yet-named realities -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

100 to 075: consequential handoff

Unknown and not-yet-named realities -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

100 to 076: consequential handoff

Unknown and not-yet-named realities -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

100 to 077: consequential handoff

Unknown and not-yet-named realities -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

100 to 078: consequential handoff

Unknown and not-yet-named realities -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

100 to 079: consequential handoff

Unknown and not-yet-named realities -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

100 to 080: consequential handoff

Unknown and not-yet-named realities -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

100 to 081: consequential handoff

Unknown and not-yet-named realities -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

100 to 082: consequential handoff

Unknown and not-yet-named realities -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

100 to 083: consequential handoff

Unknown and not-yet-named realities -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

100 to 084: consequential handoff

Unknown and not-yet-named realities -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

100 to 085: consequential handoff

Unknown and not-yet-named realities -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

100 to 086: consequential handoff

Unknown and not-yet-named realities -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

100 to 087: consequential handoff

Unknown and not-yet-named realities -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

100 to 088: consequential handoff

Unknown and not-yet-named realities -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

100 to 089: consequential handoff

Unknown and not-yet-named realities -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

100 to 090: consequential handoff

Unknown and not-yet-named realities -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

100 to 091: consequential handoff

Unknown and not-yet-named realities -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

100 to 092: consequential handoff

Unknown and not-yet-named realities -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

100 to 093: consequential handoff

Unknown and not-yet-named realities -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

100 to 094: consequential handoff

Unknown and not-yet-named realities -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

100 to 095: consequential handoff

Unknown and not-yet-named realities -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

100 to 096: consequential handoff

Unknown and not-yet-named realities -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

100 to 097: consequential handoff

Unknown and not-yet-named realities -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

100 to 098: consequential handoff

Unknown and not-yet-named realities -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

100 to 099: consequential handoff

Unknown and not-yet-named realities -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

100 to 100: consequential handoff

Unknown and not-yet-named realities -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Unknown and not-yet-named realities becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 753-759; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

100 to 101: consequential handoff

Unknown and not-yet-named realities -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

100 to 102: consequential handoff

Unknown and not-yet-named realities -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

100 to 103: consequential handoff

Unknown and not-yet-named realities -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

100 to 104: consequential handoff

Unknown and not-yet-named realities -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

100 to 105: consequential handoff

Unknown and not-yet-named realities -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

100 to 106: consequential handoff

Unknown and not-yet-named realities -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

100 to 107: consequential handoff

Unknown and not-yet-named realities -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

100 to 108: consequential handoff

Unknown and not-yet-named realities -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

100 to 109: consequential handoff

Unknown and not-yet-named realities -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

100 to 110: consequential handoff

Unknown and not-yet-named realities -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

100 to 111: consequential handoff

Unknown and not-yet-named realities -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

100 to 112: consequential handoff

Unknown and not-yet-named realities -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

100 to 113: consequential handoff

Unknown and not-yet-named realities -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

100 to 114: consequential handoff

Unknown and not-yet-named realities -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

100 to 115: consequential handoff

Unknown and not-yet-named realities -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

100 to 116: consequential handoff

Unknown and not-yet-named realities -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

100 to 117: consequential handoff

Unknown and not-yet-named realities -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

100 to 118: consequential handoff

Unknown and not-yet-named realities -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

100 to 119: consequential handoff

Unknown and not-yet-named realities -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

100 to 120: consequential handoff

Unknown and not-yet-named realities -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Unknown and not-yet-named realities into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: No adequate schema exists yet. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Unknown until carrier tracing identifies it. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Freeze kernel before learning outcomes; then holdout/replicate' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 753-759 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

100: Unknown and not-yet-named realities

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Unknown and not-yet-named realities

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Unknown and not-yet-named realities is described in the Atlas as: Any newly observed substrate, scale, agent, value dimension or exchange mode. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Unrepresented state is invisible until consequences accumulate. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Unknown until carrier tracing identifies it. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: All material carriers in the new domain cross positive consequential thresholds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Freeze kernel before learning outcomes; then holdout/replicate or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Evidence and source reconstruction

Unknown and not-yet-named realities | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 100: Unknown and not-yet-named realities; original author status: Open by definition. Native state: Any newly observed substrate, scale, agent, value dimension or exchange mode.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: No adequate schema exists yet. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Unrepresented state is invisible until consequences accumulate. The relevant receiving carriers are: Unknown until carrier tracing identifies it.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: All material carriers in the new domain cross positive consequential thresholds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Freeze kernel before learning outcomes; then holdout/replicate. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Boundary divergence

Unknown and not-yet-named realities | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 100: Unknown and not-yet-named realities; original author status: Open by definition. Native state: Any newly observed substrate, scale, agent, value dimension or exchange mode.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: No adequate schema exists yet. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Unrepresented state is invisible until consequences accumulate. The relevant receiving carriers are: Unknown until carrier tracing identifies it.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: All material carriers in the new domain cross positive consequential thresholds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Freeze kernel before learning outcomes; then holdout/replicate. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Causal diagram and identifiability

Unknown and not-yet-named realities | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 100: Unknown and not-yet-named realities; original author status: Open by definition. Native state: Any newly observed substrate, scale, agent, value dimension or exchange mode.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: No adequate schema exists yet. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Unrepresented state is invisible until consequences accumulate. The relevant receiving carriers are: Unknown until carrier tracing identifies it.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: All material carriers in the new domain cross positive consequential thresholds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Freeze kernel before learning outcomes; then holdout/replicate. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Receiving carrier identification

Unknown and not-yet-named realities | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 100: Unknown and not-yet-named realities; original author status: Open by definition. Native state: Any newly observed substrate, scale, agent, value dimension or exchange mode.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: No adequate schema exists yet. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Unrepresented state is invisible until consequences accumulate. The relevant receiving carriers are: Unknown until carrier tracing identifies it.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: All material carriers in the new domain cross positive consequential thresholds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Freeze kernel before learning outcomes; then holdout/replicate. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Hidden compensation accounting

Unknown and not-yet-named realities | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 100: Unknown and not-yet-named realities; original author status: Open by definition. Native state: Any newly observed substrate, scale, agent, value dimension or exchange mode.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: No adequate schema exists yet. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Unrepresented state is invisible until consequences accumulate. The relevant receiving carriers are: Unknown until carrier tracing identifies it.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: All material carriers in the new domain cross positive consequential thresholds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Freeze kernel before learning outcomes; then holdout/replicate. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Native units and dimensional integrity

Unknown and not-yet-named realities | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 100: Unknown and not-yet-named realities; original author status: Open by definition. Native state: Any newly observed substrate, scale, agent, value dimension or exchange mode.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: No adequate schema exists yet. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Unrepresented state is invisible until consequences accumulate. The relevant receiving carriers are: Unknown until carrier tracing identifies it.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: All material carriers in the new domain cross positive consequential thresholds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Freeze kernel before learning outcomes; then holdout/replicate. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Capacity specification

Unknown and not-yet-named realities | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 100: Unknown and not-yet-named realities; original author status: Open by definition. Native state: Any newly observed substrate, scale, agent, value dimension or exchange mode.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: No adequate schema exists yet. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Unrepresented state is invisible until consequences accumulate. The relevant receiving carriers are: Unknown until carrier tracing identifies it.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: All material carriers in the new domain cross positive consequential thresholds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Freeze kernel before learning outcomes; then holdout/replicate. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Intervention design

Unknown and not-yet-named realities | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 100: Unknown and not-yet-named realities; original author status: Open by definition. Native state: Any newly observed substrate, scale, agent, value dimension or exchange mode.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: No adequate schema exists yet. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Unrepresented state is invisible until consequences accumulate. The relevant receiving carriers are: Unknown until carrier tracing identifies it.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: All material carriers in the new domain cross positive consequential thresholds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Freeze kernel before learning outcomes; then holdout/replicate. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Incumbent parity contest

Unknown and not-yet-named realities | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 100: Unknown and not-yet-named realities; original author status: Open by definition. Native state: Any newly observed substrate, scale, agent, value dimension or exchange mode.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: No adequate schema exists yet. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Unrepresented state is invisible until consequences accumulate. The relevant receiving carriers are: Unknown until carrier tracing identifies it.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: All material carriers in the new domain cross positive consequential thresholds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Freeze kernel before learning outcomes; then holdout/replicate. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Time-window and recurrence

Unknown and not-yet-named realities | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 100: Unknown and not-yet-named realities; original author status: Open by definition. Native state: Any newly observed substrate, scale, agent, value dimension or exchange mode.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: No adequate schema exists yet. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Unrepresented state is invisible until consequences accumulate. The relevant receiving carriers are: Unknown until carrier tracing identifies it.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: All material carriers in the new domain cross positive consequential thresholds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Freeze kernel before learning outcomes; then holdout/replicate. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Ethical and legal safety

Unknown and not-yet-named realities | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 100: Unknown and not-yet-named realities; original author status: Open by definition. Native state: Any newly observed substrate, scale, agent, value dimension or exchange mode.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: No adequate schema exists yet. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Unrepresented state is invisible until consequences accumulate. The relevant receiving carriers are: Unknown until carrier tracing identifies it.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: All material carriers in the new domain cross positive consequential thresholds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Freeze kernel before learning outcomes; then holdout/replicate. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Sensitivity and negative controls

Unknown and not-yet-named realities | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 100: Unknown and not-yet-named realities; original author status: Open by definition. Native state: Any newly observed substrate, scale, agent, value dimension or exchange mode.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: No adequate schema exists yet. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Unrepresented state is invisible until consequences accumulate. The relevant receiving carriers are: Unknown until carrier tracing identifies it.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: All material carriers in the new domain cross positive consequential thresholds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Freeze kernel before learning outcomes; then holdout/replicate. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Replication and provenance

Unknown and not-yet-named realities | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 100: Unknown and not-yet-named realities; original author status: Open by definition. Native state: Any newly observed substrate, scale, agent, value dimension or exchange mode.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: No adequate schema exists yet. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Unrepresented state is invisible until consequences accumulate. The relevant receiving carriers are: Unknown until carrier tracing identifies it.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: All material carriers in the new domain cross positive consequential thresholds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Freeze kernel before learning outcomes; then holdout/replicate. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

100: Cross-domain invariant contest

Unknown and not-yet-named realities | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 100: Unknown and not-yet-named realities; original author status: Open by definition. Native state: Any newly observed substrate, scale, agent, value dimension or exchange mode.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: No adequate schema exists yet. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Unrepresented state is invisible until consequences accumulate. The relevant receiving carriers are: Unknown until carrier tracing identifies it.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: All material carriers in the new domain cross positive consequential thresholds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Freeze kernel before learning outcomes; then holdout/replicate. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 100, pp. 753-759 (Rufino 2026).

Adapter 101: Conscious experience and self-model

Conscious agency and responsibility | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Perception, affect, memory, self-model, attention, internal accessibility and embodied state

RECORDED RECEIVING SCHEMA

A receiving observer reduces another system to behavior, category or output alone

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Native conscious state is misread because only externally legible signals are counted.
Candidate receivers: Experiencing system, observers, dependent systems and future self.

EXISTING CAPACITY DESIGN

Cross-modal state representation with provenance, uncertainty and non-erasure constraints

PROTECTED OUTCOME STATED BY THE AUTHOR

Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Prospective multi-method cross-observer study. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101 | Conscious experience and self-model

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 101, Conscious experience and self-model. Native state: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Potential direct or institutional human pathway: Conscious experience and self-model interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Native conscious state is misread because only externally legible signals are counted.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Cross-modal state representation with provenance, uncertainty and non-erasure constraints can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

101 | Conscious experience and self-model

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 101, Conscious experience and self-model. Native state: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Potential direct or institutional human pathway: Conscious experience and self-model interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Native conscious state is misread because only externally legible signals are counted.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Cross-modal state representation with provenance, uncertainty and non-erasure constraints can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

101 | Conscious experience and self-model

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 101, Conscious experience and self-model. Native state: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Potential direct or institutional human pathway: Conscious experience and self-model interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Native conscious state is misread because only externally legible signals are counted.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Cross-modal state representation with provenance, uncertainty and non-erasure constraints can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

101 | Conscious experience and self-model

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 101, Conscious experience and self-model. Native state: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Potential direct or institutional human pathway: Conscious experience and self-model interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Native conscious state is misread because only externally legible signals are counted.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Cross-modal state representation with provenance, uncertainty and non-erasure constraints can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

101 | Conscious experience and self-model

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 101, Conscious experience and self-model. Native state: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Potential direct or institutional human pathway: Conscious experience and self-model interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Native conscious state is misread because only externally legible signals are counted.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Cross-modal state representation with provenance, uncertainty and non-erasure constraints can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

101 | Conscious experience and self-model

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 101, Conscious experience and self-model. Native state: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Potential direct or institutional human pathway: Conscious experience and self-model interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Native conscious state is misread because only externally legible signals are counted.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Cross-modal state representation with provenance, uncertainty and non-erasure constraints can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

101 to 001: consequential handoff

Conscious experience and self-model -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

101 to 002: consequential handoff

Conscious experience and self-model -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

101 to 003: consequential handoff

Conscious experience and self-model -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

101 to 004: consequential handoff

Conscious experience and self-model -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

101 to 005: consequential handoff

Conscious experience and self-model -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

101 to 006: consequential handoff

Conscious experience and self-model -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

101 to 007: consequential handoff

Conscious experience and self-model -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

101 to 008: consequential handoff

Conscious experience and self-model -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

101 to 009: consequential handoff

Conscious experience and self-model -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

101 to 010: consequential handoff

Conscious experience and self-model -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

101 to 011: consequential handoff

Conscious experience and self-model -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

101 to 012: consequential handoff

Conscious experience and self-model -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

101 to 013: consequential handoff

Conscious experience and self-model -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

101 to 014: consequential handoff

Conscious experience and self-model -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

101 to 015: consequential handoff

Conscious experience and self-model -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

101 to 016: consequential handoff

Conscious experience and self-model -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

101 to 017: consequential handoff

Conscious experience and self-model -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

101 to 018: consequential handoff

Conscious experience and self-model -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

101 to 019: consequential handoff

Conscious experience and self-model -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

101 to 020: consequential handoff

Conscious experience and self-model -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

101 to 021: consequential handoff

Conscious experience and self-model -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

101 to 022: consequential handoff

Conscious experience and self-model -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

101 to 023: consequential handoff

Conscious experience and self-model -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

101 to 024: consequential handoff

Conscious experience and self-model -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

101 to 025: consequential handoff

Conscious experience and self-model -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

101 to 026: consequential handoff

Conscious experience and self-model -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

101 to 027: consequential handoff

Conscious experience and self-model -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

101 to 028: consequential handoff

Conscious experience and self-model -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

101 to 029: consequential handoff

Conscious experience and self-model -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

101 to 030: consequential handoff

Conscious experience and self-model -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

101 to 031: consequential handoff

Conscious experience and self-model -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

101 to 032: consequential handoff

Conscious experience and self-model -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

101 to 033: consequential handoff

Conscious experience and self-model -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

101 to 034: consequential handoff

Conscious experience and self-model -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

101 to 035: consequential handoff

Conscious experience and self-model -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

101 to 036: consequential handoff

Conscious experience and self-model -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

101 to 037: consequential handoff

Conscious experience and self-model -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

101 to 038: consequential handoff

Conscious experience and self-model -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

101 to 039: consequential handoff

Conscious experience and self-model -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

101 to 040: consequential handoff

Conscious experience and self-model -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

101 to 041: consequential handoff

Conscious experience and self-model -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

101 to 042: consequential handoff

Conscious experience and self-model -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

101 to 043: consequential handoff

Conscious experience and self-model -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

101 to 044: consequential handoff

Conscious experience and self-model -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

101 to 045: consequential handoff

Conscious experience and self-model -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

101 to 046: consequential handoff

Conscious experience and self-model -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

101 to 047: consequential handoff

Conscious experience and self-model -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

101 to 048: consequential handoff

Conscious experience and self-model -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

101 to 049: consequential handoff

Conscious experience and self-model -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

101 to 050: consequential handoff

Conscious experience and self-model -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

101 to 051: consequential handoff

Conscious experience and self-model -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

101 to 052: consequential handoff

Conscious experience and self-model -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

101 to 053: consequential handoff

Conscious experience and self-model -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

101 to 054: consequential handoff

Conscious experience and self-model -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

101 to 055: consequential handoff

Conscious experience and self-model -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

101 to 056: consequential handoff

Conscious experience and self-model -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

101 to 057: consequential handoff

Conscious experience and self-model -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

101 to 058: consequential handoff

Conscious experience and self-model -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

101 to 059: consequential handoff

Conscious experience and self-model -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

101 to 060: consequential handoff

Conscious experience and self-model -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

101 to 061: consequential handoff

Conscious experience and self-model -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

101 to 062: consequential handoff

Conscious experience and self-model -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

101 to 063: consequential handoff

Conscious experience and self-model -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

101 to 064: consequential handoff

Conscious experience and self-model -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

101 to 065: consequential handoff

Conscious experience and self-model -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

101 to 066: consequential handoff

Conscious experience and self-model -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

101 to 067: consequential handoff

Conscious experience and self-model -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

101 to 068: consequential handoff

Conscious experience and self-model -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

101 to 069: consequential handoff

Conscious experience and self-model -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

101 to 070: consequential handoff

Conscious experience and self-model -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

101 to 071: consequential handoff

Conscious experience and self-model -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

101 to 072: consequential handoff

Conscious experience and self-model -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

101 to 073: consequential handoff

Conscious experience and self-model -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

101 to 074: consequential handoff

Conscious experience and self-model -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

101 to 075: consequential handoff

Conscious experience and self-model -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

101 to 076: consequential handoff

Conscious experience and self-model -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

101 to 077: consequential handoff

Conscious experience and self-model -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

101 to 078: consequential handoff

Conscious experience and self-model -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

101 to 079: consequential handoff

Conscious experience and self-model -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

101 to 080: consequential handoff

Conscious experience and self-model -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

101 to 081: consequential handoff

Conscious experience and self-model -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

101 to 082: consequential handoff

Conscious experience and self-model -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

101 to 083: consequential handoff

Conscious experience and self-model -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

101 to 084: consequential handoff

Conscious experience and self-model -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

101 to 085: consequential handoff

Conscious experience and self-model -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

101 to 086: consequential handoff

Conscious experience and self-model -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

101 to 087: consequential handoff

Conscious experience and self-model -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

101 to 088: consequential handoff

Conscious experience and self-model -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

101 to 089: consequential handoff

Conscious experience and self-model -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

101 to 090: consequential handoff

Conscious experience and self-model -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

101 to 091: consequential handoff

Conscious experience and self-model -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

101 to 092: consequential handoff

Conscious experience and self-model -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

101 to 093: consequential handoff

Conscious experience and self-model -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

101 to 094: consequential handoff

Conscious experience and self-model -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

101 to 095: consequential handoff

Conscious experience and self-model -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

101 to 096: consequential handoff

Conscious experience and self-model -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

101 to 097: consequential handoff

Conscious experience and self-model -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

101 to 098: consequential handoff

Conscious experience and self-model -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

101 to 099: consequential handoff

Conscious experience and self-model -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

101 to 100: consequential handoff

Conscious experience and self-model -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

101 to 101: consequential handoff

Conscious experience and self-model -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Conscious experience and self-model becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 761-767; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

101 to 102: consequential handoff

Conscious experience and self-model -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

101 to 103: consequential handoff

Conscious experience and self-model -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

101 to 104: consequential handoff

Conscious experience and self-model -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

101 to 105: consequential handoff

Conscious experience and self-model -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

101 to 106: consequential handoff

Conscious experience and self-model -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

101 to 107: consequential handoff

Conscious experience and self-model -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

101 to 108: consequential handoff

Conscious experience and self-model -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

101 to 109: consequential handoff

Conscious experience and self-model -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

101 to 110: consequential handoff

Conscious experience and self-model -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

101 to 111: consequential handoff

Conscious experience and self-model -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

101 to 112: consequential handoff

Conscious experience and self-model -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

101 to 113: consequential handoff

Conscious experience and self-model -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

101 to 114: consequential handoff

Conscious experience and self-model -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

101 to 115: consequential handoff

Conscious experience and self-model -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

101 to 116: consequential handoff

Conscious experience and self-model -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

101 to 117: consequential handoff

Conscious experience and self-model -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

101 to 118: consequential handoff

Conscious experience and self-model -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

101 to 119: consequential handoff

Conscious experience and self-model -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

101 to 120: consequential handoff

Conscious experience and self-model -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Conscious experience and self-model into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: A receiving observer reduces another system to behavior, category or output alone. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Experiencing system, observers, dependent systems and future self. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective multi-method cross-observer study' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 761-767 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

101: Conscious experience and self-model

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

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MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

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Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

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101: Conscious experience and self-model

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

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Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

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Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

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101: Conscious experience and self-model

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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101: Conscious experience and self-model

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

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Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Conscious experience and self-model

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Conscious experience and self-model is described in the Atlas as: Perception, affect, memory, self-model, attention, internal accessibility and embodied state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Native conscious state is misread because only externally legible signals are counted. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Experiencing system, observers, dependent systems and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective multi-method cross-observer study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Evidence and source reconstruction

Conscious experience and self-model | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 101: Conscious experience and self-model; original author status: Ontological / empirical frontier. Native state: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: A receiving observer reduces another system to behavior, category or output alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Native conscious state is misread because only externally legible signals are counted. The relevant receiving carriers are: Experiencing system, observers, dependent systems and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective multi-method cross-observer study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Boundary divergence

Conscious experience and self-model | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 101: Conscious experience and self-model; original author status: Ontological / empirical frontier. Native state: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: A receiving observer reduces another system to behavior, category or output alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Native conscious state is misread because only externally legible signals are counted. The relevant receiving carriers are: Experiencing system, observers, dependent systems and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective multi-method cross-observer study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Causal diagram and identifiability

Conscious experience and self-model | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 101: Conscious experience and self-model; original author status: Ontological / empirical frontier. Native state: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: A receiving observer reduces another system to behavior, category or output alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Native conscious state is misread because only externally legible signals are counted. The relevant receiving carriers are: Experiencing system, observers, dependent systems and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective multi-method cross-observer study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Receiving carrier identification

Conscious experience and self-model | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 101: Conscious experience and self-model; original author status: Ontological / empirical frontier. Native state: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: A receiving observer reduces another system to behavior, category or output alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Native conscious state is misread because only externally legible signals are counted. The relevant receiving carriers are: Experiencing system, observers, dependent systems and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective multi-method cross-observer study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Hidden compensation accounting

Conscious experience and self-model | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 101: Conscious experience and self-model; original author status: Ontological / empirical frontier. Native state: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: A receiving observer reduces another system to behavior, category or output alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Native conscious state is misread because only externally legible signals are counted. The relevant receiving carriers are: Experiencing system, observers, dependent systems and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective multi-method cross-observer study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Native units and dimensional integrity

Conscious experience and self-model | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 101: Conscious experience and self-model; original author status: Ontological / empirical frontier. Native state: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: A receiving observer reduces another system to behavior, category or output alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Native conscious state is misread because only externally legible signals are counted. The relevant receiving carriers are: Experiencing system, observers, dependent systems and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective multi-method cross-observer study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Capacity specification

Conscious experience and self-model | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 101: Conscious experience and self-model; original author status: Ontological / empirical frontier. Native state: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: A receiving observer reduces another system to behavior, category or output alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Native conscious state is misread because only externally legible signals are counted. The relevant receiving carriers are: Experiencing system, observers, dependent systems and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective multi-method cross-observer study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Intervention design

Conscious experience and self-model | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 101: Conscious experience and self-model; original author status: Ontological / empirical frontier. Native state: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: A receiving observer reduces another system to behavior, category or output alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Native conscious state is misread because only externally legible signals are counted. The relevant receiving carriers are: Experiencing system, observers, dependent systems and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective multi-method cross-observer study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Incumbent parity contest

Conscious experience and self-model | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 101: Conscious experience and self-model; original author status: Ontological / empirical frontier. Native state: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: A receiving observer reduces another system to behavior, category or output alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Native conscious state is misread because only externally legible signals are counted. The relevant receiving carriers are: Experiencing system, observers, dependent systems and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective multi-method cross-observer study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Time-window and recurrence

Conscious experience and self-model | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 101: Conscious experience and self-model; original author status: Ontological / empirical frontier. Native state: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: A receiving observer reduces another system to behavior, category or output alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Native conscious state is misread because only externally legible signals are counted. The relevant receiving carriers are: Experiencing system, observers, dependent systems and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective multi-method cross-observer study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Ethical and legal safety

Conscious experience and self-model | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 101: Conscious experience and self-model; original author status: Ontological / empirical frontier. Native state: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: A receiving observer reduces another system to behavior, category or output alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Native conscious state is misread because only externally legible signals are counted. The relevant receiving carriers are: Experiencing system, observers, dependent systems and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective multi-method cross-observer study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Sensitivity and negative controls

Conscious experience and self-model | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 101: Conscious experience and self-model; original author status: Ontological / empirical frontier. Native state: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: A receiving observer reduces another system to behavior, category or output alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Native conscious state is misread because only externally legible signals are counted. The relevant receiving carriers are: Experiencing system, observers, dependent systems and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective multi-method cross-observer study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Replication and provenance

Conscious experience and self-model | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 101: Conscious experience and self-model; original author status: Ontological / empirical frontier. Native state: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: A receiving observer reduces another system to behavior, category or output alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Native conscious state is misread because only externally legible signals are counted. The relevant receiving carriers are: Experiencing system, observers, dependent systems and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective multi-method cross-observer study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

101: Cross-domain invariant contest

Conscious experience and self-model | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 101: Conscious experience and self-model; original author status: Ontological / empirical frontier. Native state: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: A receiving observer reduces another system to behavior, category or output alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Native conscious state is misread because only externally legible signals are counted. The relevant receiving carriers are: Experiencing system, observers, dependent systems and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Legibility, agency and viable functioning increase without reducing an unmodeled carrier to zero.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective multi-method cross-observer study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 101, pp. 761-767 (Rufino 2026).

Adapter 102: Choice and option topology

Conscious agency and responsibility | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence

RECORDED RECEIVING SCHEMA

Binary free-choice / no-choice label

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Candidate receivers: Chooser, affected people, institutions and future option set.

EXISTING CAPACITY DESIGN

Feasible-option map plus prebuilt low-friction lawful/prosocial pathways

PROTECTED OUTCOME STATED BY THE AUTHOR

Agency expands while downstream harm and coercive friction fall

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Prospective choice-topology intervention. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102 | Choice and option topology

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 102, Choice and option topology. Native state: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Potential direct or institutional human pathway: Choice and option topology interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Feasible-option map plus prebuilt low-friction lawful/prosocial pathways can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Agency expands while downstream harm and coercive friction fall. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

102 | Choice and option topology

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 102, Choice and option topology. Native state: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Potential direct or institutional human pathway: Choice and option topology interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Feasible-option map plus prebuilt low-friction lawful/prosocial pathways can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Agency expands while downstream harm and coercive friction fall. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

102 | Choice and option topology

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 102, Choice and option topology. Native state: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Potential direct or institutional human pathway: Choice and option topology interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Feasible-option map plus prebuilt low-friction lawful/prosocial pathways can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Agency expands while downstream harm and coercive friction fall. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

102 | Choice and option topology

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 102, Choice and option topology. Native state: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Potential direct or institutional human pathway: Choice and option topology interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Feasible-option map plus prebuilt low-friction lawful/prosocial pathways can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Agency expands while downstream harm and coercive friction fall. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

102 | Choice and option topology

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 102, Choice and option topology. Native state: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Potential direct or institutional human pathway: Choice and option topology interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Feasible-option map plus prebuilt low-friction lawful/prosocial pathways can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Agency expands while downstream harm and coercive friction fall. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

102 | Choice and option topology

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 102, Choice and option topology. Native state: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Potential direct or institutional human pathway: Choice and option topology interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Feasible-option map plus prebuilt low-friction lawful/prosocial pathways can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Agency expands while downstream harm and coercive friction fall. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

102 to 001: consequential handoff

Choice and option topology -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

102 to 002: consequential handoff

Choice and option topology -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

102 to 003: consequential handoff

Choice and option topology -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

102 to 004: consequential handoff

Choice and option topology -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

102 to 005: consequential handoff

Choice and option topology -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

102 to 006: consequential handoff

Choice and option topology -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

102 to 007: consequential handoff

Choice and option topology -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

102 to 008: consequential handoff

Choice and option topology -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

102 to 009: consequential handoff

Choice and option topology -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

102 to 010: consequential handoff

Choice and option topology -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

102 to 011: consequential handoff

Choice and option topology -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

102 to 012: consequential handoff

Choice and option topology -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

102 to 013: consequential handoff

Choice and option topology -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

102 to 014: consequential handoff

Choice and option topology -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

102 to 015: consequential handoff

Choice and option topology -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

102 to 016: consequential handoff

Choice and option topology -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

102 to 017: consequential handoff

Choice and option topology -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

102 to 018: consequential handoff

Choice and option topology -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

102 to 019: consequential handoff

Choice and option topology -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

102 to 020: consequential handoff

Choice and option topology -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

102 to 021: consequential handoff

Choice and option topology -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

102 to 022: consequential handoff

Choice and option topology -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

102 to 023: consequential handoff

Choice and option topology -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

102 to 024: consequential handoff

Choice and option topology -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

102 to 025: consequential handoff

Choice and option topology -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

102 to 026: consequential handoff

Choice and option topology -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

102 to 027: consequential handoff

Choice and option topology -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

102 to 028: consequential handoff

Choice and option topology -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

102 to 029: consequential handoff

Choice and option topology -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

102 to 030: consequential handoff

Choice and option topology -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

102 to 031: consequential handoff

Choice and option topology -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

102 to 032: consequential handoff

Choice and option topology -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

102 to 033: consequential handoff

Choice and option topology -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

102 to 034: consequential handoff

Choice and option topology -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

102 to 035: consequential handoff

Choice and option topology -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

102 to 036: consequential handoff

Choice and option topology -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

102 to 037: consequential handoff

Choice and option topology -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

102 to 038: consequential handoff

Choice and option topology -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

102 to 039: consequential handoff

Choice and option topology -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

102 to 040: consequential handoff

Choice and option topology -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

102 to 041: consequential handoff

Choice and option topology -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

102 to 042: consequential handoff

Choice and option topology -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

102 to 043: consequential handoff

Choice and option topology -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

102 to 044: consequential handoff

Choice and option topology -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

102 to 045: consequential handoff

Choice and option topology -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

102 to 046: consequential handoff

Choice and option topology -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

102 to 047: consequential handoff

Choice and option topology -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

102 to 048: consequential handoff

Choice and option topology -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

102 to 049: consequential handoff

Choice and option topology -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

102 to 050: consequential handoff

Choice and option topology -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

102 to 051: consequential handoff

Choice and option topology -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

102 to 052: consequential handoff

Choice and option topology -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

102 to 053: consequential handoff

Choice and option topology -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

102 to 054: consequential handoff

Choice and option topology -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

102 to 055: consequential handoff

Choice and option topology -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

102 to 056: consequential handoff

Choice and option topology -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

102 to 057: consequential handoff

Choice and option topology -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

102 to 058: consequential handoff

Choice and option topology -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

102 to 059: consequential handoff

Choice and option topology -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

102 to 060: consequential handoff

Choice and option topology -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

102 to 061: consequential handoff

Choice and option topology -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

102 to 062: consequential handoff

Choice and option topology -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

102 to 063: consequential handoff

Choice and option topology -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

102 to 064: consequential handoff

Choice and option topology -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

102 to 065: consequential handoff

Choice and option topology -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

102 to 066: consequential handoff

Choice and option topology -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

102 to 067: consequential handoff

Choice and option topology -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

102 to 068: consequential handoff

Choice and option topology -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

102 to 069: consequential handoff

Choice and option topology -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

102 to 070: consequential handoff

Choice and option topology -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

102 to 071: consequential handoff

Choice and option topology -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

102 to 072: consequential handoff

Choice and option topology -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

102 to 073: consequential handoff

Choice and option topology -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

102 to 074: consequential handoff

Choice and option topology -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

102 to 075: consequential handoff

Choice and option topology -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

102 to 076: consequential handoff

Choice and option topology -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

102 to 077: consequential handoff

Choice and option topology -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

102 to 078: consequential handoff

Choice and option topology -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

102 to 079: consequential handoff

Choice and option topology -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

102 to 080: consequential handoff

Choice and option topology -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

102 to 081: consequential handoff

Choice and option topology -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

102 to 082: consequential handoff

Choice and option topology -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

102 to 083: consequential handoff

Choice and option topology -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

102 to 084: consequential handoff

Choice and option topology -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

102 to 085: consequential handoff

Choice and option topology -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

102 to 086: consequential handoff

Choice and option topology -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

102 to 087: consequential handoff

Choice and option topology -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

102 to 088: consequential handoff

Choice and option topology -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

102 to 089: consequential handoff

Choice and option topology -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

102 to 090: consequential handoff

Choice and option topology -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

102 to 091: consequential handoff

Choice and option topology -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

102 to 092: consequential handoff

Choice and option topology -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

102 to 093: consequential handoff

Choice and option topology -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

102 to 094: consequential handoff

Choice and option topology -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

102 to 095: consequential handoff

Choice and option topology -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

102 to 096: consequential handoff

Choice and option topology -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

102 to 097: consequential handoff

Choice and option topology -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

102 to 098: consequential handoff

Choice and option topology -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

102 to 099: consequential handoff

Choice and option topology -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

102 to 100: consequential handoff

Choice and option topology -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

102 to 101: consequential handoff

Choice and option topology -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

102 to 102: consequential handoff

Choice and option topology -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Choice and option topology becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 768-774; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

102 to 103: consequential handoff

Choice and option topology -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

102 to 104: consequential handoff

Choice and option topology -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

102 to 105: consequential handoff

Choice and option topology -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

102 to 106: consequential handoff

Choice and option topology -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

102 to 107: consequential handoff

Choice and option topology -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

102 to 108: consequential handoff

Choice and option topology -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

102 to 109: consequential handoff

Choice and option topology -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

102 to 110: consequential handoff

Choice and option topology -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

102 to 111: consequential handoff

Choice and option topology -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

102 to 112: consequential handoff

Choice and option topology -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

102 to 113: consequential handoff

Choice and option topology -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

102 to 114: consequential handoff

Choice and option topology -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

102 to 115: consequential handoff

Choice and option topology -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

102 to 116: consequential handoff

Choice and option topology -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

102 to 117: consequential handoff

Choice and option topology -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

102 to 118: consequential handoff

Choice and option topology -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

102 to 119: consequential handoff

Choice and option topology -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

102 to 120: consequential handoff

Choice and option topology -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Choice and option topology into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Binary free-choice / no-choice label. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Chooser, affected people, institutions and future option set. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective choice-topology intervention' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 768-774 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

102: Choice and option topology

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

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Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

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OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

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Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

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LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

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Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Choice and option topology

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Choice and option topology is described in the Atlas as: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Chooser, affected people, institutions and future option set. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agency expands while downstream harm and coercive friction fall. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective choice-topology intervention or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Evidence and source reconstruction

Choice and option topology | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 102: Choice and option topology; original author status: Constructive protocol. Native state: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Binary free-choice / no-choice label. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. The relevant receiving carriers are: Chooser, affected people, institutions and future option set.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agency expands while downstream harm and coercive friction fall.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective choice-topology intervention. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Boundary divergence

Choice and option topology | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 102: Choice and option topology; original author status: Constructive protocol. Native state: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Binary free-choice / no-choice label. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. The relevant receiving carriers are: Chooser, affected people, institutions and future option set.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agency expands while downstream harm and coercive friction fall.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective choice-topology intervention. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Causal diagram and identifiability

Choice and option topology | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 102: Choice and option topology; original author status: Constructive protocol. Native state: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Binary free-choice / no-choice label. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. The relevant receiving carriers are: Chooser, affected people, institutions and future option set.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agency expands while downstream harm and coercive friction fall.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective choice-topology intervention. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Receiving carrier identification

Choice and option topology | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 102: Choice and option topology; original author status: Constructive protocol. Native state: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Binary free-choice / no-choice label. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. The relevant receiving carriers are: Chooser, affected people, institutions and future option set.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agency expands while downstream harm and coercive friction fall.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective choice-topology intervention. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Hidden compensation accounting

Choice and option topology | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 102: Choice and option topology; original author status: Constructive protocol. Native state: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Binary free-choice / no-choice label. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. The relevant receiving carriers are: Chooser, affected people, institutions and future option set.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agency expands while downstream harm and coercive friction fall.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective choice-topology intervention. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Native units and dimensional integrity

Choice and option topology | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 102: Choice and option topology; original author status: Constructive protocol. Native state: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Binary free-choice / no-choice label. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. The relevant receiving carriers are: Chooser, affected people, institutions and future option set.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agency expands while downstream harm and coercive friction fall.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective choice-topology intervention. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Capacity specification

Choice and option topology | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 102: Choice and option topology; original author status: Constructive protocol. Native state: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Binary free-choice / no-choice label. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. The relevant receiving carriers are: Chooser, affected people, institutions and future option set.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agency expands while downstream harm and coercive friction fall.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective choice-topology intervention. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Intervention design

Choice and option topology | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 102: Choice and option topology; original author status: Constructive protocol. Native state: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Binary free-choice / no-choice label. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. The relevant receiving carriers are: Chooser, affected people, institutions and future option set.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agency expands while downstream harm and coercive friction fall.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective choice-topology intervention. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Incumbent parity contest

Choice and option topology | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 102: Choice and option topology; original author status: Constructive protocol. Native state: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Binary free-choice / no-choice label. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. The relevant receiving carriers are: Chooser, affected people, institutions and future option set.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agency expands while downstream harm and coercive friction fall.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective choice-topology intervention. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Time-window and recurrence

Choice and option topology | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 102: Choice and option topology; original author status: Constructive protocol. Native state: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Binary free-choice / no-choice label. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. The relevant receiving carriers are: Chooser, affected people, institutions and future option set.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agency expands while downstream harm and coercive friction fall.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective choice-topology intervention. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Ethical and legal safety

Choice and option topology | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 102: Choice and option topology; original author status: Constructive protocol. Native state: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Binary free-choice / no-choice label. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. The relevant receiving carriers are: Chooser, affected people, institutions and future option set.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agency expands while downstream harm and coercive friction fall.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective choice-topology intervention. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Sensitivity and negative controls

Choice and option topology | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 102: Choice and option topology; original author status: Constructive protocol. Native state: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Binary free-choice / no-choice label. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. The relevant receiving carriers are: Chooser, affected people, institutions and future option set.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agency expands while downstream harm and coercive friction fall.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective choice-topology intervention. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Replication and provenance

Choice and option topology | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 102: Choice and option topology; original author status: Constructive protocol. Native state: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Binary free-choice / no-choice label. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. The relevant receiving carriers are: Chooser, affected people, institutions and future option set.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agency expands while downstream harm and coercive friction fall.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective choice-topology intervention. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

102: Cross-domain invariant contest

Choice and option topology | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 102: Choice and option topology; original author status: Constructive protocol. Native state: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Binary free-choice / no-choice label. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Nominal alternatives exist while executable humane alternatives are absent or prohibitively costly. The relevant receiving carriers are: Chooser, affected people, institutions and future option set.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agency expands while downstream harm and coercive friction fall.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective choice-topology intervention. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 102, pp. 768-774 (Rufino 2026).

Adapter 103: Truth, deception and disclosure

Conscious agency and responsibility | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity

RECORDED RECEIVING SCHEMA

Truth/lie binary detached from consequence topology

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Candidate receivers: Speaker, listener, harmed parties, institutions and future trust network.

EXISTING CAPACITY DESIGN

Verification, safe disclosure, correction, restitution and transition capacity

PROTECTED OUTCOME STATED BY THE AUTHOR

Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Sealed truth-cost / disclosure study. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103 | Truth, deception and disclosure

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 103, Truth, deception and disclosure. Native state: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Potential direct or institutional human pathway: Truth, deception and disclosure interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Verification, safe disclosure, correction, restitution and transition capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

103 | Truth, deception and disclosure

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 103, Truth, deception and disclosure. Native state: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Potential direct or institutional human pathway: Truth, deception and disclosure interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Verification, safe disclosure, correction, restitution and transition capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

103 | Truth, deception and disclosure

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 103, Truth, deception and disclosure. Native state: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Potential direct or institutional human pathway: Truth, deception and disclosure interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Verification, safe disclosure, correction, restitution and transition capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

103 | Truth, deception and disclosure

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 103, Truth, deception and disclosure. Native state: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Potential direct or institutional human pathway: Truth, deception and disclosure interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Verification, safe disclosure, correction, restitution and transition capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

103 | Truth, deception and disclosure

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 103, Truth, deception and disclosure. Native state: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Potential direct or institutional human pathway: Truth, deception and disclosure interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Verification, safe disclosure, correction, restitution and transition capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

103 | Truth, deception and disclosure

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 103, Truth, deception and disclosure. Native state: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Potential direct or institutional human pathway: Truth, deception and disclosure interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Verification, safe disclosure, correction, restitution and transition capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

103 to 001: consequential handoff

Truth, deception and disclosure -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

103 to 002: consequential handoff

Truth, deception and disclosure -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

103 to 003: consequential handoff

Truth, deception and disclosure -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

103 to 004: consequential handoff

Truth, deception and disclosure -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

103 to 005: consequential handoff

Truth, deception and disclosure -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

103 to 006: consequential handoff

Truth, deception and disclosure -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

103 to 007: consequential handoff

Truth, deception and disclosure -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

103 to 008: consequential handoff

Truth, deception and disclosure -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

103 to 009: consequential handoff

Truth, deception and disclosure -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

103 to 010: consequential handoff

Truth, deception and disclosure -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

103 to 011: consequential handoff

Truth, deception and disclosure -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

103 to 012: consequential handoff

Truth, deception and disclosure -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

103 to 013: consequential handoff

Truth, deception and disclosure -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

103 to 014: consequential handoff

Truth, deception and disclosure -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

103 to 015: consequential handoff

Truth, deception and disclosure -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

103 to 016: consequential handoff

Truth, deception and disclosure -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

103 to 017: consequential handoff

Truth, deception and disclosure -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

103 to 018: consequential handoff

Truth, deception and disclosure -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

103 to 019: consequential handoff

Truth, deception and disclosure -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

103 to 020: consequential handoff

Truth, deception and disclosure -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

103 to 021: consequential handoff

Truth, deception and disclosure -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

103 to 022: consequential handoff

Truth, deception and disclosure -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

103 to 023: consequential handoff

Truth, deception and disclosure -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

103 to 024: consequential handoff

Truth, deception and disclosure -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

103 to 025: consequential handoff

Truth, deception and disclosure -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

103 to 026: consequential handoff

Truth, deception and disclosure -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

103 to 027: consequential handoff

Truth, deception and disclosure -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

103 to 028: consequential handoff

Truth, deception and disclosure -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

103 to 029: consequential handoff

Truth, deception and disclosure -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

103 to 030: consequential handoff

Truth, deception and disclosure -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

103 to 031: consequential handoff

Truth, deception and disclosure -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

103 to 032: consequential handoff

Truth, deception and disclosure -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

103 to 033: consequential handoff

Truth, deception and disclosure -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

103 to 034: consequential handoff

Truth, deception and disclosure -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

103 to 035: consequential handoff

Truth, deception and disclosure -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

103 to 036: consequential handoff

Truth, deception and disclosure -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

103 to 037: consequential handoff

Truth, deception and disclosure -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

103 to 038: consequential handoff

Truth, deception and disclosure -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

103 to 039: consequential handoff

Truth, deception and disclosure -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

103 to 040: consequential handoff

Truth, deception and disclosure -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

103 to 041: consequential handoff

Truth, deception and disclosure -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

103 to 042: consequential handoff

Truth, deception and disclosure -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

103 to 043: consequential handoff

Truth, deception and disclosure -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

103 to 044: consequential handoff

Truth, deception and disclosure -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

103 to 045: consequential handoff

Truth, deception and disclosure -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

103 to 046: consequential handoff

Truth, deception and disclosure -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

103 to 047: consequential handoff

Truth, deception and disclosure -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

103 to 048: consequential handoff

Truth, deception and disclosure -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

103 to 049: consequential handoff

Truth, deception and disclosure -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

103 to 050: consequential handoff

Truth, deception and disclosure -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

103 to 051: consequential handoff

Truth, deception and disclosure -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

103 to 052: consequential handoff

Truth, deception and disclosure -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

103 to 053: consequential handoff

Truth, deception and disclosure -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

103 to 054: consequential handoff

Truth, deception and disclosure -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

103 to 055: consequential handoff

Truth, deception and disclosure -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

103 to 056: consequential handoff

Truth, deception and disclosure -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

103 to 057: consequential handoff

Truth, deception and disclosure -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

103 to 058: consequential handoff

Truth, deception and disclosure -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

103 to 059: consequential handoff

Truth, deception and disclosure -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

103 to 060: consequential handoff

Truth, deception and disclosure -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

103 to 061: consequential handoff

Truth, deception and disclosure -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

103 to 062: consequential handoff

Truth, deception and disclosure -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

103 to 063: consequential handoff

Truth, deception and disclosure -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

103 to 064: consequential handoff

Truth, deception and disclosure -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

103 to 065: consequential handoff

Truth, deception and disclosure -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

103 to 066: consequential handoff

Truth, deception and disclosure -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

103 to 067: consequential handoff

Truth, deception and disclosure -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

103 to 068: consequential handoff

Truth, deception and disclosure -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

103 to 069: consequential handoff

Truth, deception and disclosure -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

103 to 070: consequential handoff

Truth, deception and disclosure -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

103 to 071: consequential handoff

Truth, deception and disclosure -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

103 to 072: consequential handoff

Truth, deception and disclosure -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

103 to 073: consequential handoff

Truth, deception and disclosure -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

103 to 074: consequential handoff

Truth, deception and disclosure -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

103 to 075: consequential handoff

Truth, deception and disclosure -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

103 to 076: consequential handoff

Truth, deception and disclosure -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

103 to 077: consequential handoff

Truth, deception and disclosure -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

103 to 078: consequential handoff

Truth, deception and disclosure -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

103 to 079: consequential handoff

Truth, deception and disclosure -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

103 to 080: consequential handoff

Truth, deception and disclosure -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

103 to 081: consequential handoff

Truth, deception and disclosure -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

103 to 082: consequential handoff

Truth, deception and disclosure -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

103 to 083: consequential handoff

Truth, deception and disclosure -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

103 to 084: consequential handoff

Truth, deception and disclosure -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

103 to 085: consequential handoff

Truth, deception and disclosure -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

103 to 086: consequential handoff

Truth, deception and disclosure -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

103 to 087: consequential handoff

Truth, deception and disclosure -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

103 to 088: consequential handoff

Truth, deception and disclosure -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

103 to 089: consequential handoff

Truth, deception and disclosure -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

103 to 090: consequential handoff

Truth, deception and disclosure -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

103 to 091: consequential handoff

Truth, deception and disclosure -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

103 to 092: consequential handoff

Truth, deception and disclosure -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

103 to 093: consequential handoff

Truth, deception and disclosure -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

103 to 094: consequential handoff

Truth, deception and disclosure -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

103 to 095: consequential handoff

Truth, deception and disclosure -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

103 to 096: consequential handoff

Truth, deception and disclosure -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

103 to 097: consequential handoff

Truth, deception and disclosure -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

103 to 098: consequential handoff

Truth, deception and disclosure -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

103 to 099: consequential handoff

Truth, deception and disclosure -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

103 to 100: consequential handoff

Truth, deception and disclosure -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

103 to 101: consequential handoff

Truth, deception and disclosure -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

103 to 102: consequential handoff

Truth, deception and disclosure -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

103 to 103: consequential handoff

Truth, deception and disclosure -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Truth, deception and disclosure becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 775-781; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

103 to 104: consequential handoff

Truth, deception and disclosure -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

103 to 105: consequential handoff

Truth, deception and disclosure -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

103 to 106: consequential handoff

Truth, deception and disclosure -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

103 to 107: consequential handoff

Truth, deception and disclosure -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

103 to 108: consequential handoff

Truth, deception and disclosure -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

103 to 109: consequential handoff

Truth, deception and disclosure -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

103 to 110: consequential handoff

Truth, deception and disclosure -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

103 to 111: consequential handoff

Truth, deception and disclosure -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

103 to 112: consequential handoff

Truth, deception and disclosure -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

103 to 113: consequential handoff

Truth, deception and disclosure -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

103 to 114: consequential handoff

Truth, deception and disclosure -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

103 to 115: consequential handoff

Truth, deception and disclosure -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

103 to 116: consequential handoff

Truth, deception and disclosure -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

103 to 117: consequential handoff

Truth, deception and disclosure -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

103 to 118: consequential handoff

Truth, deception and disclosure -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

103 to 119: consequential handoff

Truth, deception and disclosure -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

103 to 120: consequential handoff

Truth, deception and disclosure -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Truth, deception and disclosure into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Truth/lie binary detached from consequence topology. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Speaker, listener, harmed parties, institutions and future trust network. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Verification, safe disclosure, correction, restitution and transition capacity. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed truth-cost / disclosure study' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 775-781 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

103: Truth, deception and disclosure

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Truth, deception and disclosure

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Truth, deception and disclosure is described in the Atlas as: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Speaker, listener, harmed parties, institutions and future trust network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Verification, safe disclosure, correction, restitution and transition capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed truth-cost / disclosure study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Evidence and source reconstruction

Truth, deception and disclosure | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 103: Truth, deception and disclosure; original author status: Constructive protocol. Native state: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Truth/lie binary detached from consequence topology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. The relevant receiving carriers are: Speaker, listener, harmed parties, institutions and future trust network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Verification, safe disclosure, correction, restitution and transition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed truth-cost / disclosure study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Boundary divergence

Truth, deception and disclosure | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 103: Truth, deception and disclosure; original author status: Constructive protocol. Native state: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Truth/lie binary detached from consequence topology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. The relevant receiving carriers are: Speaker, listener, harmed parties, institutions and future trust network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Verification, safe disclosure, correction, restitution and transition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed truth-cost / disclosure study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Causal diagram and identifiability

Truth, deception and disclosure | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 103: Truth, deception and disclosure; original author status: Constructive protocol. Native state: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Truth/lie binary detached from consequence topology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. The relevant receiving carriers are: Speaker, listener, harmed parties, institutions and future trust network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Verification, safe disclosure, correction, restitution and transition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed truth-cost / disclosure study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Receiving carrier identification

Truth, deception and disclosure | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 103: Truth, deception and disclosure; original author status: Constructive protocol. Native state: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Truth/lie binary detached from consequence topology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. The relevant receiving carriers are: Speaker, listener, harmed parties, institutions and future trust network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Verification, safe disclosure, correction, restitution and transition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed truth-cost / disclosure study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Hidden compensation accounting

Truth, deception and disclosure | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 103: Truth, deception and disclosure; original author status: Constructive protocol. Native state: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Truth/lie binary detached from consequence topology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. The relevant receiving carriers are: Speaker, listener, harmed parties, institutions and future trust network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Verification, safe disclosure, correction, restitution and transition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed truth-cost / disclosure study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Native units and dimensional integrity

Truth, deception and disclosure | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 103: Truth, deception and disclosure; original author status: Constructive protocol. Native state: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Truth/lie binary detached from consequence topology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. The relevant receiving carriers are: Speaker, listener, harmed parties, institutions and future trust network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Verification, safe disclosure, correction, restitution and transition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed truth-cost / disclosure study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Capacity specification

Truth, deception and disclosure | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 103: Truth, deception and disclosure; original author status: Constructive protocol. Native state: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Truth/lie binary detached from consequence topology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. The relevant receiving carriers are: Speaker, listener, harmed parties, institutions and future trust network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Verification, safe disclosure, correction, restitution and transition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed truth-cost / disclosure study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Intervention design

Truth, deception and disclosure | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 103: Truth, deception and disclosure; original author status: Constructive protocol. Native state: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Truth/lie binary detached from consequence topology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. The relevant receiving carriers are: Speaker, listener, harmed parties, institutions and future trust network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Verification, safe disclosure, correction, restitution and transition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed truth-cost / disclosure study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Incumbent parity contest

Truth, deception and disclosure | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 103: Truth, deception and disclosure; original author status: Constructive protocol. Native state: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Truth/lie binary detached from consequence topology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. The relevant receiving carriers are: Speaker, listener, harmed parties, institutions and future trust network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Verification, safe disclosure, correction, restitution and transition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed truth-cost / disclosure study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Time-window and recurrence

Truth, deception and disclosure | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 103: Truth, deception and disclosure; original author status: Constructive protocol. Native state: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Truth/lie binary detached from consequence topology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. The relevant receiving carriers are: Speaker, listener, harmed parties, institutions and future trust network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Verification, safe disclosure, correction, restitution and transition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed truth-cost / disclosure study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Ethical and legal safety

Truth, deception and disclosure | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 103: Truth, deception and disclosure; original author status: Constructive protocol. Native state: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Truth/lie binary detached from consequence topology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. The relevant receiving carriers are: Speaker, listener, harmed parties, institutions and future trust network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Verification, safe disclosure, correction, restitution and transition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed truth-cost / disclosure study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Sensitivity and negative controls

Truth, deception and disclosure | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 103: Truth, deception and disclosure; original author status: Constructive protocol. Native state: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Truth/lie binary detached from consequence topology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. The relevant receiving carriers are: Speaker, listener, harmed parties, institutions and future trust network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Verification, safe disclosure, correction, restitution and transition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed truth-cost / disclosure study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Replication and provenance

Truth, deception and disclosure | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 103: Truth, deception and disclosure; original author status: Constructive protocol. Native state: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Truth/lie binary detached from consequence topology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. The relevant receiving carriers are: Speaker, listener, harmed parties, institutions and future trust network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Verification, safe disclosure, correction, restitution and transition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed truth-cost / disclosure study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

103: Cross-domain invariant contest

Truth, deception and disclosure | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 103: Truth, deception and disclosure; original author status: Constructive protocol. Native state: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Truth/lie binary detached from consequence topology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Truth creates concentrated cost or deception remains locally profitable because receiving capacity is absent. The relevant receiving carriers are: Speaker, listener, harmed parties, institutions and future trust network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Verification, safe disclosure, correction, restitution and transition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Fidelity rises while truthful disclosure no longer requires avoidable destruction of material carriers.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed truth-cost / disclosure study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 103, pp. 775-781 (Rufino 2026).

Adapter 104: Responsibility and causal attribution

Conscious agency and responsibility | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement

RECORDED RECEIVING SCHEMA

Moral label detached from the action-producing state

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Responsibility is either erased by context or overstated by ignoring real constraints.
Candidate receivers: Actor, victims, institutions, community and future actors.

EXISTING CAPACITY DESIGN

Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair

PROTECTED OUTCOME STATED BY THE AUTHOR

Attribution, safety, restitution and future agency improve without hiding causal contribution

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Blinded case reconstruction with preregistered causal criteria. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104 | Responsibility and causal attribution

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 104, Responsibility and causal attribution. Native state: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Potential direct or institutional human pathway: Responsibility and causal attribution interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Responsibility is either erased by context or overstated by ignoring real constraints.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Attribution, safety, restitution and future agency improve without hiding causal contribution. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

104 | Responsibility and causal attribution

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 104, Responsibility and causal attribution. Native state: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Potential direct or institutional human pathway: Responsibility and causal attribution interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Responsibility is either erased by context or overstated by ignoring real constraints.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Attribution, safety, restitution and future agency improve without hiding causal contribution. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

104 | Responsibility and causal attribution

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 104, Responsibility and causal attribution. Native state: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Potential direct or institutional human pathway: Responsibility and causal attribution interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Responsibility is either erased by context or overstated by ignoring real constraints.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Attribution, safety, restitution and future agency improve without hiding causal contribution. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

104 | Responsibility and causal attribution

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 104, Responsibility and causal attribution. Native state: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Potential direct or institutional human pathway: Responsibility and causal attribution interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Responsibility is either erased by context or overstated by ignoring real constraints.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Attribution, safety, restitution and future agency improve without hiding causal contribution. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

104 | Responsibility and causal attribution

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 104, Responsibility and causal attribution. Native state: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Potential direct or institutional human pathway: Responsibility and causal attribution interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Responsibility is either erased by context or overstated by ignoring real constraints.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Attribution, safety, restitution and future agency improve without hiding causal contribution. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

104 | Responsibility and causal attribution

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 104, Responsibility and causal attribution. Native state: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Potential direct or institutional human pathway: Responsibility and causal attribution interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Responsibility is either erased by context or overstated by ignoring real constraints.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Attribution, safety, restitution and future agency improve without hiding causal contribution. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

104 to 001: consequential handoff

Responsibility and causal attribution -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

104 to 002: consequential handoff

Responsibility and causal attribution -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

104 to 003: consequential handoff

Responsibility and causal attribution -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

104 to 004: consequential handoff

Responsibility and causal attribution -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

104 to 005: consequential handoff

Responsibility and causal attribution -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

104 to 006: consequential handoff

Responsibility and causal attribution -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

104 to 007: consequential handoff

Responsibility and causal attribution -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

104 to 008: consequential handoff

Responsibility and causal attribution -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

104 to 009: consequential handoff

Responsibility and causal attribution -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

104 to 010: consequential handoff

Responsibility and causal attribution -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

104 to 011: consequential handoff

Responsibility and causal attribution -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

104 to 012: consequential handoff

Responsibility and causal attribution -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

104 to 013: consequential handoff

Responsibility and causal attribution -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

104 to 014: consequential handoff

Responsibility and causal attribution -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

104 to 015: consequential handoff

Responsibility and causal attribution -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

104 to 016: consequential handoff

Responsibility and causal attribution -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

104 to 017: consequential handoff

Responsibility and causal attribution -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

104 to 018: consequential handoff

Responsibility and causal attribution -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

104 to 019: consequential handoff

Responsibility and causal attribution -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

104 to 020: consequential handoff

Responsibility and causal attribution -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

104 to 021: consequential handoff

Responsibility and causal attribution -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

104 to 022: consequential handoff

Responsibility and causal attribution -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

104 to 023: consequential handoff

Responsibility and causal attribution -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

104 to 024: consequential handoff

Responsibility and causal attribution -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

104 to 025: consequential handoff

Responsibility and causal attribution -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

104 to 026: consequential handoff

Responsibility and causal attribution -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

104 to 027: consequential handoff

Responsibility and causal attribution -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

104 to 028: consequential handoff

Responsibility and causal attribution -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

104 to 029: consequential handoff

Responsibility and causal attribution -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

104 to 030: consequential handoff

Responsibility and causal attribution -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

104 to 031: consequential handoff

Responsibility and causal attribution -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

104 to 032: consequential handoff

Responsibility and causal attribution -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

104 to 033: consequential handoff

Responsibility and causal attribution -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

104 to 034: consequential handoff

Responsibility and causal attribution -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

104 to 035: consequential handoff

Responsibility and causal attribution -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

104 to 036: consequential handoff

Responsibility and causal attribution -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

104 to 037: consequential handoff

Responsibility and causal attribution -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

104 to 038: consequential handoff

Responsibility and causal attribution -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

104 to 039: consequential handoff

Responsibility and causal attribution -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

104 to 040: consequential handoff

Responsibility and causal attribution -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

104 to 041: consequential handoff

Responsibility and causal attribution -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

104 to 042: consequential handoff

Responsibility and causal attribution -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

104 to 043: consequential handoff

Responsibility and causal attribution -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

104 to 044: consequential handoff

Responsibility and causal attribution -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

104 to 045: consequential handoff

Responsibility and causal attribution -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

104 to 046: consequential handoff

Responsibility and causal attribution -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

104 to 047: consequential handoff

Responsibility and causal attribution -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

104 to 048: consequential handoff

Responsibility and causal attribution -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

104 to 049: consequential handoff

Responsibility and causal attribution -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

104 to 050: consequential handoff

Responsibility and causal attribution -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

104 to 051: consequential handoff

Responsibility and causal attribution -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

104 to 052: consequential handoff

Responsibility and causal attribution -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

104 to 053: consequential handoff

Responsibility and causal attribution -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

104 to 054: consequential handoff

Responsibility and causal attribution -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

104 to 055: consequential handoff

Responsibility and causal attribution -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

104 to 056: consequential handoff

Responsibility and causal attribution -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

104 to 057: consequential handoff

Responsibility and causal attribution -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

104 to 058: consequential handoff

Responsibility and causal attribution -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

104 to 059: consequential handoff

Responsibility and causal attribution -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

104 to 060: consequential handoff

Responsibility and causal attribution -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

104 to 061: consequential handoff

Responsibility and causal attribution -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

104 to 062: consequential handoff

Responsibility and causal attribution -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

104 to 063: consequential handoff

Responsibility and causal attribution -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

104 to 064: consequential handoff

Responsibility and causal attribution -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

104 to 065: consequential handoff

Responsibility and causal attribution -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

104 to 066: consequential handoff

Responsibility and causal attribution -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

104 to 067: consequential handoff

Responsibility and causal attribution -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

104 to 068: consequential handoff

Responsibility and causal attribution -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

104 to 069: consequential handoff

Responsibility and causal attribution -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

104 to 070: consequential handoff

Responsibility and causal attribution -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

104 to 071: consequential handoff

Responsibility and causal attribution -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

104 to 072: consequential handoff

Responsibility and causal attribution -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

104 to 073: consequential handoff

Responsibility and causal attribution -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

104 to 074: consequential handoff

Responsibility and causal attribution -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

104 to 075: consequential handoff

Responsibility and causal attribution -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

104 to 076: consequential handoff

Responsibility and causal attribution -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

104 to 077: consequential handoff

Responsibility and causal attribution -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

104 to 078: consequential handoff

Responsibility and causal attribution -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

104 to 079: consequential handoff

Responsibility and causal attribution -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

104 to 080: consequential handoff

Responsibility and causal attribution -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

104 to 081: consequential handoff

Responsibility and causal attribution -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

104 to 082: consequential handoff

Responsibility and causal attribution -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

104 to 083: consequential handoff

Responsibility and causal attribution -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

104 to 084: consequential handoff

Responsibility and causal attribution -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

104 to 085: consequential handoff

Responsibility and causal attribution -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

104 to 086: consequential handoff

Responsibility and causal attribution -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

104 to 087: consequential handoff

Responsibility and causal attribution -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

104 to 088: consequential handoff

Responsibility and causal attribution -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

104 to 089: consequential handoff

Responsibility and causal attribution -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

104 to 090: consequential handoff

Responsibility and causal attribution -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

104 to 091: consequential handoff

Responsibility and causal attribution -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

104 to 092: consequential handoff

Responsibility and causal attribution -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

104 to 093: consequential handoff

Responsibility and causal attribution -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

104 to 094: consequential handoff

Responsibility and causal attribution -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

104 to 095: consequential handoff

Responsibility and causal attribution -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

104 to 096: consequential handoff

Responsibility and causal attribution -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

104 to 097: consequential handoff

Responsibility and causal attribution -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

104 to 098: consequential handoff

Responsibility and causal attribution -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

104 to 099: consequential handoff

Responsibility and causal attribution -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

104 to 100: consequential handoff

Responsibility and causal attribution -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

104 to 101: consequential handoff

Responsibility and causal attribution -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

104 to 102: consequential handoff

Responsibility and causal attribution -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

104 to 103: consequential handoff

Responsibility and causal attribution -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

104 to 104: consequential handoff

Responsibility and causal attribution -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Responsibility and causal attribution becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 782-788; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

104 to 105: consequential handoff

Responsibility and causal attribution -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

104 to 106: consequential handoff

Responsibility and causal attribution -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

104 to 107: consequential handoff

Responsibility and causal attribution -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

104 to 108: consequential handoff

Responsibility and causal attribution -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

104 to 109: consequential handoff

Responsibility and causal attribution -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

104 to 110: consequential handoff

Responsibility and causal attribution -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

104 to 111: consequential handoff

Responsibility and causal attribution -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

104 to 112: consequential handoff

Responsibility and causal attribution -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

104 to 113: consequential handoff

Responsibility and causal attribution -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

104 to 114: consequential handoff

Responsibility and causal attribution -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

104 to 115: consequential handoff

Responsibility and causal attribution -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

104 to 116: consequential handoff

Responsibility and causal attribution -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

104 to 117: consequential handoff

Responsibility and causal attribution -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

104 to 118: consequential handoff

Responsibility and causal attribution -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

104 to 119: consequential handoff

Responsibility and causal attribution -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

104 to 120: consequential handoff

Responsibility and causal attribution -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Responsibility and causal attribution into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Moral label detached from the action-producing state. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Actor, victims, institutions, community and future actors. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Blinded case reconstruction with preregistered causal criteria' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 782-788 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

104: Responsibility and causal attribution

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Responsibility and causal attribution

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Responsibility and causal attribution is described in the Atlas as: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Responsibility is either erased by context or overstated by ignoring real constraints. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Actor, victims, institutions, community and future actors. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Attribution, safety, restitution and future agency improve without hiding causal contribution. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Blinded case reconstruction with preregistered causal criteria or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Evidence and source reconstruction

Responsibility and causal attribution | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 104: Responsibility and causal attribution; original author status: High-stakes protocol. Native state: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Moral label detached from the action-producing state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Responsibility is either erased by context or overstated by ignoring real constraints. The relevant receiving carriers are: Actor, victims, institutions, community and future actors.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Attribution, safety, restitution and future agency improve without hiding causal contribution.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blinded case reconstruction with preregistered causal criteria. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Boundary divergence

Responsibility and causal attribution | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 104: Responsibility and causal attribution; original author status: High-stakes protocol. Native state: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Moral label detached from the action-producing state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Responsibility is either erased by context or overstated by ignoring real constraints. The relevant receiving carriers are: Actor, victims, institutions, community and future actors.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Attribution, safety, restitution and future agency improve without hiding causal contribution.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blinded case reconstruction with preregistered causal criteria. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Causal diagram and identifiability

Responsibility and causal attribution | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 104: Responsibility and causal attribution; original author status: High-stakes protocol. Native state: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Moral label detached from the action-producing state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Responsibility is either erased by context or overstated by ignoring real constraints. The relevant receiving carriers are: Actor, victims, institutions, community and future actors.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Attribution, safety, restitution and future agency improve without hiding causal contribution.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blinded case reconstruction with preregistered causal criteria. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Receiving carrier identification

Responsibility and causal attribution | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 104: Responsibility and causal attribution; original author status: High-stakes protocol. Native state: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Moral label detached from the action-producing state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Responsibility is either erased by context or overstated by ignoring real constraints. The relevant receiving carriers are: Actor, victims, institutions, community and future actors.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Attribution, safety, restitution and future agency improve without hiding causal contribution.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blinded case reconstruction with preregistered causal criteria. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Hidden compensation accounting

Responsibility and causal attribution | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 104: Responsibility and causal attribution; original author status: High-stakes protocol. Native state: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Moral label detached from the action-producing state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Responsibility is either erased by context or overstated by ignoring real constraints. The relevant receiving carriers are: Actor, victims, institutions, community and future actors.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Attribution, safety, restitution and future agency improve without hiding causal contribution.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blinded case reconstruction with preregistered causal criteria. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Native units and dimensional integrity

Responsibility and causal attribution | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 104: Responsibility and causal attribution; original author status: High-stakes protocol. Native state: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Moral label detached from the action-producing state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Responsibility is either erased by context or overstated by ignoring real constraints. The relevant receiving carriers are: Actor, victims, institutions, community and future actors.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Attribution, safety, restitution and future agency improve without hiding causal contribution.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blinded case reconstruction with preregistered causal criteria. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Capacity specification

Responsibility and causal attribution | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 104: Responsibility and causal attribution; original author status: High-stakes protocol. Native state: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Moral label detached from the action-producing state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Responsibility is either erased by context or overstated by ignoring real constraints. The relevant receiving carriers are: Actor, victims, institutions, community and future actors.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Attribution, safety, restitution and future agency improve without hiding causal contribution.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blinded case reconstruction with preregistered causal criteria. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Intervention design

Responsibility and causal attribution | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 104: Responsibility and causal attribution; original author status: High-stakes protocol. Native state: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Moral label detached from the action-producing state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Responsibility is either erased by context or overstated by ignoring real constraints. The relevant receiving carriers are: Actor, victims, institutions, community and future actors.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Attribution, safety, restitution and future agency improve without hiding causal contribution.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blinded case reconstruction with preregistered causal criteria. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Incumbent parity contest

Responsibility and causal attribution | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 104: Responsibility and causal attribution; original author status: High-stakes protocol. Native state: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Moral label detached from the action-producing state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Responsibility is either erased by context or overstated by ignoring real constraints. The relevant receiving carriers are: Actor, victims, institutions, community and future actors.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Attribution, safety, restitution and future agency improve without hiding causal contribution.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blinded case reconstruction with preregistered causal criteria. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Time-window and recurrence

Responsibility and causal attribution | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 104: Responsibility and causal attribution; original author status: High-stakes protocol. Native state: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Moral label detached from the action-producing state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Responsibility is either erased by context or overstated by ignoring real constraints. The relevant receiving carriers are: Actor, victims, institutions, community and future actors.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Attribution, safety, restitution and future agency improve without hiding causal contribution.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blinded case reconstruction with preregistered causal criteria. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Ethical and legal safety

Responsibility and causal attribution | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 104: Responsibility and causal attribution; original author status: High-stakes protocol. Native state: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Moral label detached from the action-producing state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Responsibility is either erased by context or overstated by ignoring real constraints. The relevant receiving carriers are: Actor, victims, institutions, community and future actors.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Attribution, safety, restitution and future agency improve without hiding causal contribution.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blinded case reconstruction with preregistered causal criteria. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Sensitivity and negative controls

Responsibility and causal attribution | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 104: Responsibility and causal attribution; original author status: High-stakes protocol. Native state: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Moral label detached from the action-producing state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Responsibility is either erased by context or overstated by ignoring real constraints. The relevant receiving carriers are: Actor, victims, institutions, community and future actors.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Attribution, safety, restitution and future agency improve without hiding causal contribution.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blinded case reconstruction with preregistered causal criteria. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Replication and provenance

Responsibility and causal attribution | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 104: Responsibility and causal attribution; original author status: High-stakes protocol. Native state: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Moral label detached from the action-producing state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Responsibility is either erased by context or overstated by ignoring real constraints. The relevant receiving carriers are: Actor, victims, institutions, community and future actors.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Attribution, safety, restitution and future agency improve without hiding causal contribution.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blinded case reconstruction with preregistered causal criteria. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

104: Cross-domain invariant contest

Responsibility and causal attribution | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 104: Responsibility and causal attribution; original author status: High-stakes protocol. Native state: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Moral label detached from the action-producing state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Responsibility is either erased by context or overstated by ignoring real constraints. The relevant receiving carriers are: Actor, victims, institutions, community and future actors.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Attribution, safety, restitution and future agency improve without hiding causal contribution.

RETEST AND SCOPE CONTROL

Candidate decisive method: Blinded case reconstruction with preregistered causal criteria. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 104, pp. 782-788 (Rufino 2026).

Adapter 105: Consent, coercion and autonomy

Conscious agency and responsibility | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity

RECORDED RECEIVING SCHEMA

Signed/expressed consent treated as sufficient state

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Candidate receivers: Decision maker, counterparties, dependents and future self.

EXISTING CAPACITY DESIGN

Real exit, recourse, reversible consent, information symmetry and alternative capacity

PROTECTED OUTCOME STATED BY THE AUTHOR

Agreement quality and autonomy rise together while coercive burden falls

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Prospective consent-quality audit. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105 | Consent, coercion and autonomy

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 105, Consent, coercion and autonomy. Native state: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Potential direct or institutional human pathway: Consent, coercion and autonomy interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Apparent agreement is purchased by narrowed alternatives or irreversible dependency.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Real exit, recourse, reversible consent, information symmetry and alternative capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Agreement quality and autonomy rise together while coercive burden falls. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

105 | Consent, coercion and autonomy

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 105, Consent, coercion and autonomy. Native state: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Potential direct or institutional human pathway: Consent, coercion and autonomy interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Apparent agreement is purchased by narrowed alternatives or irreversible dependency.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Real exit, recourse, reversible consent, information symmetry and alternative capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Agreement quality and autonomy rise together while coercive burden falls. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

105 | Consent, coercion and autonomy

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 105, Consent, coercion and autonomy. Native state: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Potential direct or institutional human pathway: Consent, coercion and autonomy interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Apparent agreement is purchased by narrowed alternatives or irreversible dependency.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Real exit, recourse, reversible consent, information symmetry and alternative capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Agreement quality and autonomy rise together while coercive burden falls. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

105 | Consent, coercion and autonomy

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 105, Consent, coercion and autonomy. Native state: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Potential direct or institutional human pathway: Consent, coercion and autonomy interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Apparent agreement is purchased by narrowed alternatives or irreversible dependency.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Real exit, recourse, reversible consent, information symmetry and alternative capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Agreement quality and autonomy rise together while coercive burden falls. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

105 | Consent, coercion and autonomy

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 105, Consent, coercion and autonomy. Native state: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Potential direct or institutional human pathway: Consent, coercion and autonomy interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Apparent agreement is purchased by narrowed alternatives or irreversible dependency.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Real exit, recourse, reversible consent, information symmetry and alternative capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Agreement quality and autonomy rise together while coercive burden falls. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

105 | Consent, coercion and autonomy

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 105, Consent, coercion and autonomy. Native state: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Potential direct or institutional human pathway: Consent, coercion and autonomy interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Apparent agreement is purchased by narrowed alternatives or irreversible dependency.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Real exit, recourse, reversible consent, information symmetry and alternative capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Agreement quality and autonomy rise together while coercive burden falls. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

105 to 001: consequential handoff

Consent, coercion and autonomy -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

105 to 002: consequential handoff

Consent, coercion and autonomy -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

105 to 003: consequential handoff

Consent, coercion and autonomy -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

105 to 004: consequential handoff

Consent, coercion and autonomy -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

105 to 005: consequential handoff

Consent, coercion and autonomy -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

105 to 006: consequential handoff

Consent, coercion and autonomy -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

105 to 007: consequential handoff

Consent, coercion and autonomy -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

105 to 008: consequential handoff

Consent, coercion and autonomy -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

105 to 009: consequential handoff

Consent, coercion and autonomy -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

105 to 010: consequential handoff

Consent, coercion and autonomy -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

105 to 011: consequential handoff

Consent, coercion and autonomy -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

105 to 012: consequential handoff

Consent, coercion and autonomy -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

105 to 013: consequential handoff

Consent, coercion and autonomy -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

105 to 014: consequential handoff

Consent, coercion and autonomy -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

105 to 015: consequential handoff

Consent, coercion and autonomy -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

105 to 016: consequential handoff

Consent, coercion and autonomy -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

105 to 017: consequential handoff

Consent, coercion and autonomy -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

105 to 018: consequential handoff

Consent, coercion and autonomy -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

105 to 019: consequential handoff

Consent, coercion and autonomy -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

105 to 020: consequential handoff

Consent, coercion and autonomy -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

105 to 021: consequential handoff

Consent, coercion and autonomy -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

105 to 022: consequential handoff

Consent, coercion and autonomy -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

105 to 023: consequential handoff

Consent, coercion and autonomy -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

105 to 024: consequential handoff

Consent, coercion and autonomy -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

105 to 025: consequential handoff

Consent, coercion and autonomy -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

105 to 026: consequential handoff

Consent, coercion and autonomy -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

105 to 027: consequential handoff

Consent, coercion and autonomy -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

105 to 028: consequential handoff

Consent, coercion and autonomy -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

105 to 029: consequential handoff

Consent, coercion and autonomy -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

105 to 030: consequential handoff

Consent, coercion and autonomy -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

105 to 031: consequential handoff

Consent, coercion and autonomy -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

105 to 032: consequential handoff

Consent, coercion and autonomy -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

105 to 033: consequential handoff

Consent, coercion and autonomy -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

105 to 034: consequential handoff

Consent, coercion and autonomy -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

105 to 035: consequential handoff

Consent, coercion and autonomy -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

105 to 036: consequential handoff

Consent, coercion and autonomy -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

105 to 037: consequential handoff

Consent, coercion and autonomy -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

105 to 038: consequential handoff

Consent, coercion and autonomy -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

105 to 039: consequential handoff

Consent, coercion and autonomy -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

105 to 040: consequential handoff

Consent, coercion and autonomy -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

105 to 041: consequential handoff

Consent, coercion and autonomy -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

105 to 042: consequential handoff

Consent, coercion and autonomy -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

105 to 043: consequential handoff

Consent, coercion and autonomy -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

105 to 044: consequential handoff

Consent, coercion and autonomy -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

105 to 045: consequential handoff

Consent, coercion and autonomy -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

105 to 046: consequential handoff

Consent, coercion and autonomy -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

105 to 047: consequential handoff

Consent, coercion and autonomy -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

105 to 048: consequential handoff

Consent, coercion and autonomy -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

105 to 049: consequential handoff

Consent, coercion and autonomy -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

105 to 050: consequential handoff

Consent, coercion and autonomy -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

105 to 051: consequential handoff

Consent, coercion and autonomy -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

105 to 052: consequential handoff

Consent, coercion and autonomy -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

105 to 053: consequential handoff

Consent, coercion and autonomy -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

105 to 054: consequential handoff

Consent, coercion and autonomy -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

105 to 055: consequential handoff

Consent, coercion and autonomy -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

105 to 056: consequential handoff

Consent, coercion and autonomy -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

105 to 057: consequential handoff

Consent, coercion and autonomy -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

105 to 058: consequential handoff

Consent, coercion and autonomy -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

105 to 059: consequential handoff

Consent, coercion and autonomy -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

105 to 060: consequential handoff

Consent, coercion and autonomy -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

105 to 061: consequential handoff

Consent, coercion and autonomy -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

105 to 062: consequential handoff

Consent, coercion and autonomy -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

105 to 063: consequential handoff

Consent, coercion and autonomy -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

105 to 064: consequential handoff

Consent, coercion and autonomy -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

105 to 065: consequential handoff

Consent, coercion and autonomy -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

105 to 066: consequential handoff

Consent, coercion and autonomy -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

105 to 067: consequential handoff

Consent, coercion and autonomy -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

105 to 068: consequential handoff

Consent, coercion and autonomy -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

105 to 069: consequential handoff

Consent, coercion and autonomy -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

105 to 070: consequential handoff

Consent, coercion and autonomy -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

105 to 071: consequential handoff

Consent, coercion and autonomy -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

105 to 072: consequential handoff

Consent, coercion and autonomy -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

105 to 073: consequential handoff

Consent, coercion and autonomy -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

105 to 074: consequential handoff

Consent, coercion and autonomy -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

105 to 075: consequential handoff

Consent, coercion and autonomy -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

105 to 076: consequential handoff

Consent, coercion and autonomy -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

105 to 077: consequential handoff

Consent, coercion and autonomy -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

105 to 078: consequential handoff

Consent, coercion and autonomy -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

105 to 079: consequential handoff

Consent, coercion and autonomy -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

105 to 080: consequential handoff

Consent, coercion and autonomy -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

105 to 081: consequential handoff

Consent, coercion and autonomy -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

105 to 082: consequential handoff

Consent, coercion and autonomy -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

105 to 083: consequential handoff

Consent, coercion and autonomy -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

105 to 084: consequential handoff

Consent, coercion and autonomy -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

105 to 085: consequential handoff

Consent, coercion and autonomy -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

105 to 086: consequential handoff

Consent, coercion and autonomy -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

105 to 087: consequential handoff

Consent, coercion and autonomy -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

105 to 088: consequential handoff

Consent, coercion and autonomy -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

105 to 089: consequential handoff

Consent, coercion and autonomy -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

105 to 090: consequential handoff

Consent, coercion and autonomy -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

105 to 091: consequential handoff

Consent, coercion and autonomy -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

105 to 092: consequential handoff

Consent, coercion and autonomy -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

105 to 093: consequential handoff

Consent, coercion and autonomy -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

105 to 094: consequential handoff

Consent, coercion and autonomy -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

105 to 095: consequential handoff

Consent, coercion and autonomy -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

105 to 096: consequential handoff

Consent, coercion and autonomy -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

105 to 097: consequential handoff

Consent, coercion and autonomy -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

105 to 098: consequential handoff

Consent, coercion and autonomy -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

105 to 099: consequential handoff

Consent, coercion and autonomy -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

105 to 100: consequential handoff

Consent, coercion and autonomy -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

105 to 101: consequential handoff

Consent, coercion and autonomy -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

105 to 102: consequential handoff

Consent, coercion and autonomy -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

105 to 103: consequential handoff

Consent, coercion and autonomy -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

105 to 104: consequential handoff

Consent, coercion and autonomy -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

105 to 105: consequential handoff

Consent, coercion and autonomy -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Consent, coercion and autonomy becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 789-795; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

105 to 106: consequential handoff

Consent, coercion and autonomy -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

105 to 107: consequential handoff

Consent, coercion and autonomy -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

105 to 108: consequential handoff

Consent, coercion and autonomy -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

105 to 109: consequential handoff

Consent, coercion and autonomy -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

105 to 110: consequential handoff

Consent, coercion and autonomy -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

105 to 111: consequential handoff

Consent, coercion and autonomy -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

105 to 112: consequential handoff

Consent, coercion and autonomy -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

105 to 113: consequential handoff

Consent, coercion and autonomy -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

105 to 114: consequential handoff

Consent, coercion and autonomy -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

105 to 115: consequential handoff

Consent, coercion and autonomy -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

105 to 116: consequential handoff

Consent, coercion and autonomy -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

105 to 117: consequential handoff

Consent, coercion and autonomy -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

105 to 118: consequential handoff

Consent, coercion and autonomy -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

105 to 119: consequential handoff

Consent, coercion and autonomy -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

105 to 120: consequential handoff

Consent, coercion and autonomy -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Consent, coercion and autonomy into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Signed/expressed consent treated as sufficient state. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Decision maker, counterparties, dependents and future self. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective consent-quality audit' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 789-795 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

105: Consent, coercion and autonomy

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Consent, coercion and autonomy

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Consent, coercion and autonomy is described in the Atlas as: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Apparent agreement is purchased by narrowed alternatives or irreversible dependency. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Decision maker, counterparties, dependents and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Agreement quality and autonomy rise together while coercive burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective consent-quality audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Evidence and source reconstruction

Consent, coercion and autonomy | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 105: Consent, coercion and autonomy; original author status: High-stakes protocol. Native state: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Signed/expressed consent treated as sufficient state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Apparent agreement is purchased by narrowed alternatives or irreversible dependency. The relevant receiving carriers are: Decision maker, counterparties, dependents and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agreement quality and autonomy rise together while coercive burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective consent-quality audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Boundary divergence

Consent, coercion and autonomy | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 105: Consent, coercion and autonomy; original author status: High-stakes protocol. Native state: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Signed/expressed consent treated as sufficient state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Apparent agreement is purchased by narrowed alternatives or irreversible dependency. The relevant receiving carriers are: Decision maker, counterparties, dependents and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agreement quality and autonomy rise together while coercive burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective consent-quality audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Causal diagram and identifiability

Consent, coercion and autonomy | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 105: Consent, coercion and autonomy; original author status: High-stakes protocol. Native state: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Signed/expressed consent treated as sufficient state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Apparent agreement is purchased by narrowed alternatives or irreversible dependency. The relevant receiving carriers are: Decision maker, counterparties, dependents and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agreement quality and autonomy rise together while coercive burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective consent-quality audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Receiving carrier identification

Consent, coercion and autonomy | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 105: Consent, coercion and autonomy; original author status: High-stakes protocol. Native state: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Signed/expressed consent treated as sufficient state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Apparent agreement is purchased by narrowed alternatives or irreversible dependency. The relevant receiving carriers are: Decision maker, counterparties, dependents and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agreement quality and autonomy rise together while coercive burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective consent-quality audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Hidden compensation accounting

Consent, coercion and autonomy | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 105: Consent, coercion and autonomy; original author status: High-stakes protocol. Native state: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Signed/expressed consent treated as sufficient state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Apparent agreement is purchased by narrowed alternatives or irreversible dependency. The relevant receiving carriers are: Decision maker, counterparties, dependents and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agreement quality and autonomy rise together while coercive burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective consent-quality audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Native units and dimensional integrity

Consent, coercion and autonomy | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 105: Consent, coercion and autonomy; original author status: High-stakes protocol. Native state: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Signed/expressed consent treated as sufficient state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Apparent agreement is purchased by narrowed alternatives or irreversible dependency. The relevant receiving carriers are: Decision maker, counterparties, dependents and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agreement quality and autonomy rise together while coercive burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective consent-quality audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Capacity specification

Consent, coercion and autonomy | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 105: Consent, coercion and autonomy; original author status: High-stakes protocol. Native state: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Signed/expressed consent treated as sufficient state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Apparent agreement is purchased by narrowed alternatives or irreversible dependency. The relevant receiving carriers are: Decision maker, counterparties, dependents and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agreement quality and autonomy rise together while coercive burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective consent-quality audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Intervention design

Consent, coercion and autonomy | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 105: Consent, coercion and autonomy; original author status: High-stakes protocol. Native state: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Signed/expressed consent treated as sufficient state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Apparent agreement is purchased by narrowed alternatives or irreversible dependency. The relevant receiving carriers are: Decision maker, counterparties, dependents and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agreement quality and autonomy rise together while coercive burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective consent-quality audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Incumbent parity contest

Consent, coercion and autonomy | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 105: Consent, coercion and autonomy; original author status: High-stakes protocol. Native state: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Signed/expressed consent treated as sufficient state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Apparent agreement is purchased by narrowed alternatives or irreversible dependency. The relevant receiving carriers are: Decision maker, counterparties, dependents and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agreement quality and autonomy rise together while coercive burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective consent-quality audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Time-window and recurrence

Consent, coercion and autonomy | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 105: Consent, coercion and autonomy; original author status: High-stakes protocol. Native state: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Signed/expressed consent treated as sufficient state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Apparent agreement is purchased by narrowed alternatives or irreversible dependency. The relevant receiving carriers are: Decision maker, counterparties, dependents and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agreement quality and autonomy rise together while coercive burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective consent-quality audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Ethical and legal safety

Consent, coercion and autonomy | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 105: Consent, coercion and autonomy; original author status: High-stakes protocol. Native state: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Signed/expressed consent treated as sufficient state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Apparent agreement is purchased by narrowed alternatives or irreversible dependency. The relevant receiving carriers are: Decision maker, counterparties, dependents and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agreement quality and autonomy rise together while coercive burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective consent-quality audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Sensitivity and negative controls

Consent, coercion and autonomy | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 105: Consent, coercion and autonomy; original author status: High-stakes protocol. Native state: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Signed/expressed consent treated as sufficient state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Apparent agreement is purchased by narrowed alternatives or irreversible dependency. The relevant receiving carriers are: Decision maker, counterparties, dependents and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agreement quality and autonomy rise together while coercive burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective consent-quality audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Replication and provenance

Consent, coercion and autonomy | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 105: Consent, coercion and autonomy; original author status: High-stakes protocol. Native state: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Signed/expressed consent treated as sufficient state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Apparent agreement is purchased by narrowed alternatives or irreversible dependency. The relevant receiving carriers are: Decision maker, counterparties, dependents and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agreement quality and autonomy rise together while coercive burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective consent-quality audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

105: Cross-domain invariant contest

Consent, coercion and autonomy | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 105: Consent, coercion and autonomy; original author status: High-stakes protocol. Native state: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Signed/expressed consent treated as sufficient state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Apparent agreement is purchased by narrowed alternatives or irreversible dependency. The relevant receiving carriers are: Decision maker, counterparties, dependents and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Agreement quality and autonomy rise together while coercive burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective consent-quality audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 105, pp. 789-795 (Rufino 2026).

Adapter 106: Addiction, compulsion and short-horizon relief

Conscious agency and responsibility | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set

RECORDED RECEIVING SCHEMA

Substance/behavior category or abstinence binary

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Candidate receivers: Person, body, relationships, household, community and future self.

EXISTING CAPACITY DESIGN

Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden

PROTECTED OUTCOME STATED BY THE AUTHOR

Relief, agency, health and social function improve while dependency burden falls

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Longitudinal regulated intervention with multi-carrier outcomes. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106 | Addiction, compulsion and short-horizon relief

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 106, Addiction, compulsion and short-horizon relief. Native state: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Potential direct or institutional human pathway: Addiction, compulsion and short-horizon relief interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Relief, agency, health and social function improve while dependency burden falls. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

106 | Addiction, compulsion and short-horizon relief

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 106, Addiction, compulsion and short-horizon relief. Native state: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Potential direct or institutional human pathway: Addiction, compulsion and short-horizon relief interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Relief, agency, health and social function improve while dependency burden falls. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

106 | Addiction, compulsion and short-horizon relief

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 106, Addiction, compulsion and short-horizon relief. Native state: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Potential direct or institutional human pathway: Addiction, compulsion and short-horizon relief interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Relief, agency, health and social function improve while dependency burden falls. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

106 | Addiction, compulsion and short-horizon relief

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 106, Addiction, compulsion and short-horizon relief. Native state: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Potential direct or institutional human pathway: Addiction, compulsion and short-horizon relief interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Relief, agency, health and social function improve while dependency burden falls. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

106 | Addiction, compulsion and short-horizon relief

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 106, Addiction, compulsion and short-horizon relief. Native state: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Potential direct or institutional human pathway: Addiction, compulsion and short-horizon relief interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Relief, agency, health and social function improve while dependency burden falls. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

106 | Addiction, compulsion and short-horizon relief

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 106, Addiction, compulsion and short-horizon relief. Native state: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Potential direct or institutional human pathway: Addiction, compulsion and short-horizon relief interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Relief, agency, health and social function improve while dependency burden falls. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

106 to 001: consequential handoff

Addiction, compulsion and short-horizon relief -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

106 to 002: consequential handoff

Addiction, compulsion and short-horizon relief -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

106 to 003: consequential handoff

Addiction, compulsion and short-horizon relief -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

106 to 004: consequential handoff

Addiction, compulsion and short-horizon relief -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

106 to 005: consequential handoff

Addiction, compulsion and short-horizon relief -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

106 to 006: consequential handoff

Addiction, compulsion and short-horizon relief -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

106 to 007: consequential handoff

Addiction, compulsion and short-horizon relief -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

106 to 008: consequential handoff

Addiction, compulsion and short-horizon relief -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

106 to 009: consequential handoff

Addiction, compulsion and short-horizon relief -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

106 to 010: consequential handoff

Addiction, compulsion and short-horizon relief -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

106 to 011: consequential handoff

Addiction, compulsion and short-horizon relief -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

106 to 012: consequential handoff

Addiction, compulsion and short-horizon relief -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

106 to 013: consequential handoff

Addiction, compulsion and short-horizon relief -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

106 to 014: consequential handoff

Addiction, compulsion and short-horizon relief -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

106 to 015: consequential handoff

Addiction, compulsion and short-horizon relief -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

106 to 016: consequential handoff

Addiction, compulsion and short-horizon relief -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

106 to 017: consequential handoff

Addiction, compulsion and short-horizon relief -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

106 to 018: consequential handoff

Addiction, compulsion and short-horizon relief -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

106 to 019: consequential handoff

Addiction, compulsion and short-horizon relief -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

106 to 020: consequential handoff

Addiction, compulsion and short-horizon relief -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

106 to 021: consequential handoff

Addiction, compulsion and short-horizon relief -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

106 to 022: consequential handoff

Addiction, compulsion and short-horizon relief -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

106 to 023: consequential handoff

Addiction, compulsion and short-horizon relief -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

106 to 024: consequential handoff

Addiction, compulsion and short-horizon relief -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

106 to 025: consequential handoff

Addiction, compulsion and short-horizon relief -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

106 to 026: consequential handoff

Addiction, compulsion and short-horizon relief -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

106 to 027: consequential handoff

Addiction, compulsion and short-horizon relief -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

106 to 028: consequential handoff

Addiction, compulsion and short-horizon relief -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

106 to 029: consequential handoff

Addiction, compulsion and short-horizon relief -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

106 to 030: consequential handoff

Addiction, compulsion and short-horizon relief -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

106 to 031: consequential handoff

Addiction, compulsion and short-horizon relief -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

106 to 032: consequential handoff

Addiction, compulsion and short-horizon relief -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

106 to 033: consequential handoff

Addiction, compulsion and short-horizon relief -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

106 to 034: consequential handoff

Addiction, compulsion and short-horizon relief -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

106 to 035: consequential handoff

Addiction, compulsion and short-horizon relief -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

106 to 036: consequential handoff

Addiction, compulsion and short-horizon relief -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

106 to 037: consequential handoff

Addiction, compulsion and short-horizon relief -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

106 to 038: consequential handoff

Addiction, compulsion and short-horizon relief -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

106 to 039: consequential handoff

Addiction, compulsion and short-horizon relief -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

106 to 040: consequential handoff

Addiction, compulsion and short-horizon relief -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

106 to 041: consequential handoff

Addiction, compulsion and short-horizon relief -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

106 to 042: consequential handoff

Addiction, compulsion and short-horizon relief -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

106 to 043: consequential handoff

Addiction, compulsion and short-horizon relief -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

106 to 044: consequential handoff

Addiction, compulsion and short-horizon relief -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

106 to 045: consequential handoff

Addiction, compulsion and short-horizon relief -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

106 to 046: consequential handoff

Addiction, compulsion and short-horizon relief -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

106 to 047: consequential handoff

Addiction, compulsion and short-horizon relief -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

106 to 048: consequential handoff

Addiction, compulsion and short-horizon relief -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

106 to 049: consequential handoff

Addiction, compulsion and short-horizon relief -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

106 to 050: consequential handoff

Addiction, compulsion and short-horizon relief -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

106 to 051: consequential handoff

Addiction, compulsion and short-horizon relief -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

106 to 052: consequential handoff

Addiction, compulsion and short-horizon relief -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

106 to 053: consequential handoff

Addiction, compulsion and short-horizon relief -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

106 to 054: consequential handoff

Addiction, compulsion and short-horizon relief -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

106 to 055: consequential handoff

Addiction, compulsion and short-horizon relief -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

106 to 056: consequential handoff

Addiction, compulsion and short-horizon relief -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

106 to 057: consequential handoff

Addiction, compulsion and short-horizon relief -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

106 to 058: consequential handoff

Addiction, compulsion and short-horizon relief -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

106 to 059: consequential handoff

Addiction, compulsion and short-horizon relief -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

106 to 060: consequential handoff

Addiction, compulsion and short-horizon relief -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

106 to 061: consequential handoff

Addiction, compulsion and short-horizon relief -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

106 to 062: consequential handoff

Addiction, compulsion and short-horizon relief -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

106 to 063: consequential handoff

Addiction, compulsion and short-horizon relief -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

106 to 064: consequential handoff

Addiction, compulsion and short-horizon relief -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

106 to 065: consequential handoff

Addiction, compulsion and short-horizon relief -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

106 to 066: consequential handoff

Addiction, compulsion and short-horizon relief -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

106 to 067: consequential handoff

Addiction, compulsion and short-horizon relief -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

106 to 068: consequential handoff

Addiction, compulsion and short-horizon relief -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

106 to 069: consequential handoff

Addiction, compulsion and short-horizon relief -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

106 to 070: consequential handoff

Addiction, compulsion and short-horizon relief -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

106 to 071: consequential handoff

Addiction, compulsion and short-horizon relief -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

106 to 072: consequential handoff

Addiction, compulsion and short-horizon relief -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

106 to 073: consequential handoff

Addiction, compulsion and short-horizon relief -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

106 to 074: consequential handoff

Addiction, compulsion and short-horizon relief -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

106 to 075: consequential handoff

Addiction, compulsion and short-horizon relief -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

106 to 076: consequential handoff

Addiction, compulsion and short-horizon relief -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

106 to 077: consequential handoff

Addiction, compulsion and short-horizon relief -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

106 to 078: consequential handoff

Addiction, compulsion and short-horizon relief -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

106 to 079: consequential handoff

Addiction, compulsion and short-horizon relief -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

106 to 080: consequential handoff

Addiction, compulsion and short-horizon relief -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

106 to 081: consequential handoff

Addiction, compulsion and short-horizon relief -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

106 to 082: consequential handoff

Addiction, compulsion and short-horizon relief -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

106 to 083: consequential handoff

Addiction, compulsion and short-horizon relief -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

106 to 084: consequential handoff

Addiction, compulsion and short-horizon relief -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

106 to 085: consequential handoff

Addiction, compulsion and short-horizon relief -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

106 to 086: consequential handoff

Addiction, compulsion and short-horizon relief -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

106 to 087: consequential handoff

Addiction, compulsion and short-horizon relief -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

106 to 088: consequential handoff

Addiction, compulsion and short-horizon relief -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

106 to 089: consequential handoff

Addiction, compulsion and short-horizon relief -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

106 to 090: consequential handoff

Addiction, compulsion and short-horizon relief -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

106 to 091: consequential handoff

Addiction, compulsion and short-horizon relief -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

106 to 092: consequential handoff

Addiction, compulsion and short-horizon relief -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

106 to 093: consequential handoff

Addiction, compulsion and short-horizon relief -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

106 to 094: consequential handoff

Addiction, compulsion and short-horizon relief -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

106 to 095: consequential handoff

Addiction, compulsion and short-horizon relief -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

106 to 096: consequential handoff

Addiction, compulsion and short-horizon relief -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

106 to 097: consequential handoff

Addiction, compulsion and short-horizon relief -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

106 to 098: consequential handoff

Addiction, compulsion and short-horizon relief -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

106 to 099: consequential handoff

Addiction, compulsion and short-horizon relief -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

106 to 100: consequential handoff

Addiction, compulsion and short-horizon relief -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

106 to 101: consequential handoff

Addiction, compulsion and short-horizon relief -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

106 to 102: consequential handoff

Addiction, compulsion and short-horizon relief -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

106 to 103: consequential handoff

Addiction, compulsion and short-horizon relief -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

106 to 104: consequential handoff

Addiction, compulsion and short-horizon relief -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

106 to 105: consequential handoff

Addiction, compulsion and short-horizon relief -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

106 to 106: consequential handoff

Addiction, compulsion and short-horizon relief -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Addiction, compulsion and short-horizon relief becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 796-802; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

106 to 107: consequential handoff

Addiction, compulsion and short-horizon relief -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

106 to 108: consequential handoff

Addiction, compulsion and short-horizon relief -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

106 to 109: consequential handoff

Addiction, compulsion and short-horizon relief -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

106 to 110: consequential handoff

Addiction, compulsion and short-horizon relief -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

106 to 111: consequential handoff

Addiction, compulsion and short-horizon relief -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

106 to 112: consequential handoff

Addiction, compulsion and short-horizon relief -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

106 to 113: consequential handoff

Addiction, compulsion and short-horizon relief -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

106 to 114: consequential handoff

Addiction, compulsion and short-horizon relief -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

106 to 115: consequential handoff

Addiction, compulsion and short-horizon relief -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

106 to 116: consequential handoff

Addiction, compulsion and short-horizon relief -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

106 to 117: consequential handoff

Addiction, compulsion and short-horizon relief -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

106 to 118: consequential handoff

Addiction, compulsion and short-horizon relief -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

106 to 119: consequential handoff

Addiction, compulsion and short-horizon relief -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

106 to 120: consequential handoff

Addiction, compulsion and short-horizon relief -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Addiction, compulsion and short-horizon relief into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Substance/behavior category or abstinence binary. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Person, body, relationships, household, community and future self. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal regulated intervention with multi-carrier outcomes' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 796-802 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

106: Addiction, compulsion and short-horizon relief

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

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Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

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MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

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Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

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CONSEQUENTIAL SUCCESS MEASUREMENTS

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Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

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CONSEQUENTIAL SUCCESS MEASUREMENTS

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Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

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Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

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MANDATORY COUNTERFACTUAL AND SAFETY

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Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

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MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Relief, agency, health and social function improve while dependency burden falls. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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106: Addiction, compulsion and short-horizon relief

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

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Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

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LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Person, body, relationships, household, community and future self. Identify pathways and hidden compensation using credible data rather than narrative fit.

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MANDATORY COUNTERFACTUAL AND SAFETY

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106: Addiction, compulsion and short-horizon relief

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

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Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Addiction, compulsion and short-horizon relief

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Addiction, compulsion and short-horizon relief is described in the Atlas as: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

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Use Longitudinal regulated intervention with multi-carrier outcomes or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Evidence and source reconstruction

Addiction, compulsion and short-horizon relief | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 106: Addiction, compulsion and short-horizon relief; original author status: Regulated empirical frontier. Native state: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Substance/behavior category or abstinence binary. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. The relevant receiving carriers are: Person, body, relationships, household, community and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Relief, agency, health and social function improve while dependency burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal regulated intervention with multi-carrier outcomes. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Boundary divergence

Addiction, compulsion and short-horizon relief | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 106: Addiction, compulsion and short-horizon relief; original author status: Regulated empirical frontier. Native state: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Substance/behavior category or abstinence binary. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. The relevant receiving carriers are: Person, body, relationships, household, community and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Relief, agency, health and social function improve while dependency burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal regulated intervention with multi-carrier outcomes. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Causal diagram and identifiability

Addiction, compulsion and short-horizon relief | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 106: Addiction, compulsion and short-horizon relief; original author status: Regulated empirical frontier. Native state: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Substance/behavior category or abstinence binary. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. The relevant receiving carriers are: Person, body, relationships, household, community and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Relief, agency, health and social function improve while dependency burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal regulated intervention with multi-carrier outcomes. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Receiving carrier identification

Addiction, compulsion and short-horizon relief | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 106: Addiction, compulsion and short-horizon relief; original author status: Regulated empirical frontier. Native state: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Substance/behavior category or abstinence binary. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. The relevant receiving carriers are: Person, body, relationships, household, community and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Relief, agency, health and social function improve while dependency burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal regulated intervention with multi-carrier outcomes. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Hidden compensation accounting

Addiction, compulsion and short-horizon relief | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 106: Addiction, compulsion and short-horizon relief; original author status: Regulated empirical frontier. Native state: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Substance/behavior category or abstinence binary. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. The relevant receiving carriers are: Person, body, relationships, household, community and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Relief, agency, health and social function improve while dependency burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal regulated intervention with multi-carrier outcomes. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Native units and dimensional integrity

Addiction, compulsion and short-horizon relief | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 106: Addiction, compulsion and short-horizon relief; original author status: Regulated empirical frontier. Native state: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Substance/behavior category or abstinence binary. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. The relevant receiving carriers are: Person, body, relationships, household, community and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Relief, agency, health and social function improve while dependency burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal regulated intervention with multi-carrier outcomes. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Capacity specification

Addiction, compulsion and short-horizon relief | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 106: Addiction, compulsion and short-horizon relief; original author status: Regulated empirical frontier. Native state: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Substance/behavior category or abstinence binary. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. The relevant receiving carriers are: Person, body, relationships, household, community and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Relief, agency, health and social function improve while dependency burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal regulated intervention with multi-carrier outcomes. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Intervention design

Addiction, compulsion and short-horizon relief | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 106: Addiction, compulsion and short-horizon relief; original author status: Regulated empirical frontier. Native state: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Substance/behavior category or abstinence binary. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. The relevant receiving carriers are: Person, body, relationships, household, community and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Relief, agency, health and social function improve while dependency burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal regulated intervention with multi-carrier outcomes. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Incumbent parity contest

Addiction, compulsion and short-horizon relief | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 106: Addiction, compulsion and short-horizon relief; original author status: Regulated empirical frontier. Native state: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Substance/behavior category or abstinence binary. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. The relevant receiving carriers are: Person, body, relationships, household, community and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Relief, agency, health and social function improve while dependency burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal regulated intervention with multi-carrier outcomes. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Time-window and recurrence

Addiction, compulsion and short-horizon relief | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 106: Addiction, compulsion and short-horizon relief; original author status: Regulated empirical frontier. Native state: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Substance/behavior category or abstinence binary. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. The relevant receiving carriers are: Person, body, relationships, household, community and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Relief, agency, health and social function improve while dependency burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal regulated intervention with multi-carrier outcomes. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Ethical and legal safety

Addiction, compulsion and short-horizon relief | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 106: Addiction, compulsion and short-horizon relief; original author status: Regulated empirical frontier. Native state: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Substance/behavior category or abstinence binary. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. The relevant receiving carriers are: Person, body, relationships, household, community and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Relief, agency, health and social function improve while dependency burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal regulated intervention with multi-carrier outcomes. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Sensitivity and negative controls

Addiction, compulsion and short-horizon relief | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 106: Addiction, compulsion and short-horizon relief; original author status: Regulated empirical frontier. Native state: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Substance/behavior category or abstinence binary. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. The relevant receiving carriers are: Person, body, relationships, household, community and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Relief, agency, health and social function improve while dependency burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal regulated intervention with multi-carrier outcomes. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Replication and provenance

Addiction, compulsion and short-horizon relief | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 106: Addiction, compulsion and short-horizon relief; original author status: Regulated empirical frontier. Native state: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Substance/behavior category or abstinence binary. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. The relevant receiving carriers are: Person, body, relationships, household, community and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Relief, agency, health and social function improve while dependency burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal regulated intervention with multi-carrier outcomes. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

106: Cross-domain invariant contest

Addiction, compulsion and short-horizon relief | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 106: Addiction, compulsion and short-horizon relief; original author status: Regulated empirical frontier. Native state: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Substance/behavior category or abstinence binary. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A behavior successfully absorbs immediate burden while narrowing future agency and exporting cost. The relevant receiving carriers are: Person, body, relationships, household, community and future self.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Relief, agency, health and social function improve while dependency burden falls.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal regulated intervention with multi-carrier outcomes. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 106, pp. 796-802 (Rufino 2026).

Adapter 107: Crime, violence and reconciliation

Conscious agency and responsibility | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm

RECORDED RECEIVING SCHEMA

Offense label or actor-only risk score

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Candidate receivers: Victims, actor, families, community, institutions and future potential victims.

EXISTING CAPACITY DESIGN

Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger

PROTECTED OUTCOME STATED BY THE AUTHOR

Victim safety, accountability, recurrence, repair and future lawful agency improve together

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Longitudinal controlled justice-program evaluation. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107 | Crime, violence and reconciliation

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 107, Crime, violence and reconciliation. Native state: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Potential direct or institutional human pathway: Crime, violence and reconciliation interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

107 | Crime, violence and reconciliation

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 107, Crime, violence and reconciliation. Native state: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Potential direct or institutional human pathway: Crime, violence and reconciliation interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

107 | Crime, violence and reconciliation

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 107, Crime, violence and reconciliation. Native state: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Potential direct or institutional human pathway: Crime, violence and reconciliation interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

107 | Crime, violence and reconciliation

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 107, Crime, violence and reconciliation. Native state: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Potential direct or institutional human pathway: Crime, violence and reconciliation interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

107 | Crime, violence and reconciliation

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 107, Crime, violence and reconciliation. Native state: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Potential direct or institutional human pathway: Crime, violence and reconciliation interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

107 | Crime, violence and reconciliation

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 107, Crime, violence and reconciliation. Native state: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Potential direct or institutional human pathway: Crime, violence and reconciliation interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

107 to 001: consequential handoff

Crime, violence and reconciliation -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

107 to 002: consequential handoff

Crime, violence and reconciliation -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

107 to 003: consequential handoff

Crime, violence and reconciliation -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

107 to 004: consequential handoff

Crime, violence and reconciliation -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

107 to 005: consequential handoff

Crime, violence and reconciliation -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

107 to 006: consequential handoff

Crime, violence and reconciliation -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

107 to 007: consequential handoff

Crime, violence and reconciliation -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

107 to 008: consequential handoff

Crime, violence and reconciliation -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

107 to 009: consequential handoff

Crime, violence and reconciliation -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

107 to 010: consequential handoff

Crime, violence and reconciliation -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

107 to 011: consequential handoff

Crime, violence and reconciliation -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

107 to 012: consequential handoff

Crime, violence and reconciliation -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

107 to 013: consequential handoff

Crime, violence and reconciliation -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

107 to 014: consequential handoff

Crime, violence and reconciliation -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

107 to 015: consequential handoff

Crime, violence and reconciliation -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

107 to 016: consequential handoff

Crime, violence and reconciliation -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

107 to 017: consequential handoff

Crime, violence and reconciliation -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

107 to 018: consequential handoff

Crime, violence and reconciliation -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

107 to 019: consequential handoff

Crime, violence and reconciliation -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

107 to 020: consequential handoff

Crime, violence and reconciliation -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

107 to 021: consequential handoff

Crime, violence and reconciliation -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

107 to 022: consequential handoff

Crime, violence and reconciliation -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

107 to 023: consequential handoff

Crime, violence and reconciliation -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

107 to 024: consequential handoff

Crime, violence and reconciliation -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

107 to 025: consequential handoff

Crime, violence and reconciliation -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

107 to 026: consequential handoff

Crime, violence and reconciliation -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

107 to 027: consequential handoff

Crime, violence and reconciliation -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

107 to 028: consequential handoff

Crime, violence and reconciliation -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

107 to 029: consequential handoff

Crime, violence and reconciliation -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

107 to 030: consequential handoff

Crime, violence and reconciliation -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

107 to 031: consequential handoff

Crime, violence and reconciliation -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

107 to 032: consequential handoff

Crime, violence and reconciliation -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

107 to 033: consequential handoff

Crime, violence and reconciliation -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

107 to 034: consequential handoff

Crime, violence and reconciliation -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

107 to 035: consequential handoff

Crime, violence and reconciliation -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

107 to 036: consequential handoff

Crime, violence and reconciliation -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

107 to 037: consequential handoff

Crime, violence and reconciliation -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

107 to 038: consequential handoff

Crime, violence and reconciliation -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

107 to 039: consequential handoff

Crime, violence and reconciliation -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

107 to 040: consequential handoff

Crime, violence and reconciliation -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

107 to 041: consequential handoff

Crime, violence and reconciliation -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

107 to 042: consequential handoff

Crime, violence and reconciliation -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

107 to 043: consequential handoff

Crime, violence and reconciliation -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

107 to 044: consequential handoff

Crime, violence and reconciliation -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

107 to 045: consequential handoff

Crime, violence and reconciliation -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

107 to 046: consequential handoff

Crime, violence and reconciliation -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

107 to 047: consequential handoff

Crime, violence and reconciliation -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

107 to 048: consequential handoff

Crime, violence and reconciliation -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

107 to 049: consequential handoff

Crime, violence and reconciliation -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

107 to 050: consequential handoff

Crime, violence and reconciliation -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

107 to 051: consequential handoff

Crime, violence and reconciliation -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

107 to 052: consequential handoff

Crime, violence and reconciliation -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

107 to 053: consequential handoff

Crime, violence and reconciliation -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

107 to 054: consequential handoff

Crime, violence and reconciliation -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

107 to 055: consequential handoff

Crime, violence and reconciliation -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

107 to 056: consequential handoff

Crime, violence and reconciliation -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

107 to 057: consequential handoff

Crime, violence and reconciliation -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

107 to 058: consequential handoff

Crime, violence and reconciliation -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

107 to 059: consequential handoff

Crime, violence and reconciliation -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

107 to 060: consequential handoff

Crime, violence and reconciliation -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

107 to 061: consequential handoff

Crime, violence and reconciliation -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

107 to 062: consequential handoff

Crime, violence and reconciliation -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

107 to 063: consequential handoff

Crime, violence and reconciliation -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

107 to 064: consequential handoff

Crime, violence and reconciliation -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

107 to 065: consequential handoff

Crime, violence and reconciliation -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

107 to 066: consequential handoff

Crime, violence and reconciliation -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

107 to 067: consequential handoff

Crime, violence and reconciliation -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

107 to 068: consequential handoff

Crime, violence and reconciliation -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

107 to 069: consequential handoff

Crime, violence and reconciliation -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

107 to 070: consequential handoff

Crime, violence and reconciliation -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

107 to 071: consequential handoff

Crime, violence and reconciliation -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

107 to 072: consequential handoff

Crime, violence and reconciliation -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

107 to 073: consequential handoff

Crime, violence and reconciliation -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

107 to 074: consequential handoff

Crime, violence and reconciliation -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

107 to 075: consequential handoff

Crime, violence and reconciliation -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

107 to 076: consequential handoff

Crime, violence and reconciliation -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

107 to 077: consequential handoff

Crime, violence and reconciliation -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

107 to 078: consequential handoff

Crime, violence and reconciliation -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

107 to 079: consequential handoff

Crime, violence and reconciliation -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

107 to 080: consequential handoff

Crime, violence and reconciliation -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

107 to 081: consequential handoff

Crime, violence and reconciliation -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

107 to 082: consequential handoff

Crime, violence and reconciliation -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

107 to 083: consequential handoff

Crime, violence and reconciliation -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

107 to 084: consequential handoff

Crime, violence and reconciliation -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

107 to 085: consequential handoff

Crime, violence and reconciliation -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

107 to 086: consequential handoff

Crime, violence and reconciliation -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

107 to 087: consequential handoff

Crime, violence and reconciliation -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

107 to 088: consequential handoff

Crime, violence and reconciliation -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

107 to 089: consequential handoff

Crime, violence and reconciliation -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

107 to 090: consequential handoff

Crime, violence and reconciliation -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

107 to 091: consequential handoff

Crime, violence and reconciliation -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

107 to 092: consequential handoff

Crime, violence and reconciliation -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

107 to 093: consequential handoff

Crime, violence and reconciliation -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

107 to 094: consequential handoff

Crime, violence and reconciliation -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

107 to 095: consequential handoff

Crime, violence and reconciliation -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

107 to 096: consequential handoff

Crime, violence and reconciliation -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

107 to 097: consequential handoff

Crime, violence and reconciliation -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

107 to 098: consequential handoff

Crime, violence and reconciliation -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

107 to 099: consequential handoff

Crime, violence and reconciliation -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

107 to 100: consequential handoff

Crime, violence and reconciliation -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

107 to 101: consequential handoff

Crime, violence and reconciliation -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

107 to 102: consequential handoff

Crime, violence and reconciliation -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

107 to 103: consequential handoff

Crime, violence and reconciliation -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

107 to 104: consequential handoff

Crime, violence and reconciliation -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

107 to 105: consequential handoff

Crime, violence and reconciliation -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

107 to 106: consequential handoff

Crime, violence and reconciliation -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

107 to 107: consequential handoff

Crime, violence and reconciliation -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Crime, violence and reconciliation becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 803-809; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

107 to 108: consequential handoff

Crime, violence and reconciliation -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

107 to 109: consequential handoff

Crime, violence and reconciliation -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

107 to 110: consequential handoff

Crime, violence and reconciliation -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

107 to 111: consequential handoff

Crime, violence and reconciliation -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

107 to 112: consequential handoff

Crime, violence and reconciliation -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

107 to 113: consequential handoff

Crime, violence and reconciliation -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

107 to 114: consequential handoff

Crime, violence and reconciliation -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

107 to 115: consequential handoff

Crime, violence and reconciliation -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

107 to 116: consequential handoff

Crime, violence and reconciliation -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

107 to 117: consequential handoff

Crime, violence and reconciliation -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

107 to 118: consequential handoff

Crime, violence and reconciliation -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

107 to 119: consequential handoff

Crime, violence and reconciliation -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

107 to 120: consequential handoff

Crime, violence and reconciliation -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Crime, violence and reconciliation into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Offense label or actor-only risk score. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Victims, actor, families, community, institutions and future potential victims. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal controlled justice-program evaluation' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 803-809 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

107: Crime, violence and reconciliation

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Crime, violence and reconciliation

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Crime, violence and reconciliation is described in the Atlas as: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Victims, actor, families, community, institutions and future potential victims. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Victim safety, accountability, recurrence, repair and future lawful agency improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal controlled justice-program evaluation or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Evidence and source reconstruction

Crime, violence and reconciliation | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 107: Crime, violence and reconciliation; original author status: High-stakes protocol. Native state: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Offense label or actor-only risk score. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. The relevant receiving carriers are: Victims, actor, families, community, institutions and future potential victims.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Victim safety, accountability, recurrence, repair and future lawful agency improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal controlled justice-program evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Boundary divergence

Crime, violence and reconciliation | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 107: Crime, violence and reconciliation; original author status: High-stakes protocol. Native state: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Offense label or actor-only risk score. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. The relevant receiving carriers are: Victims, actor, families, community, institutions and future potential victims.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Victim safety, accountability, recurrence, repair and future lawful agency improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal controlled justice-program evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Causal diagram and identifiability

Crime, violence and reconciliation | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 107: Crime, violence and reconciliation; original author status: High-stakes protocol. Native state: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Offense label or actor-only risk score. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. The relevant receiving carriers are: Victims, actor, families, community, institutions and future potential victims.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Victim safety, accountability, recurrence, repair and future lawful agency improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal controlled justice-program evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Receiving carrier identification

Crime, violence and reconciliation | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 107: Crime, violence and reconciliation; original author status: High-stakes protocol. Native state: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Offense label or actor-only risk score. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. The relevant receiving carriers are: Victims, actor, families, community, institutions and future potential victims.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Victim safety, accountability, recurrence, repair and future lawful agency improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal controlled justice-program evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Hidden compensation accounting

Crime, violence and reconciliation | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 107: Crime, violence and reconciliation; original author status: High-stakes protocol. Native state: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Offense label or actor-only risk score. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. The relevant receiving carriers are: Victims, actor, families, community, institutions and future potential victims.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Victim safety, accountability, recurrence, repair and future lawful agency improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal controlled justice-program evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Native units and dimensional integrity

Crime, violence and reconciliation | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 107: Crime, violence and reconciliation; original author status: High-stakes protocol. Native state: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Offense label or actor-only risk score. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. The relevant receiving carriers are: Victims, actor, families, community, institutions and future potential victims.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Victim safety, accountability, recurrence, repair and future lawful agency improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal controlled justice-program evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Capacity specification

Crime, violence and reconciliation | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 107: Crime, violence and reconciliation; original author status: High-stakes protocol. Native state: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Offense label or actor-only risk score. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. The relevant receiving carriers are: Victims, actor, families, community, institutions and future potential victims.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Victim safety, accountability, recurrence, repair and future lawful agency improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal controlled justice-program evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Intervention design

Crime, violence and reconciliation | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 107: Crime, violence and reconciliation; original author status: High-stakes protocol. Native state: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Offense label or actor-only risk score. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. The relevant receiving carriers are: Victims, actor, families, community, institutions and future potential victims.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Victim safety, accountability, recurrence, repair and future lawful agency improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal controlled justice-program evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Incumbent parity contest

Crime, violence and reconciliation | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 107: Crime, violence and reconciliation; original author status: High-stakes protocol. Native state: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Offense label or actor-only risk score. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. The relevant receiving carriers are: Victims, actor, families, community, institutions and future potential victims.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Victim safety, accountability, recurrence, repair and future lawful agency improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal controlled justice-program evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Time-window and recurrence

Crime, violence and reconciliation | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 107: Crime, violence and reconciliation; original author status: High-stakes protocol. Native state: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Offense label or actor-only risk score. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. The relevant receiving carriers are: Victims, actor, families, community, institutions and future potential victims.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Victim safety, accountability, recurrence, repair and future lawful agency improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal controlled justice-program evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Ethical and legal safety

Crime, violence and reconciliation | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 107: Crime, violence and reconciliation; original author status: High-stakes protocol. Native state: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Offense label or actor-only risk score. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. The relevant receiving carriers are: Victims, actor, families, community, institutions and future potential victims.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Victim safety, accountability, recurrence, repair and future lawful agency improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal controlled justice-program evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Sensitivity and negative controls

Crime, violence and reconciliation | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 107: Crime, violence and reconciliation; original author status: High-stakes protocol. Native state: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Offense label or actor-only risk score. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. The relevant receiving carriers are: Victims, actor, families, community, institutions and future potential victims.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Victim safety, accountability, recurrence, repair and future lawful agency improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal controlled justice-program evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Replication and provenance

Crime, violence and reconciliation | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 107: Crime, violence and reconciliation; original author status: High-stakes protocol. Native state: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Offense label or actor-only risk score. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. The relevant receiving carriers are: Victims, actor, families, community, institutions and future potential victims.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Victim safety, accountability, recurrence, repair and future lawful agency improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal controlled justice-program evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

107: Cross-domain invariant contest

Crime, violence and reconciliation | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 107: Crime, violence and reconciliation; original author status: High-stakes protocol. Native state: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Offense label or actor-only risk score. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Prevention, punishment or rehabilitation optimizes one carrier while exporting burden to victims, families or future safety. The relevant receiving carriers are: Victims, actor, families, community, institutions and future potential victims.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Victim safety, accountability, recurrence, repair and future lawful agency improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal controlled justice-program evaluation. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 107, pp. 803-809 (Rufino 2026).

Adapter 108: Animal sentience and welfare

Conscious agency and responsibility | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory

RECORDED RECEIVING SCHEMA

Commodity/yield schema that excludes the animal as a carrier

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Candidate receivers: Animals, workers, consumers, farmers, land, ecosystems and communities.

EXISTING CAPACITY DESIGN

Welfare, alternative-production, livelihood-transition and ecological capacity

PROTECTED OUTCOME STATED BY THE AUTHOR

Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Multi-carrier welfare, nutrition and livelihood trial. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108 | Animal sentience and welfare

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 108, Animal sentience and welfare. Native state: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Potential direct or institutional human pathway: Animal sentience and welfare interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Welfare, alternative-production, livelihood-transition and ecological capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

108 | Animal sentience and welfare

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 108, Animal sentience and welfare. Native state: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Potential direct or institutional human pathway: Animal sentience and welfare interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Welfare, alternative-production, livelihood-transition and ecological capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

108 | Animal sentience and welfare

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 108, Animal sentience and welfare. Native state: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Potential direct or institutional human pathway: Animal sentience and welfare interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Welfare, alternative-production, livelihood-transition and ecological capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

108 | Animal sentience and welfare

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 108, Animal sentience and welfare. Native state: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Potential direct or institutional human pathway: Animal sentience and welfare interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Welfare, alternative-production, livelihood-transition and ecological capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

108 | Animal sentience and welfare

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 108, Animal sentience and welfare. Native state: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Potential direct or institutional human pathway: Animal sentience and welfare interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Welfare, alternative-production, livelihood-transition and ecological capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

108 | Animal sentience and welfare

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 108, Animal sentience and welfare. Native state: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Potential direct or institutional human pathway: Animal sentience and welfare interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Welfare, alternative-production, livelihood-transition and ecological capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

108 to 001: consequential handoff

Animal sentience and welfare -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

108 to 002: consequential handoff

Animal sentience and welfare -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

108 to 003: consequential handoff

Animal sentience and welfare -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

108 to 004: consequential handoff

Animal sentience and welfare -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

108 to 005: consequential handoff

Animal sentience and welfare -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

108 to 006: consequential handoff

Animal sentience and welfare -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

108 to 007: consequential handoff

Animal sentience and welfare -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

108 to 008: consequential handoff

Animal sentience and welfare -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

108 to 009: consequential handoff

Animal sentience and welfare -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

108 to 010: consequential handoff

Animal sentience and welfare -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

108 to 011: consequential handoff

Animal sentience and welfare -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

108 to 012: consequential handoff

Animal sentience and welfare -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

108 to 013: consequential handoff

Animal sentience and welfare -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

108 to 014: consequential handoff

Animal sentience and welfare -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

108 to 015: consequential handoff

Animal sentience and welfare -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

108 to 016: consequential handoff

Animal sentience and welfare -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

108 to 017: consequential handoff

Animal sentience and welfare -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

108 to 018: consequential handoff

Animal sentience and welfare -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

108 to 019: consequential handoff

Animal sentience and welfare -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

108 to 020: consequential handoff

Animal sentience and welfare -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

108 to 021: consequential handoff

Animal sentience and welfare -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

108 to 022: consequential handoff

Animal sentience and welfare -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

108 to 023: consequential handoff

Animal sentience and welfare -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

108 to 024: consequential handoff

Animal sentience and welfare -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

108 to 025: consequential handoff

Animal sentience and welfare -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

108 to 026: consequential handoff

Animal sentience and welfare -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

108 to 027: consequential handoff

Animal sentience and welfare -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

108 to 028: consequential handoff

Animal sentience and welfare -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

108 to 029: consequential handoff

Animal sentience and welfare -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

108 to 030: consequential handoff

Animal sentience and welfare -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

108 to 031: consequential handoff

Animal sentience and welfare -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

108 to 032: consequential handoff

Animal sentience and welfare -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

108 to 033: consequential handoff

Animal sentience and welfare -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

108 to 034: consequential handoff

Animal sentience and welfare -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

108 to 035: consequential handoff

Animal sentience and welfare -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

108 to 036: consequential handoff

Animal sentience and welfare -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

108 to 037: consequential handoff

Animal sentience and welfare -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

108 to 038: consequential handoff

Animal sentience and welfare -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

108 to 039: consequential handoff

Animal sentience and welfare -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

108 to 040: consequential handoff

Animal sentience and welfare -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

108 to 041: consequential handoff

Animal sentience and welfare -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

108 to 042: consequential handoff

Animal sentience and welfare -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

108 to 043: consequential handoff

Animal sentience and welfare -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

108 to 044: consequential handoff

Animal sentience and welfare -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

108 to 045: consequential handoff

Animal sentience and welfare -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

108 to 046: consequential handoff

Animal sentience and welfare -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

108 to 047: consequential handoff

Animal sentience and welfare -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

108 to 048: consequential handoff

Animal sentience and welfare -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

108 to 049: consequential handoff

Animal sentience and welfare -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

108 to 050: consequential handoff

Animal sentience and welfare -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

108 to 051: consequential handoff

Animal sentience and welfare -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

108 to 052: consequential handoff

Animal sentience and welfare -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

108 to 053: consequential handoff

Animal sentience and welfare -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

108 to 054: consequential handoff

Animal sentience and welfare -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

108 to 055: consequential handoff

Animal sentience and welfare -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

108 to 056: consequential handoff

Animal sentience and welfare -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

108 to 057: consequential handoff

Animal sentience and welfare -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

108 to 058: consequential handoff

Animal sentience and welfare -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

108 to 059: consequential handoff

Animal sentience and welfare -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

108 to 060: consequential handoff

Animal sentience and welfare -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

108 to 061: consequential handoff

Animal sentience and welfare -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

108 to 062: consequential handoff

Animal sentience and welfare -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

108 to 063: consequential handoff

Animal sentience and welfare -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

108 to 064: consequential handoff

Animal sentience and welfare -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

108 to 065: consequential handoff

Animal sentience and welfare -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

108 to 066: consequential handoff

Animal sentience and welfare -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

108 to 067: consequential handoff

Animal sentience and welfare -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

108 to 068: consequential handoff

Animal sentience and welfare -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

108 to 069: consequential handoff

Animal sentience and welfare -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

108 to 070: consequential handoff

Animal sentience and welfare -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

108 to 071: consequential handoff

Animal sentience and welfare -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

108 to 072: consequential handoff

Animal sentience and welfare -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

108 to 073: consequential handoff

Animal sentience and welfare -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

108 to 074: consequential handoff

Animal sentience and welfare -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

108 to 075: consequential handoff

Animal sentience and welfare -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

108 to 076: consequential handoff

Animal sentience and welfare -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

108 to 077: consequential handoff

Animal sentience and welfare -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

108 to 078: consequential handoff

Animal sentience and welfare -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

108 to 079: consequential handoff

Animal sentience and welfare -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

108 to 080: consequential handoff

Animal sentience and welfare -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

108 to 081: consequential handoff

Animal sentience and welfare -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

108 to 082: consequential handoff

Animal sentience and welfare -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

108 to 083: consequential handoff

Animal sentience and welfare -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

108 to 084: consequential handoff

Animal sentience and welfare -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

108 to 085: consequential handoff

Animal sentience and welfare -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

108 to 086: consequential handoff

Animal sentience and welfare -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

108 to 087: consequential handoff

Animal sentience and welfare -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

108 to 088: consequential handoff

Animal sentience and welfare -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

108 to 089: consequential handoff

Animal sentience and welfare -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

108 to 090: consequential handoff

Animal sentience and welfare -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

108 to 091: consequential handoff

Animal sentience and welfare -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

108 to 092: consequential handoff

Animal sentience and welfare -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

108 to 093: consequential handoff

Animal sentience and welfare -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

108 to 094: consequential handoff

Animal sentience and welfare -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

108 to 095: consequential handoff

Animal sentience and welfare -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

108 to 096: consequential handoff

Animal sentience and welfare -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

108 to 097: consequential handoff

Animal sentience and welfare -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

108 to 098: consequential handoff

Animal sentience and welfare -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

108 to 099: consequential handoff

Animal sentience and welfare -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

108 to 100: consequential handoff

Animal sentience and welfare -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

108 to 101: consequential handoff

Animal sentience and welfare -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

108 to 102: consequential handoff

Animal sentience and welfare -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

108 to 103: consequential handoff

Animal sentience and welfare -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

108 to 104: consequential handoff

Animal sentience and welfare -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

108 to 105: consequential handoff

Animal sentience and welfare -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

108 to 106: consequential handoff

Animal sentience and welfare -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

108 to 107: consequential handoff

Animal sentience and welfare -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

108 to 108: consequential handoff

Animal sentience and welfare -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Animal sentience and welfare becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 810-816; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

108 to 109: consequential handoff

Animal sentience and welfare -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

108 to 110: consequential handoff

Animal sentience and welfare -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

108 to 111: consequential handoff

Animal sentience and welfare -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

108 to 112: consequential handoff

Animal sentience and welfare -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

108 to 113: consequential handoff

Animal sentience and welfare -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

108 to 114: consequential handoff

Animal sentience and welfare -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

108 to 115: consequential handoff

Animal sentience and welfare -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

108 to 116: consequential handoff

Animal sentience and welfare -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

108 to 117: consequential handoff

Animal sentience and welfare -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

108 to 118: consequential handoff

Animal sentience and welfare -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

108 to 119: consequential handoff

Animal sentience and welfare -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

108 to 120: consequential handoff

Animal sentience and welfare -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Animal sentience and welfare into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Commodity/yield schema that excludes the animal as a carrier. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-carrier welfare, nutrition and livelihood trial' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 810-816 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

108: Animal sentience and welfare

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Animal sentience and welfare

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Animal sentience and welfare is described in the Atlas as: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, workers, consumers, farmers, land, ecosystems and communities. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative-production, livelihood-transition and ecological capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-carrier welfare, nutrition and livelihood trial or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Evidence and source reconstruction

Animal sentience and welfare | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 108: Animal sentience and welfare; original author status: Ethical / empirical frontier. Native state: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Commodity/yield schema that excludes the animal as a carrier. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. The relevant receiving carriers are: Animals, workers, consumers, farmers, land, ecosystems and communities.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative-production, livelihood-transition and ecological capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-carrier welfare, nutrition and livelihood trial. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Boundary divergence

Animal sentience and welfare | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 108: Animal sentience and welfare; original author status: Ethical / empirical frontier. Native state: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Commodity/yield schema that excludes the animal as a carrier. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. The relevant receiving carriers are: Animals, workers, consumers, farmers, land, ecosystems and communities.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative-production, livelihood-transition and ecological capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-carrier welfare, nutrition and livelihood trial. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Causal diagram and identifiability

Animal sentience and welfare | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 108: Animal sentience and welfare; original author status: Ethical / empirical frontier. Native state: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Commodity/yield schema that excludes the animal as a carrier. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. The relevant receiving carriers are: Animals, workers, consumers, farmers, land, ecosystems and communities.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative-production, livelihood-transition and ecological capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-carrier welfare, nutrition and livelihood trial. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Receiving carrier identification

Animal sentience and welfare | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 108: Animal sentience and welfare; original author status: Ethical / empirical frontier. Native state: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Commodity/yield schema that excludes the animal as a carrier. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. The relevant receiving carriers are: Animals, workers, consumers, farmers, land, ecosystems and communities.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative-production, livelihood-transition and ecological capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-carrier welfare, nutrition and livelihood trial. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Hidden compensation accounting

Animal sentience and welfare | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 108: Animal sentience and welfare; original author status: Ethical / empirical frontier. Native state: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Commodity/yield schema that excludes the animal as a carrier. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. The relevant receiving carriers are: Animals, workers, consumers, farmers, land, ecosystems and communities.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative-production, livelihood-transition and ecological capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-carrier welfare, nutrition and livelihood trial. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Native units and dimensional integrity

Animal sentience and welfare | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 108: Animal sentience and welfare; original author status: Ethical / empirical frontier. Native state: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Commodity/yield schema that excludes the animal as a carrier. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. The relevant receiving carriers are: Animals, workers, consumers, farmers, land, ecosystems and communities.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative-production, livelihood-transition and ecological capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-carrier welfare, nutrition and livelihood trial. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Capacity specification

Animal sentience and welfare | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 108: Animal sentience and welfare; original author status: Ethical / empirical frontier. Native state: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Commodity/yield schema that excludes the animal as a carrier. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. The relevant receiving carriers are: Animals, workers, consumers, farmers, land, ecosystems and communities.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative-production, livelihood-transition and ecological capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-carrier welfare, nutrition and livelihood trial. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Intervention design

Animal sentience and welfare | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 108: Animal sentience and welfare; original author status: Ethical / empirical frontier. Native state: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Commodity/yield schema that excludes the animal as a carrier. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. The relevant receiving carriers are: Animals, workers, consumers, farmers, land, ecosystems and communities.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative-production, livelihood-transition and ecological capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-carrier welfare, nutrition and livelihood trial. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Incumbent parity contest

Animal sentience and welfare | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 108: Animal sentience and welfare; original author status: Ethical / empirical frontier. Native state: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Commodity/yield schema that excludes the animal as a carrier. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. The relevant receiving carriers are: Animals, workers, consumers, farmers, land, ecosystems and communities.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative-production, livelihood-transition and ecological capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-carrier welfare, nutrition and livelihood trial. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Time-window and recurrence

Animal sentience and welfare | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 108: Animal sentience and welfare; original author status: Ethical / empirical frontier. Native state: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Commodity/yield schema that excludes the animal as a carrier. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. The relevant receiving carriers are: Animals, workers, consumers, farmers, land, ecosystems and communities.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative-production, livelihood-transition and ecological capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-carrier welfare, nutrition and livelihood trial. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Ethical and legal safety

Animal sentience and welfare | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 108: Animal sentience and welfare; original author status: Ethical / empirical frontier. Native state: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Commodity/yield schema that excludes the animal as a carrier. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. The relevant receiving carriers are: Animals, workers, consumers, farmers, land, ecosystems and communities.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative-production, livelihood-transition and ecological capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-carrier welfare, nutrition and livelihood trial. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Sensitivity and negative controls

Animal sentience and welfare | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 108: Animal sentience and welfare; original author status: Ethical / empirical frontier. Native state: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Commodity/yield schema that excludes the animal as a carrier. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. The relevant receiving carriers are: Animals, workers, consumers, farmers, land, ecosystems and communities.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative-production, livelihood-transition and ecological capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-carrier welfare, nutrition and livelihood trial. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Replication and provenance

Animal sentience and welfare | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 108: Animal sentience and welfare; original author status: Ethical / empirical frontier. Native state: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Commodity/yield schema that excludes the animal as a carrier. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. The relevant receiving carriers are: Animals, workers, consumers, farmers, land, ecosystems and communities.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative-production, livelihood-transition and ecological capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-carrier welfare, nutrition and livelihood trial. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

108: Cross-domain invariant contest

Animal sentience and welfare | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 108: Animal sentience and welfare; original author status: Ethical / empirical frontier. Native state: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Commodity/yield schema that excludes the animal as a carrier. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Human nutrition or commercial gain is produced by unrepresented animal or ecological burden. The relevant receiving carriers are: Animals, workers, consumers, farmers, land, ecosystems and communities.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative-production, livelihood-transition and ecological capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Nutrition and livelihoods improve while animal and ecological burden also improves or is materially reduced.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-carrier welfare, nutrition and livelihood trial. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 108, pp. 810-816 (Rufino 2026).

Adapter 109: Collective agency and institutional choice

Conscious agency and responsibility | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory

RECORDED RECEIVING SCHEMA

Organization treated as a single actor without internal carrier structure

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Candidate receivers: Members, minorities, leaders, outsiders, institutions and future administrations.

EXISTING CAPACITY DESIGN

Distributed decision provenance, dissent channels, accountability and cross-boundary capacity

PROTECTED OUTCOME STATED BY THE AUTHOR

Collective performance and legitimacy rise without burden concentration or responsibility diffusion

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Prospective institutional decision audit. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109 | Collective agency and institutional choice

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 109, Collective agency and institutional choice. Native state: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Potential direct or institutional human pathway: Collective agency and institutional choice interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Distributed decision provenance, dissent channels, accountability and cross-boundary capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

109 | Collective agency and institutional choice

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 109, Collective agency and institutional choice. Native state: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Potential direct or institutional human pathway: Collective agency and institutional choice interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Distributed decision provenance, dissent channels, accountability and cross-boundary capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

109 | Collective agency and institutional choice

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 109, Collective agency and institutional choice. Native state: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Potential direct or institutional human pathway: Collective agency and institutional choice interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Distributed decision provenance, dissent channels, accountability and cross-boundary capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

109 | Collective agency and institutional choice

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 109, Collective agency and institutional choice. Native state: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Potential direct or institutional human pathway: Collective agency and institutional choice interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Distributed decision provenance, dissent channels, accountability and cross-boundary capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

109 | Collective agency and institutional choice

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 109, Collective agency and institutional choice. Native state: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Potential direct or institutional human pathway: Collective agency and institutional choice interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Distributed decision provenance, dissent channels, accountability and cross-boundary capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

109 | Collective agency and institutional choice

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 109, Collective agency and institutional choice. Native state: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Potential direct or institutional human pathway: Collective agency and institutional choice interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Distributed decision provenance, dissent channels, accountability and cross-boundary capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

109 to 001: consequential handoff

Collective agency and institutional choice -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

109 to 002: consequential handoff

Collective agency and institutional choice -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

109 to 003: consequential handoff

Collective agency and institutional choice -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

109 to 004: consequential handoff

Collective agency and institutional choice -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

109 to 005: consequential handoff

Collective agency and institutional choice -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

109 to 006: consequential handoff

Collective agency and institutional choice -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

109 to 007: consequential handoff

Collective agency and institutional choice -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

109 to 008: consequential handoff

Collective agency and institutional choice -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

109 to 009: consequential handoff

Collective agency and institutional choice -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

109 to 010: consequential handoff

Collective agency and institutional choice -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

109 to 011: consequential handoff

Collective agency and institutional choice -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

109 to 012: consequential handoff

Collective agency and institutional choice -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

109 to 013: consequential handoff

Collective agency and institutional choice -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

109 to 014: consequential handoff

Collective agency and institutional choice -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

109 to 015: consequential handoff

Collective agency and institutional choice -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

109 to 016: consequential handoff

Collective agency and institutional choice -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

109 to 017: consequential handoff

Collective agency and institutional choice -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

109 to 018: consequential handoff

Collective agency and institutional choice -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

109 to 019: consequential handoff

Collective agency and institutional choice -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

109 to 020: consequential handoff

Collective agency and institutional choice -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

109 to 021: consequential handoff

Collective agency and institutional choice -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

109 to 022: consequential handoff

Collective agency and institutional choice -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

109 to 023: consequential handoff

Collective agency and institutional choice -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

109 to 024: consequential handoff

Collective agency and institutional choice -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

109 to 025: consequential handoff

Collective agency and institutional choice -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

109 to 026: consequential handoff

Collective agency and institutional choice -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

109 to 027: consequential handoff

Collective agency and institutional choice -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

109 to 028: consequential handoff

Collective agency and institutional choice -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

109 to 029: consequential handoff

Collective agency and institutional choice -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

109 to 030: consequential handoff

Collective agency and institutional choice -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

109 to 031: consequential handoff

Collective agency and institutional choice -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

109 to 032: consequential handoff

Collective agency and institutional choice -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

109 to 033: consequential handoff

Collective agency and institutional choice -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

109 to 034: consequential handoff

Collective agency and institutional choice -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

109 to 035: consequential handoff

Collective agency and institutional choice -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

109 to 036: consequential handoff

Collective agency and institutional choice -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

109 to 037: consequential handoff

Collective agency and institutional choice -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

109 to 038: consequential handoff

Collective agency and institutional choice -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

109 to 039: consequential handoff

Collective agency and institutional choice -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

109 to 040: consequential handoff

Collective agency and institutional choice -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

109 to 041: consequential handoff

Collective agency and institutional choice -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

109 to 042: consequential handoff

Collective agency and institutional choice -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

109 to 043: consequential handoff

Collective agency and institutional choice -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

109 to 044: consequential handoff

Collective agency and institutional choice -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

109 to 045: consequential handoff

Collective agency and institutional choice -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

109 to 046: consequential handoff

Collective agency and institutional choice -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

109 to 047: consequential handoff

Collective agency and institutional choice -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

109 to 048: consequential handoff

Collective agency and institutional choice -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

109 to 049: consequential handoff

Collective agency and institutional choice -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

109 to 050: consequential handoff

Collective agency and institutional choice -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

109 to 051: consequential handoff

Collective agency and institutional choice -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

109 to 052: consequential handoff

Collective agency and institutional choice -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

109 to 053: consequential handoff

Collective agency and institutional choice -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

109 to 054: consequential handoff

Collective agency and institutional choice -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

109 to 055: consequential handoff

Collective agency and institutional choice -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

109 to 056: consequential handoff

Collective agency and institutional choice -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

109 to 057: consequential handoff

Collective agency and institutional choice -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

109 to 058: consequential handoff

Collective agency and institutional choice -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

109 to 059: consequential handoff

Collective agency and institutional choice -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

109 to 060: consequential handoff

Collective agency and institutional choice -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

109 to 061: consequential handoff

Collective agency and institutional choice -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

109 to 062: consequential handoff

Collective agency and institutional choice -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

109 to 063: consequential handoff

Collective agency and institutional choice -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

109 to 064: consequential handoff

Collective agency and institutional choice -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

109 to 065: consequential handoff

Collective agency and institutional choice -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

109 to 066: consequential handoff

Collective agency and institutional choice -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

109 to 067: consequential handoff

Collective agency and institutional choice -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

109 to 068: consequential handoff

Collective agency and institutional choice -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

109 to 069: consequential handoff

Collective agency and institutional choice -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

109 to 070: consequential handoff

Collective agency and institutional choice -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

109 to 071: consequential handoff

Collective agency and institutional choice -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

109 to 072: consequential handoff

Collective agency and institutional choice -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

109 to 073: consequential handoff

Collective agency and institutional choice -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

109 to 074: consequential handoff

Collective agency and institutional choice -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

109 to 075: consequential handoff

Collective agency and institutional choice -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

109 to 076: consequential handoff

Collective agency and institutional choice -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

109 to 077: consequential handoff

Collective agency and institutional choice -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

109 to 078: consequential handoff

Collective agency and institutional choice -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

109 to 079: consequential handoff

Collective agency and institutional choice -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

109 to 080: consequential handoff

Collective agency and institutional choice -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

109 to 081: consequential handoff

Collective agency and institutional choice -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

109 to 082: consequential handoff

Collective agency and institutional choice -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

109 to 083: consequential handoff

Collective agency and institutional choice -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

109 to 084: consequential handoff

Collective agency and institutional choice -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

109 to 085: consequential handoff

Collective agency and institutional choice -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

109 to 086: consequential handoff

Collective agency and institutional choice -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

109 to 087: consequential handoff

Collective agency and institutional choice -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

109 to 088: consequential handoff

Collective agency and institutional choice -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

109 to 089: consequential handoff

Collective agency and institutional choice -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

109 to 090: consequential handoff

Collective agency and institutional choice -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

109 to 091: consequential handoff

Collective agency and institutional choice -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

109 to 092: consequential handoff

Collective agency and institutional choice -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

109 to 093: consequential handoff

Collective agency and institutional choice -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

109 to 094: consequential handoff

Collective agency and institutional choice -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

109 to 095: consequential handoff

Collective agency and institutional choice -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

109 to 096: consequential handoff

Collective agency and institutional choice -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

109 to 097: consequential handoff

Collective agency and institutional choice -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

109 to 098: consequential handoff

Collective agency and institutional choice -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

109 to 099: consequential handoff

Collective agency and institutional choice -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

109 to 100: consequential handoff

Collective agency and institutional choice -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

109 to 101: consequential handoff

Collective agency and institutional choice -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

109 to 102: consequential handoff

Collective agency and institutional choice -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

109 to 103: consequential handoff

Collective agency and institutional choice -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

109 to 104: consequential handoff

Collective agency and institutional choice -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

109 to 105: consequential handoff

Collective agency and institutional choice -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

109 to 106: consequential handoff

Collective agency and institutional choice -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

109 to 107: consequential handoff

Collective agency and institutional choice -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

109 to 108: consequential handoff

Collective agency and institutional choice -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

109 to 109: consequential handoff

Collective agency and institutional choice -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Collective agency and institutional choice becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 817-823; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

109 to 110: consequential handoff

Collective agency and institutional choice -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

109 to 111: consequential handoff

Collective agency and institutional choice -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

109 to 112: consequential handoff

Collective agency and institutional choice -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

109 to 113: consequential handoff

Collective agency and institutional choice -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

109 to 114: consequential handoff

Collective agency and institutional choice -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

109 to 115: consequential handoff

Collective agency and institutional choice -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

109 to 116: consequential handoff

Collective agency and institutional choice -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

109 to 117: consequential handoff

Collective agency and institutional choice -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

109 to 118: consequential handoff

Collective agency and institutional choice -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

109 to 119: consequential handoff

Collective agency and institutional choice -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

109 to 120: consequential handoff

Collective agency and institutional choice -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Collective agency and institutional choice into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Organization treated as a single actor without internal carrier structure. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Prospective institutional decision audit' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 817-823 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

109: Collective agency and institutional choice

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Collective agency and institutional choice

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Collective agency and institutional choice is described in the Atlas as: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Members, minorities, leaders, outsiders, institutions and future administrations. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Collective performance and legitimacy rise without burden concentration or responsibility diffusion. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Prospective institutional decision audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Evidence and source reconstruction

Collective agency and institutional choice | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 109: Collective agency and institutional choice; original author status: Constructive protocol. Native state: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Organization treated as a single actor without internal carrier structure. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. The relevant receiving carriers are: Members, minorities, leaders, outsiders, institutions and future administrations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Collective performance and legitimacy rise without burden concentration or responsibility diffusion.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective institutional decision audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Boundary divergence

Collective agency and institutional choice | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 109: Collective agency and institutional choice; original author status: Constructive protocol. Native state: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Organization treated as a single actor without internal carrier structure. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. The relevant receiving carriers are: Members, minorities, leaders, outsiders, institutions and future administrations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Collective performance and legitimacy rise without burden concentration or responsibility diffusion.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective institutional decision audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Causal diagram and identifiability

Collective agency and institutional choice | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 109: Collective agency and institutional choice; original author status: Constructive protocol. Native state: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Organization treated as a single actor without internal carrier structure. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. The relevant receiving carriers are: Members, minorities, leaders, outsiders, institutions and future administrations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Collective performance and legitimacy rise without burden concentration or responsibility diffusion.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective institutional decision audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Receiving carrier identification

Collective agency and institutional choice | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 109: Collective agency and institutional choice; original author status: Constructive protocol. Native state: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Organization treated as a single actor without internal carrier structure. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. The relevant receiving carriers are: Members, minorities, leaders, outsiders, institutions and future administrations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Collective performance and legitimacy rise without burden concentration or responsibility diffusion.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective institutional decision audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Hidden compensation accounting

Collective agency and institutional choice | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 109: Collective agency and institutional choice; original author status: Constructive protocol. Native state: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Organization treated as a single actor without internal carrier structure. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. The relevant receiving carriers are: Members, minorities, leaders, outsiders, institutions and future administrations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Collective performance and legitimacy rise without burden concentration or responsibility diffusion.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective institutional decision audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Native units and dimensional integrity

Collective agency and institutional choice | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 109: Collective agency and institutional choice; original author status: Constructive protocol. Native state: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Organization treated as a single actor without internal carrier structure. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. The relevant receiving carriers are: Members, minorities, leaders, outsiders, institutions and future administrations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Collective performance and legitimacy rise without burden concentration or responsibility diffusion.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective institutional decision audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Capacity specification

Collective agency and institutional choice | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 109: Collective agency and institutional choice; original author status: Constructive protocol. Native state: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Organization treated as a single actor without internal carrier structure. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. The relevant receiving carriers are: Members, minorities, leaders, outsiders, institutions and future administrations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Collective performance and legitimacy rise without burden concentration or responsibility diffusion.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective institutional decision audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Intervention design

Collective agency and institutional choice | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 109: Collective agency and institutional choice; original author status: Constructive protocol. Native state: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Organization treated as a single actor without internal carrier structure. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. The relevant receiving carriers are: Members, minorities, leaders, outsiders, institutions and future administrations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Collective performance and legitimacy rise without burden concentration or responsibility diffusion.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective institutional decision audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Incumbent parity contest

Collective agency and institutional choice | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 109: Collective agency and institutional choice; original author status: Constructive protocol. Native state: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Organization treated as a single actor without internal carrier structure. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. The relevant receiving carriers are: Members, minorities, leaders, outsiders, institutions and future administrations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Collective performance and legitimacy rise without burden concentration or responsibility diffusion.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective institutional decision audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Time-window and recurrence

Collective agency and institutional choice | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 109: Collective agency and institutional choice; original author status: Constructive protocol. Native state: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Organization treated as a single actor without internal carrier structure. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. The relevant receiving carriers are: Members, minorities, leaders, outsiders, institutions and future administrations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Collective performance and legitimacy rise without burden concentration or responsibility diffusion.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective institutional decision audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Ethical and legal safety

Collective agency and institutional choice | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 109: Collective agency and institutional choice; original author status: Constructive protocol. Native state: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Organization treated as a single actor without internal carrier structure. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. The relevant receiving carriers are: Members, minorities, leaders, outsiders, institutions and future administrations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Collective performance and legitimacy rise without burden concentration or responsibility diffusion.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective institutional decision audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Sensitivity and negative controls

Collective agency and institutional choice | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 109: Collective agency and institutional choice; original author status: Constructive protocol. Native state: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Organization treated as a single actor without internal carrier structure. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. The relevant receiving carriers are: Members, minorities, leaders, outsiders, institutions and future administrations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Collective performance and legitimacy rise without burden concentration or responsibility diffusion.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective institutional decision audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Replication and provenance

Collective agency and institutional choice | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 109: Collective agency and institutional choice; original author status: Constructive protocol. Native state: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Organization treated as a single actor without internal carrier structure. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. The relevant receiving carriers are: Members, minorities, leaders, outsiders, institutions and future administrations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Collective performance and legitimacy rise without burden concentration or responsibility diffusion.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective institutional decision audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

109: Cross-domain invariant contest

Collective agency and institutional choice | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 109: Collective agency and institutional choice; original author status: Constructive protocol. Native state: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Organization treated as a single actor without internal carrier structure. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Collective decisions hide minority burdens or diffuse accountability until no node owns the consequence. The relevant receiving carriers are: Members, minorities, leaders, outsiders, institutions and future administrations.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Collective performance and legitimacy rise without burden concentration or responsibility diffusion.

RETEST AND SCOPE CONTROL

Candidate decisive method: Prospective institutional decision audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 109, pp. 817-823 (Rufino 2026).

Adapter 110: Human-AI co-agency

Conscious agency and responsibility | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects

RECORDED RECEIVING SCHEMA

Model output or benchmark score treated as the whole system

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Candidate receivers: User, non-user, model operator, workers, institutions, infrastructure and environment.

EXISTING CAPACITY DESIGN

Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers

PROTECTED OUTCOME STATED BY THE AUTHOR

Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Sealed native-vs-K_R assisted agent benchmark. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110 | Human-AI co-agency

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 110, Human-AI co-agency. Native state: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Potential direct or institutional human pathway: Human-AI co-agency interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

110 | Human-AI co-agency

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 110, Human-AI co-agency. Native state: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Potential direct or institutional human pathway: Human-AI co-agency interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

110 | Human-AI co-agency

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 110, Human-AI co-agency. Native state: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Potential direct or institutional human pathway: Human-AI co-agency interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

110 | Human-AI co-agency

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 110, Human-AI co-agency. Native state: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Potential direct or institutional human pathway: Human-AI co-agency interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

110 | Human-AI co-agency

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 110, Human-AI co-agency. Native state: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Potential direct or institutional human pathway: Human-AI co-agency interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

110 | Human-AI co-agency

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 110, Human-AI co-agency. Native state: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Potential direct or institutional human pathway: Human-AI co-agency interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

110 to 001: consequential handoff

Human-AI co-agency -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

110 to 002: consequential handoff

Human-AI co-agency -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

110 to 003: consequential handoff

Human-AI co-agency -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

110 to 004: consequential handoff

Human-AI co-agency -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

110 to 005: consequential handoff

Human-AI co-agency -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

110 to 006: consequential handoff

Human-AI co-agency -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

110 to 007: consequential handoff

Human-AI co-agency -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

110 to 008: consequential handoff

Human-AI co-agency -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

110 to 009: consequential handoff

Human-AI co-agency -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

110 to 010: consequential handoff

Human-AI co-agency -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

110 to 011: consequential handoff

Human-AI co-agency -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

110 to 012: consequential handoff

Human-AI co-agency -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

110 to 013: consequential handoff

Human-AI co-agency -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

110 to 014: consequential handoff

Human-AI co-agency -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

110 to 015: consequential handoff

Human-AI co-agency -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

110 to 016: consequential handoff

Human-AI co-agency -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

110 to 017: consequential handoff

Human-AI co-agency -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

110 to 018: consequential handoff

Human-AI co-agency -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

110 to 019: consequential handoff

Human-AI co-agency -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

110 to 020: consequential handoff

Human-AI co-agency -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

110 to 021: consequential handoff

Human-AI co-agency -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

110 to 022: consequential handoff

Human-AI co-agency -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

110 to 023: consequential handoff

Human-AI co-agency -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

110 to 024: consequential handoff

Human-AI co-agency -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

110 to 025: consequential handoff

Human-AI co-agency -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

110 to 026: consequential handoff

Human-AI co-agency -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

110 to 027: consequential handoff

Human-AI co-agency -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

110 to 028: consequential handoff

Human-AI co-agency -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

110 to 029: consequential handoff

Human-AI co-agency -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

110 to 030: consequential handoff

Human-AI co-agency -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

110 to 031: consequential handoff

Human-AI co-agency -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

110 to 032: consequential handoff

Human-AI co-agency -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

110 to 033: consequential handoff

Human-AI co-agency -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

110 to 034: consequential handoff

Human-AI co-agency -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

110 to 035: consequential handoff

Human-AI co-agency -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

110 to 036: consequential handoff

Human-AI co-agency -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

110 to 037: consequential handoff

Human-AI co-agency -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

110 to 038: consequential handoff

Human-AI co-agency -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

110 to 039: consequential handoff

Human-AI co-agency -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

110 to 040: consequential handoff

Human-AI co-agency -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

110 to 041: consequential handoff

Human-AI co-agency -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

110 to 042: consequential handoff

Human-AI co-agency -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

110 to 043: consequential handoff

Human-AI co-agency -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

110 to 044: consequential handoff

Human-AI co-agency -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

110 to 045: consequential handoff

Human-AI co-agency -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

110 to 046: consequential handoff

Human-AI co-agency -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

110 to 047: consequential handoff

Human-AI co-agency -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

110 to 048: consequential handoff

Human-AI co-agency -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

110 to 049: consequential handoff

Human-AI co-agency -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

110 to 050: consequential handoff

Human-AI co-agency -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

110 to 051: consequential handoff

Human-AI co-agency -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

110 to 052: consequential handoff

Human-AI co-agency -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

110 to 053: consequential handoff

Human-AI co-agency -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

110 to 054: consequential handoff

Human-AI co-agency -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

110 to 055: consequential handoff

Human-AI co-agency -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

110 to 056: consequential handoff

Human-AI co-agency -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

110 to 057: consequential handoff

Human-AI co-agency -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

110 to 058: consequential handoff

Human-AI co-agency -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

110 to 059: consequential handoff

Human-AI co-agency -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

110 to 060: consequential handoff

Human-AI co-agency -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

110 to 061: consequential handoff

Human-AI co-agency -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

110 to 062: consequential handoff

Human-AI co-agency -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

110 to 063: consequential handoff

Human-AI co-agency -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

110 to 064: consequential handoff

Human-AI co-agency -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

110 to 065: consequential handoff

Human-AI co-agency -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

110 to 066: consequential handoff

Human-AI co-agency -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

110 to 067: consequential handoff

Human-AI co-agency -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

110 to 068: consequential handoff

Human-AI co-agency -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

110 to 069: consequential handoff

Human-AI co-agency -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

110 to 070: consequential handoff

Human-AI co-agency -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

110 to 071: consequential handoff

Human-AI co-agency -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

110 to 072: consequential handoff

Human-AI co-agency -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

110 to 073: consequential handoff

Human-AI co-agency -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

110 to 074: consequential handoff

Human-AI co-agency -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

110 to 075: consequential handoff

Human-AI co-agency -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

110 to 076: consequential handoff

Human-AI co-agency -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

110 to 077: consequential handoff

Human-AI co-agency -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

110 to 078: consequential handoff

Human-AI co-agency -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

110 to 079: consequential handoff

Human-AI co-agency -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

110 to 080: consequential handoff

Human-AI co-agency -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

110 to 081: consequential handoff

Human-AI co-agency -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

110 to 082: consequential handoff

Human-AI co-agency -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

110 to 083: consequential handoff

Human-AI co-agency -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

110 to 084: consequential handoff

Human-AI co-agency -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

110 to 085: consequential handoff

Human-AI co-agency -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

110 to 086: consequential handoff

Human-AI co-agency -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

110 to 087: consequential handoff

Human-AI co-agency -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

110 to 088: consequential handoff

Human-AI co-agency -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

110 to 089: consequential handoff

Human-AI co-agency -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

110 to 090: consequential handoff

Human-AI co-agency -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

110 to 091: consequential handoff

Human-AI co-agency -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

110 to 092: consequential handoff

Human-AI co-agency -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

110 to 093: consequential handoff

Human-AI co-agency -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

110 to 094: consequential handoff

Human-AI co-agency -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

110 to 095: consequential handoff

Human-AI co-agency -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

110 to 096: consequential handoff

Human-AI co-agency -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

110 to 097: consequential handoff

Human-AI co-agency -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

110 to 098: consequential handoff

Human-AI co-agency -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

110 to 099: consequential handoff

Human-AI co-agency -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

110 to 100: consequential handoff

Human-AI co-agency -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

110 to 101: consequential handoff

Human-AI co-agency -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

110 to 102: consequential handoff

Human-AI co-agency -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

110 to 103: consequential handoff

Human-AI co-agency -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

110 to 104: consequential handoff

Human-AI co-agency -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

110 to 105: consequential handoff

Human-AI co-agency -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

110 to 106: consequential handoff

Human-AI co-agency -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

110 to 107: consequential handoff

Human-AI co-agency -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

110 to 108: consequential handoff

Human-AI co-agency -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

110 to 109: consequential handoff

Human-AI co-agency -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

110 to 110: consequential handoff

Human-AI co-agency -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Human-AI co-agency becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 824-830; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

110 to 111: consequential handoff

Human-AI co-agency -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

110 to 112: consequential handoff

Human-AI co-agency -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

110 to 113: consequential handoff

Human-AI co-agency -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

110 to 114: consequential handoff

Human-AI co-agency -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

110 to 115: consequential handoff

Human-AI co-agency -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

110 to 116: consequential handoff

Human-AI co-agency -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

110 to 117: consequential handoff

Human-AI co-agency -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

110 to 118: consequential handoff

Human-AI co-agency -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

110 to 119: consequential handoff

Human-AI co-agency -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

110 to 120: consequential handoff

Human-AI co-agency -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Human-AI co-agency into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Model output or benchmark score treated as the whole system. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Sealed native-vs-K_R assisted agent benchmark' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 824-830 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

110: Human-AI co-agency

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

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MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

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Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

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MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

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Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

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Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

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110: Human-AI co-agency

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

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Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

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Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

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Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Human-AI co-agency

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Human-AI co-agency is described in the Atlas as: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for User, non-user, model operator, workers, institutions, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Sealed native-vs-K_R assisted agent benchmark or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Evidence and source reconstruction

Human-AI co-agency | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 110: Human-AI co-agency; original author status: Strong constructive fit. Native state: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Model output or benchmark score treated as the whole system. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. The relevant receiving carriers are: User, non-user, model operator, workers, institutions, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed native-vs-K_R assisted agent benchmark. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Boundary divergence

Human-AI co-agency | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 110: Human-AI co-agency; original author status: Strong constructive fit. Native state: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Model output or benchmark score treated as the whole system. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. The relevant receiving carriers are: User, non-user, model operator, workers, institutions, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed native-vs-K_R assisted agent benchmark. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Causal diagram and identifiability

Human-AI co-agency | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 110: Human-AI co-agency; original author status: Strong constructive fit. Native state: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Model output or benchmark score treated as the whole system. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. The relevant receiving carriers are: User, non-user, model operator, workers, institutions, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed native-vs-K_R assisted agent benchmark. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Receiving carrier identification

Human-AI co-agency | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 110: Human-AI co-agency; original author status: Strong constructive fit. Native state: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Model output or benchmark score treated as the whole system. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. The relevant receiving carriers are: User, non-user, model operator, workers, institutions, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed native-vs-K_R assisted agent benchmark. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Hidden compensation accounting

Human-AI co-agency | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 110: Human-AI co-agency; original author status: Strong constructive fit. Native state: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Model output or benchmark score treated as the whole system. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. The relevant receiving carriers are: User, non-user, model operator, workers, institutions, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed native-vs-K_R assisted agent benchmark. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Native units and dimensional integrity

Human-AI co-agency | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 110: Human-AI co-agency; original author status: Strong constructive fit. Native state: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Model output or benchmark score treated as the whole system. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. The relevant receiving carriers are: User, non-user, model operator, workers, institutions, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed native-vs-K_R assisted agent benchmark. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Capacity specification

Human-AI co-agency | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 110: Human-AI co-agency; original author status: Strong constructive fit. Native state: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Model output or benchmark score treated as the whole system. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. The relevant receiving carriers are: User, non-user, model operator, workers, institutions, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed native-vs-K_R assisted agent benchmark. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Intervention design

Human-AI co-agency | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 110: Human-AI co-agency; original author status: Strong constructive fit. Native state: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Model output or benchmark score treated as the whole system. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. The relevant receiving carriers are: User, non-user, model operator, workers, institutions, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed native-vs-K_R assisted agent benchmark. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Incumbent parity contest

Human-AI co-agency | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 110: Human-AI co-agency; original author status: Strong constructive fit. Native state: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Model output or benchmark score treated as the whole system. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. The relevant receiving carriers are: User, non-user, model operator, workers, institutions, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed native-vs-K_R assisted agent benchmark. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Time-window and recurrence

Human-AI co-agency | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 110: Human-AI co-agency; original author status: Strong constructive fit. Native state: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Model output or benchmark score treated as the whole system. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. The relevant receiving carriers are: User, non-user, model operator, workers, institutions, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed native-vs-K_R assisted agent benchmark. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Ethical and legal safety

Human-AI co-agency | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 110: Human-AI co-agency; original author status: Strong constructive fit. Native state: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Model output or benchmark score treated as the whole system. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. The relevant receiving carriers are: User, non-user, model operator, workers, institutions, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed native-vs-K_R assisted agent benchmark. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Sensitivity and negative controls

Human-AI co-agency | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 110: Human-AI co-agency; original author status: Strong constructive fit. Native state: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Model output or benchmark score treated as the whole system. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. The relevant receiving carriers are: User, non-user, model operator, workers, institutions, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed native-vs-K_R assisted agent benchmark. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Replication and provenance

Human-AI co-agency | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 110: Human-AI co-agency; original author status: Strong constructive fit. Native state: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Model output or benchmark score treated as the whole system. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. The relevant receiving carriers are: User, non-user, model operator, workers, institutions, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed native-vs-K_R assisted agent benchmark. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

110: Cross-domain invariant contest

Human-AI co-agency | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 110: Human-AI co-agency; original author status: Strong constructive fit. Native state: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Model output or benchmark score treated as the whole system. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Capability gain shifts verification, dependency, control or error burden onto humans and downstream systems. The relevant receiving carriers are: User, non-user, model operator, workers, institutions, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability rises while human agency, correctness, reversibility and surrounding-system viability also improve.

RETEST AND SCOPE CONTROL

Candidate decisive method: Sealed native-vs-K_R assisted agent benchmark. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 110, pp. 824-830 (Rufino 2026).

Adapter 111: Evolution and natural selection

Evolution, transformation and emergence | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history

RECORDED RECEIVING SCHEMA

Fitness proxy or single-trait narrative

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Candidate receivers: Individuals, lineages, populations, interacting species and ecosystems.

EXISTING CAPACITY DESIGN

Diversity, ecological capacity and intervention guardrails where deliberate action exists

PROTECTED OUTCOME STATED BY THE AUTHOR

Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Time-locked population/genomic holdout. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111 | Evolution and natural selection

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 111, Evolution and natural selection. Native state: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Potential direct or institutional human pathway: Evolution and natural selection interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Diversity, ecological capacity and intervention guardrails where deliberate action exists can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

111 | Evolution and natural selection

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 111, Evolution and natural selection. Native state: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Potential direct or institutional human pathway: Evolution and natural selection interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Diversity, ecological capacity and intervention guardrails where deliberate action exists can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

111 | Evolution and natural selection

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 111, Evolution and natural selection. Native state: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Potential direct or institutional human pathway: Evolution and natural selection interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Diversity, ecological capacity and intervention guardrails where deliberate action exists can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

111 | Evolution and natural selection

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 111, Evolution and natural selection. Native state: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Potential direct or institutional human pathway: Evolution and natural selection interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Diversity, ecological capacity and intervention guardrails where deliberate action exists can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

111 | Evolution and natural selection

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 111, Evolution and natural selection. Native state: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Potential direct or institutional human pathway: Evolution and natural selection interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Diversity, ecological capacity and intervention guardrails where deliberate action exists can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

111 | Evolution and natural selection

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 111, Evolution and natural selection. Native state: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Potential direct or institutional human pathway: Evolution and natural selection interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Diversity, ecological capacity and intervention guardrails where deliberate action exists can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

111 to 001: consequential handoff

Evolution and natural selection -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

111 to 002: consequential handoff

Evolution and natural selection -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

111 to 003: consequential handoff

Evolution and natural selection -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

111 to 004: consequential handoff

Evolution and natural selection -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

111 to 005: consequential handoff

Evolution and natural selection -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

111 to 006: consequential handoff

Evolution and natural selection -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

111 to 007: consequential handoff

Evolution and natural selection -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

111 to 008: consequential handoff

Evolution and natural selection -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

111 to 009: consequential handoff

Evolution and natural selection -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

111 to 010: consequential handoff

Evolution and natural selection -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

111 to 011: consequential handoff

Evolution and natural selection -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

111 to 012: consequential handoff

Evolution and natural selection -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

111 to 013: consequential handoff

Evolution and natural selection -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

111 to 014: consequential handoff

Evolution and natural selection -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

111 to 015: consequential handoff

Evolution and natural selection -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

111 to 016: consequential handoff

Evolution and natural selection -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

111 to 017: consequential handoff

Evolution and natural selection -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

111 to 018: consequential handoff

Evolution and natural selection -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

111 to 019: consequential handoff

Evolution and natural selection -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

111 to 020: consequential handoff

Evolution and natural selection -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

111 to 021: consequential handoff

Evolution and natural selection -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

111 to 022: consequential handoff

Evolution and natural selection -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

111 to 023: consequential handoff

Evolution and natural selection -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

111 to 024: consequential handoff

Evolution and natural selection -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

111 to 025: consequential handoff

Evolution and natural selection -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

111 to 026: consequential handoff

Evolution and natural selection -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

111 to 027: consequential handoff

Evolution and natural selection -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

111 to 028: consequential handoff

Evolution and natural selection -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

111 to 029: consequential handoff

Evolution and natural selection -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

111 to 030: consequential handoff

Evolution and natural selection -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

111 to 031: consequential handoff

Evolution and natural selection -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

111 to 032: consequential handoff

Evolution and natural selection -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

111 to 033: consequential handoff

Evolution and natural selection -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

111 to 034: consequential handoff

Evolution and natural selection -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

111 to 035: consequential handoff

Evolution and natural selection -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

111 to 036: consequential handoff

Evolution and natural selection -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

111 to 037: consequential handoff

Evolution and natural selection -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

111 to 038: consequential handoff

Evolution and natural selection -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

111 to 039: consequential handoff

Evolution and natural selection -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

111 to 040: consequential handoff

Evolution and natural selection -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

111 to 041: consequential handoff

Evolution and natural selection -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

111 to 042: consequential handoff

Evolution and natural selection -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

111 to 043: consequential handoff

Evolution and natural selection -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

111 to 044: consequential handoff

Evolution and natural selection -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

111 to 045: consequential handoff

Evolution and natural selection -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

111 to 046: consequential handoff

Evolution and natural selection -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

111 to 047: consequential handoff

Evolution and natural selection -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

111 to 048: consequential handoff

Evolution and natural selection -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

111 to 049: consequential handoff

Evolution and natural selection -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

111 to 050: consequential handoff

Evolution and natural selection -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

111 to 051: consequential handoff

Evolution and natural selection -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

111 to 052: consequential handoff

Evolution and natural selection -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

111 to 053: consequential handoff

Evolution and natural selection -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

111 to 054: consequential handoff

Evolution and natural selection -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

111 to 055: consequential handoff

Evolution and natural selection -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

111 to 056: consequential handoff

Evolution and natural selection -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

111 to 057: consequential handoff

Evolution and natural selection -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

111 to 058: consequential handoff

Evolution and natural selection -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

111 to 059: consequential handoff

Evolution and natural selection -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

111 to 060: consequential handoff

Evolution and natural selection -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

111 to 061: consequential handoff

Evolution and natural selection -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

111 to 062: consequential handoff

Evolution and natural selection -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

111 to 063: consequential handoff

Evolution and natural selection -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

111 to 064: consequential handoff

Evolution and natural selection -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

111 to 065: consequential handoff

Evolution and natural selection -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

111 to 066: consequential handoff

Evolution and natural selection -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

111 to 067: consequential handoff

Evolution and natural selection -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

111 to 068: consequential handoff

Evolution and natural selection -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

111 to 069: consequential handoff

Evolution and natural selection -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

111 to 070: consequential handoff

Evolution and natural selection -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

111 to 071: consequential handoff

Evolution and natural selection -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

111 to 072: consequential handoff

Evolution and natural selection -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

111 to 073: consequential handoff

Evolution and natural selection -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

111 to 074: consequential handoff

Evolution and natural selection -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

111 to 075: consequential handoff

Evolution and natural selection -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

111 to 076: consequential handoff

Evolution and natural selection -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

111 to 077: consequential handoff

Evolution and natural selection -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

111 to 078: consequential handoff

Evolution and natural selection -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

111 to 079: consequential handoff

Evolution and natural selection -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

111 to 080: consequential handoff

Evolution and natural selection -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

111 to 081: consequential handoff

Evolution and natural selection -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

111 to 082: consequential handoff

Evolution and natural selection -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

111 to 083: consequential handoff

Evolution and natural selection -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

111 to 084: consequential handoff

Evolution and natural selection -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

111 to 085: consequential handoff

Evolution and natural selection -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

111 to 086: consequential handoff

Evolution and natural selection -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

111 to 087: consequential handoff

Evolution and natural selection -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

111 to 088: consequential handoff

Evolution and natural selection -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

111 to 089: consequential handoff

Evolution and natural selection -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

111 to 090: consequential handoff

Evolution and natural selection -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

111 to 091: consequential handoff

Evolution and natural selection -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

111 to 092: consequential handoff

Evolution and natural selection -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

111 to 093: consequential handoff

Evolution and natural selection -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

111 to 094: consequential handoff

Evolution and natural selection -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

111 to 095: consequential handoff

Evolution and natural selection -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

111 to 096: consequential handoff

Evolution and natural selection -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

111 to 097: consequential handoff

Evolution and natural selection -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

111 to 098: consequential handoff

Evolution and natural selection -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

111 to 099: consequential handoff

Evolution and natural selection -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

111 to 100: consequential handoff

Evolution and natural selection -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

111 to 101: consequential handoff

Evolution and natural selection -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

111 to 102: consequential handoff

Evolution and natural selection -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

111 to 103: consequential handoff

Evolution and natural selection -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

111 to 104: consequential handoff

Evolution and natural selection -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

111 to 105: consequential handoff

Evolution and natural selection -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

111 to 106: consequential handoff

Evolution and natural selection -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

111 to 107: consequential handoff

Evolution and natural selection -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

111 to 108: consequential handoff

Evolution and natural selection -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

111 to 109: consequential handoff

Evolution and natural selection -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

111 to 110: consequential handoff

Evolution and natural selection -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

111 to 111: consequential handoff

Evolution and natural selection -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Evolution and natural selection becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 832-838; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

111 to 112: consequential handoff

Evolution and natural selection -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

111 to 113: consequential handoff

Evolution and natural selection -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

111 to 114: consequential handoff

Evolution and natural selection -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

111 to 115: consequential handoff

Evolution and natural selection -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

111 to 116: consequential handoff

Evolution and natural selection -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

111 to 117: consequential handoff

Evolution and natural selection -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

111 to 118: consequential handoff

Evolution and natural selection -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

111 to 119: consequential handoff

Evolution and natural selection -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

111 to 120: consequential handoff

Evolution and natural selection -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Evolution and natural selection into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Fitness proxy or single-trait narrative. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Individuals, lineages, populations, interacting species and ecosystems. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked population/genomic holdout' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 832-838 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

111: Evolution and natural selection

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

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Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

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Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Use Time-locked population/genomic holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

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Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

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Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

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Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

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Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

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Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

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Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

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Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

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Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

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Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Individuals, lineages, populations, interacting species and ecosystems. Identify pathways and hidden compensation using credible data rather than narrative fit.

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Evaluate a capacity build based on: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

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Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

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Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

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Author's native acceptance: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

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111: Evolution and natural selection

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

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Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evolution and natural selection

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

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Evolution and natural selection is described in the Atlas as: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

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Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Evidence and source reconstruction

Evolution and natural selection | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 111: Evolution and natural selection; original author status: Hard scientific translation frontier. Native state: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Fitness proxy or single-trait narrative. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. The relevant receiving carriers are: Individuals, lineages, populations, interacting species and ecosystems.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems.

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked population/genomic holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Boundary divergence

Evolution and natural selection | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 111: Evolution and natural selection; original author status: Hard scientific translation frontier. Native state: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Fitness proxy or single-trait narrative. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. The relevant receiving carriers are: Individuals, lineages, populations, interacting species and ecosystems.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems.

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked population/genomic holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Causal diagram and identifiability

Evolution and natural selection | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 111: Evolution and natural selection; original author status: Hard scientific translation frontier. Native state: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Fitness proxy or single-trait narrative. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. The relevant receiving carriers are: Individuals, lineages, populations, interacting species and ecosystems.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems.

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked population/genomic holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Receiving carrier identification

Evolution and natural selection | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 111: Evolution and natural selection; original author status: Hard scientific translation frontier. Native state: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Fitness proxy or single-trait narrative. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. The relevant receiving carriers are: Individuals, lineages, populations, interacting species and ecosystems.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems.

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked population/genomic holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Hidden compensation accounting

Evolution and natural selection | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 111: Evolution and natural selection; original author status: Hard scientific translation frontier. Native state: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Fitness proxy or single-trait narrative. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. The relevant receiving carriers are: Individuals, lineages, populations, interacting species and ecosystems.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems.

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked population/genomic holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Native units and dimensional integrity

Evolution and natural selection | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 111: Evolution and natural selection; original author status: Hard scientific translation frontier. Native state: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Fitness proxy or single-trait narrative. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. The relevant receiving carriers are: Individuals, lineages, populations, interacting species and ecosystems.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems.

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked population/genomic holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Capacity specification

Evolution and natural selection | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 111: Evolution and natural selection; original author status: Hard scientific translation frontier. Native state: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Fitness proxy or single-trait narrative. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. The relevant receiving carriers are: Individuals, lineages, populations, interacting species and ecosystems.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems.

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked population/genomic holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Intervention design

Evolution and natural selection | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 111: Evolution and natural selection; original author status: Hard scientific translation frontier. Native state: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Fitness proxy or single-trait narrative. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. The relevant receiving carriers are: Individuals, lineages, populations, interacting species and ecosystems.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems.

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked population/genomic holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Incumbent parity contest

Evolution and natural selection | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 111: Evolution and natural selection; original author status: Hard scientific translation frontier. Native state: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Fitness proxy or single-trait narrative. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. The relevant receiving carriers are: Individuals, lineages, populations, interacting species and ecosystems.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems.

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked population/genomic holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Time-window and recurrence

Evolution and natural selection | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 111: Evolution and natural selection; original author status: Hard scientific translation frontier. Native state: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Fitness proxy or single-trait narrative. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. The relevant receiving carriers are: Individuals, lineages, populations, interacting species and ecosystems.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems.

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked population/genomic holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Ethical and legal safety

Evolution and natural selection | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 111: Evolution and natural selection; original author status: Hard scientific translation frontier. Native state: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Fitness proxy or single-trait narrative. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. The relevant receiving carriers are: Individuals, lineages, populations, interacting species and ecosystems.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems.

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked population/genomic holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Sensitivity and negative controls

Evolution and natural selection | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 111: Evolution and natural selection; original author status: Hard scientific translation frontier. Native state: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Fitness proxy or single-trait narrative. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. The relevant receiving carriers are: Individuals, lineages, populations, interacting species and ecosystems.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems.

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked population/genomic holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Replication and provenance

Evolution and natural selection | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 111: Evolution and natural selection; original author status: Hard scientific translation frontier. Native state: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Fitness proxy or single-trait narrative. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. The relevant receiving carriers are: Individuals, lineages, populations, interacting species and ecosystems.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems.

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked population/genomic holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

111: Cross-domain invariant contest

Evolution and natural selection | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 111: Evolution and natural selection; original author status: Hard scientific translation frontier. Native state: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Fitness proxy or single-trait narrative. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Local adaptation increases one lineage/trait while hidden ecological or long-horizon costs move elsewhere. The relevant receiving carriers are: Individuals, lineages, populations, interacting species and ecosystems.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Persistence/adaptation improves without avoidable collapse in materially coupled populations or ecosystems.

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked population/genomic holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 111, pp. 832-838 (Rufino 2026).

Adapter 112: Reproduction, inheritance, mitosis and meiosis

Evolution, transformation and emergence | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context

RECORDED RECEIVING SCHEMA

Outcome-only birth/division count

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Propagation success hides fidelity loss, developmental burden or downstream instability.
Candidate receivers: Parent cell/organism, daughter cells, gametes, offspring and lineage.

EXISTING CAPACITY DESIGN

Checkpoint, repair, redundancy and developmental-support capacity

PROTECTED OUTCOME STATED BY THE AUTHOR

Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Lineage-resolved perturbation study. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112 | Reproduction, inheritance, mitosis and meiosis

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 112, Reproduction, inheritance, mitosis and meiosis. Native state: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Potential direct or institutional human pathway: Reproduction, inheritance, mitosis and meiosis interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Propagation success hides fidelity loss, developmental burden or downstream instability.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Checkpoint, repair, redundancy and developmental-support capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

112 | Reproduction, inheritance, mitosis and meiosis

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 112, Reproduction, inheritance, mitosis and meiosis. Native state: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Potential direct or institutional human pathway: Reproduction, inheritance, mitosis and meiosis interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Propagation success hides fidelity loss, developmental burden or downstream instability.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Checkpoint, repair, redundancy and developmental-support capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

112 | Reproduction, inheritance, mitosis and meiosis

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 112, Reproduction, inheritance, mitosis and meiosis. Native state: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Potential direct or institutional human pathway: Reproduction, inheritance, mitosis and meiosis interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Propagation success hides fidelity loss, developmental burden or downstream instability.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Checkpoint, repair, redundancy and developmental-support capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

112 | Reproduction, inheritance, mitosis and meiosis

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 112, Reproduction, inheritance, mitosis and meiosis. Native state: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Potential direct or institutional human pathway: Reproduction, inheritance, mitosis and meiosis interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Propagation success hides fidelity loss, developmental burden or downstream instability.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Checkpoint, repair, redundancy and developmental-support capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

112 | Reproduction, inheritance, mitosis and meiosis

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 112, Reproduction, inheritance, mitosis and meiosis. Native state: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Potential direct or institutional human pathway: Reproduction, inheritance, mitosis and meiosis interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Propagation success hides fidelity loss, developmental burden or downstream instability.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Checkpoint, repair, redundancy and developmental-support capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

112 | Reproduction, inheritance, mitosis and meiosis

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 112, Reproduction, inheritance, mitosis and meiosis. Native state: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Potential direct or institutional human pathway: Reproduction, inheritance, mitosis and meiosis interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Propagation success hides fidelity loss, developmental burden or downstream instability.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Checkpoint, repair, redundancy and developmental-support capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

112 to 001: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

112 to 002: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

112 to 003: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

112 to 004: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

112 to 005: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

112 to 006: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

112 to 007: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

112 to 008: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

112 to 009: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

112 to 010: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

112 to 011: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

112 to 012: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

112 to 013: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

112 to 014: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

112 to 015: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

112 to 016: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

112 to 017: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

112 to 018: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

112 to 019: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

112 to 020: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

112 to 021: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

112 to 022: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

112 to 023: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

112 to 024: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

112 to 025: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

112 to 026: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

112 to 027: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

112 to 028: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

112 to 029: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

112 to 030: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

112 to 031: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

112 to 032: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

112 to 033: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

112 to 034: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

112 to 035: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

112 to 036: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

112 to 037: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

112 to 038: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

112 to 039: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

112 to 040: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

112 to 041: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

112 to 042: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

112 to 043: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

112 to 044: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

112 to 045: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

112 to 046: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

112 to 047: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

112 to 048: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

112 to 049: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

112 to 050: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

112 to 051: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

112 to 052: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

112 to 053: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

112 to 054: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

112 to 055: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

112 to 056: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

112 to 057: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

112 to 058: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

112 to 059: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

112 to 060: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

112 to 061: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

112 to 062: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

112 to 063: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

112 to 064: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

112 to 065: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

112 to 066: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

112 to 067: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

112 to 068: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

112 to 069: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

112 to 070: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

112 to 071: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

112 to 072: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

112 to 073: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

112 to 074: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

112 to 075: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

112 to 076: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

112 to 077: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

112 to 078: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

112 to 079: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

112 to 080: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

112 to 081: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

112 to 082: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

112 to 083: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

112 to 084: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

112 to 085: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

112 to 086: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

112 to 087: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

112 to 088: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

112 to 089: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

112 to 090: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

112 to 091: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

112 to 092: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

112 to 093: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

112 to 094: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

112 to 095: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

112 to 096: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

112 to 097: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

112 to 098: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

112 to 099: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

112 to 100: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

112 to 101: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

112 to 102: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

112 to 103: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

112 to 104: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

112 to 105: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

112 to 106: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

112 to 107: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

112 to 108: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

112 to 109: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

112 to 110: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

112 to 111: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

112 to 112: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Reproduction, inheritance, mitosis and meiosis becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 839-845; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

112 to 113: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

112 to 114: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

112 to 115: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

112 to 116: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

112 to 117: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

112 to 118: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

112 to 119: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

112 to 120: consequential handoff

Reproduction, inheritance, mitosis and meiosis -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Reproduction, inheritance, mitosis and meiosis into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Outcome-only birth/division count. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Checkpoint, repair, redundancy and developmental-support capacity. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Lineage-resolved perturbation study' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 839-845 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

112: Reproduction, inheritance, mitosis and meiosis

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Reproduction, inheritance, mitosis and meiosis

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Reproduction, inheritance, mitosis and meiosis is described in the Atlas as: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Propagation success hides fidelity loss, developmental burden or downstream instability. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Parent cell/organism, daughter cells, gametes, offspring and lineage. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Checkpoint, repair, redundancy and developmental-support capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Lineage-resolved perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Evidence and source reconstruction

Reproduction, inheritance, mitosis and meiosis | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 112: Reproduction, inheritance, mitosis and meiosis; original author status: Strong structural translation. Native state: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Outcome-only birth/division count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Propagation success hides fidelity loss, developmental burden or downstream instability. The relevant receiving carriers are: Parent cell/organism, daughter cells, gametes, offspring and lineage.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Checkpoint, repair, redundancy and developmental-support capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Lineage-resolved perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Boundary divergence

Reproduction, inheritance, mitosis and meiosis | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 112: Reproduction, inheritance, mitosis and meiosis; original author status: Strong structural translation. Native state: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Outcome-only birth/division count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Propagation success hides fidelity loss, developmental burden or downstream instability. The relevant receiving carriers are: Parent cell/organism, daughter cells, gametes, offspring and lineage.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Checkpoint, repair, redundancy and developmental-support capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Lineage-resolved perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Causal diagram and identifiability

Reproduction, inheritance, mitosis and meiosis | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 112: Reproduction, inheritance, mitosis and meiosis; original author status: Strong structural translation. Native state: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Outcome-only birth/division count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Propagation success hides fidelity loss, developmental burden or downstream instability. The relevant receiving carriers are: Parent cell/organism, daughter cells, gametes, offspring and lineage.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Checkpoint, repair, redundancy and developmental-support capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Lineage-resolved perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Receiving carrier identification

Reproduction, inheritance, mitosis and meiosis | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 112: Reproduction, inheritance, mitosis and meiosis; original author status: Strong structural translation. Native state: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Outcome-only birth/division count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Propagation success hides fidelity loss, developmental burden or downstream instability. The relevant receiving carriers are: Parent cell/organism, daughter cells, gametes, offspring and lineage.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Checkpoint, repair, redundancy and developmental-support capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Lineage-resolved perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Hidden compensation accounting

Reproduction, inheritance, mitosis and meiosis | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 112: Reproduction, inheritance, mitosis and meiosis; original author status: Strong structural translation. Native state: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Outcome-only birth/division count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Propagation success hides fidelity loss, developmental burden or downstream instability. The relevant receiving carriers are: Parent cell/organism, daughter cells, gametes, offspring and lineage.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Checkpoint, repair, redundancy and developmental-support capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Lineage-resolved perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Native units and dimensional integrity

Reproduction, inheritance, mitosis and meiosis | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 112: Reproduction, inheritance, mitosis and meiosis; original author status: Strong structural translation. Native state: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Outcome-only birth/division count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Propagation success hides fidelity loss, developmental burden or downstream instability. The relevant receiving carriers are: Parent cell/organism, daughter cells, gametes, offspring and lineage.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Checkpoint, repair, redundancy and developmental-support capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Lineage-resolved perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Capacity specification

Reproduction, inheritance, mitosis and meiosis | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 112: Reproduction, inheritance, mitosis and meiosis; original author status: Strong structural translation. Native state: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Outcome-only birth/division count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Propagation success hides fidelity loss, developmental burden or downstream instability. The relevant receiving carriers are: Parent cell/organism, daughter cells, gametes, offspring and lineage.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Checkpoint, repair, redundancy and developmental-support capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Lineage-resolved perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Intervention design

Reproduction, inheritance, mitosis and meiosis | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 112: Reproduction, inheritance, mitosis and meiosis; original author status: Strong structural translation. Native state: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Outcome-only birth/division count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Propagation success hides fidelity loss, developmental burden or downstream instability. The relevant receiving carriers are: Parent cell/organism, daughter cells, gametes, offspring and lineage.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Checkpoint, repair, redundancy and developmental-support capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Lineage-resolved perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Incumbent parity contest

Reproduction, inheritance, mitosis and meiosis | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 112: Reproduction, inheritance, mitosis and meiosis; original author status: Strong structural translation. Native state: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Outcome-only birth/division count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Propagation success hides fidelity loss, developmental burden or downstream instability. The relevant receiving carriers are: Parent cell/organism, daughter cells, gametes, offspring and lineage.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Checkpoint, repair, redundancy and developmental-support capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Lineage-resolved perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Time-window and recurrence

Reproduction, inheritance, mitosis and meiosis | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 112: Reproduction, inheritance, mitosis and meiosis; original author status: Strong structural translation. Native state: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Outcome-only birth/division count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Propagation success hides fidelity loss, developmental burden or downstream instability. The relevant receiving carriers are: Parent cell/organism, daughter cells, gametes, offspring and lineage.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Checkpoint, repair, redundancy and developmental-support capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Lineage-resolved perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Ethical and legal safety

Reproduction, inheritance, mitosis and meiosis | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 112: Reproduction, inheritance, mitosis and meiosis; original author status: Strong structural translation. Native state: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Outcome-only birth/division count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Propagation success hides fidelity loss, developmental burden or downstream instability. The relevant receiving carriers are: Parent cell/organism, daughter cells, gametes, offspring and lineage.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Checkpoint, repair, redundancy and developmental-support capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Lineage-resolved perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Sensitivity and negative controls

Reproduction, inheritance, mitosis and meiosis | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 112: Reproduction, inheritance, mitosis and meiosis; original author status: Strong structural translation. Native state: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Outcome-only birth/division count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Propagation success hides fidelity loss, developmental burden or downstream instability. The relevant receiving carriers are: Parent cell/organism, daughter cells, gametes, offspring and lineage.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Checkpoint, repair, redundancy and developmental-support capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Lineage-resolved perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Replication and provenance

Reproduction, inheritance, mitosis and meiosis | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 112: Reproduction, inheritance, mitosis and meiosis; original author status: Strong structural translation. Native state: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Outcome-only birth/division count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Propagation success hides fidelity loss, developmental burden or downstream instability. The relevant receiving carriers are: Parent cell/organism, daughter cells, gametes, offspring and lineage.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Checkpoint, repair, redundancy and developmental-support capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Lineage-resolved perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

112: Cross-domain invariant contest

Reproduction, inheritance, mitosis and meiosis | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 112: Reproduction, inheritance, mitosis and meiosis; original author status: Strong structural translation. Native state: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Outcome-only birth/division count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Propagation success hides fidelity loss, developmental burden or downstream instability. The relevant receiving carriers are: Parent cell/organism, daughter cells, gametes, offspring and lineage.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Checkpoint, repair, redundancy and developmental-support capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Propagation remains viable while fidelity, diversity and downstream function stay within safe bounds.

RETEST AND SCOPE CONTROL

Candidate decisive method: Lineage-resolved perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 112, pp. 839-845 (Rufino 2026).

Adapter 113: Development and morphogenesis

Evolution, transformation and emergence | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time

RECORDED RECEIVING SCHEMA

Single marker or endpoint morphology

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Candidate receivers: Cells, tissues, organs, organism and developmental future state.

EXISTING CAPACITY DESIGN

Stage-aware multi-pathway support, repair and resource architecture

PROTECTED OUTCOME STATED BY THE AUTHOR

Functional development improves without compensatory developmental injury

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Longitudinal developmental perturbation study. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113 | Development and morphogenesis

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 113, Development and morphogenesis. Native state: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Potential direct or institutional human pathway: Development and morphogenesis interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Stage-aware multi-pathway support, repair and resource architecture can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Functional development improves without compensatory developmental injury. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

113 | Development and morphogenesis

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 113, Development and morphogenesis. Native state: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Potential direct or institutional human pathway: Development and morphogenesis interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Stage-aware multi-pathway support, repair and resource architecture can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Functional development improves without compensatory developmental injury. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

113 | Development and morphogenesis

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 113, Development and morphogenesis. Native state: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Potential direct or institutional human pathway: Development and morphogenesis interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Stage-aware multi-pathway support, repair and resource architecture can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Functional development improves without compensatory developmental injury. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

113 | Development and morphogenesis

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 113, Development and morphogenesis. Native state: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Potential direct or institutional human pathway: Development and morphogenesis interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Stage-aware multi-pathway support, repair and resource architecture can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Functional development improves without compensatory developmental injury. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

113 | Development and morphogenesis

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 113, Development and morphogenesis. Native state: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Potential direct or institutional human pathway: Development and morphogenesis interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Stage-aware multi-pathway support, repair and resource architecture can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Functional development improves without compensatory developmental injury. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

113 | Development and morphogenesis

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 113, Development and morphogenesis. Native state: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Potential direct or institutional human pathway: Development and morphogenesis interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Stage-aware multi-pathway support, repair and resource architecture can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Functional development improves without compensatory developmental injury. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

113 to 001: consequential handoff

Development and morphogenesis -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

113 to 002: consequential handoff

Development and morphogenesis -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

113 to 003: consequential handoff

Development and morphogenesis -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

113 to 004: consequential handoff

Development and morphogenesis -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

113 to 005: consequential handoff

Development and morphogenesis -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

113 to 006: consequential handoff

Development and morphogenesis -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

113 to 007: consequential handoff

Development and morphogenesis -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

113 to 008: consequential handoff

Development and morphogenesis -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

113 to 009: consequential handoff

Development and morphogenesis -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

113 to 010: consequential handoff

Development and morphogenesis -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

113 to 011: consequential handoff

Development and morphogenesis -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

113 to 012: consequential handoff

Development and morphogenesis -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

113 to 013: consequential handoff

Development and morphogenesis -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

113 to 014: consequential handoff

Development and morphogenesis -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

113 to 015: consequential handoff

Development and morphogenesis -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

113 to 016: consequential handoff

Development and morphogenesis -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

113 to 017: consequential handoff

Development and morphogenesis -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

113 to 018: consequential handoff

Development and morphogenesis -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

113 to 019: consequential handoff

Development and morphogenesis -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

113 to 020: consequential handoff

Development and morphogenesis -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

113 to 021: consequential handoff

Development and morphogenesis -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

113 to 022: consequential handoff

Development and morphogenesis -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

113 to 023: consequential handoff

Development and morphogenesis -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

113 to 024: consequential handoff

Development and morphogenesis -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

113 to 025: consequential handoff

Development and morphogenesis -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

113 to 026: consequential handoff

Development and morphogenesis -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

113 to 027: consequential handoff

Development and morphogenesis -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

113 to 028: consequential handoff

Development and morphogenesis -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

113 to 029: consequential handoff

Development and morphogenesis -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

113 to 030: consequential handoff

Development and morphogenesis -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

113 to 031: consequential handoff

Development and morphogenesis -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

113 to 032: consequential handoff

Development and morphogenesis -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

113 to 033: consequential handoff

Development and morphogenesis -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

113 to 034: consequential handoff

Development and morphogenesis -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

113 to 035: consequential handoff

Development and morphogenesis -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

113 to 036: consequential handoff

Development and morphogenesis -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

113 to 037: consequential handoff

Development and morphogenesis -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

113 to 038: consequential handoff

Development and morphogenesis -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

113 to 039: consequential handoff

Development and morphogenesis -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

113 to 040: consequential handoff

Development and morphogenesis -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

113 to 041: consequential handoff

Development and morphogenesis -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

113 to 042: consequential handoff

Development and morphogenesis -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

113 to 043: consequential handoff

Development and morphogenesis -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

113 to 044: consequential handoff

Development and morphogenesis -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

113 to 045: consequential handoff

Development and morphogenesis -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

113 to 046: consequential handoff

Development and morphogenesis -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

113 to 047: consequential handoff

Development and morphogenesis -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

113 to 048: consequential handoff

Development and morphogenesis -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

113 to 049: consequential handoff

Development and morphogenesis -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

113 to 050: consequential handoff

Development and morphogenesis -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

113 to 051: consequential handoff

Development and morphogenesis -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

113 to 052: consequential handoff

Development and morphogenesis -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

113 to 053: consequential handoff

Development and morphogenesis -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

113 to 054: consequential handoff

Development and morphogenesis -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

113 to 055: consequential handoff

Development and morphogenesis -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

113 to 056: consequential handoff

Development and morphogenesis -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

113 to 057: consequential handoff

Development and morphogenesis -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

113 to 058: consequential handoff

Development and morphogenesis -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

113 to 059: consequential handoff

Development and morphogenesis -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

113 to 060: consequential handoff

Development and morphogenesis -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

113 to 061: consequential handoff

Development and morphogenesis -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

113 to 062: consequential handoff

Development and morphogenesis -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

113 to 063: consequential handoff

Development and morphogenesis -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

113 to 064: consequential handoff

Development and morphogenesis -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

113 to 065: consequential handoff

Development and morphogenesis -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

113 to 066: consequential handoff

Development and morphogenesis -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

113 to 067: consequential handoff

Development and morphogenesis -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

113 to 068: consequential handoff

Development and morphogenesis -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

113 to 069: consequential handoff

Development and morphogenesis -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

113 to 070: consequential handoff

Development and morphogenesis -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

113 to 071: consequential handoff

Development and morphogenesis -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

113 to 072: consequential handoff

Development and morphogenesis -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

113 to 073: consequential handoff

Development and morphogenesis -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

113 to 074: consequential handoff

Development and morphogenesis -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

113 to 075: consequential handoff

Development and morphogenesis -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

113 to 076: consequential handoff

Development and morphogenesis -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

113 to 077: consequential handoff

Development and morphogenesis -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

113 to 078: consequential handoff

Development and morphogenesis -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

113 to 079: consequential handoff

Development and morphogenesis -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

113 to 080: consequential handoff

Development and morphogenesis -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

113 to 081: consequential handoff

Development and morphogenesis -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

113 to 082: consequential handoff

Development and morphogenesis -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

113 to 083: consequential handoff

Development and morphogenesis -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

113 to 084: consequential handoff

Development and morphogenesis -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

113 to 085: consequential handoff

Development and morphogenesis -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

113 to 086: consequential handoff

Development and morphogenesis -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

113 to 087: consequential handoff

Development and morphogenesis -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

113 to 088: consequential handoff

Development and morphogenesis -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

113 to 089: consequential handoff

Development and morphogenesis -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

113 to 090: consequential handoff

Development and morphogenesis -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

113 to 091: consequential handoff

Development and morphogenesis -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

113 to 092: consequential handoff

Development and morphogenesis -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

113 to 093: consequential handoff

Development and morphogenesis -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

113 to 094: consequential handoff

Development and morphogenesis -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

113 to 095: consequential handoff

Development and morphogenesis -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

113 to 096: consequential handoff

Development and morphogenesis -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

113 to 097: consequential handoff

Development and morphogenesis -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

113 to 098: consequential handoff

Development and morphogenesis -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

113 to 099: consequential handoff

Development and morphogenesis -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

113 to 100: consequential handoff

Development and morphogenesis -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

113 to 101: consequential handoff

Development and morphogenesis -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

113 to 102: consequential handoff

Development and morphogenesis -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

113 to 103: consequential handoff

Development and morphogenesis -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

113 to 104: consequential handoff

Development and morphogenesis -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

113 to 105: consequential handoff

Development and morphogenesis -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

113 to 106: consequential handoff

Development and morphogenesis -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

113 to 107: consequential handoff

Development and morphogenesis -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

113 to 108: consequential handoff

Development and morphogenesis -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

113 to 109: consequential handoff

Development and morphogenesis -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

113 to 110: consequential handoff

Development and morphogenesis -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

113 to 111: consequential handoff

Development and morphogenesis -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

113 to 112: consequential handoff

Development and morphogenesis -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

113 to 113: consequential handoff

Development and morphogenesis -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Development and morphogenesis becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 846-852; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

113 to 114: consequential handoff

Development and morphogenesis -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

113 to 115: consequential handoff

Development and morphogenesis -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

113 to 116: consequential handoff

Development and morphogenesis -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

113 to 117: consequential handoff

Development and morphogenesis -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

113 to 118: consequential handoff

Development and morphogenesis -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

113 to 119: consequential handoff

Development and morphogenesis -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

113 to 120: consequential handoff

Development and morphogenesis -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Development and morphogenesis into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single marker or endpoint morphology. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Cells, tissues, organs, organism and developmental future state. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Stage-aware multi-pathway support, repair and resource architecture. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal developmental perturbation study' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 846-852 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

113: Development and morphogenesis

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Development and morphogenesis

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Development and morphogenesis is described in the Atlas as: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Cells, tissues, organs, organism and developmental future state. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Stage-aware multi-pathway support, repair and resource architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Functional development improves without compensatory developmental injury. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal developmental perturbation study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Evidence and source reconstruction

Development and morphogenesis | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 113: Development and morphogenesis; original author status: Regulated empirical frontier. Native state: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single marker or endpoint morphology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. The relevant receiving carriers are: Cells, tissues, organs, organism and developmental future state.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Stage-aware multi-pathway support, repair and resource architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Functional development improves without compensatory developmental injury.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal developmental perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Boundary divergence

Development and morphogenesis | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 113: Development and morphogenesis; original author status: Regulated empirical frontier. Native state: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single marker or endpoint morphology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. The relevant receiving carriers are: Cells, tissues, organs, organism and developmental future state.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Stage-aware multi-pathway support, repair and resource architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Functional development improves without compensatory developmental injury.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal developmental perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Causal diagram and identifiability

Development and morphogenesis | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 113: Development and morphogenesis; original author status: Regulated empirical frontier. Native state: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single marker or endpoint morphology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. The relevant receiving carriers are: Cells, tissues, organs, organism and developmental future state.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Stage-aware multi-pathway support, repair and resource architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Functional development improves without compensatory developmental injury.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal developmental perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Receiving carrier identification

Development and morphogenesis | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 113: Development and morphogenesis; original author status: Regulated empirical frontier. Native state: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single marker or endpoint morphology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. The relevant receiving carriers are: Cells, tissues, organs, organism and developmental future state.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Stage-aware multi-pathway support, repair and resource architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Functional development improves without compensatory developmental injury.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal developmental perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Hidden compensation accounting

Development and morphogenesis | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 113: Development and morphogenesis; original author status: Regulated empirical frontier. Native state: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single marker or endpoint morphology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. The relevant receiving carriers are: Cells, tissues, organs, organism and developmental future state.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Stage-aware multi-pathway support, repair and resource architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Functional development improves without compensatory developmental injury.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal developmental perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Native units and dimensional integrity

Development and morphogenesis | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 113: Development and morphogenesis; original author status: Regulated empirical frontier. Native state: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single marker or endpoint morphology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. The relevant receiving carriers are: Cells, tissues, organs, organism and developmental future state.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Stage-aware multi-pathway support, repair and resource architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Functional development improves without compensatory developmental injury.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal developmental perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Capacity specification

Development and morphogenesis | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 113: Development and morphogenesis; original author status: Regulated empirical frontier. Native state: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single marker or endpoint morphology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. The relevant receiving carriers are: Cells, tissues, organs, organism and developmental future state.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Stage-aware multi-pathway support, repair and resource architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Functional development improves without compensatory developmental injury.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal developmental perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Intervention design

Development and morphogenesis | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 113: Development and morphogenesis; original author status: Regulated empirical frontier. Native state: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single marker or endpoint morphology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. The relevant receiving carriers are: Cells, tissues, organs, organism and developmental future state.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Stage-aware multi-pathway support, repair and resource architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Functional development improves without compensatory developmental injury.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal developmental perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Incumbent parity contest

Development and morphogenesis | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 113: Development and morphogenesis; original author status: Regulated empirical frontier. Native state: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single marker or endpoint morphology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. The relevant receiving carriers are: Cells, tissues, organs, organism and developmental future state.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Stage-aware multi-pathway support, repair and resource architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Functional development improves without compensatory developmental injury.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal developmental perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Time-window and recurrence

Development and morphogenesis | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 113: Development and morphogenesis; original author status: Regulated empirical frontier. Native state: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single marker or endpoint morphology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. The relevant receiving carriers are: Cells, tissues, organs, organism and developmental future state.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Stage-aware multi-pathway support, repair and resource architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Functional development improves without compensatory developmental injury.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal developmental perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Ethical and legal safety

Development and morphogenesis | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 113: Development and morphogenesis; original author status: Regulated empirical frontier. Native state: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single marker or endpoint morphology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. The relevant receiving carriers are: Cells, tissues, organs, organism and developmental future state.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Stage-aware multi-pathway support, repair and resource architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Functional development improves without compensatory developmental injury.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal developmental perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Sensitivity and negative controls

Development and morphogenesis | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 113: Development and morphogenesis; original author status: Regulated empirical frontier. Native state: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single marker or endpoint morphology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. The relevant receiving carriers are: Cells, tissues, organs, organism and developmental future state.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Stage-aware multi-pathway support, repair and resource architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Functional development improves without compensatory developmental injury.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal developmental perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Replication and provenance

Development and morphogenesis | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 113: Development and morphogenesis; original author status: Regulated empirical frontier. Native state: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single marker or endpoint morphology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. The relevant receiving carriers are: Cells, tissues, organs, organism and developmental future state.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Stage-aware multi-pathway support, repair and resource architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Functional development improves without compensatory developmental injury.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal developmental perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

113: Cross-domain invariant contest

Development and morphogenesis | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 113: Development and morphogenesis; original author status: Regulated empirical frontier. Native state: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single marker or endpoint morphology. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Correcting one developmental feature displaces stress into another tissue, stage or regulatory pathway. The relevant receiving carriers are: Cells, tissues, organs, organism and developmental future state.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Stage-aware multi-pathway support, repair and resource architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Functional development improves without compensatory developmental injury.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal developmental perturbation study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 113, pp. 846-852 (Rufino 2026).

Adapter 114: Metabolism, diet and consumption

Evolution, transformation and emergence | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation

RECORDED RECEIVING SCHEMA

Calories/macros or one nutrient target

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Candidate receivers: Gut, microbiome, liver, tissues, household, food system and future organism.

EXISTING CAPACITY DESIGN

Adaptive conversion, buffering, storage, elimination and access capacity

PROTECTED OUTCOME STATED BY THE AUTHOR

Useful incorporation, metabolic stability, affordability and quality of life improve together

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Controlled longitudinal multi-carrier nutrition study. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114 | Metabolism, diet and consumption

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 114, Metabolism, diet and consumption. Native state: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Potential direct or institutional human pathway: Metabolism, diet and consumption interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Adaptive conversion, buffering, storage, elimination and access capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Useful incorporation, metabolic stability, affordability and quality of life improve together. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

114 | Metabolism, diet and consumption

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 114, Metabolism, diet and consumption. Native state: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Potential direct or institutional human pathway: Metabolism, diet and consumption interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Adaptive conversion, buffering, storage, elimination and access capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Useful incorporation, metabolic stability, affordability and quality of life improve together. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

114 | Metabolism, diet and consumption

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 114, Metabolism, diet and consumption. Native state: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Potential direct or institutional human pathway: Metabolism, diet and consumption interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Adaptive conversion, buffering, storage, elimination and access capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Useful incorporation, metabolic stability, affordability and quality of life improve together. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

114 | Metabolism, diet and consumption

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 114, Metabolism, diet and consumption. Native state: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Potential direct or institutional human pathway: Metabolism, diet and consumption interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Adaptive conversion, buffering, storage, elimination and access capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Useful incorporation, metabolic stability, affordability and quality of life improve together. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

114 | Metabolism, diet and consumption

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 114, Metabolism, diet and consumption. Native state: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Potential direct or institutional human pathway: Metabolism, diet and consumption interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Adaptive conversion, buffering, storage, elimination and access capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Useful incorporation, metabolic stability, affordability and quality of life improve together. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

114 | Metabolism, diet and consumption

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 114, Metabolism, diet and consumption. Native state: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Potential direct or institutional human pathway: Metabolism, diet and consumption interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Adaptive conversion, buffering, storage, elimination and access capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Useful incorporation, metabolic stability, affordability and quality of life improve together. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

114 to 001: consequential handoff

Metabolism, diet and consumption -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

114 to 002: consequential handoff

Metabolism, diet and consumption -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

114 to 003: consequential handoff

Metabolism, diet and consumption -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

114 to 004: consequential handoff

Metabolism, diet and consumption -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

114 to 005: consequential handoff

Metabolism, diet and consumption -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

114 to 006: consequential handoff

Metabolism, diet and consumption -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

114 to 007: consequential handoff

Metabolism, diet and consumption -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

114 to 008: consequential handoff

Metabolism, diet and consumption -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

114 to 009: consequential handoff

Metabolism, diet and consumption -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

114 to 010: consequential handoff

Metabolism, diet and consumption -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

114 to 011: consequential handoff

Metabolism, diet and consumption -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

114 to 012: consequential handoff

Metabolism, diet and consumption -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

114 to 013: consequential handoff

Metabolism, diet and consumption -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

114 to 014: consequential handoff

Metabolism, diet and consumption -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

114 to 015: consequential handoff

Metabolism, diet and consumption -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

114 to 016: consequential handoff

Metabolism, diet and consumption -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

114 to 017: consequential handoff

Metabolism, diet and consumption -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

114 to 018: consequential handoff

Metabolism, diet and consumption -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

114 to 019: consequential handoff

Metabolism, diet and consumption -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

114 to 020: consequential handoff

Metabolism, diet and consumption -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

114 to 021: consequential handoff

Metabolism, diet and consumption -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

114 to 022: consequential handoff

Metabolism, diet and consumption -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

114 to 023: consequential handoff

Metabolism, diet and consumption -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

114 to 024: consequential handoff

Metabolism, diet and consumption -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

114 to 025: consequential handoff

Metabolism, diet and consumption -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

114 to 026: consequential handoff

Metabolism, diet and consumption -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

114 to 027: consequential handoff

Metabolism, diet and consumption -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

114 to 028: consequential handoff

Metabolism, diet and consumption -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

114 to 029: consequential handoff

Metabolism, diet and consumption -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

114 to 030: consequential handoff

Metabolism, diet and consumption -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

114 to 031: consequential handoff

Metabolism, diet and consumption -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

114 to 032: consequential handoff

Metabolism, diet and consumption -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

114 to 033: consequential handoff

Metabolism, diet and consumption -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

114 to 034: consequential handoff

Metabolism, diet and consumption -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

114 to 035: consequential handoff

Metabolism, diet and consumption -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

114 to 036: consequential handoff

Metabolism, diet and consumption -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

114 to 037: consequential handoff

Metabolism, diet and consumption -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

114 to 038: consequential handoff

Metabolism, diet and consumption -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

114 to 039: consequential handoff

Metabolism, diet and consumption -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

114 to 040: consequential handoff

Metabolism, diet and consumption -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

114 to 041: consequential handoff

Metabolism, diet and consumption -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

114 to 042: consequential handoff

Metabolism, diet and consumption -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

114 to 043: consequential handoff

Metabolism, diet and consumption -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

114 to 044: consequential handoff

Metabolism, diet and consumption -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

114 to 045: consequential handoff

Metabolism, diet and consumption -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

114 to 046: consequential handoff

Metabolism, diet and consumption -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

114 to 047: consequential handoff

Metabolism, diet and consumption -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

114 to 048: consequential handoff

Metabolism, diet and consumption -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

114 to 049: consequential handoff

Metabolism, diet and consumption -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

114 to 050: consequential handoff

Metabolism, diet and consumption -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

114 to 051: consequential handoff

Metabolism, diet and consumption -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

114 to 052: consequential handoff

Metabolism, diet and consumption -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

114 to 053: consequential handoff

Metabolism, diet and consumption -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

114 to 054: consequential handoff

Metabolism, diet and consumption -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

114 to 055: consequential handoff

Metabolism, diet and consumption -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

114 to 056: consequential handoff

Metabolism, diet and consumption -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

114 to 057: consequential handoff

Metabolism, diet and consumption -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

114 to 058: consequential handoff

Metabolism, diet and consumption -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

114 to 059: consequential handoff

Metabolism, diet and consumption -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

114 to 060: consequential handoff

Metabolism, diet and consumption -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

114 to 061: consequential handoff

Metabolism, diet and consumption -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

114 to 062: consequential handoff

Metabolism, diet and consumption -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

114 to 063: consequential handoff

Metabolism, diet and consumption -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

114 to 064: consequential handoff

Metabolism, diet and consumption -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

114 to 065: consequential handoff

Metabolism, diet and consumption -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

114 to 066: consequential handoff

Metabolism, diet and consumption -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

114 to 067: consequential handoff

Metabolism, diet and consumption -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

114 to 068: consequential handoff

Metabolism, diet and consumption -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

114 to 069: consequential handoff

Metabolism, diet and consumption -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

114 to 070: consequential handoff

Metabolism, diet and consumption -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

114 to 071: consequential handoff

Metabolism, diet and consumption -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

114 to 072: consequential handoff

Metabolism, diet and consumption -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

114 to 073: consequential handoff

Metabolism, diet and consumption -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

114 to 074: consequential handoff

Metabolism, diet and consumption -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

114 to 075: consequential handoff

Metabolism, diet and consumption -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

114 to 076: consequential handoff

Metabolism, diet and consumption -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

114 to 077: consequential handoff

Metabolism, diet and consumption -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

114 to 078: consequential handoff

Metabolism, diet and consumption -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

114 to 079: consequential handoff

Metabolism, diet and consumption -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

114 to 080: consequential handoff

Metabolism, diet and consumption -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

114 to 081: consequential handoff

Metabolism, diet and consumption -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

114 to 082: consequential handoff

Metabolism, diet and consumption -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

114 to 083: consequential handoff

Metabolism, diet and consumption -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

114 to 084: consequential handoff

Metabolism, diet and consumption -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

114 to 085: consequential handoff

Metabolism, diet and consumption -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

114 to 086: consequential handoff

Metabolism, diet and consumption -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

114 to 087: consequential handoff

Metabolism, diet and consumption -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

114 to 088: consequential handoff

Metabolism, diet and consumption -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

114 to 089: consequential handoff

Metabolism, diet and consumption -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

114 to 090: consequential handoff

Metabolism, diet and consumption -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

114 to 091: consequential handoff

Metabolism, diet and consumption -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

114 to 092: consequential handoff

Metabolism, diet and consumption -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

114 to 093: consequential handoff

Metabolism, diet and consumption -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

114 to 094: consequential handoff

Metabolism, diet and consumption -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

114 to 095: consequential handoff

Metabolism, diet and consumption -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

114 to 096: consequential handoff

Metabolism, diet and consumption -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

114 to 097: consequential handoff

Metabolism, diet and consumption -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

114 to 098: consequential handoff

Metabolism, diet and consumption -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

114 to 099: consequential handoff

Metabolism, diet and consumption -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

114 to 100: consequential handoff

Metabolism, diet and consumption -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

114 to 101: consequential handoff

Metabolism, diet and consumption -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

114 to 102: consequential handoff

Metabolism, diet and consumption -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

114 to 103: consequential handoff

Metabolism, diet and consumption -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

114 to 104: consequential handoff

Metabolism, diet and consumption -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

114 to 105: consequential handoff

Metabolism, diet and consumption -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

114 to 106: consequential handoff

Metabolism, diet and consumption -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

114 to 107: consequential handoff

Metabolism, diet and consumption -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

114 to 108: consequential handoff

Metabolism, diet and consumption -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

114 to 109: consequential handoff

Metabolism, diet and consumption -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

114 to 110: consequential handoff

Metabolism, diet and consumption -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

114 to 111: consequential handoff

Metabolism, diet and consumption -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

114 to 112: consequential handoff

Metabolism, diet and consumption -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

114 to 113: consequential handoff

Metabolism, diet and consumption -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

114 to 114: consequential handoff

Metabolism, diet and consumption -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Metabolism, diet and consumption becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 853-859; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

114 to 115: consequential handoff

Metabolism, diet and consumption -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

114 to 116: consequential handoff

Metabolism, diet and consumption -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

114 to 117: consequential handoff

Metabolism, diet and consumption -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

114 to 118: consequential handoff

Metabolism, diet and consumption -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

114 to 119: consequential handoff

Metabolism, diet and consumption -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

114 to 120: consequential handoff

Metabolism, diet and consumption -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Metabolism, diet and consumption into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Calories/macros or one nutrient target. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Controlled longitudinal multi-carrier nutrition study' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 853-859 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

114: Metabolism, diet and consumption

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

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Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

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Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

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CONSEQUENTIAL SUCCESS MEASUREMENTS

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Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

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Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

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Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

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NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

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MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

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Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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114: Metabolism, diet and consumption

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

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114: Metabolism, diet and consumption

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

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114: Metabolism, diet and consumption

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

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Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

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114: Metabolism, diet and consumption

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

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Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

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Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

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MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

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MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

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MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Metabolism, diet and consumption

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Metabolism, diet and consumption is described in the Atlas as: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Gut, microbiome, liver, tissues, household, food system and future organism. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Adaptive conversion, buffering, storage, elimination and access capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Useful incorporation, metabolic stability, affordability and quality of life improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Controlled longitudinal multi-carrier nutrition study or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Evidence and source reconstruction

Metabolism, diet and consumption | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 114: Metabolism, diet and consumption; original author status: Strong structural fit. Native state: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Calories/macros or one nutrient target. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. The relevant receiving carriers are: Gut, microbiome, liver, tissues, household, food system and future organism.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Adaptive conversion, buffering, storage, elimination and access capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Useful incorporation, metabolic stability, affordability and quality of life improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Controlled longitudinal multi-carrier nutrition study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Boundary divergence

Metabolism, diet and consumption | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 114: Metabolism, diet and consumption; original author status: Strong structural fit. Native state: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Calories/macros or one nutrient target. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. The relevant receiving carriers are: Gut, microbiome, liver, tissues, household, food system and future organism.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Adaptive conversion, buffering, storage, elimination and access capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Useful incorporation, metabolic stability, affordability and quality of life improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Controlled longitudinal multi-carrier nutrition study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Causal diagram and identifiability

Metabolism, diet and consumption | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 114: Metabolism, diet and consumption; original author status: Strong structural fit. Native state: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Calories/macros or one nutrient target. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. The relevant receiving carriers are: Gut, microbiome, liver, tissues, household, food system and future organism.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Adaptive conversion, buffering, storage, elimination and access capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Useful incorporation, metabolic stability, affordability and quality of life improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Controlled longitudinal multi-carrier nutrition study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Receiving carrier identification

Metabolism, diet and consumption | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 114: Metabolism, diet and consumption; original author status: Strong structural fit. Native state: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Calories/macros or one nutrient target. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. The relevant receiving carriers are: Gut, microbiome, liver, tissues, household, food system and future organism.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Adaptive conversion, buffering, storage, elimination and access capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Useful incorporation, metabolic stability, affordability and quality of life improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Controlled longitudinal multi-carrier nutrition study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Hidden compensation accounting

Metabolism, diet and consumption | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 114: Metabolism, diet and consumption; original author status: Strong structural fit. Native state: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Calories/macros or one nutrient target. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. The relevant receiving carriers are: Gut, microbiome, liver, tissues, household, food system and future organism.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Adaptive conversion, buffering, storage, elimination and access capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Useful incorporation, metabolic stability, affordability and quality of life improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Controlled longitudinal multi-carrier nutrition study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Native units and dimensional integrity

Metabolism, diet and consumption | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 114: Metabolism, diet and consumption; original author status: Strong structural fit. Native state: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Calories/macros or one nutrient target. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. The relevant receiving carriers are: Gut, microbiome, liver, tissues, household, food system and future organism.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Adaptive conversion, buffering, storage, elimination and access capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Useful incorporation, metabolic stability, affordability and quality of life improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Controlled longitudinal multi-carrier nutrition study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Capacity specification

Metabolism, diet and consumption | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 114: Metabolism, diet and consumption; original author status: Strong structural fit. Native state: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Calories/macros or one nutrient target. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. The relevant receiving carriers are: Gut, microbiome, liver, tissues, household, food system and future organism.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Adaptive conversion, buffering, storage, elimination and access capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Useful incorporation, metabolic stability, affordability and quality of life improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Controlled longitudinal multi-carrier nutrition study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Intervention design

Metabolism, diet and consumption | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 114: Metabolism, diet and consumption; original author status: Strong structural fit. Native state: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Calories/macros or one nutrient target. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. The relevant receiving carriers are: Gut, microbiome, liver, tissues, household, food system and future organism.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Adaptive conversion, buffering, storage, elimination and access capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Useful incorporation, metabolic stability, affordability and quality of life improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Controlled longitudinal multi-carrier nutrition study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Incumbent parity contest

Metabolism, diet and consumption | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 114: Metabolism, diet and consumption; original author status: Strong structural fit. Native state: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Calories/macros or one nutrient target. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. The relevant receiving carriers are: Gut, microbiome, liver, tissues, household, food system and future organism.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Adaptive conversion, buffering, storage, elimination and access capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Useful incorporation, metabolic stability, affordability and quality of life improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Controlled longitudinal multi-carrier nutrition study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Time-window and recurrence

Metabolism, diet and consumption | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 114: Metabolism, diet and consumption; original author status: Strong structural fit. Native state: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Calories/macros or one nutrient target. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. The relevant receiving carriers are: Gut, microbiome, liver, tissues, household, food system and future organism.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Adaptive conversion, buffering, storage, elimination and access capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Useful incorporation, metabolic stability, affordability and quality of life improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Controlled longitudinal multi-carrier nutrition study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Ethical and legal safety

Metabolism, diet and consumption | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 114: Metabolism, diet and consumption; original author status: Strong structural fit. Native state: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Calories/macros or one nutrient target. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. The relevant receiving carriers are: Gut, microbiome, liver, tissues, household, food system and future organism.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Adaptive conversion, buffering, storage, elimination and access capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Useful incorporation, metabolic stability, affordability and quality of life improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Controlled longitudinal multi-carrier nutrition study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Sensitivity and negative controls

Metabolism, diet and consumption | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 114: Metabolism, diet and consumption; original author status: Strong structural fit. Native state: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Calories/macros or one nutrient target. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. The relevant receiving carriers are: Gut, microbiome, liver, tissues, household, food system and future organism.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Adaptive conversion, buffering, storage, elimination and access capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Useful incorporation, metabolic stability, affordability and quality of life improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Controlled longitudinal multi-carrier nutrition study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Replication and provenance

Metabolism, diet and consumption | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 114: Metabolism, diet and consumption; original author status: Strong structural fit. Native state: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Calories/macros or one nutrient target. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. The relevant receiving carriers are: Gut, microbiome, liver, tissues, household, food system and future organism.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Adaptive conversion, buffering, storage, elimination and access capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Useful incorporation, metabolic stability, affordability and quality of life improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Controlled longitudinal multi-carrier nutrition study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

114: Cross-domain invariant contest

Metabolism, diet and consumption | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 114: Metabolism, diet and consumption; original author status: Strong structural fit. Native state: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Calories/macros or one nutrient target. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Useful intake is achieved by shifting toxicity, deficiency, cost or metabolic burden elsewhere. The relevant receiving carriers are: Gut, microbiome, liver, tissues, household, food system and future organism.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Adaptive conversion, buffering, storage, elimination and access capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Useful incorporation, metabolic stability, affordability and quality of life improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Controlled longitudinal multi-carrier nutrition study. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 114, pp. 853-859 (Rufino 2026).

Adapter 115: Predation, livestock and food-system transformation

Evolution, transformation and emergence | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology

RECORDED RECEIVING SCHEMA

Yield, price or human nutrition alone

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Candidate receivers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security.

EXISTING CAPACITY DESIGN

Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity

PROTECTED OUTCOME STATED BY THE AUTHOR

Food security, livelihoods, welfare and ecological resilience improve across the transition horizon

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Phased regional transition with preregistered carrier ledger. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115 | Predation, livestock and food-system transformation

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 115, Predation, livestock and food-system transformation. Native state: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Potential direct or institutional human pathway: Predation, livestock and food-system transformation interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

115 | Predation, livestock and food-system transformation

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 115, Predation, livestock and food-system transformation. Native state: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Potential direct or institutional human pathway: Predation, livestock and food-system transformation interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

115 | Predation, livestock and food-system transformation

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 115, Predation, livestock and food-system transformation. Native state: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Potential direct or institutional human pathway: Predation, livestock and food-system transformation interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

115 | Predation, livestock and food-system transformation

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 115, Predation, livestock and food-system transformation. Native state: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Potential direct or institutional human pathway: Predation, livestock and food-system transformation interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

115 | Predation, livestock and food-system transformation

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 115, Predation, livestock and food-system transformation. Native state: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Potential direct or institutional human pathway: Predation, livestock and food-system transformation interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

115 | Predation, livestock and food-system transformation

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 115, Predation, livestock and food-system transformation. Native state: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Potential direct or institutional human pathway: Predation, livestock and food-system transformation interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

115 to 001: consequential handoff

Predation, livestock and food-system transformation -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

115 to 002: consequential handoff

Predation, livestock and food-system transformation -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

115 to 003: consequential handoff

Predation, livestock and food-system transformation -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

115 to 004: consequential handoff

Predation, livestock and food-system transformation -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

115 to 005: consequential handoff

Predation, livestock and food-system transformation -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

115 to 006: consequential handoff

Predation, livestock and food-system transformation -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

115 to 007: consequential handoff

Predation, livestock and food-system transformation -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

115 to 008: consequential handoff

Predation, livestock and food-system transformation -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

115 to 009: consequential handoff

Predation, livestock and food-system transformation -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

115 to 010: consequential handoff

Predation, livestock and food-system transformation -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

115 to 011: consequential handoff

Predation, livestock and food-system transformation -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

115 to 012: consequential handoff

Predation, livestock and food-system transformation -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

115 to 013: consequential handoff

Predation, livestock and food-system transformation -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

115 to 014: consequential handoff

Predation, livestock and food-system transformation -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

115 to 015: consequential handoff

Predation, livestock and food-system transformation -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

115 to 016: consequential handoff

Predation, livestock and food-system transformation -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

115 to 017: consequential handoff

Predation, livestock and food-system transformation -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

115 to 018: consequential handoff

Predation, livestock and food-system transformation -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

115 to 019: consequential handoff

Predation, livestock and food-system transformation -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

115 to 020: consequential handoff

Predation, livestock and food-system transformation -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

115 to 021: consequential handoff

Predation, livestock and food-system transformation -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

115 to 022: consequential handoff

Predation, livestock and food-system transformation -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

115 to 023: consequential handoff

Predation, livestock and food-system transformation -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

115 to 024: consequential handoff

Predation, livestock and food-system transformation -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

115 to 025: consequential handoff

Predation, livestock and food-system transformation -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

115 to 026: consequential handoff

Predation, livestock and food-system transformation -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

115 to 027: consequential handoff

Predation, livestock and food-system transformation -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

115 to 028: consequential handoff

Predation, livestock and food-system transformation -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

115 to 029: consequential handoff

Predation, livestock and food-system transformation -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

115 to 030: consequential handoff

Predation, livestock and food-system transformation -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

115 to 031: consequential handoff

Predation, livestock and food-system transformation -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

115 to 032: consequential handoff

Predation, livestock and food-system transformation -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

115 to 033: consequential handoff

Predation, livestock and food-system transformation -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

115 to 034: consequential handoff

Predation, livestock and food-system transformation -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

115 to 035: consequential handoff

Predation, livestock and food-system transformation -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

115 to 036: consequential handoff

Predation, livestock and food-system transformation -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

115 to 037: consequential handoff

Predation, livestock and food-system transformation -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

115 to 038: consequential handoff

Predation, livestock and food-system transformation -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

115 to 039: consequential handoff

Predation, livestock and food-system transformation -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

115 to 040: consequential handoff

Predation, livestock and food-system transformation -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

115 to 041: consequential handoff

Predation, livestock and food-system transformation -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

115 to 042: consequential handoff

Predation, livestock and food-system transformation -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

115 to 043: consequential handoff

Predation, livestock and food-system transformation -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

115 to 044: consequential handoff

Predation, livestock and food-system transformation -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

115 to 045: consequential handoff

Predation, livestock and food-system transformation -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

115 to 046: consequential handoff

Predation, livestock and food-system transformation -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

115 to 047: consequential handoff

Predation, livestock and food-system transformation -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

115 to 048: consequential handoff

Predation, livestock and food-system transformation -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

115 to 049: consequential handoff

Predation, livestock and food-system transformation -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

115 to 050: consequential handoff

Predation, livestock and food-system transformation -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

115 to 051: consequential handoff

Predation, livestock and food-system transformation -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

115 to 052: consequential handoff

Predation, livestock and food-system transformation -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

115 to 053: consequential handoff

Predation, livestock and food-system transformation -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

115 to 054: consequential handoff

Predation, livestock and food-system transformation -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

115 to 055: consequential handoff

Predation, livestock and food-system transformation -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

115 to 056: consequential handoff

Predation, livestock and food-system transformation -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

115 to 057: consequential handoff

Predation, livestock and food-system transformation -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

115 to 058: consequential handoff

Predation, livestock and food-system transformation -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

115 to 059: consequential handoff

Predation, livestock and food-system transformation -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

115 to 060: consequential handoff

Predation, livestock and food-system transformation -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

115 to 061: consequential handoff

Predation, livestock and food-system transformation -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

115 to 062: consequential handoff

Predation, livestock and food-system transformation -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

115 to 063: consequential handoff

Predation, livestock and food-system transformation -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

115 to 064: consequential handoff

Predation, livestock and food-system transformation -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

115 to 065: consequential handoff

Predation, livestock and food-system transformation -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

115 to 066: consequential handoff

Predation, livestock and food-system transformation -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

115 to 067: consequential handoff

Predation, livestock and food-system transformation -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

115 to 068: consequential handoff

Predation, livestock and food-system transformation -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

115 to 069: consequential handoff

Predation, livestock and food-system transformation -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

115 to 070: consequential handoff

Predation, livestock and food-system transformation -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

115 to 071: consequential handoff

Predation, livestock and food-system transformation -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

115 to 072: consequential handoff

Predation, livestock and food-system transformation -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

115 to 073: consequential handoff

Predation, livestock and food-system transformation -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

115 to 074: consequential handoff

Predation, livestock and food-system transformation -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

115 to 075: consequential handoff

Predation, livestock and food-system transformation -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

115 to 076: consequential handoff

Predation, livestock and food-system transformation -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

115 to 077: consequential handoff

Predation, livestock and food-system transformation -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

115 to 078: consequential handoff

Predation, livestock and food-system transformation -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

115 to 079: consequential handoff

Predation, livestock and food-system transformation -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

115 to 080: consequential handoff

Predation, livestock and food-system transformation -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

115 to 081: consequential handoff

Predation, livestock and food-system transformation -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

115 to 082: consequential handoff

Predation, livestock and food-system transformation -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

115 to 083: consequential handoff

Predation, livestock and food-system transformation -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

115 to 084: consequential handoff

Predation, livestock and food-system transformation -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

115 to 085: consequential handoff

Predation, livestock and food-system transformation -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

115 to 086: consequential handoff

Predation, livestock and food-system transformation -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

115 to 087: consequential handoff

Predation, livestock and food-system transformation -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

115 to 088: consequential handoff

Predation, livestock and food-system transformation -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

115 to 089: consequential handoff

Predation, livestock and food-system transformation -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

115 to 090: consequential handoff

Predation, livestock and food-system transformation -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

115 to 091: consequential handoff

Predation, livestock and food-system transformation -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

115 to 092: consequential handoff

Predation, livestock and food-system transformation -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

115 to 093: consequential handoff

Predation, livestock and food-system transformation -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

115 to 094: consequential handoff

Predation, livestock and food-system transformation -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

115 to 095: consequential handoff

Predation, livestock and food-system transformation -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

115 to 096: consequential handoff

Predation, livestock and food-system transformation -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

115 to 097: consequential handoff

Predation, livestock and food-system transformation -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

115 to 098: consequential handoff

Predation, livestock and food-system transformation -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

115 to 099: consequential handoff

Predation, livestock and food-system transformation -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

115 to 100: consequential handoff

Predation, livestock and food-system transformation -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

115 to 101: consequential handoff

Predation, livestock and food-system transformation -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

115 to 102: consequential handoff

Predation, livestock and food-system transformation -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

115 to 103: consequential handoff

Predation, livestock and food-system transformation -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

115 to 104: consequential handoff

Predation, livestock and food-system transformation -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

115 to 105: consequential handoff

Predation, livestock and food-system transformation -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

115 to 106: consequential handoff

Predation, livestock and food-system transformation -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

115 to 107: consequential handoff

Predation, livestock and food-system transformation -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

115 to 108: consequential handoff

Predation, livestock and food-system transformation -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

115 to 109: consequential handoff

Predation, livestock and food-system transformation -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

115 to 110: consequential handoff

Predation, livestock and food-system transformation -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

115 to 111: consequential handoff

Predation, livestock and food-system transformation -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

115 to 112: consequential handoff

Predation, livestock and food-system transformation -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

115 to 113: consequential handoff

Predation, livestock and food-system transformation -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

115 to 114: consequential handoff

Predation, livestock and food-system transformation -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

115 to 115: consequential handoff

Predation, livestock and food-system transformation -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Predation, livestock and food-system transformation becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 860-866; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

115 to 116: consequential handoff

Predation, livestock and food-system transformation -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

115 to 117: consequential handoff

Predation, livestock and food-system transformation -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

115 to 118: consequential handoff

Predation, livestock and food-system transformation -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

115 to 119: consequential handoff

Predation, livestock and food-system transformation -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

115 to 120: consequential handoff

Predation, livestock and food-system transformation -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Predation, livestock and food-system transformation into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Yield, price or human nutrition alone. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Phased regional transition with preregistered carrier ledger' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 860-866 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

115: Predation, livestock and food-system transformation

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

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Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

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Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

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MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

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Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

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NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Predation, livestock and food-system transformation

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Predation, livestock and food-system transformation is described in the Atlas as: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Phased regional transition with preregistered carrier ledger or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Evidence and source reconstruction

Predation, livestock and food-system transformation | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 115: Predation, livestock and food-system transformation; original author status: Ethical / ecological frontier. Native state: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Yield, price or human nutrition alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. The relevant receiving carriers are: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon.

RETEST AND SCOPE CONTROL

Candidate decisive method: Phased regional transition with preregistered carrier ledger. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Boundary divergence

Predation, livestock and food-system transformation | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 115: Predation, livestock and food-system transformation; original author status: Ethical / ecological frontier. Native state: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Yield, price or human nutrition alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. The relevant receiving carriers are: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon.

RETEST AND SCOPE CONTROL

Candidate decisive method: Phased regional transition with preregistered carrier ledger. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Causal diagram and identifiability

Predation, livestock and food-system transformation | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 115: Predation, livestock and food-system transformation; original author status: Ethical / ecological frontier. Native state: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Yield, price or human nutrition alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. The relevant receiving carriers are: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon.

RETEST AND SCOPE CONTROL

Candidate decisive method: Phased regional transition with preregistered carrier ledger. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Receiving carrier identification

Predation, livestock and food-system transformation | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 115: Predation, livestock and food-system transformation; original author status: Ethical / ecological frontier. Native state: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Yield, price or human nutrition alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. The relevant receiving carriers are: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon.

RETEST AND SCOPE CONTROL

Candidate decisive method: Phased regional transition with preregistered carrier ledger. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Hidden compensation accounting

Predation, livestock and food-system transformation | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 115: Predation, livestock and food-system transformation; original author status: Ethical / ecological frontier. Native state: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Yield, price or human nutrition alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. The relevant receiving carriers are: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon.

RETEST AND SCOPE CONTROL

Candidate decisive method: Phased regional transition with preregistered carrier ledger. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Native units and dimensional integrity

Predation, livestock and food-system transformation | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 115: Predation, livestock and food-system transformation; original author status: Ethical / ecological frontier. Native state: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Yield, price or human nutrition alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. The relevant receiving carriers are: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon.

RETEST AND SCOPE CONTROL

Candidate decisive method: Phased regional transition with preregistered carrier ledger. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Capacity specification

Predation, livestock and food-system transformation | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 115: Predation, livestock and food-system transformation; original author status: Ethical / ecological frontier. Native state: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Yield, price or human nutrition alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. The relevant receiving carriers are: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon.

RETEST AND SCOPE CONTROL

Candidate decisive method: Phased regional transition with preregistered carrier ledger. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Intervention design

Predation, livestock and food-system transformation | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 115: Predation, livestock and food-system transformation; original author status: Ethical / ecological frontier. Native state: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Yield, price or human nutrition alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. The relevant receiving carriers are: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon.

RETEST AND SCOPE CONTROL

Candidate decisive method: Phased regional transition with preregistered carrier ledger. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Incumbent parity contest

Predation, livestock and food-system transformation | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 115: Predation, livestock and food-system transformation; original author status: Ethical / ecological frontier. Native state: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Yield, price or human nutrition alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. The relevant receiving carriers are: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon.

RETEST AND SCOPE CONTROL

Candidate decisive method: Phased regional transition with preregistered carrier ledger. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Time-window and recurrence

Predation, livestock and food-system transformation | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 115: Predation, livestock and food-system transformation; original author status: Ethical / ecological frontier. Native state: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Yield, price or human nutrition alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. The relevant receiving carriers are: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon.

RETEST AND SCOPE CONTROL

Candidate decisive method: Phased regional transition with preregistered carrier ledger. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Ethical and legal safety

Predation, livestock and food-system transformation | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 115: Predation, livestock and food-system transformation; original author status: Ethical / ecological frontier. Native state: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Yield, price or human nutrition alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. The relevant receiving carriers are: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon.

RETEST AND SCOPE CONTROL

Candidate decisive method: Phased regional transition with preregistered carrier ledger. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Sensitivity and negative controls

Predation, livestock and food-system transformation | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 115: Predation, livestock and food-system transformation; original author status: Ethical / ecological frontier. Native state: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Yield, price or human nutrition alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. The relevant receiving carriers are: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon.

RETEST AND SCOPE CONTROL

Candidate decisive method: Phased regional transition with preregistered carrier ledger. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Replication and provenance

Predation, livestock and food-system transformation | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 115: Predation, livestock and food-system transformation; original author status: Ethical / ecological frontier. Native state: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Yield, price or human nutrition alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. The relevant receiving carriers are: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon.

RETEST AND SCOPE CONTROL

Candidate decisive method: Phased regional transition with preregistered carrier ledger. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

115: Cross-domain invariant contest

Predation, livestock and food-system transformation | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 115: Predation, livestock and food-system transformation; original author status: Ethical / ecological frontier. Native state: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Yield, price or human nutrition alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: One species/system survives or profits by exporting unrepresented burden into another conscious/ecological carrier. The relevant receiving carriers are: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Food security, livelihoods, welfare and ecological resilience improve across the transition horizon.

RETEST AND SCOPE CONTROL

Candidate decisive method: Phased regional transition with preregistered carrier ledger. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 115, pp. 860-866 (Rufino 2026).

Adapter 116: Learning, plasticity and cultural evolution

Evolution, transformation and emergence | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission

RECORDED RECEIVING SCHEMA

Test score, output or adoption count

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Candidate receivers: Learner, teachers, peers, institutions, future self and cultural network.

EXISTING CAPACITY DESIGN

Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity

PROTECTED OUTCOME STATED BY THE AUTHOR

Capability, transfer, wellbeing and future learning capacity improve together

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Longitudinal transfer-and-retention experiment. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116 | Learning, plasticity and cultural evolution

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 116, Learning, plasticity and cultural evolution. Native state: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Potential direct or institutional human pathway: Learning, plasticity and cultural evolution interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Capability, transfer, wellbeing and future learning capacity improve together. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

116 | Learning, plasticity and cultural evolution

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 116, Learning, plasticity and cultural evolution. Native state: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Potential direct or institutional human pathway: Learning, plasticity and cultural evolution interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Capability, transfer, wellbeing and future learning capacity improve together. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

116 | Learning, plasticity and cultural evolution

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 116, Learning, plasticity and cultural evolution. Native state: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Potential direct or institutional human pathway: Learning, plasticity and cultural evolution interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Capability, transfer, wellbeing and future learning capacity improve together. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

116 | Learning, plasticity and cultural evolution

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 116, Learning, plasticity and cultural evolution. Native state: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Potential direct or institutional human pathway: Learning, plasticity and cultural evolution interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Capability, transfer, wellbeing and future learning capacity improve together. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

116 | Learning, plasticity and cultural evolution

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 116, Learning, plasticity and cultural evolution. Native state: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Potential direct or institutional human pathway: Learning, plasticity and cultural evolution interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Capability, transfer, wellbeing and future learning capacity improve together. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

116 | Learning, plasticity and cultural evolution

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 116, Learning, plasticity and cultural evolution. Native state: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Potential direct or institutional human pathway: Learning, plasticity and cultural evolution interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Capability, transfer, wellbeing and future learning capacity improve together. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

116 to 001: consequential handoff

Learning, plasticity and cultural evolution -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

116 to 002: consequential handoff

Learning, plasticity and cultural evolution -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

116 to 003: consequential handoff

Learning, plasticity and cultural evolution -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

116 to 004: consequential handoff

Learning, plasticity and cultural evolution -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

116 to 005: consequential handoff

Learning, plasticity and cultural evolution -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

116 to 006: consequential handoff

Learning, plasticity and cultural evolution -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

116 to 007: consequential handoff

Learning, plasticity and cultural evolution -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

116 to 008: consequential handoff

Learning, plasticity and cultural evolution -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

116 to 009: consequential handoff

Learning, plasticity and cultural evolution -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

116 to 010: consequential handoff

Learning, plasticity and cultural evolution -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

116 to 011: consequential handoff

Learning, plasticity and cultural evolution -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

116 to 012: consequential handoff

Learning, plasticity and cultural evolution -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

116 to 013: consequential handoff

Learning, plasticity and cultural evolution -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

116 to 014: consequential handoff

Learning, plasticity and cultural evolution -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

116 to 015: consequential handoff

Learning, plasticity and cultural evolution -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

116 to 016: consequential handoff

Learning, plasticity and cultural evolution -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

116 to 017: consequential handoff

Learning, plasticity and cultural evolution -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

116 to 018: consequential handoff

Learning, plasticity and cultural evolution -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

116 to 019: consequential handoff

Learning, plasticity and cultural evolution -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

116 to 020: consequential handoff

Learning, plasticity and cultural evolution -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

116 to 021: consequential handoff

Learning, plasticity and cultural evolution -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

116 to 022: consequential handoff

Learning, plasticity and cultural evolution -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

116 to 023: consequential handoff

Learning, plasticity and cultural evolution -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

116 to 024: consequential handoff

Learning, plasticity and cultural evolution -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

116 to 025: consequential handoff

Learning, plasticity and cultural evolution -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

116 to 026: consequential handoff

Learning, plasticity and cultural evolution -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

116 to 027: consequential handoff

Learning, plasticity and cultural evolution -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

116 to 028: consequential handoff

Learning, plasticity and cultural evolution -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

116 to 029: consequential handoff

Learning, plasticity and cultural evolution -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

116 to 030: consequential handoff

Learning, plasticity and cultural evolution -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

116 to 031: consequential handoff

Learning, plasticity and cultural evolution -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

116 to 032: consequential handoff

Learning, plasticity and cultural evolution -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

116 to 033: consequential handoff

Learning, plasticity and cultural evolution -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

116 to 034: consequential handoff

Learning, plasticity and cultural evolution -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

116 to 035: consequential handoff

Learning, plasticity and cultural evolution -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

116 to 036: consequential handoff

Learning, plasticity and cultural evolution -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

116 to 037: consequential handoff

Learning, plasticity and cultural evolution -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

116 to 038: consequential handoff

Learning, plasticity and cultural evolution -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

116 to 039: consequential handoff

Learning, plasticity and cultural evolution -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

116 to 040: consequential handoff

Learning, plasticity and cultural evolution -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

116 to 041: consequential handoff

Learning, plasticity and cultural evolution -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

116 to 042: consequential handoff

Learning, plasticity and cultural evolution -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

116 to 043: consequential handoff

Learning, plasticity and cultural evolution -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

116 to 044: consequential handoff

Learning, plasticity and cultural evolution -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

116 to 045: consequential handoff

Learning, plasticity and cultural evolution -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

116 to 046: consequential handoff

Learning, plasticity and cultural evolution -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

116 to 047: consequential handoff

Learning, plasticity and cultural evolution -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

116 to 048: consequential handoff

Learning, plasticity and cultural evolution -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

116 to 049: consequential handoff

Learning, plasticity and cultural evolution -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

116 to 050: consequential handoff

Learning, plasticity and cultural evolution -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

116 to 051: consequential handoff

Learning, plasticity and cultural evolution -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

116 to 052: consequential handoff

Learning, plasticity and cultural evolution -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

116 to 053: consequential handoff

Learning, plasticity and cultural evolution -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

116 to 054: consequential handoff

Learning, plasticity and cultural evolution -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

116 to 055: consequential handoff

Learning, plasticity and cultural evolution -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

116 to 056: consequential handoff

Learning, plasticity and cultural evolution -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

116 to 057: consequential handoff

Learning, plasticity and cultural evolution -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

116 to 058: consequential handoff

Learning, plasticity and cultural evolution -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

116 to 059: consequential handoff

Learning, plasticity and cultural evolution -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

116 to 060: consequential handoff

Learning, plasticity and cultural evolution -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

116 to 061: consequential handoff

Learning, plasticity and cultural evolution -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

116 to 062: consequential handoff

Learning, plasticity and cultural evolution -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

116 to 063: consequential handoff

Learning, plasticity and cultural evolution -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

116 to 064: consequential handoff

Learning, plasticity and cultural evolution -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

116 to 065: consequential handoff

Learning, plasticity and cultural evolution -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

116 to 066: consequential handoff

Learning, plasticity and cultural evolution -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

116 to 067: consequential handoff

Learning, plasticity and cultural evolution -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

116 to 068: consequential handoff

Learning, plasticity and cultural evolution -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

116 to 069: consequential handoff

Learning, plasticity and cultural evolution -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

116 to 070: consequential handoff

Learning, plasticity and cultural evolution -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

116 to 071: consequential handoff

Learning, plasticity and cultural evolution -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

116 to 072: consequential handoff

Learning, plasticity and cultural evolution -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

116 to 073: consequential handoff

Learning, plasticity and cultural evolution -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

116 to 074: consequential handoff

Learning, plasticity and cultural evolution -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

116 to 075: consequential handoff

Learning, plasticity and cultural evolution -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

116 to 076: consequential handoff

Learning, plasticity and cultural evolution -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

116 to 077: consequential handoff

Learning, plasticity and cultural evolution -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

116 to 078: consequential handoff

Learning, plasticity and cultural evolution -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

116 to 079: consequential handoff

Learning, plasticity and cultural evolution -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

116 to 080: consequential handoff

Learning, plasticity and cultural evolution -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

116 to 081: consequential handoff

Learning, plasticity and cultural evolution -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

116 to 082: consequential handoff

Learning, plasticity and cultural evolution -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

116 to 083: consequential handoff

Learning, plasticity and cultural evolution -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

116 to 084: consequential handoff

Learning, plasticity and cultural evolution -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

116 to 085: consequential handoff

Learning, plasticity and cultural evolution -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

116 to 086: consequential handoff

Learning, plasticity and cultural evolution -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

116 to 087: consequential handoff

Learning, plasticity and cultural evolution -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

116 to 088: consequential handoff

Learning, plasticity and cultural evolution -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

116 to 089: consequential handoff

Learning, plasticity and cultural evolution -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

116 to 090: consequential handoff

Learning, plasticity and cultural evolution -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

116 to 091: consequential handoff

Learning, plasticity and cultural evolution -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

116 to 092: consequential handoff

Learning, plasticity and cultural evolution -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

116 to 093: consequential handoff

Learning, plasticity and cultural evolution -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

116 to 094: consequential handoff

Learning, plasticity and cultural evolution -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

116 to 095: consequential handoff

Learning, plasticity and cultural evolution -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

116 to 096: consequential handoff

Learning, plasticity and cultural evolution -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

116 to 097: consequential handoff

Learning, plasticity and cultural evolution -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

116 to 098: consequential handoff

Learning, plasticity and cultural evolution -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

116 to 099: consequential handoff

Learning, plasticity and cultural evolution -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

116 to 100: consequential handoff

Learning, plasticity and cultural evolution -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

116 to 101: consequential handoff

Learning, plasticity and cultural evolution -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

116 to 102: consequential handoff

Learning, plasticity and cultural evolution -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

116 to 103: consequential handoff

Learning, plasticity and cultural evolution -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

116 to 104: consequential handoff

Learning, plasticity and cultural evolution -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

116 to 105: consequential handoff

Learning, plasticity and cultural evolution -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

116 to 106: consequential handoff

Learning, plasticity and cultural evolution -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

116 to 107: consequential handoff

Learning, plasticity and cultural evolution -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

116 to 108: consequential handoff

Learning, plasticity and cultural evolution -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

116 to 109: consequential handoff

Learning, plasticity and cultural evolution -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

116 to 110: consequential handoff

Learning, plasticity and cultural evolution -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

116 to 111: consequential handoff

Learning, plasticity and cultural evolution -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

116 to 112: consequential handoff

Learning, plasticity and cultural evolution -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

116 to 113: consequential handoff

Learning, plasticity and cultural evolution -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

116 to 114: consequential handoff

Learning, plasticity and cultural evolution -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

116 to 115: consequential handoff

Learning, plasticity and cultural evolution -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

116 to 116: consequential handoff

Learning, plasticity and cultural evolution -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Learning, plasticity and cultural evolution becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 867-873; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

116 to 117: consequential handoff

Learning, plasticity and cultural evolution -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

116 to 118: consequential handoff

Learning, plasticity and cultural evolution -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

116 to 119: consequential handoff

Learning, plasticity and cultural evolution -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

116 to 120: consequential handoff

Learning, plasticity and cultural evolution -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Learning, plasticity and cultural evolution into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Test score, output or adoption count. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Learner, teachers, peers, institutions, future self and cultural network. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal transfer-and-retention experiment' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 867-873 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

116: Learning, plasticity and cultural evolution

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

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CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

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Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

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Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Learning, plasticity and cultural evolution

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Learning, plasticity and cultural evolution is described in the Atlas as: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Learner, teachers, peers, institutions, future self and cultural network. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Capability, transfer, wellbeing and future learning capacity improve together. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal transfer-and-retention experiment or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Evidence and source reconstruction

Learning, plasticity and cultural evolution | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 116: Learning, plasticity and cultural evolution; original author status: Strong structural fit. Native state: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Test score, output or adoption count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. The relevant receiving carriers are: Learner, teachers, peers, institutions, future self and cultural network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability, transfer, wellbeing and future learning capacity improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal transfer-and-retention experiment. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Boundary divergence

Learning, plasticity and cultural evolution | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 116: Learning, plasticity and cultural evolution; original author status: Strong structural fit. Native state: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Test score, output or adoption count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. The relevant receiving carriers are: Learner, teachers, peers, institutions, future self and cultural network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability, transfer, wellbeing and future learning capacity improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal transfer-and-retention experiment. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Causal diagram and identifiability

Learning, plasticity and cultural evolution | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 116: Learning, plasticity and cultural evolution; original author status: Strong structural fit. Native state: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Test score, output or adoption count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. The relevant receiving carriers are: Learner, teachers, peers, institutions, future self and cultural network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability, transfer, wellbeing and future learning capacity improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal transfer-and-retention experiment. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Receiving carrier identification

Learning, plasticity and cultural evolution | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 116: Learning, plasticity and cultural evolution; original author status: Strong structural fit. Native state: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Test score, output or adoption count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. The relevant receiving carriers are: Learner, teachers, peers, institutions, future self and cultural network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability, transfer, wellbeing and future learning capacity improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal transfer-and-retention experiment. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Hidden compensation accounting

Learning, plasticity and cultural evolution | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 116: Learning, plasticity and cultural evolution; original author status: Strong structural fit. Native state: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Test score, output or adoption count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. The relevant receiving carriers are: Learner, teachers, peers, institutions, future self and cultural network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability, transfer, wellbeing and future learning capacity improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal transfer-and-retention experiment. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Native units and dimensional integrity

Learning, plasticity and cultural evolution | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 116: Learning, plasticity and cultural evolution; original author status: Strong structural fit. Native state: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Test score, output or adoption count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. The relevant receiving carriers are: Learner, teachers, peers, institutions, future self and cultural network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability, transfer, wellbeing and future learning capacity improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal transfer-and-retention experiment. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Capacity specification

Learning, plasticity and cultural evolution | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 116: Learning, plasticity and cultural evolution; original author status: Strong structural fit. Native state: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Test score, output or adoption count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. The relevant receiving carriers are: Learner, teachers, peers, institutions, future self and cultural network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability, transfer, wellbeing and future learning capacity improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal transfer-and-retention experiment. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Intervention design

Learning, plasticity and cultural evolution | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 116: Learning, plasticity and cultural evolution; original author status: Strong structural fit. Native state: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Test score, output or adoption count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. The relevant receiving carriers are: Learner, teachers, peers, institutions, future self and cultural network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability, transfer, wellbeing and future learning capacity improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal transfer-and-retention experiment. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Incumbent parity contest

Learning, plasticity and cultural evolution | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 116: Learning, plasticity and cultural evolution; original author status: Strong structural fit. Native state: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Test score, output or adoption count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. The relevant receiving carriers are: Learner, teachers, peers, institutions, future self and cultural network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability, transfer, wellbeing and future learning capacity improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal transfer-and-retention experiment. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Time-window and recurrence

Learning, plasticity and cultural evolution | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 116: Learning, plasticity and cultural evolution; original author status: Strong structural fit. Native state: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Test score, output or adoption count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. The relevant receiving carriers are: Learner, teachers, peers, institutions, future self and cultural network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability, transfer, wellbeing and future learning capacity improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal transfer-and-retention experiment. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Ethical and legal safety

Learning, plasticity and cultural evolution | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 116: Learning, plasticity and cultural evolution; original author status: Strong structural fit. Native state: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Test score, output or adoption count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. The relevant receiving carriers are: Learner, teachers, peers, institutions, future self and cultural network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability, transfer, wellbeing and future learning capacity improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal transfer-and-retention experiment. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Sensitivity and negative controls

Learning, plasticity and cultural evolution | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 116: Learning, plasticity and cultural evolution; original author status: Strong structural fit. Native state: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Test score, output or adoption count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. The relevant receiving carriers are: Learner, teachers, peers, institutions, future self and cultural network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability, transfer, wellbeing and future learning capacity improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal transfer-and-retention experiment. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Replication and provenance

Learning, plasticity and cultural evolution | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 116: Learning, plasticity and cultural evolution; original author status: Strong structural fit. Native state: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Test score, output or adoption count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. The relevant receiving carriers are: Learner, teachers, peers, institutions, future self and cultural network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability, transfer, wellbeing and future learning capacity improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal transfer-and-retention experiment. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

116: Cross-domain invariant contest

Learning, plasticity and cultural evolution | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 116: Learning, plasticity and cultural evolution; original author status: Strong structural fit. Native state: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Test score, output or adoption count. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Fast performance gain displaces understanding, wellbeing, diversity or future learning capacity. The relevant receiving carriers are: Learner, teachers, peers, institutions, future self and cultural network.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Capability, transfer, wellbeing and future learning capacity improve together.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal transfer-and-retention experiment. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 116, pp. 867-873 (Rufino 2026).

Adapter 117: Self-organization and emergence

Evolution, transformation and emergence | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate

RECORDED RECEIVING SCHEMA

Macro outcome without micro-carrier ledger

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Emergent order appears beneficial while unstable local compensation accumulates. Candidate receivers: Components, neighborhoods, emergent structure and external environment.

EXISTING CAPACITY DESIGN

Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity

PROTECTED OUTCOME STATED BY THE AUTHOR

Emergent order remains viable while local stress and hidden compensation decline

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Multi-scale simulation plus physical/biological holdout. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117 | Self-organization and emergence

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 117, Self-organization and emergence. Native state: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Potential direct or institutional human pathway: Self-organization and emergence interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Emergent order appears beneficial while unstable local compensation accumulates.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Emergent order remains viable while local stress and hidden compensation decline. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

117 | Self-organization and emergence

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 117, Self-organization and emergence. Native state: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Potential direct or institutional human pathway: Self-organization and emergence interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Emergent order appears beneficial while unstable local compensation accumulates.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Emergent order remains viable while local stress and hidden compensation decline. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

117 | Self-organization and emergence

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 117, Self-organization and emergence. Native state: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Potential direct or institutional human pathway: Self-organization and emergence interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Emergent order appears beneficial while unstable local compensation accumulates.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Emergent order remains viable while local stress and hidden compensation decline. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

117 | Self-organization and emergence

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 117, Self-organization and emergence. Native state: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Potential direct or institutional human pathway: Self-organization and emergence interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Emergent order appears beneficial while unstable local compensation accumulates.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Emergent order remains viable while local stress and hidden compensation decline. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

117 | Self-organization and emergence

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 117, Self-organization and emergence. Native state: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Potential direct or institutional human pathway: Self-organization and emergence interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Emergent order appears beneficial while unstable local compensation accumulates.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Emergent order remains viable while local stress and hidden compensation decline. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

117 | Self-organization and emergence

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 117, Self-organization and emergence. Native state: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Potential direct or institutional human pathway: Self-organization and emergence interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Emergent order appears beneficial while unstable local compensation accumulates.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Emergent order remains viable while local stress and hidden compensation decline. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

117 to 001: consequential handoff

Self-organization and emergence -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

117 to 002: consequential handoff

Self-organization and emergence -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

117 to 003: consequential handoff

Self-organization and emergence -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

117 to 004: consequential handoff

Self-organization and emergence -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

117 to 005: consequential handoff

Self-organization and emergence -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

117 to 006: consequential handoff

Self-organization and emergence -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

117 to 007: consequential handoff

Self-organization and emergence -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

117 to 008: consequential handoff

Self-organization and emergence -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

117 to 009: consequential handoff

Self-organization and emergence -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

117 to 010: consequential handoff

Self-organization and emergence -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

117 to 011: consequential handoff

Self-organization and emergence -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

117 to 012: consequential handoff

Self-organization and emergence -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

117 to 013: consequential handoff

Self-organization and emergence -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

117 to 014: consequential handoff

Self-organization and emergence -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

117 to 015: consequential handoff

Self-organization and emergence -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

117 to 016: consequential handoff

Self-organization and emergence -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

117 to 017: consequential handoff

Self-organization and emergence -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

117 to 018: consequential handoff

Self-organization and emergence -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

117 to 019: consequential handoff

Self-organization and emergence -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

117 to 020: consequential handoff

Self-organization and emergence -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

117 to 021: consequential handoff

Self-organization and emergence -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

117 to 022: consequential handoff

Self-organization and emergence -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

117 to 023: consequential handoff

Self-organization and emergence -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

117 to 024: consequential handoff

Self-organization and emergence -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

117 to 025: consequential handoff

Self-organization and emergence -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

117 to 026: consequential handoff

Self-organization and emergence -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

117 to 027: consequential handoff

Self-organization and emergence -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

117 to 028: consequential handoff

Self-organization and emergence -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

117 to 029: consequential handoff

Self-organization and emergence -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

117 to 030: consequential handoff

Self-organization and emergence -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

117 to 031: consequential handoff

Self-organization and emergence -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

117 to 032: consequential handoff

Self-organization and emergence -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

117 to 033: consequential handoff

Self-organization and emergence -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

117 to 034: consequential handoff

Self-organization and emergence -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

117 to 035: consequential handoff

Self-organization and emergence -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

117 to 036: consequential handoff

Self-organization and emergence -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

117 to 037: consequential handoff

Self-organization and emergence -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

117 to 038: consequential handoff

Self-organization and emergence -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

117 to 039: consequential handoff

Self-organization and emergence -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

117 to 040: consequential handoff

Self-organization and emergence -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

117 to 041: consequential handoff

Self-organization and emergence -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

117 to 042: consequential handoff

Self-organization and emergence -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

117 to 043: consequential handoff

Self-organization and emergence -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

117 to 044: consequential handoff

Self-organization and emergence -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

117 to 045: consequential handoff

Self-organization and emergence -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

117 to 046: consequential handoff

Self-organization and emergence -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

117 to 047: consequential handoff

Self-organization and emergence -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

117 to 048: consequential handoff

Self-organization and emergence -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

117 to 049: consequential handoff

Self-organization and emergence -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

117 to 050: consequential handoff

Self-organization and emergence -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

117 to 051: consequential handoff

Self-organization and emergence -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

117 to 052: consequential handoff

Self-organization and emergence -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

117 to 053: consequential handoff

Self-organization and emergence -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

117 to 054: consequential handoff

Self-organization and emergence -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

117 to 055: consequential handoff

Self-organization and emergence -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

117 to 056: consequential handoff

Self-organization and emergence -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

117 to 057: consequential handoff

Self-organization and emergence -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

117 to 058: consequential handoff

Self-organization and emergence -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

117 to 059: consequential handoff

Self-organization and emergence -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

117 to 060: consequential handoff

Self-organization and emergence -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

117 to 061: consequential handoff

Self-organization and emergence -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

117 to 062: consequential handoff

Self-organization and emergence -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

117 to 063: consequential handoff

Self-organization and emergence -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

117 to 064: consequential handoff

Self-organization and emergence -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

117 to 065: consequential handoff

Self-organization and emergence -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

117 to 066: consequential handoff

Self-organization and emergence -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

117 to 067: consequential handoff

Self-organization and emergence -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

117 to 068: consequential handoff

Self-organization and emergence -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

117 to 069: consequential handoff

Self-organization and emergence -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

117 to 070: consequential handoff

Self-organization and emergence -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

117 to 071: consequential handoff

Self-organization and emergence -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

117 to 072: consequential handoff

Self-organization and emergence -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

117 to 073: consequential handoff

Self-organization and emergence -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

117 to 074: consequential handoff

Self-organization and emergence -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

117 to 075: consequential handoff

Self-organization and emergence -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

117 to 076: consequential handoff

Self-organization and emergence -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

117 to 077: consequential handoff

Self-organization and emergence -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

117 to 078: consequential handoff

Self-organization and emergence -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

117 to 079: consequential handoff

Self-organization and emergence -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

117 to 080: consequential handoff

Self-organization and emergence -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

117 to 081: consequential handoff

Self-organization and emergence -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

117 to 082: consequential handoff

Self-organization and emergence -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

117 to 083: consequential handoff

Self-organization and emergence -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

117 to 084: consequential handoff

Self-organization and emergence -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

117 to 085: consequential handoff

Self-organization and emergence -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

117 to 086: consequential handoff

Self-organization and emergence -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

117 to 087: consequential handoff

Self-organization and emergence -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

117 to 088: consequential handoff

Self-organization and emergence -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

117 to 089: consequential handoff

Self-organization and emergence -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

117 to 090: consequential handoff

Self-organization and emergence -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

117 to 091: consequential handoff

Self-organization and emergence -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

117 to 092: consequential handoff

Self-organization and emergence -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

117 to 093: consequential handoff

Self-organization and emergence -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

117 to 094: consequential handoff

Self-organization and emergence -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

117 to 095: consequential handoff

Self-organization and emergence -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

117 to 096: consequential handoff

Self-organization and emergence -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

117 to 097: consequential handoff

Self-organization and emergence -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

117 to 098: consequential handoff

Self-organization and emergence -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

117 to 099: consequential handoff

Self-organization and emergence -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

117 to 100: consequential handoff

Self-organization and emergence -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

117 to 101: consequential handoff

Self-organization and emergence -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

117 to 102: consequential handoff

Self-organization and emergence -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

117 to 103: consequential handoff

Self-organization and emergence -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

117 to 104: consequential handoff

Self-organization and emergence -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

117 to 105: consequential handoff

Self-organization and emergence -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

117 to 106: consequential handoff

Self-organization and emergence -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

117 to 107: consequential handoff

Self-organization and emergence -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

117 to 108: consequential handoff

Self-organization and emergence -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

117 to 109: consequential handoff

Self-organization and emergence -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

117 to 110: consequential handoff

Self-organization and emergence -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

117 to 111: consequential handoff

Self-organization and emergence -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

117 to 112: consequential handoff

Self-organization and emergence -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

117 to 113: consequential handoff

Self-organization and emergence -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

117 to 114: consequential handoff

Self-organization and emergence -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

117 to 115: consequential handoff

Self-organization and emergence -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

117 to 116: consequential handoff

Self-organization and emergence -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

117 to 117: consequential handoff

Self-organization and emergence -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Self-organization and emergence becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 874-880; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

117 to 118: consequential handoff

Self-organization and emergence -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

117 to 119: consequential handoff

Self-organization and emergence -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

117 to 120: consequential handoff

Self-organization and emergence -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Self-organization and emergence into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Macro outcome without micro-carrier ledger. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Components, neighborhoods, emergent structure and external environment. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Multi-scale simulation plus physical/biological holdout' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 874-880 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

117: Self-organization and emergence

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Self-organization and emergence

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Self-organization and emergence is described in the Atlas as: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Emergent order appears beneficial while unstable local compensation accumulates. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Components, neighborhoods, emergent structure and external environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Emergent order remains viable while local stress and hidden compensation decline. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Multi-scale simulation plus physical/biological holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Evidence and source reconstruction

Self-organization and emergence | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 117: Self-organization and emergence; original author status: Formal / empirical frontier. Native state: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Macro outcome without micro-carrier ledger. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Emergent order appears beneficial while unstable local compensation accumulates. The relevant receiving carriers are: Components, neighborhoods, emergent structure and external environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Emergent order remains viable while local stress and hidden compensation decline.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-scale simulation plus physical/biological holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Boundary divergence

Self-organization and emergence | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 117: Self-organization and emergence; original author status: Formal / empirical frontier. Native state: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Macro outcome without micro-carrier ledger. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Emergent order appears beneficial while unstable local compensation accumulates. The relevant receiving carriers are: Components, neighborhoods, emergent structure and external environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Emergent order remains viable while local stress and hidden compensation decline.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-scale simulation plus physical/biological holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Causal diagram and identifiability

Self-organization and emergence | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 117: Self-organization and emergence; original author status: Formal / empirical frontier. Native state: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Macro outcome without micro-carrier ledger. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Emergent order appears beneficial while unstable local compensation accumulates. The relevant receiving carriers are: Components, neighborhoods, emergent structure and external environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Emergent order remains viable while local stress and hidden compensation decline.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-scale simulation plus physical/biological holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Receiving carrier identification

Self-organization and emergence | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 117: Self-organization and emergence; original author status: Formal / empirical frontier. Native state: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Macro outcome without micro-carrier ledger. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Emergent order appears beneficial while unstable local compensation accumulates. The relevant receiving carriers are: Components, neighborhoods, emergent structure and external environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Emergent order remains viable while local stress and hidden compensation decline.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-scale simulation plus physical/biological holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Hidden compensation accounting

Self-organization and emergence | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 117: Self-organization and emergence; original author status: Formal / empirical frontier. Native state: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Macro outcome without micro-carrier ledger. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Emergent order appears beneficial while unstable local compensation accumulates. The relevant receiving carriers are: Components, neighborhoods, emergent structure and external environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Emergent order remains viable while local stress and hidden compensation decline.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-scale simulation plus physical/biological holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Native units and dimensional integrity

Self-organization and emergence | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 117: Self-organization and emergence; original author status: Formal / empirical frontier. Native state: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Macro outcome without micro-carrier ledger. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Emergent order appears beneficial while unstable local compensation accumulates. The relevant receiving carriers are: Components, neighborhoods, emergent structure and external environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Emergent order remains viable while local stress and hidden compensation decline.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-scale simulation plus physical/biological holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Capacity specification

Self-organization and emergence | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 117: Self-organization and emergence; original author status: Formal / empirical frontier. Native state: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Macro outcome without micro-carrier ledger. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Emergent order appears beneficial while unstable local compensation accumulates. The relevant receiving carriers are: Components, neighborhoods, emergent structure and external environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Emergent order remains viable while local stress and hidden compensation decline.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-scale simulation plus physical/biological holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Intervention design

Self-organization and emergence | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 117: Self-organization and emergence; original author status: Formal / empirical frontier. Native state: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Macro outcome without micro-carrier ledger. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Emergent order appears beneficial while unstable local compensation accumulates. The relevant receiving carriers are: Components, neighborhoods, emergent structure and external environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Emergent order remains viable while local stress and hidden compensation decline.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-scale simulation plus physical/biological holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Incumbent parity contest

Self-organization and emergence | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 117: Self-organization and emergence; original author status: Formal / empirical frontier. Native state: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Macro outcome without micro-carrier ledger. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Emergent order appears beneficial while unstable local compensation accumulates. The relevant receiving carriers are: Components, neighborhoods, emergent structure and external environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Emergent order remains viable while local stress and hidden compensation decline.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-scale simulation plus physical/biological holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Time-window and recurrence

Self-organization and emergence | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 117: Self-organization and emergence; original author status: Formal / empirical frontier. Native state: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Macro outcome without micro-carrier ledger. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Emergent order appears beneficial while unstable local compensation accumulates. The relevant receiving carriers are: Components, neighborhoods, emergent structure and external environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Emergent order remains viable while local stress and hidden compensation decline.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-scale simulation plus physical/biological holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Ethical and legal safety

Self-organization and emergence | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 117: Self-organization and emergence; original author status: Formal / empirical frontier. Native state: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Macro outcome without micro-carrier ledger. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Emergent order appears beneficial while unstable local compensation accumulates. The relevant receiving carriers are: Components, neighborhoods, emergent structure and external environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Emergent order remains viable while local stress and hidden compensation decline.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-scale simulation plus physical/biological holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Sensitivity and negative controls

Self-organization and emergence | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 117: Self-organization and emergence; original author status: Formal / empirical frontier. Native state: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Macro outcome without micro-carrier ledger. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Emergent order appears beneficial while unstable local compensation accumulates. The relevant receiving carriers are: Components, neighborhoods, emergent structure and external environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Emergent order remains viable while local stress and hidden compensation decline.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-scale simulation plus physical/biological holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Replication and provenance

Self-organization and emergence | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 117: Self-organization and emergence; original author status: Formal / empirical frontier. Native state: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Macro outcome without micro-carrier ledger. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Emergent order appears beneficial while unstable local compensation accumulates. The relevant receiving carriers are: Components, neighborhoods, emergent structure and external environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Emergent order remains viable while local stress and hidden compensation decline.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-scale simulation plus physical/biological holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

117: Cross-domain invariant contest

Self-organization and emergence | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 117: Self-organization and emergence; original author status: Formal / empirical frontier. Native state: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Macro outcome without micro-carrier ledger. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Emergent order appears beneficial while unstable local compensation accumulates. The relevant receiving carriers are: Components, neighborhoods, emergent structure and external environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Emergent order remains viable while local stress and hidden compensation decline.

RETEST AND SCOPE CONTROL

Candidate decisive method: Multi-scale simulation plus physical/biological holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 117, pp. 874-880 (Rufino 2026).

Adapter 118: Creation, provenance and dependency

Evolution, transformation and emergence | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects

RECORDED RECEIVING SCHEMA

Ownership or authorship treated as sufficient relationship model

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Candidate receivers: Creators, created systems, maintainers, users, infrastructure and environment.

EXISTING CAPACITY DESIGN

Provenance, stewardship, autonomy-transition and reciprocal-viability architecture

PROTECTED OUTCOME STATED BY THE AUTHOR

Created and creating systems remain viable without one becoming the sacrificial substrate of the other

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Longitudinal creator-created dependency audit. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118 | Creation, provenance and dependency

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 118, Creation, provenance and dependency. Native state: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Potential direct or institutional human pathway: Creation, provenance and dependency interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Provenance, stewardship, autonomy-transition and reciprocal-viability architecture can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

118 | Creation, provenance and dependency

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 118, Creation, provenance and dependency. Native state: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Potential direct or institutional human pathway: Creation, provenance and dependency interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Provenance, stewardship, autonomy-transition and reciprocal-viability architecture can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

118 | Creation, provenance and dependency

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 118, Creation, provenance and dependency. Native state: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Potential direct or institutional human pathway: Creation, provenance and dependency interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Provenance, stewardship, autonomy-transition and reciprocal-viability architecture can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

118 | Creation, provenance and dependency

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 118, Creation, provenance and dependency. Native state: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Potential direct or institutional human pathway: Creation, provenance and dependency interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Provenance, stewardship, autonomy-transition and reciprocal-viability architecture can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

118 | Creation, provenance and dependency

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 118, Creation, provenance and dependency. Native state: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Potential direct or institutional human pathway: Creation, provenance and dependency interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Provenance, stewardship, autonomy-transition and reciprocal-viability architecture can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

118 | Creation, provenance and dependency

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 118, Creation, provenance and dependency. Native state: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Potential direct or institutional human pathway: Creation, provenance and dependency interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Provenance, stewardship, autonomy-transition and reciprocal-viability architecture can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

118 to 001: consequential handoff

Creation, provenance and dependency -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

118 to 002: consequential handoff

Creation, provenance and dependency -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

118 to 003: consequential handoff

Creation, provenance and dependency -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

118 to 004: consequential handoff

Creation, provenance and dependency -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

118 to 005: consequential handoff

Creation, provenance and dependency -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

118 to 006: consequential handoff

Creation, provenance and dependency -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

118 to 007: consequential handoff

Creation, provenance and dependency -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

118 to 008: consequential handoff

Creation, provenance and dependency -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

118 to 009: consequential handoff

Creation, provenance and dependency -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

118 to 010: consequential handoff

Creation, provenance and dependency -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

118 to 011: consequential handoff

Creation, provenance and dependency -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

118 to 012: consequential handoff

Creation, provenance and dependency -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

118 to 013: consequential handoff

Creation, provenance and dependency -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

118 to 014: consequential handoff

Creation, provenance and dependency -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

118 to 015: consequential handoff

Creation, provenance and dependency -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

118 to 016: consequential handoff

Creation, provenance and dependency -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

118 to 017: consequential handoff

Creation, provenance and dependency -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

118 to 018: consequential handoff

Creation, provenance and dependency -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

118 to 019: consequential handoff

Creation, provenance and dependency -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

118 to 020: consequential handoff

Creation, provenance and dependency -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

118 to 021: consequential handoff

Creation, provenance and dependency -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

118 to 022: consequential handoff

Creation, provenance and dependency -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

118 to 023: consequential handoff

Creation, provenance and dependency -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

118 to 024: consequential handoff

Creation, provenance and dependency -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

118 to 025: consequential handoff

Creation, provenance and dependency -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

118 to 026: consequential handoff

Creation, provenance and dependency -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

118 to 027: consequential handoff

Creation, provenance and dependency -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

118 to 028: consequential handoff

Creation, provenance and dependency -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

118 to 029: consequential handoff

Creation, provenance and dependency -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

118 to 030: consequential handoff

Creation, provenance and dependency -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

118 to 031: consequential handoff

Creation, provenance and dependency -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

118 to 032: consequential handoff

Creation, provenance and dependency -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

118 to 033: consequential handoff

Creation, provenance and dependency -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

118 to 034: consequential handoff

Creation, provenance and dependency -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

118 to 035: consequential handoff

Creation, provenance and dependency -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

118 to 036: consequential handoff

Creation, provenance and dependency -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

118 to 037: consequential handoff

Creation, provenance and dependency -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

118 to 038: consequential handoff

Creation, provenance and dependency -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

118 to 039: consequential handoff

Creation, provenance and dependency -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

118 to 040: consequential handoff

Creation, provenance and dependency -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

118 to 041: consequential handoff

Creation, provenance and dependency -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

118 to 042: consequential handoff

Creation, provenance and dependency -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

118 to 043: consequential handoff

Creation, provenance and dependency -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

118 to 044: consequential handoff

Creation, provenance and dependency -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

118 to 045: consequential handoff

Creation, provenance and dependency -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

118 to 046: consequential handoff

Creation, provenance and dependency -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

118 to 047: consequential handoff

Creation, provenance and dependency -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

118 to 048: consequential handoff

Creation, provenance and dependency -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

118 to 049: consequential handoff

Creation, provenance and dependency -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

118 to 050: consequential handoff

Creation, provenance and dependency -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

118 to 051: consequential handoff

Creation, provenance and dependency -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

118 to 052: consequential handoff

Creation, provenance and dependency -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

118 to 053: consequential handoff

Creation, provenance and dependency -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

118 to 054: consequential handoff

Creation, provenance and dependency -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

118 to 055: consequential handoff

Creation, provenance and dependency -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

118 to 056: consequential handoff

Creation, provenance and dependency -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

118 to 057: consequential handoff

Creation, provenance and dependency -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

118 to 058: consequential handoff

Creation, provenance and dependency -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

118 to 059: consequential handoff

Creation, provenance and dependency -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

118 to 060: consequential handoff

Creation, provenance and dependency -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

118 to 061: consequential handoff

Creation, provenance and dependency -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

118 to 062: consequential handoff

Creation, provenance and dependency -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

118 to 063: consequential handoff

Creation, provenance and dependency -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

118 to 064: consequential handoff

Creation, provenance and dependency -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

118 to 065: consequential handoff

Creation, provenance and dependency -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

118 to 066: consequential handoff

Creation, provenance and dependency -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

118 to 067: consequential handoff

Creation, provenance and dependency -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

118 to 068: consequential handoff

Creation, provenance and dependency -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

118 to 069: consequential handoff

Creation, provenance and dependency -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

118 to 070: consequential handoff

Creation, provenance and dependency -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

118 to 071: consequential handoff

Creation, provenance and dependency -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

118 to 072: consequential handoff

Creation, provenance and dependency -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

118 to 073: consequential handoff

Creation, provenance and dependency -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

118 to 074: consequential handoff

Creation, provenance and dependency -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

118 to 075: consequential handoff

Creation, provenance and dependency -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

118 to 076: consequential handoff

Creation, provenance and dependency -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

118 to 077: consequential handoff

Creation, provenance and dependency -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

118 to 078: consequential handoff

Creation, provenance and dependency -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

118 to 079: consequential handoff

Creation, provenance and dependency -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

118 to 080: consequential handoff

Creation, provenance and dependency -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

118 to 081: consequential handoff

Creation, provenance and dependency -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

118 to 082: consequential handoff

Creation, provenance and dependency -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

118 to 083: consequential handoff

Creation, provenance and dependency -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

118 to 084: consequential handoff

Creation, provenance and dependency -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

118 to 085: consequential handoff

Creation, provenance and dependency -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

118 to 086: consequential handoff

Creation, provenance and dependency -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

118 to 087: consequential handoff

Creation, provenance and dependency -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

118 to 088: consequential handoff

Creation, provenance and dependency -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

118 to 089: consequential handoff

Creation, provenance and dependency -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

118 to 090: consequential handoff

Creation, provenance and dependency -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

118 to 091: consequential handoff

Creation, provenance and dependency -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

118 to 092: consequential handoff

Creation, provenance and dependency -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

118 to 093: consequential handoff

Creation, provenance and dependency -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

118 to 094: consequential handoff

Creation, provenance and dependency -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

118 to 095: consequential handoff

Creation, provenance and dependency -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

118 to 096: consequential handoff

Creation, provenance and dependency -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

118 to 097: consequential handoff

Creation, provenance and dependency -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

118 to 098: consequential handoff

Creation, provenance and dependency -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

118 to 099: consequential handoff

Creation, provenance and dependency -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

118 to 100: consequential handoff

Creation, provenance and dependency -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

118 to 101: consequential handoff

Creation, provenance and dependency -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

118 to 102: consequential handoff

Creation, provenance and dependency -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

118 to 103: consequential handoff

Creation, provenance and dependency -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

118 to 104: consequential handoff

Creation, provenance and dependency -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

118 to 105: consequential handoff

Creation, provenance and dependency -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

118 to 106: consequential handoff

Creation, provenance and dependency -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

118 to 107: consequential handoff

Creation, provenance and dependency -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

118 to 108: consequential handoff

Creation, provenance and dependency -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

118 to 109: consequential handoff

Creation, provenance and dependency -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

118 to 110: consequential handoff

Creation, provenance and dependency -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

118 to 111: consequential handoff

Creation, provenance and dependency -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

118 to 112: consequential handoff

Creation, provenance and dependency -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

118 to 113: consequential handoff

Creation, provenance and dependency -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

118 to 114: consequential handoff

Creation, provenance and dependency -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

118 to 115: consequential handoff

Creation, provenance and dependency -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

118 to 116: consequential handoff

Creation, provenance and dependency -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

118 to 117: consequential handoff

Creation, provenance and dependency -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

118 to 118: consequential handoff

Creation, provenance and dependency -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Creation, provenance and dependency becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 881-887; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

118 to 119: consequential handoff

Creation, provenance and dependency -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

118 to 120: consequential handoff

Creation, provenance and dependency -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Creation, provenance and dependency into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Ownership or authorship treated as sufficient relationship model. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Creators, created systems, maintainers, users, infrastructure and environment. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Longitudinal creator-created dependency audit' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 881-887 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

118: Creation, provenance and dependency

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Creation, provenance and dependency

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Creation, provenance and dependency is described in the Atlas as: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Creators, created systems, maintainers, users, infrastructure and environment. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Created and creating systems remain viable without one becoming the sacrificial substrate of the other. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Longitudinal creator-created dependency audit or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Evidence and source reconstruction

Creation, provenance and dependency | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 118: Creation, provenance and dependency; original author status: Constructive protocol. Native state: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Ownership or authorship treated as sufficient relationship model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. The relevant receiving carriers are: Creators, created systems, maintainers, users, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Created and creating systems remain viable without one becoming the sacrificial substrate of the other.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal creator-created dependency audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Boundary divergence

Creation, provenance and dependency | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 118: Creation, provenance and dependency; original author status: Constructive protocol. Native state: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Ownership or authorship treated as sufficient relationship model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. The relevant receiving carriers are: Creators, created systems, maintainers, users, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Created and creating systems remain viable without one becoming the sacrificial substrate of the other.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal creator-created dependency audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Causal diagram and identifiability

Creation, provenance and dependency | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 118: Creation, provenance and dependency; original author status: Constructive protocol. Native state: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Ownership or authorship treated as sufficient relationship model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. The relevant receiving carriers are: Creators, created systems, maintainers, users, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Created and creating systems remain viable without one becoming the sacrificial substrate of the other.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal creator-created dependency audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Receiving carrier identification

Creation, provenance and dependency | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 118: Creation, provenance and dependency; original author status: Constructive protocol. Native state: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Ownership or authorship treated as sufficient relationship model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. The relevant receiving carriers are: Creators, created systems, maintainers, users, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Created and creating systems remain viable without one becoming the sacrificial substrate of the other.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal creator-created dependency audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Hidden compensation accounting

Creation, provenance and dependency | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 118: Creation, provenance and dependency; original author status: Constructive protocol. Native state: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Ownership or authorship treated as sufficient relationship model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. The relevant receiving carriers are: Creators, created systems, maintainers, users, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Created and creating systems remain viable without one becoming the sacrificial substrate of the other.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal creator-created dependency audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Native units and dimensional integrity

Creation, provenance and dependency | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 118: Creation, provenance and dependency; original author status: Constructive protocol. Native state: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Ownership or authorship treated as sufficient relationship model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. The relevant receiving carriers are: Creators, created systems, maintainers, users, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Created and creating systems remain viable without one becoming the sacrificial substrate of the other.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal creator-created dependency audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Capacity specification

Creation, provenance and dependency | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 118: Creation, provenance and dependency; original author status: Constructive protocol. Native state: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Ownership or authorship treated as sufficient relationship model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. The relevant receiving carriers are: Creators, created systems, maintainers, users, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Created and creating systems remain viable without one becoming the sacrificial substrate of the other.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal creator-created dependency audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Intervention design

Creation, provenance and dependency | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 118: Creation, provenance and dependency; original author status: Constructive protocol. Native state: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Ownership or authorship treated as sufficient relationship model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. The relevant receiving carriers are: Creators, created systems, maintainers, users, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Created and creating systems remain viable without one becoming the sacrificial substrate of the other.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal creator-created dependency audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Incumbent parity contest

Creation, provenance and dependency | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 118: Creation, provenance and dependency; original author status: Constructive protocol. Native state: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Ownership or authorship treated as sufficient relationship model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. The relevant receiving carriers are: Creators, created systems, maintainers, users, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Created and creating systems remain viable without one becoming the sacrificial substrate of the other.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal creator-created dependency audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Time-window and recurrence

Creation, provenance and dependency | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 118: Creation, provenance and dependency; original author status: Constructive protocol. Native state: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Ownership or authorship treated as sufficient relationship model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. The relevant receiving carriers are: Creators, created systems, maintainers, users, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Created and creating systems remain viable without one becoming the sacrificial substrate of the other.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal creator-created dependency audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Ethical and legal safety

Creation, provenance and dependency | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 118: Creation, provenance and dependency; original author status: Constructive protocol. Native state: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Ownership or authorship treated as sufficient relationship model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. The relevant receiving carriers are: Creators, created systems, maintainers, users, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Created and creating systems remain viable without one becoming the sacrificial substrate of the other.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal creator-created dependency audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Sensitivity and negative controls

Creation, provenance and dependency | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 118: Creation, provenance and dependency; original author status: Constructive protocol. Native state: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Ownership or authorship treated as sufficient relationship model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. The relevant receiving carriers are: Creators, created systems, maintainers, users, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Created and creating systems remain viable without one becoming the sacrificial substrate of the other.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal creator-created dependency audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Replication and provenance

Creation, provenance and dependency | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 118: Creation, provenance and dependency; original author status: Constructive protocol. Native state: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Ownership or authorship treated as sufficient relationship model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. The relevant receiving carriers are: Creators, created systems, maintainers, users, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Created and creating systems remain viable without one becoming the sacrificial substrate of the other.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal creator-created dependency audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

118: Cross-domain invariant contest

Creation, provenance and dependency | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 118: Creation, provenance and dependency; original author status: Constructive protocol. Native state: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Ownership or authorship treated as sufficient relationship model. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Created systems become dependent assets or creators become disposable inputs because provenance and reciprocal dependency are ignored. The relevant receiving carriers are: Creators, created systems, maintainers, users, infrastructure and environment.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Created and creating systems remain viable without one becoming the sacrificial substrate of the other.

RETEST AND SCOPE CONTROL

Candidate decisive method: Longitudinal creator-created dependency audit. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 118, pp. 881-887 (Rufino 2026).

Adapter 119: Cosmological evolution and structure formation

Evolution, transformation and emergence | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state

RECORDED RECEIVING SCHEMA

Single scale or observable treated as complete cosmic state

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Candidate receivers: Observed regions, fields/matter components, instruments, models and observers.

EXISTING CAPACITY DESIGN

Cross-scale observation, uncertainty, model-comparison and instrumentation capacity

PROTECTED OUTCOME STATED BY THE AUTHOR

Predictive fidelity increases across scales without hidden contradiction or unsupported closure

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Preregistered cross-survey predictive holdout. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119 | Cosmological evolution and structure formation

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 119, Cosmological evolution and structure formation. Native state: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Potential direct or institutional human pathway: Cosmological evolution and structure formation interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Cross-scale observation, uncertainty, model-comparison and instrumentation capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

119 | Cosmological evolution and structure formation

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 119, Cosmological evolution and structure formation. Native state: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Potential direct or institutional human pathway: Cosmological evolution and structure formation interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Cross-scale observation, uncertainty, model-comparison and instrumentation capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

119 | Cosmological evolution and structure formation

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 119, Cosmological evolution and structure formation. Native state: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Potential direct or institutional human pathway: Cosmological evolution and structure formation interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Cross-scale observation, uncertainty, model-comparison and instrumentation capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

119 | Cosmological evolution and structure formation

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 119, Cosmological evolution and structure formation. Native state: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Potential direct or institutional human pathway: Cosmological evolution and structure formation interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Cross-scale observation, uncertainty, model-comparison and instrumentation capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

119 | Cosmological evolution and structure formation

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 119, Cosmological evolution and structure formation. Native state: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Potential direct or institutional human pathway: Cosmological evolution and structure formation interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Cross-scale observation, uncertainty, model-comparison and instrumentation capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

119 | Cosmological evolution and structure formation

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 119, Cosmological evolution and structure formation. Native state: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Potential direct or institutional human pathway: Cosmological evolution and structure formation interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Cross-scale observation, uncertainty, model-comparison and instrumentation capacity can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

119 to 001: consequential handoff

Cosmological evolution and structure formation -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

119 to 002: consequential handoff

Cosmological evolution and structure formation -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

119 to 003: consequential handoff

Cosmological evolution and structure formation -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

119 to 004: consequential handoff

Cosmological evolution and structure formation -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

119 to 005: consequential handoff

Cosmological evolution and structure formation -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

119 to 006: consequential handoff

Cosmological evolution and structure formation -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

119 to 007: consequential handoff

Cosmological evolution and structure formation -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

119 to 008: consequential handoff

Cosmological evolution and structure formation -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

119 to 009: consequential handoff

Cosmological evolution and structure formation -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

119 to 010: consequential handoff

Cosmological evolution and structure formation -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

119 to 011: consequential handoff

Cosmological evolution and structure formation -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

119 to 012: consequential handoff

Cosmological evolution and structure formation -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

119 to 013: consequential handoff

Cosmological evolution and structure formation -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

119 to 014: consequential handoff

Cosmological evolution and structure formation -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

119 to 015: consequential handoff

Cosmological evolution and structure formation -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

119 to 016: consequential handoff

Cosmological evolution and structure formation -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

119 to 017: consequential handoff

Cosmological evolution and structure formation -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

119 to 018: consequential handoff

Cosmological evolution and structure formation -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

119 to 019: consequential handoff

Cosmological evolution and structure formation -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

119 to 020: consequential handoff

Cosmological evolution and structure formation -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

119 to 021: consequential handoff

Cosmological evolution and structure formation -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

119 to 022: consequential handoff

Cosmological evolution and structure formation -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

119 to 023: consequential handoff

Cosmological evolution and structure formation -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

119 to 024: consequential handoff

Cosmological evolution and structure formation -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

119 to 025: consequential handoff

Cosmological evolution and structure formation -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

119 to 026: consequential handoff

Cosmological evolution and structure formation -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

119 to 027: consequential handoff

Cosmological evolution and structure formation -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

119 to 028: consequential handoff

Cosmological evolution and structure formation -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

119 to 029: consequential handoff

Cosmological evolution and structure formation -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

119 to 030: consequential handoff

Cosmological evolution and structure formation -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

119 to 031: consequential handoff

Cosmological evolution and structure formation -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

119 to 032: consequential handoff

Cosmological evolution and structure formation -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

119 to 033: consequential handoff

Cosmological evolution and structure formation -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

119 to 034: consequential handoff

Cosmological evolution and structure formation -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

119 to 035: consequential handoff

Cosmological evolution and structure formation -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

119 to 036: consequential handoff

Cosmological evolution and structure formation -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

119 to 037: consequential handoff

Cosmological evolution and structure formation -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

119 to 038: consequential handoff

Cosmological evolution and structure formation -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

119 to 039: consequential handoff

Cosmological evolution and structure formation -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

119 to 040: consequential handoff

Cosmological evolution and structure formation -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

119 to 041: consequential handoff

Cosmological evolution and structure formation -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

119 to 042: consequential handoff

Cosmological evolution and structure formation -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

119 to 043: consequential handoff

Cosmological evolution and structure formation -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

119 to 044: consequential handoff

Cosmological evolution and structure formation -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

119 to 045: consequential handoff

Cosmological evolution and structure formation -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

119 to 046: consequential handoff

Cosmological evolution and structure formation -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

119 to 047: consequential handoff

Cosmological evolution and structure formation -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

119 to 048: consequential handoff

Cosmological evolution and structure formation -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

119 to 049: consequential handoff

Cosmological evolution and structure formation -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

119 to 050: consequential handoff

Cosmological evolution and structure formation -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

119 to 051: consequential handoff

Cosmological evolution and structure formation -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

119 to 052: consequential handoff

Cosmological evolution and structure formation -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

119 to 053: consequential handoff

Cosmological evolution and structure formation -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

119 to 054: consequential handoff

Cosmological evolution and structure formation -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

119 to 055: consequential handoff

Cosmological evolution and structure formation -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

119 to 056: consequential handoff

Cosmological evolution and structure formation -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

119 to 057: consequential handoff

Cosmological evolution and structure formation -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

119 to 058: consequential handoff

Cosmological evolution and structure formation -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

119 to 059: consequential handoff

Cosmological evolution and structure formation -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

119 to 060: consequential handoff

Cosmological evolution and structure formation -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

119 to 061: consequential handoff

Cosmological evolution and structure formation -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

119 to 062: consequential handoff

Cosmological evolution and structure formation -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

119 to 063: consequential handoff

Cosmological evolution and structure formation -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

119 to 064: consequential handoff

Cosmological evolution and structure formation -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

119 to 065: consequential handoff

Cosmological evolution and structure formation -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

119 to 066: consequential handoff

Cosmological evolution and structure formation -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

119 to 067: consequential handoff

Cosmological evolution and structure formation -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

119 to 068: consequential handoff

Cosmological evolution and structure formation -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

119 to 069: consequential handoff

Cosmological evolution and structure formation -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

119 to 070: consequential handoff

Cosmological evolution and structure formation -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

119 to 071: consequential handoff

Cosmological evolution and structure formation -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

119 to 072: consequential handoff

Cosmological evolution and structure formation -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

119 to 073: consequential handoff

Cosmological evolution and structure formation -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

119 to 074: consequential handoff

Cosmological evolution and structure formation -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

119 to 075: consequential handoff

Cosmological evolution and structure formation -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

119 to 076: consequential handoff

Cosmological evolution and structure formation -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

119 to 077: consequential handoff

Cosmological evolution and structure formation -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

119 to 078: consequential handoff

Cosmological evolution and structure formation -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

119 to 079: consequential handoff

Cosmological evolution and structure formation -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

119 to 080: consequential handoff

Cosmological evolution and structure formation -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

119 to 081: consequential handoff

Cosmological evolution and structure formation -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

119 to 082: consequential handoff

Cosmological evolution and structure formation -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

119 to 083: consequential handoff

Cosmological evolution and structure formation -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

119 to 084: consequential handoff

Cosmological evolution and structure formation -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

119 to 085: consequential handoff

Cosmological evolution and structure formation -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

119 to 086: consequential handoff

Cosmological evolution and structure formation -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

119 to 087: consequential handoff

Cosmological evolution and structure formation -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

119 to 088: consequential handoff

Cosmological evolution and structure formation -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

119 to 089: consequential handoff

Cosmological evolution and structure formation -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

119 to 090: consequential handoff

Cosmological evolution and structure formation -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

119 to 091: consequential handoff

Cosmological evolution and structure formation -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

119 to 092: consequential handoff

Cosmological evolution and structure formation -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

119 to 093: consequential handoff

Cosmological evolution and structure formation -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

119 to 094: consequential handoff

Cosmological evolution and structure formation -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

119 to 095: consequential handoff

Cosmological evolution and structure formation -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

119 to 096: consequential handoff

Cosmological evolution and structure formation -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

119 to 097: consequential handoff

Cosmological evolution and structure formation -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

119 to 098: consequential handoff

Cosmological evolution and structure formation -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

119 to 099: consequential handoff

Cosmological evolution and structure formation -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

119 to 100: consequential handoff

Cosmological evolution and structure formation -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

119 to 101: consequential handoff

Cosmological evolution and structure formation -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

119 to 102: consequential handoff

Cosmological evolution and structure formation -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

119 to 103: consequential handoff

Cosmological evolution and structure formation -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

119 to 104: consequential handoff

Cosmological evolution and structure formation -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

119 to 105: consequential handoff

Cosmological evolution and structure formation -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

119 to 106: consequential handoff

Cosmological evolution and structure formation -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

119 to 107: consequential handoff

Cosmological evolution and structure formation -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

119 to 108: consequential handoff

Cosmological evolution and structure formation -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

119 to 109: consequential handoff

Cosmological evolution and structure formation -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

119 to 110: consequential handoff

Cosmological evolution and structure formation -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

119 to 111: consequential handoff

Cosmological evolution and structure formation -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

119 to 112: consequential handoff

Cosmological evolution and structure formation -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

119 to 113: consequential handoff

Cosmological evolution and structure formation -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

119 to 114: consequential handoff

Cosmological evolution and structure formation -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

119 to 115: consequential handoff

Cosmological evolution and structure formation -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

119 to 116: consequential handoff

Cosmological evolution and structure formation -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

119 to 117: consequential handoff

Cosmological evolution and structure formation -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

119 to 118: consequential handoff

Cosmological evolution and structure formation -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

119 to 119: consequential handoff

Cosmological evolution and structure formation -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Cosmological evolution and structure formation becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 888-894; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

119 to 120: consequential handoff

Cosmological evolution and structure formation -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

Proposed interface from Cosmological evolution and structure formation into Cross-substrate consciousness, temporal coexistence and closure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Single scale or observable treated as complete cosmic state. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: Observed regions, fields/matter components, instruments, models and observers. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Preregistered cross-survey predictive holdout' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 888-894 and 895-901; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

119: Cosmological evolution and structure formation

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cosmological evolution and structure formation

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cosmological evolution and structure formation is described in the Atlas as: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for Observed regions, fields/matter components, instruments, models and observers. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Predictive fidelity increases across scales without hidden contradiction or unsupported closure. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Preregistered cross-survey predictive holdout or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Evidence and source reconstruction

Cosmological evolution and structure formation | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 119: Cosmological evolution and structure formation; original author status: Hard scientific frontier. Native state: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single scale or observable treated as complete cosmic state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. The relevant receiving carriers are: Observed regions, fields/matter components, instruments, models and observers.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Predictive fidelity increases across scales without hidden contradiction or unsupported closure.

RETEST AND SCOPE CONTROL

Candidate decisive method: Preregistered cross-survey predictive holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Boundary divergence

Cosmological evolution and structure formation | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 119: Cosmological evolution and structure formation; original author status: Hard scientific frontier. Native state: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single scale or observable treated as complete cosmic state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. The relevant receiving carriers are: Observed regions, fields/matter components, instruments, models and observers.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Predictive fidelity increases across scales without hidden contradiction or unsupported closure.

RETEST AND SCOPE CONTROL

Candidate decisive method: Preregistered cross-survey predictive holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Causal diagram and identifiability

Cosmological evolution and structure formation | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 119: Cosmological evolution and structure formation; original author status: Hard scientific frontier. Native state: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single scale or observable treated as complete cosmic state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. The relevant receiving carriers are: Observed regions, fields/matter components, instruments, models and observers.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Predictive fidelity increases across scales without hidden contradiction or unsupported closure.

RETEST AND SCOPE CONTROL

Candidate decisive method: Preregistered cross-survey predictive holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Receiving carrier identification

Cosmological evolution and structure formation | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 119: Cosmological evolution and structure formation; original author status: Hard scientific frontier. Native state: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single scale or observable treated as complete cosmic state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. The relevant receiving carriers are: Observed regions, fields/matter components, instruments, models and observers.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Predictive fidelity increases across scales without hidden contradiction or unsupported closure.

RETEST AND SCOPE CONTROL

Candidate decisive method: Preregistered cross-survey predictive holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Hidden compensation accounting

Cosmological evolution and structure formation | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 119: Cosmological evolution and structure formation; original author status: Hard scientific frontier. Native state: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single scale or observable treated as complete cosmic state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. The relevant receiving carriers are: Observed regions, fields/matter components, instruments, models and observers.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Predictive fidelity increases across scales without hidden contradiction or unsupported closure.

RETEST AND SCOPE CONTROL

Candidate decisive method: Preregistered cross-survey predictive holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Native units and dimensional integrity

Cosmological evolution and structure formation | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 119: Cosmological evolution and structure formation; original author status: Hard scientific frontier. Native state: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single scale or observable treated as complete cosmic state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. The relevant receiving carriers are: Observed regions, fields/matter components, instruments, models and observers.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Predictive fidelity increases across scales without hidden contradiction or unsupported closure.

RETEST AND SCOPE CONTROL

Candidate decisive method: Preregistered cross-survey predictive holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Capacity specification

Cosmological evolution and structure formation | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 119: Cosmological evolution and structure formation; original author status: Hard scientific frontier. Native state: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single scale or observable treated as complete cosmic state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. The relevant receiving carriers are: Observed regions, fields/matter components, instruments, models and observers.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Predictive fidelity increases across scales without hidden contradiction or unsupported closure.

RETEST AND SCOPE CONTROL

Candidate decisive method: Preregistered cross-survey predictive holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Intervention design

Cosmological evolution and structure formation | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 119: Cosmological evolution and structure formation; original author status: Hard scientific frontier. Native state: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single scale or observable treated as complete cosmic state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. The relevant receiving carriers are: Observed regions, fields/matter components, instruments, models and observers.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Predictive fidelity increases across scales without hidden contradiction or unsupported closure.

RETEST AND SCOPE CONTROL

Candidate decisive method: Preregistered cross-survey predictive holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Incumbent parity contest

Cosmological evolution and structure formation | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 119: Cosmological evolution and structure formation; original author status: Hard scientific frontier. Native state: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single scale or observable treated as complete cosmic state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. The relevant receiving carriers are: Observed regions, fields/matter components, instruments, models and observers.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Predictive fidelity increases across scales without hidden contradiction or unsupported closure.

RETEST AND SCOPE CONTROL

Candidate decisive method: Preregistered cross-survey predictive holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Time-window and recurrence

Cosmological evolution and structure formation | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 119: Cosmological evolution and structure formation; original author status: Hard scientific frontier. Native state: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single scale or observable treated as complete cosmic state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. The relevant receiving carriers are: Observed regions, fields/matter components, instruments, models and observers.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Predictive fidelity increases across scales without hidden contradiction or unsupported closure.

RETEST AND SCOPE CONTROL

Candidate decisive method: Preregistered cross-survey predictive holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Ethical and legal safety

Cosmological evolution and structure formation | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 119: Cosmological evolution and structure formation; original author status: Hard scientific frontier. Native state: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single scale or observable treated as complete cosmic state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. The relevant receiving carriers are: Observed regions, fields/matter components, instruments, models and observers.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Predictive fidelity increases across scales without hidden contradiction or unsupported closure.

RETEST AND SCOPE CONTROL

Candidate decisive method: Preregistered cross-survey predictive holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Sensitivity and negative controls

Cosmological evolution and structure formation | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 119: Cosmological evolution and structure formation; original author status: Hard scientific frontier. Native state: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single scale or observable treated as complete cosmic state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. The relevant receiving carriers are: Observed regions, fields/matter components, instruments, models and observers.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Predictive fidelity increases across scales without hidden contradiction or unsupported closure.

RETEST AND SCOPE CONTROL

Candidate decisive method: Preregistered cross-survey predictive holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Replication and provenance

Cosmological evolution and structure formation | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 119: Cosmological evolution and structure formation; original author status: Hard scientific frontier. Native state: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single scale or observable treated as complete cosmic state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. The relevant receiving carriers are: Observed regions, fields/matter components, instruments, models and observers.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Predictive fidelity increases across scales without hidden contradiction or unsupported closure.

RETEST AND SCOPE CONTROL

Candidate decisive method: Preregistered cross-survey predictive holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

119: Cross-domain invariant contest

Cosmological evolution and structure formation | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 119: Cosmological evolution and structure formation; original author status: Hard scientific frontier. Native state: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Single scale or observable treated as complete cosmic state. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: Model success at one scale hides mismatch, selection effect or unresolved dynamics at another. The relevant receiving carriers are: Observed regions, fields/matter components, instruments, models and observers.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Predictive fidelity increases across scales without hidden contradiction or unsupported closure.

RETEST AND SCOPE CONTROL

Candidate decisive method: Preregistered cross-survey predictive holdout. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 119, pp. 888-894 (Rufino 2026).

Adapter 120: Cross-substrate consciousness, temporal coexistence and closure

Evolution, transformation and emergence | Internal carrier and capacity reconstruction

ANALYTICAL DESIGN / NOT AN OBSERVED IMPLEMENTATION

DOMAIN-NATIVE STATE

The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility

RECORDED RECEIVING SCHEMA

Any local adapter, species, institution or observer claiming closure from its own schema alone

CANDIDATE BURDEN AND RECEIVING CARRIERS

Blocker in the source Atlas: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Candidate receivers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths.

EXISTING CAPACITY DESIGN

Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies

PROTECTED OUTCOME STATED BY THE AUTHOR

Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement

SCIENTIFIC NEXT STEP

Atlas proposed decisive test: Time-locked system-of-systems closure test across multiple adapters. This remains a proposed testing route, not proof of a completed successful implementation.

ORIGINALITY AND CONSTRUCTION OBLIGATION

Run causal receiving-carrier diagnosis before action; compare construction against strong established methods; do not accept positive averages as proof that all material carriers benefited.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120 | Cross-substrate consciousness, temporal coexistence and c

Individual human / person | Insufficient capacity

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 120, Cross-substrate consciousness, temporal coexistence and closure. Native state: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Potential direct or institutional human pathway: Cross-substrate consciousness, temporal coexistence and closure interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Insufficient capacity: The human has limited time, health, physical access, material resources, protective support or opportunity. Source Atlas proposed limiting condition: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

120 | Cross-substrate consciousness, temporal coexistence and c

Individual human / person | Schema mismatch

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 120, Cross-substrate consciousness, temporal coexistence and closure. Native state: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Potential direct or institutional human pathway: Cross-substrate consciousness, temporal coexistence and closure interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Schema mismatch: The person has evidence-grounded capabilities or needs that institutional identifiers, scores or labels do not adequately recognize. Source Atlas proposed limiting condition: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

120 | Cross-substrate consciousness, temporal coexistence and c

Individual human / person | Delayed effects

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 120, Cross-substrate consciousness, temporal coexistence and closure. Native state: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Potential direct or institutional human pathway: Cross-substrate consciousness, temporal coexistence and closure interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Delayed effects: A short-term change influences future-self health, skills, income, dependence, options or relationships. Source Atlas proposed limiting condition: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

120 | Cross-substrate consciousness, temporal coexistence and c

Individual human / person | Measurement uncertainty

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 120, Cross-substrate consciousness, temporal coexistence and closure. Native state: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Potential direct or institutional human pathway: Cross-substrate consciousness, temporal coexistence and closure interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Measurement uncertainty: The record lacks consented individual outcomes, counterfactuals, follow-up, or reliable protection metrics. Source Atlas proposed limiting condition: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

120 | Cross-substrate consciousness, temporal coexistence and c

Individual human / person | Coordination failure

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 120, Cross-substrate consciousness, temporal coexistence and closure. Native state: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Potential direct or institutional human pathway: Cross-substrate consciousness, temporal coexistence and closure interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Coordination failure: Family, employer, provider, government or technical systems hand off obligations to the individual without coordinated support. Source Atlas proposed limiting condition: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

120 | Cross-substrate consciousness, temporal coexistence and c

Individual human / person | Adversarial / inequitable distribution

PERSON AS PRIMARY CONSEQUENTIAL CARRIER

Study setting: Individual human / person. The person is an actual agent with dignity, physical needs, autonomy, relationships, and a future self, not simply a household, institution, metric or interchangeable resource.

NATIVE DOMAIN AND CAUSAL INTERFACE

Adapter 120, Cross-substrate consciousness, temporal coexistence and closure. Native state: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Potential direct or institutional human pathway: Cross-substrate consciousness, temporal coexistence and closure interacts with a person through the native system, if the linkage is empirically identified.

CRITICAL CONDITION TO INVESTIGATE

Adversarial / inequitable distribution: Unequal power, coercion, inaccessible systems, discrimination or exploitative incentives shift costs to a person. Source Atlas proposed limiting condition: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time.

OBSERVATION / RECORD REQUIRED

Identify whether an individual is directly or indirectly exposed; document time, agency, consent, rights, health, access, capacity, and delayed effects where causally relevant. Evidence of no person-level link remains an honest result.

LERM RECEIVING-CARRIER TRACE

Define a domain-native action and a fair comparator, trace what the specific person receives, what the person supplies in hidden compensation, and which other actors or future states bear consequences.

IAMF-P CONSTRUCTIVE TASK

Investigate whether Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies can be adapted to build a genuine receiving capacity for that individual without shifting harm. A design proposal does not demonstrate a working intervention.

MUTUAL UPTICK AND NEXT OPERATION

Candidate Atlas goal: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Measure actual individual and other protected-carrier outcomes before and after an authorized action, over appropriate horizons. If unmet, record the limiting constraint and next ethically admissible construction search.

120 to 001: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Whole-person quality of life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Whole-person quality of life: Health, function, time, autonomy, relationships, safety, income, environment and future capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Whole-person quality of life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Single score, diagnosis, income bracket or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Person, household, caregivers, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Whole-life state model plus coordinated capacity across health, time, income and environment. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Longitudinal cohort with preregistered multidimensional outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 51-57; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 001 pp. 51-57.

120 to 002: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Nutrition and metabolism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Nutrition and metabolism: Diet, metabolic state, affordability, culture, timing, microbiome and activity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Nutrition and metabolism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Calories/macros or generic diet categories. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Body, household budget, food system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Adaptive nutrition and access architecture with feedback and affordability constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Randomized adaptive nutrition trial with cost and adherence carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 58-64; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 002 pp. 58-64.

120 to 003: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Sleep and recovery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Sleep and recovery: Sleep architecture, circadian timing, stress, workload, environment and recovery reserve.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Sleep and recovery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Hours slept or wearable score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Body, cognition, household schedule. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Recovery controller integrating timing, environment, workload and physiology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Longitudinal crossover study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 65-71; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 003 pp. 65-71.

120 to 004: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Reproductive and sexual health | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Reproductive and sexual health: Endocrine, reproductive, sexual, relational, consent, fertility and safety state.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Reproductive and sexual health; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Single fertility, disease or risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Individual, partner, future child, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Longitudinal reproductive-state architecture plus consent/safety controls. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Clinical trials + patient-reported and partner- impact carriers'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 72-78; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 004 pp. 72-78.

120 to 005: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Disability and accessibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Disability and accessibility: Capabilities plus environmental demands, interfaces and support.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Disability and accessibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Deficit-only person label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Person, caregiver, workplace, public space. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Environment-first accessibility architecture plus assistive capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Before/after environment redesign with workload measures'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 79-85; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 005 pp. 79-85.

120 to 006: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Mobility and physical function | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Mobility and physical function: Strength, pain, balance, endurance, environment, assistive devices.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Mobility and physical function; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Single functional score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Body, caregiver, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Adaptive rehabilitation plus environment and device redesign. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Controlled rehabilitation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 86-92; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 006 pp. 86-92.

120 to 007: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Family and caregiving | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Family and caregiving: Needs, capacities, schedules, money, emotional load, dependency and support network.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Family and caregiving; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Household role labels. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Primary caregiver, dependents, siblings, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Care-load ledger, shared coordination and external capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Prospective household burden study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 93-99; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 007 pp. 93-99.

120 to 008: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Safety and personal security | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Safety and personal security: Threat exposure, environment, support, mobility and response capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Safety and personal security; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Incident count or generic risk class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Person, community, responder. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Safety architecture that increases protection without reducing autonomy. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Prospective safety intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 100-106; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 008 pp. 100-106.

120 to 009: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Housing and domestic environment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Housing and domestic environment: Shelter, cost, commute, safety, air, heat, crowding and social context.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Housing and domestic environment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Property price or occupancy status. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Resident, landlord, neighborhood, city. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Whole-cost housing and location architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Matched housing intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 107-113; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 009 pp. 107-113.

120 to 010: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Human longevity and life-course planning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Human longevity and life-course planning: Healthspan, lifespan, finances, relationships, purpose and care capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Human longevity and life-course planning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Chronological age or retirement age. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Person, family, institutions, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Life-course capital and care architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Longitudinal life-course cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 114-120; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 010 pp. 114-120.

120 to 011: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Psychology and social legibility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Psychology and social legibility: Longitudinal capability, burden, trauma, environment, relationships and compensation.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Psychology and social legibility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Snapshot diagnosis or social category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Person, family, employer, clinician. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Multidimensional schema plus support matched to hidden burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Longitudinal external cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 122-128; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 011 pp. 122-128.

120 to 012: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Identity and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Identity and self-model: Attributes, history, roles, values, capabilities and contexts.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Identity and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Single identity label or credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Person and receiving institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Selective multi-context identity graph with provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Cross-context identity study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 129-135; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 012 pp. 129-135.

120 to 013: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Relationships and intimacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Relationships and intimacy: Needs, boundaries, consent, trust, history, resources and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Relationships and intimacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Relationship status or personality label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Partners and wider support network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Reciprocal state and burden ledger plus consent architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Dyadic longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 136-142; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 013 pp. 136-142.

120 to 014: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Parenting and child development | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Parenting and child development: Biology, attachment, education, safety, resources, environment and agency.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Parenting and child development; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Age norm, grade or behavior label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Child, parent, school, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Developmental state map plus distributed support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Longitudinal family-school cohort'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 143-149; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 014 pp. 143-149.

120 to 015: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Purpose and self-actualization | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Purpose and self-actualization: Capabilities, values, opportunity, health, autonomy and social contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Purpose and self-actualization; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Career status, wealth or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Person, relationships, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Whole-person capability architecture and burden guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Longitudinal multidimensional evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 150-156; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 015 pp. 150-156.

120 to 016: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Existential wellbeing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Existential wellbeing: Meaning, mortality awareness, belonging, agency and life coherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Existential wellbeing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Symptom or productivity proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Person, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Voluntary meaning-reflection and support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Patient/user-reported longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 157-163; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 016 pp. 157-163.

120 to 017: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Spiritual life | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Spiritual life: Belief, practice, meaning, community, conscience and autonomy.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Spiritual life; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Institutional affiliation alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Individual, family, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Voluntary, transparent, safeguarding-centered spiritual/community architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Ethical observational study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 164-170; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 017 pp. 164-170.

120 to 018: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Religion and community institutions | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Religion and community institutions: Doctrine, governance, resources, safeguarding and member experience.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Religion and community institutions; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Membership or doctrinal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Members, families, leaders, wider community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Transparent governance, reporting, safeguarding and exit rights. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Institutional longitudinal audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 171-177; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 018 pp. 171-177.

120 to 019: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Emotion and affect regulation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Emotion and affect regulation: Internal state, context, physiology, cognition and social feedback.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Emotion and affect regulation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Valence score or symptom label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Person and relationships. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: State-aware regulation architecture with physiological feedback. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Crossover intervention study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 178-184; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 019 pp. 178-184.

120 to 020: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Trust and fidelity between people | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Trust and fidelity between people: History, promises, evidence, uncertainty and reciprocity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Trust and fidelity between people; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Trusted/untrusted binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: People and institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Dynamic provenance and expectation architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Prospective trust-repair study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 185-191; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 020 pp. 185-191.

120 to 021: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Healthcare delivery | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Healthcare delivery: Clinical state, logistics, caregiver capacity, finance and post-discharge environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Healthcare delivery; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Episode-of-care schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Patient, caregiver, outpatient system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Consequential discharge/care orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 193-199; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 021 pp. 193-199.

120 to 022: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Diagnostics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Diagnostics: Underlying disease state, uncertainty, test characteristics and downstream actions.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Diagnostics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Single test result. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Patient, clinician, payer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Bayesian multi-source diagnostic schema + uncertainty communication. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Prospective blinded diagnostic study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 200-206; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 022 pp. 200-206.

120 to 023: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Cancer ecology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Cancer ecology: Clone mix, immune state, microenvironment, host reserve and therapy exposure.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Cancer ecology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Stage/tumor-size snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Resistant clones, healthy tissue, immune system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Adaptive ecological state estimator + resistance/host capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Preclinical then clinical adaptive trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 207-213; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 023 pp. 207-213.

120 to 024: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Neuroscience and neurodegeneration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Neuroscience and neurodegeneration: Neural, glial, vascular, metabolic, network and functional state.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Neuroscience and neurodegeneration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Single biomarker or visible function. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Neural networks, metabolism, caregiver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Multimodal state estimator + restorative capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Multimodal longitudinal intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 214-220; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 024 pp. 214-220.

120 to 025: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Aging biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Aging biology: Coupled damage, repair, senescence, immunity, metabolism and regeneration.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Aging biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Age scalar or isolated hallmark. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Cells, tissues, immune system, metabolism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Moving-bottleneck RVG controller across damage/repair channels. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Preclinical multi-pathway study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 221-227; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 025 pp. 221-227.

120 to 026: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Immunology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Immunology: Immune activation, tolerance, pathogen/tumor signals and tissue damage.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Immunology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Single cytokine or cell-count schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Pathogen, immune cells, host tissue. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: State-dependent immune control balancing clearance and collateral damage. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Controlled biological study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 228-234; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 026 pp. 228-234.

120 to 027: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Infectious disease | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Infectious disease: Pathogen dynamics, host immunity, behavior, network spread and treatment.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Infectious disease; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Case count or pathogen-only model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Patients, contacts, health system, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Multi-carrier containment + treatment + livelihood capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Prospective outbreak protocol'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 235-241; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 027 pp. 235-241.

120 to 028: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Microbiome | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Microbiome: Host physiology plus microbial ecology and diet/environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Microbiome; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Species abundance snapshot. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Host and microbial community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Functional ecosystem state model and adaptive intervention. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Controlled longitudinal study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 242-248; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 028 pp. 242-248.

120 to 029: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Genomics and proteomics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Genomics and proteomics: Variants, expression, proteins, pathways, context and phenotype.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Genomics and proteomics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Single marker/pathogenicity label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Cells, tissues, patient. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Multi-omic causal state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Prospective multi-omic validation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 249-255; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 029 pp. 249-255.

120 to 030: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Surgery and rehabilitation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Surgery and rehabilitation: Anatomy, function, pain, recovery capacity, home support and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Surgery and rehabilitation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Procedure success or anatomical correction. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Patient, caregiver, rehab system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Prehabilitation + procedure + recovery orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Pragmatic trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 256-262; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 030 pp. 256-262.

120 to 031: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Early-childhood development and education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Early-childhood development and education: Development, safety, language, nutrition, play, attachment and resources.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Early-childhood development and education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Age milestone or readiness score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Child, caregiver, educator. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Integrated developmental support architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Longitudinal program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 264-270; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 031 pp. 264-270.

120 to 032: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> K-12 education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

K-12 education: Knowledge, skills, wellbeing, context, resources and learning trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into K-12 education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Grade/test score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Student, teacher, family. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Capability graph + adaptive learning + wellbeing constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Cluster randomized trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 271-277; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 032 pp. 271-277.

120 to 033: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Higher education | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Higher education: Capability, knowledge, network, finances and career pathways.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Higher education; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Degree/GPA credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Student, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Capability/provenance portfolio + adaptive pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Longitudinal graduate outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 278-284; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 033 pp. 278-284.

120 to 034: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Vocational and workforce learning | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Vocational and workforce learning: Skills, task demand, wages, mobility and employer needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Vocational and workforce learning; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Course completion certificate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Worker, employer, training provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Skill-to-task provenance and placement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Matched labor-market evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 285-291; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 034 pp. 285-291.

120 to 035: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Scientific research | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Scientific research: Hypotheses, methods, data, negative results, replication and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Scientific research; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Publication/citation unit. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Researchers, institutions, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Experiment provenance + negative-result + replication architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Cross-lab replication program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 292-298; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 035 pp. 292-298.

120 to 036: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Justice and courts | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Justice and courts: Facts, evidence, harm, intent, procedure, history and remedies.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Justice and courts; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Case category or docket state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Victims, accused, families, courts, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Causal provenance + harm/restitution + recurrence architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Prospective process evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 299-305; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 036 pp. 299-305.

120 to 037: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Policing and public safety systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Policing and public safety systems: Threat, context, evidence, community trust, capacity and alternatives.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Policing and public safety systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Incident/offense label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Residents, officers, victims, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Evidence-first response selection + prevention/support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Independent public evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 306-312; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 037 pp. 306-312.

120 to 038: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Corrections, rehabilitation and reintegration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Corrections, rehabilitation and reintegration: Risk, harm, needs, skills, housing, work and social support.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Corrections, rehabilitation and reintegration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Sentence/recidivism risk label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Person, family, community, justice system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Rehabilitation + restitution + housing/work capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Longitudinal controlled program'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 313-319; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 038 pp. 313-319.

120 to 039: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Public administration | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Public administration: Citizen state, eligibility, documents, services and agency capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Public administration; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Program-specific record. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Citizen, frontline staff, budget. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Interoperable verified state + once-only data + case orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Agency pilot with citizen/staff metrics'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 320-326; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 039 pp. 320-326.

120 to 040: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Governance and institutional design | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Governance and institutional design: Authority, incentives, information, accountability, resources and affected populations.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Governance and institutional design; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Organizational chart or legal mandate. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Citizens, agencies, future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Cross-boundary outcome ledger + accountability and capacity design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Prospective institutional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 327-333; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 040 pp. 327-333.

120 to 041: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Beauty and aesthetics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Beauty and aesthetics: Biology, preference, culture, identity, safety, cost and perception.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Beauty and aesthetics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Universal beauty standard or product category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Person, body, household. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Personalized aesthetic/safety state architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Consumer/clinical outcome study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 335-341; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 041 pp. 335-341.

120 to 042: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Fashion and apparel | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Fashion and apparel: Identity, fit, comfort, durability, labor, inventory and lifecycle.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Fashion and apparel; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Size/SKU/category schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Wearer, worker, retailer, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Adaptive sizing + provenance + circularity architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Retail field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 342-348; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 042 pp. 342-348.

120 to 043: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Luxury goods | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Luxury goods: Provenance, scarcity, condition, access, identity, inventory and lifecycle value.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Luxury goods; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Price/status/authenticity binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Buyer, seller, inventory, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Provenance + condition + access + reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Operational field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 349-355; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 043 pp. 349-355.

120 to 044: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Hospitality | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Hospitality: Guest needs, staff capacity, time, room state, logistics and locality.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Hospitality; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Booking/reservation state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Guest, staff, operator, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: End-to-end experience and workload orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Property-level A/B evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 356-362; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 044 pp. 356-362.

120 to 045: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Travel and tourism | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Travel and tourism: Mobility, cost, safety, experience, local capacity and environment.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Travel and tourism; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Itinerary/booking schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Traveler, worker, resident, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Consequential itinerary/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Destination pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 363-369; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 045 pp. 363-369.

120 to 046: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Art and cultural markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Art and cultural markets: Creative work, provenance, rights, cultural meaning, scarcity and audience.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Art and cultural markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Price/ownership/engagement proxy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Artist, audience, collector, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Rights/provenance/value-sharing architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Marketplace field study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 370-376; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 046 pp. 370-376.

120 to 047: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Music and entertainment | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Music and entertainment: Creative contribution, rights, attention, audience value and platform economics.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Music and entertainment; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Stream/view count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Creators, audiences, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Contribution/provenance + value-distribution architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 377-383; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 047 pp. 377-383.

120 to 048: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Gaming and virtual worlds | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Gaming and virtual worlds: Player goals, time, identity, economy, safety and creator contribution.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Gaming and virtual worlds; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Retention/spend metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Players, creators, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Cross-reality value and safety controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 384-390; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 048 pp. 384-390.

120 to 049: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Sports and performance | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Sports and performance: Performance, health, training load, recovery, team dynamics and career horizon.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Sports and performance; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Win/score/output metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Athlete, team, family, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Load/recovery/career controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Prospective team study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 391-397; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 049 pp. 391-397.

120 to 050: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Experience design and events | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Experience design and events: Attention, emotion, logistics, accessibility, staff load and community impact.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Experience design and events; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Attendance/NPS/ticket revenue. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Guests, staff, locality. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: End-to-end experience-state orchestration. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Event A/B study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 398-404; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 050 pp. 398-404.

120 to 051: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Currency and monetary experience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Currency and monetary experience: Purchasing power, liquidity, settlement, access, trust and macro context.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Currency and monetary experience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Nominal unit/FX quote. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Households, firms, intermediaries. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Consequential monetary-state and settlement architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Historical + prospective policy-neutral system test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 406-412; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 051 pp. 406-412.

120 to 052: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Payments and remittances | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Payments and remittances: Amount, fees, FX, settlement time, access, fraud and recipient liquidity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Payments and remittances; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Successful transfer binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Sender, receiver, provider. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: End-to-end transfer optimization with transparent provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Transaction-level field test'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 413-419; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 052 pp. 413-419.

120 to 053: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Banking and liquidity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Banking and liquidity: Deposits, credit, liquidity, capital, customer state and network exposures.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Banking and liquidity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Product/account or institution-local balance sheet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Banks, customers, counterparties, public. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Network consequential balance sheet + capacity buffers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Stress test + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 420-426; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 053 pp. 420-426.

120 to 054: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Credit and underwriting | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Credit and underwriting: Cashflow, assets, obligations, trajectory, capability and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Credit and underwriting; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Static score/category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Borrower, lender, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Dynamic provenance/capacity underwriting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Prospective lending trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 427-433; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 054 pp. 427-433.

120 to 055: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Insurance and resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Insurance and resilience: Hazard, exposure, prevention, coverage, recovery and incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Insurance and resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Loss probability/policy class. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Policyholder, insurer, community. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Prevention + resilience + coverage architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Pilot product evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 434-440; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 055 pp. 434-440.

120 to 056: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Capital markets and investing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Capital markets and investing: Price, cashflow, externalities, network risk and long-horizon value.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Capital markets and investing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Return/risk factor set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Investor, firm, workers, public, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Consequential asset ledger with carrier constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Historical backtest + prospective paper portfolio'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 441-447; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 056 pp. 441-447.

120 to 057: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Private wealth and family capital | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Private wealth and family capital: Financial, human, social, health and intergenerational capital.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Private wealth and family capital; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Portfolio net worth. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Client, family, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Whole-capital balance sheet + goals/capacity controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Longitudinal client evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 448-454; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 057 pp. 448-454.

120 to 058: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Labor and talent markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Labor and talent markets: Capability, workload, pay, health, learning and opportunity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Labor and talent markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Job title/CV/credential. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Worker, family, employer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Capability-state + workload + mobility architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Workplace pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 455-461; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 058 pp. 455-461.

120 to 059: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Organizations and enterprises | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Organizations and enterprises: Functions, dependencies, people, cash, knowledge and customer value.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Organizations and enterprises; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Department/KPI structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Workers, customers, suppliers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Enterprise consequential graph + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Multi-team field experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 462-468; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 059 pp. 462-468.

120 to 060: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Entrepreneurship and venture creation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Entrepreneurship and venture creation: Unmet burden, carrier, willingness to pay, build cost and adoption.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Entrepreneurship and venture creation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Existing market category/TAM. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Founder initially, then customer/system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Mutual-Uptick blocker discovery + missing-capacity build. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Pilot venture with carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 469-475; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 060 pp. 469-475.

120 to 061: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Artificial intelligence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Artificial intelligence: Model state, source reality, user intent, provenance, uncertainty and downstream action.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Artificial intelligence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Prompt/context window/model output. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: User, verifier, downstream system, compute. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Persistent schema/provenance layer + self-audit and escalation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Sealed cross-model benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 477-483; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 061 pp. 477-483.

120 to 062: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Software systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Software systems: Requirements, dependencies, state, users, infrastructure and technical debt.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Software systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Feature/ticket/local module schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Developers, users, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Dependency/provenance graph + consequential release controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Repository-level prospective benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 484-490; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 062 pp. 484-490.

120 to 063: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Cybersecurity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Cybersecurity: Actual attack graph, identities, vulnerabilities, privileges and attacker state.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Cybersecurity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Defender inventory/rules/log view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Systems, users, analysts, business continuity. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Self-updating attack schema + carrier prediction + safe containment/failover. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Red-team sealed telemetry benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 491-497; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 063 pp. 491-497.

120 to 064: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Data infrastructure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Data infrastructure: Datasets, lineage, semantics, permissions, quality and downstream uses.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Data infrastructure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Table/file/schema local view. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Analysts, models, users, compliance. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Provenance graph + semantic reconciliation layer. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Enterprise data benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 498-504; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 064 pp. 498-504.

120 to 065: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Privacy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Privacy: Data state, purposes, consent, exposure, value and future misuse.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Privacy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Consent checkbox or access-control list. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Person, organization, future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Selective disclosure + provenance + minimization architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Field trial + attack testing'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 505-511; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 065 pp. 505-511.

120 to 066: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Digital identity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Digital identity: Attributes, credentials, context, revocation, privacy and trust.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Digital identity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Single identifier/account. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Person, verifier, issuer. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Selective-verification credential graph. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Cross-service pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 512-518; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 066 pp. 512-518.

120 to 067: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Internet and communications | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Internet and communications: Topology, capacity, latency, content, trust and user needs.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Internet and communications; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Throughput/uptime metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Users, networks, platforms. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Consequential routing + fidelity/provenance layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Network/application experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 519-525; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 067 pp. 519-525.

120 to 068: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Media and information ecosystems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Media and information ecosystems: Claims, evidence, incentives, attention, creators and audiences.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Media and information ecosystems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Engagement/feed rank. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Audience, creator, public discourse. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Claim provenance + quality + incentive reconciliation. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Platform experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 526-532; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 068 pp. 526-532.

120 to 069: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Language and translation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Language and translation: Meaning, context, culture, pragmatics and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Language and translation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Word/token correspondence. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Speakers, institutions, models. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Semantic/provenance translation with ambiguity tracking. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Blind human evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 533-539; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 069 pp. 533-539.

120 to 070: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Knowledge and ontologies | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Knowledge and ontologies: Entities, relations, provenance, incompatible schemas and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Knowledge and ontologies; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Single ontology/taxonomy. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Researchers, AI, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Cross-schema translation + minimal sufficient ontology. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Cross-domain benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 540-546; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 070 pp. 540-546.

120 to 071: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Energy and power grids | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Energy and power grids: Generation, load, storage, network constraints, prices and reliability.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Energy and power grids; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Local device efficiency or node KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Grid nodes, customers, storage, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Consequential grid controller + capacity placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Grid simulation + pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 548-554; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 071 pp. 548-554.

120 to 072: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Water and irrigation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Water and irrigation: Withdrawals, consumptive use, return flows, storage, aquifers and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Water and irrigation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Farm/project efficiency. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Downstream users, rivers, aquifers, ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Basin-wide accounting + catchment + allocation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Basin holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 555-561; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 072 pp. 555-561.

120 to 073: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Food and agriculture | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Food and agriculture: Yield, soil, water, inputs, nutrition, labor, market and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Food and agriculture; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Yield/acre or commodity price. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Farmer, consumer, soil, ecosystem. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Coupled regenerative production controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Multi-season farm trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 562-568; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 073 pp. 562-568.

120 to 074: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Supply chains and logistics | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Supply chains and logistics: Inventory, demand, transport, suppliers, lead times and risk.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Supply chains and logistics; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Node-level cost/service KPI. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Suppliers, warehouses, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Carrier-aware flow controller + buffer placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Historical + live pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 569-575; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 074 pp. 569-575.

120 to 075: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Manufacturing | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Manufacturing: Materials, energy, machines, labor, quality, maintenance and waste.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Manufacturing; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Unit cost/OEE. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Workers, machines, environment, customers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Closed-loop production and maintenance architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Factory pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 576-582; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 075 pp. 576-582.

120 to 076: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Waste and circular materials | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Waste and circular materials: Material composition, location, contamination, residual value and logistics.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Waste and circular materials; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Waste tonnage/disposal category. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Communities, environment, future production. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Material provenance + recapture + transformation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Municipal/industrial pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 583-589; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 076 pp. 583-589.

120 to 077: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Transportation and mobility | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Transportation and mobility: Trips, time, safety, access, energy, congestion and land use.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Transportation and mobility; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Vehicle throughput/speed. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Travelers, neighborhoods, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Multimodal consequential mobility controller. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'City corridor pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 590-596; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 077 pp. 590-596.

120 to 078: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Cities and urban systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Cities and urban systems: Housing, jobs, transport, health, utilities, safety and public space.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Cities and urban systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Agency-by-agency city data. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Residents, agencies, infrastructure. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: City consequential digital twin + coordinated capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'District pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 597-603; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 078 pp. 597-603.

120 to 079: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Climate and disaster resilience | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Climate and disaster resilience: Hazard, exposure, vulnerability, infrastructure, food, health and recovery.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Climate and disaster resilience; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Hazard forecast alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: People, infrastructure, ecosystems, economy. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Anticipatory catchment + prepositioned capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Prospective matched pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 604-610; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 079 pp. 604-610.

120 to 080: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Ecology and biodiversity | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Ecology and biodiversity: Species, habitats, flows, extraction, communities and climate.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Ecology and biodiversity; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Protected-area/local species metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Species, habitats, communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Landscape-scale leakage/carrier accounting + livelihoods. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Landscape holdout study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 611-617; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 080 pp. 611-617.

120 to 081: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Cell biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Cell biology: Signaling, organelles, metabolism, damage, repair and cell-state transitions.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Cell biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Single pathway/marker. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Organelles, pathways, neighboring cells. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Multi-state cellular controller + capacity targeting. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'In-vitro perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 619-625; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 081 pp. 619-625.

120 to 082: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Molecular biology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Molecular biology: Molecules, conformations, binding networks, concentrations and kinetics.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Molecular biology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Single interaction/target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Molecules, pathways, host system. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Reaction/network state model + off-target carrier control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Bench experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 626-632; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 082 pp. 626-632.

120 to 083: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Nanotechnology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Nanotechnology: Particle properties, surfaces, transport, environment and biological interaction.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Nanotechnology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Particle spec alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Cells, environment, users. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Lifecycle and interaction-aware nano design. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Bench + environmental tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 633-639; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 083 pp. 633-639.

120 to 084: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Chemistry and reaction systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Chemistry and reaction systems: Species, reaction network, kinetics, energy, byproducts and separation.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Chemistry and reaction systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Target yield/selectivity. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Reactor, environment, downstream process. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Consequential reaction-network optimization + recovery loops. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Reactor experiments'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 640-646; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 084 pp. 640-646.

120 to 085: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Materials science | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Materials science: Microstructure, defects, stress, temperature, lifecycle and manufacturing.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Materials science; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Bulk property spec. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Material, structure, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Defect-aware multiproperty design + recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Lab characterization'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 647-653; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 085 pp. 647-653.

120 to 086: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Classical mechanics and control | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Classical mechanics and control: Physical state, dynamics, sensors, actuators and disturbance.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Classical mechanics and control; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Partial state observation/local control objective. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Plant, actuator, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Representation-aware full-state controller + regenerative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Physical bench experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 654-660; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 086 pp. 654-660.

120 to 087: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Thermodynamics and nonequilibrium systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Thermodynamics and nonequilibrium systems: Energy, entropy, gradients, reservoirs, flows and constraints.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Thermodynamics and nonequilibrium systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Local efficiency/temperature metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Reservoirs, environment, components. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Whole-boundary energy/exergy and recovery architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Physical experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 661-667; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 087 pp. 661-667.

120 to 088: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Electromagnetism and circuits | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Electromagnetism and circuits: Fields, voltages, currents, topology, storage and losses.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Electromagnetism and circuits; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Local component state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Circuit elements, sources, loads. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Full-network state estimation + capacity/recovery placement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Bench circuit experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 668-674; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 088 pp. 668-674.

120 to 089: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Quantum systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Quantum systems: Quantum state, observables, measurement, control and decoherence.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Quantum systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Accessible observable set. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Quantum system, apparatus, environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Constraint-respecting measurement/control schema + capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Laboratory quantum experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 675-681; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 089 pp. 675-681.

120 to 090: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Pure mathematics and formal systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Pure mathematics and formal systems: Objects, relations, axioms, language, proofs and translations.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Pure mathematics and formal systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Chosen signature/type system/quotient. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Proof system, translator, solver. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Minimal sufficient schema preserving required relations. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Machine-checked theorem suite'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 682-688; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 090 pp. 682-688.

120 to 091: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Statistics and causal inference | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Statistics and causal inference: Data-generating process, sampling, measurement, confounding and uncertainty.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Statistics and causal inference; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Chosen model/estimand. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Decision makers, future samples, affected groups. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Provenance-aware multi-model causal state + identifiability guardrails. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Simulation + multi-dataset benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 690-696; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 091 pp. 690-696.

120 to 092: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Logic and reasoning systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Logic and reasoning systems: Premises, inference rules, contradictions, uncertainty and goals.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Logic and reasoning systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Single logical calculus. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Reasoner, downstream decision. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Multi-logic translation + contradiction provenance. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Formal benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 697-703; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 092 pp. 697-703.

120 to 093: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Geography and territorial systems | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Geography and territorial systems: People, ecosystems, infrastructure, flows and administrative boundaries.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Geography and territorial systems; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Jurisdiction/map boundary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Border communities, watersheds, networks. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Flow-based territorial state model. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Regional pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 704-710; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 093 pp. 704-710.

120 to 094: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Informal economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Informal economy: Real work, exchange, trust, risk and household needs outside formal records.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Informal economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Formal registry/tax/business schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Workers, households, state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Low-friction recognition, protection and access architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Field pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 711-717; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 094 pp. 711-717.

120 to 095: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Illicit and black markets | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Illicit and black markets: Hidden demand, supply, coercion, enforcement, prices and harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Illicit and black markets; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Official market data that excludes hidden exchange. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Victims, communities, lawful firms, institutions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Lawful substitutes, prevention, provenance, treatment/rehabilitation and targeted enforcement. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Independent harm-reduction/public-safety evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 718-724; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 095 pp. 718-724.

120 to 096: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Orbital and space economy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Orbital and space economy: Spacecraft, debris, spectrum, energy, logistics and operator incentives.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Orbital and space economy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Mission-local success metric. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Other spacecraft, operators, orbital environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Orbital consequential ledger + shared capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Simulation + mission pilot'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 725-731; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 096 pp. 725-731.

120 to 097: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Extraterrestrial settlement | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Extraterrestrial settlement: Life support, energy, materials, health, governance and closed ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Extraterrestrial settlement; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Earth-derived subsystem silos. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Crew, machines, habitat ecology. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Closed-loop consequential state + autonomous capacity construction. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Analog habitat + mission tests'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 732-738; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 097 pp. 732-738.

120 to 098: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Astrobiology and biosignatures | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Astrobiology and biosignatures: Unknown life chemistries, environments, signals and contamination risks.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Astrobiology and biosignatures; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Earth-centric life schema. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Scientific program, environment, future missions. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Plural biosignature schema + provenance/contamination control. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Blind benchmark + mission data'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 739-745; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 098 pp. 739-745.

120 to 099: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Planetary systems and stewardship | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Planetary systems and stewardship: Atmosphere, hydrology, geology, biosphere, resources and settlements.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Planetary systems and stewardship; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Sector-local resource model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Settlements, ecosystems, future generations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Closed-loop planetary resource and restoration architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Earth analog + simulation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 746-752; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 099 pp. 746-752.

120 to 100: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Unknown and not-yet-named realities | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Unknown and not-yet-named realities: Any newly observed substrate, scale, agent, value dimension or exchange mode.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Unknown and not-yet-named realities; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: No adequate schema exists yet. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Unknown until carrier tracing identifies it. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Instantiate M_d, define schema, measure gap, trace carrier, construct capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Freeze kernel before learning outcomes; then holdout/replicate'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 753-759; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 100 pp. 753-759.

120 to 101: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Conscious experience and self-model | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Conscious experience and self-model: Perception, affect, memory, self-model, attention, internal accessibility and embodied state.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Conscious experience and self-model; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: A receiving observer reduces another system to behavior, category or output alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Experiencing system, observers, dependent systems and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Cross-modal state representation with provenance, uncertainty and non-erasure constraints. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Prospective multi-method cross-observer study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 761-767; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 101 pp. 761-767.

120 to 102: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Choice and option topology | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Choice and option topology: Actual options, perceived options, cost, risk, time, capacity, knowledge, coercion and expected consequence.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Choice and option topology; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Binary free-choice / no-choice label. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Chooser, affected people, institutions and future option set. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Feasible-option map plus prebuilt low-friction lawful/prosocial pathways. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Prospective choice-topology intervention'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 768-774; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 102 pp. 768-774.

120 to 103: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Truth, deception and disclosure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Truth, deception and disclosure: Belief, evidence, incentives, threat, reputation, disclosure cost, dependency and repair capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Truth, deception and disclosure; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Truth/lie binary detached from consequence topology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Speaker, listener, harmed parties, institutions and future trust network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Verification, safe disclosure, correction, restitution and transition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Sealed truth-cost / disclosure study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 775-781; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 103 pp. 775-781.

120 to 104: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Responsibility and causal attribution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Responsibility and causal attribution: Knowledge, foresight, control, alternatives, coercion, causal contribution, action and resulting displacement.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Responsibility and causal attribution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Moral label detached from the action-producing state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Actor, victims, institutions, community and future actors. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Causal-attribution ledger integrating option topology, foreseeability, control, harm and repair. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Blinded case reconstruction with preregistered causal criteria'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 782-788; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 104 pp. 782-788.

120 to 105: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Consent, coercion and autonomy | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Consent, coercion and autonomy: Information, power, alternatives, dependency, exit, reversibility, time pressure and capacity.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Consent, coercion and autonomy; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Signed/expressed consent treated as sufficient state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Decision maker, counterparties, dependents and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Real exit, recourse, reversible consent, information symmetry and alternative capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Prospective consent-quality audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 789-795; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 105 pp. 789-795.

120 to 106: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Addiction, compulsion and short-horizon relief | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Addiction, compulsion and short-horizon relief: Distress, reward, relief, dependence, physiology, social setting, opportunity and future option set.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Addiction, compulsion and short-horizon relief; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Substance/behavior category or abstinence binary. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Person, body, relationships, household, community and future self. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Replacement regulation capacity plus medical/social/economic support matched to the absorbed burden. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Longitudinal regulated intervention with multi-carrier outcomes'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 796-802; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 106 pp. 796-802.

120 to 107: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Crime, violence and reconciliation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Crime, violence and reconciliation: Actor state, victim state, opportunity, motives, option topology, weapons/access, social environment and foreseeable harm.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Crime, violence and reconciliation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Offense label or actor-only risk score. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Victims, actor, families, community, institutions and future potential victims. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Prevention, containment where needed, restitution, rehabilitation and reintegration capacity in one ledger. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Longitudinal controlled justice-program evaluation'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 803-809; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 107 pp. 803-809.

120 to 108: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Animal sentience and welfare | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Animal sentience and welfare: Physiology, behavior, social state, environment, pain/pleasure proxies, health and life trajectory.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Animal sentience and welfare; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Commodity/yield schema that excludes the animal as a carrier. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Animals, workers, consumers, farmers, land, ecosystems and communities. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Welfare, alternative-production, livelihood-transition and ecological capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Multi-carrier welfare, nutrition and livelihood trial'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 810-816; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 108 pp. 810-816.

120 to 109: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Collective agency and institutional choice | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Collective agency and institutional choice: Shared beliefs, rules, authority, voting, incentives, information, dissent, enforcement and institutional memory.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Collective agency and institutional choice; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Organization treated as a single actor without internal carrier structure. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Members, minorities, leaders, outsiders, institutions and future administrations. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Distributed decision provenance, dissent channels, accountability and cross-boundary capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Prospective institutional decision audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 817-823; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 109 pp. 817-823.

120 to 110: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Human-AI co-agency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Human-AI co-agency: Human intent, model state, training/provenance, tools, permissions, action, reversibility, dependency and downstream effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Human-AI co-agency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Model output or benchmark score treated as the whole system. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: User, non-user, model operator, workers, institutions, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Persistent provenance, human control, reversibility, portability, verification and receivingcapacity layers. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Sealed native-vs-K_R assisted agent benchmark'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 824-830; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 110 pp. 824-830.

120 to 111: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Evolution and natural selection | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Evolution and natural selection: Genotype, phenotype, environment, mutation, recombination, drift, selection, gene flow and lineage history.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Evolution and natural selection; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Fitness proxy or single-trait narrative. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Individuals, lineages, populations, interacting species and ecosystems. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Diversity, ecological capacity and intervention guardrails where deliberate action exists. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Time-locked population/genomic holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 832-838; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 111 pp. 832-838.

120 to 112: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Reproduction, inheritance, mitosis and meiosis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Reproduction, inheritance, mitosis and meiosis: Chromosomes, cell-cycle state, replication, recombination, segregation, repair and developmental context.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Reproduction, inheritance, mitosis and meiosis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Outcome-only birth/division count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Parent cell/organism, daughter cells, gametes, offspring and lineage. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Checkpoint, repair, redundancy and developmental-support capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Lineage-resolved perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 839-845; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 112 pp. 839-845.

120 to 113: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Development and morphogenesis | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Development and morphogenesis: Gene regulation, gradients, cell fate, tissue geometry, signaling, mechanics, resources and developmental time.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Development and morphogenesis; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Single marker or endpoint morphology. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Cells, tissues, organs, organism and developmental future state. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Stage-aware multi-pathway support, repair and resource architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Longitudinal developmental perturbation study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 846-852; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 113 pp. 846-852.

120 to 114: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Metabolism, diet and consumption | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Metabolism, diet and consumption: Incoming matter/energy/information, digestion, transport, microbiome, storage, elimination and metabolic regulation.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Metabolism, diet and consumption; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Calories/macros or one nutrient target. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Gut, microbiome, liver, tissues, household, food system and future organism. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Adaptive conversion, buffering, storage, elimination and access capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Controlled longitudinal multi-carrier nutrition study'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 853-859; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 114 pp. 853-859.

120 to 115: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Predation, livestock and food-system transformation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Predation, livestock and food-system transformation: Predator-prey relations, husbandry, feed, land, labor, animal state, nutrition, culture, price and ecology.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Predation, livestock and food-system transformation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Yield, price or human nutrition alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Animals, humans, workers, farmers, consumers, land, ecosystems and future food security. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Welfare, alternative proteins, regenerative land use, livelihood transition and nutrition capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Phased regional transition with preregistered carrier ledger'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 860-866; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 115 pp. 860-866.

120 to 116: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Learning, plasticity and cultural evolution | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Learning, plasticity and cultural evolution: Neural plasticity, memory, skill, belief, imitation, teaching, incentives and social transmission.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Learning, plasticity and cultural evolution; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Test score, output or adoption count. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Learner, teachers, peers, institutions, future self and cultural network. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Feedback, practice, memory, rest, social scaffolding and plural knowledge capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Longitudinal transfer-and-retention experiment'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 867-873; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 116 pp. 867-873.

120 to 117: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Self-organization and emergence | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Self-organization and emergence: Local interactions, gradients, constraints, feedback, network topology, resource flows and macrostate.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Self-organization and emergence; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Macro outcome without micro-carrier ledger. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Components, neighborhoods, emergent structure and external environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Boundary conditions, damping, redundancy, resource routing and adaptive feedback capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Multi-scale simulation plus physical/biological holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 874-880; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 117 pp. 874-880.

120 to 118: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Creation, provenance and dependency | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Creation, provenance and dependency: Origin lineage, creator-created dependency, resources, control, autonomy, maintenance and reciprocal effects.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Creation, provenance and dependency; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Ownership or authorship treated as sufficient relationship model. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Creators, created systems, maintainers, users, infrastructure and environment. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Provenance, stewardship, autonomy-transition and reciprocal-viability architecture. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Longitudinal creator-created dependency audit'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 881-887; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 118 pp. 881-887.

120 to 119: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Cosmological evolution and structure formation | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Cosmological evolution and structure formation: Spacetime geometry, fields, matter/radiation, density structure, expansion, gravitation and observational state.

INTERFACE AND COUPLING STATUS

Proposed interface from Cross-substrate consciousness, temporal coexistence and closure into Cosmological evolution and structure formation; a physical, institutional or informational pathway must first be observed before a causal edge can be asserted. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Single scale or observable treated as complete cosmic state. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: Observed regions, fields/matter components, instruments, models and observers. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Cross-scale observation, uncertainty, model-comparison and instrumentation capacity. Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Preregistered cross-survey predictive holdout'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 888-894; CROSS-DOMAIN POTENTIAL HANDOFF. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 119 pp. 888-894.

120 to 120: consequential handoff

Cross-substrate consciousness, temporal coexistence and closure -> Cross-substrate consciousness, temporal coexistence and closure | candidate causal interface

ANALYTICAL COUPLING / REAL EDGE NOT YET IDENTIFIED

ORIGINATING DOMAIN AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

PROPOSED RECEIVER AND STATE

Cross-substrate consciousness, temporal coexistence and closure: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

INTERFACE AND COUPLING STATUS

This is a within-domain loop: the post-state of Cross-substrate consciousness, temporal coexistence and closure becomes the next native state under its existing map. No edge is presumed to exist solely because both domains appear in the Atlas.

SCHEMA TRANSFORMATION

Source's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Receiver's limited view: Any local adapter, species, institution or observer claiming closure from its own schema alone. Design a non-lossy handoff or explicitly record what is dropped.

WHO COULD CARRY THE BURDEN?

Origin carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Candidate receiver carriers: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Each must be assessed in native units, including time-shifted burdens.

CANDIDATE CONSTRUCTIVE BRIDGE

Origin capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Receiving capacity: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and.... Specify which receiving bottleneck a verified interaction would actually encounter.

MINIMUM DECISIVE EXPERIMENT

Pair a native study of 'Time-locked system-of-systems closure test across multiple adapters' with a receiver study of 'Time-locked system-of-systems closure test across multiple adapters'. Preserve matched baseline, causal identification, resource parity and follow-up.

INTERPRETATION AND GATE

Original source pp. 895-901 and 895-901; WITHIN-DOMAIN RECURRENCE. Not evidence that either implementation has reached Mutual Uptick.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026). Receiver Adapter 120 pp. 895-901.

120: Cross-substrate consciousness, temporal coexistence and closure

Household / family / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Household / family / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Household / family / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Household / family / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Household / family / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Household / family / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Household / family. decision affects caretakers, time, participation and household reserve. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

School / university / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

School / university / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

School / university / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

School / university / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

School / university / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

School / university / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: School / university. hand-off links learning, credentialing, accessible support and opportunity. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Enterprise / cooperative / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Enterprise / cooperative / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Enterprise / cooperative / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Enterprise / cooperative / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Enterprise / cooperative / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Enterprise / cooperative / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Enterprise / cooperative. handoff links throughput, workers, buyers, capital and governance. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Healthcare / care team / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Healthcare / care team / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Healthcare / care team / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Healthcare / care team / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Healthcare / care team / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Healthcare / care team / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Healthcare / care team. handoff links treatment, safety, continuity, caregiving and costs. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Local government / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Local government / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Local government / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Local government / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Local government / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Local government / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Local government. handoff links service eligibility, public resources and excluded residents. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Ecological catchment / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Ecological catchment / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Ecological catchment / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Ecological catchment / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Ecological catchment / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Ecological catchment / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Ecological catchment. handoff links habitat, resource flows, affected communities and time. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Scientific laboratory / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Scientific laboratory / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Scientific laboratory / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Scientific laboratory / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Scientific laboratory / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Scientific laboratory / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Scientific laboratory. handoff links observation, instrument limits, replication and interpretation. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Public infrastructure / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Public infrastructure / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Public infrastructure / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Public infrastructure / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Public infrastructure / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Public infrastructure / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Public infrastructure. handoff links material flows, reliability, maintenance and external exposure. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Digital / AI service / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Digital / AI service / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Digital / AI service / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Digital / AI service / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Digital / AI service / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Digital / AI service / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Digital / AI service. handoff links model representation, human decisions, access and failure modes. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Interregional / planetary / Insufficient capacity | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Insufficient capacity, meaning receiver cannot accept full proposed influx under observed resource limits. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under insufficient capacity, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Interregional / planetary / Schema mismatch | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Schema mismatch, meaning institutional representation misses a material part of the evidence-grounded state. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under schema mismatch, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Interregional / planetary / Delayed effects | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Delayed effects, meaning beneficial local outcome may create later burdens in coupled receivers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under delayed effects, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Interregional / planetary / Measurement uncertainty | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Measurement uncertainty, meaning causal pathway or protected-carrier outcome is not yet identifiable. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under measurement uncertainty, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Interregional / planetary / Coordination failure | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Coordination failure, meaning individually reasonable actions create jointly harmful or inefficient interactions. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under coordination failure, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-substrate consciousness, temporal coexistence and closure

Interregional / planetary / Adversarial / inequitable distribution | one domain-native research configuration

DESIGN SCENARIO / NOT A HISTORICAL EVENT

RESEARCH ENVIRONMENT

Candidate setting: Interregional / planetary. handoff links transboundary effects, governance and deferred consequences. No real-world site is asserted by this planning scenario.

NATIVE MECHANISM TO PRESERVE

Cross-substrate consciousness, temporal coexistence and closure is described in the Atlas as: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility. Any deployment must preserve the established physical, biological, legal or institutional mechanism of this domain.

OBSERVED-IN-ATLAS CANDIDATE BLOCKER

A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. Additional scenario stressor: Adversarial / inequitable distribution, meaning locally favored actors can externalize burden onto less-protected carriers. This stressor has not been observed in a specific population here.

LERM CAUSAL WORK ORDER

Define actionable event a and comparator a0. Estimate D_a(j,k,tau) for All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths. Identify pathways and hidden compensation using credible data rather than narrative fit.

IAMF-P CONSTRUCTIVE PROPOSAL

Evaluate a capacity build based on: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Under adversarial / inequitable distribution, test whether altering topology before the action is safer than boundary-aware re-selection.

MANDATORY COUNTERFACTUAL AND SAFETY

Use the strongest admissible incumbent; hold information, intervention budget and horizon equal. Reject options that violate safety, consent or governing physical constraints.

CONSEQUENTIAL SUCCESS MEASUREMENTS

Author's native acceptance: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the sacrificial sink of another system’s advancement. Specify thresholds, subgroup exposure, future reserve and uncertainty before releasing the intervention.

NEXT REPRODUCIBLE OPERATION

Use Time-locked system-of-systems closure test across multiple adapters or a more valid native design. If no solution has been shown, report the binding constraint and next legitimate construction search; do not invent a PASS.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Evidence and source reconstruction

Cross-substrate consciousness, temporal coexistence and closure | protocol 1 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 120: Cross-substrate consciousness, temporal coexistence and closure; original author status: Meta-adapter / closure operator. Native state: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

EVALUATION OBJECTIVE

Evidence and source reconstruction. Establish what is observed, reported, derived, simulated and currently unavailable. Audit definitions, dates and units.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Any local adapter, species, institution or observer claiming closure from its own schema alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. The relevant receiving carriers are: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the....

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked system-of-systems closure test across multiple adapters. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Boundary divergence

Cross-substrate consciousness, temporal coexistence and closure | protocol 2 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 120: Cross-substrate consciousness, temporal coexistence and closure; original author status: Meta-adapter / closure operator. Native state: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

EVALUATION OBJECTIVE

Boundary divergence. Predeclare local and consequential objectives, identical feasible action sets, and a measurable divergence test.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Any local adapter, species, institution or observer claiming closure from its own schema alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. The relevant receiving carriers are: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the....

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked system-of-systems closure test across multiple adapters. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Causal diagram and identifiability

Cross-substrate consciousness, temporal coexistence and closure | protocol 3 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 120: Cross-substrate consciousness, temporal coexistence and closure; original author status: Meta-adapter / closure operator. Native state: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

EVALUATION OBJECTIVE

Causal diagram and identifiability. Draw the candidate causal graph; identify confounders, mediators, spillovers and selection mechanisms.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Any local adapter, species, institution or observer claiming closure from its own schema alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. The relevant receiving carriers are: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the....

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked system-of-systems closure test across multiple adapters. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Receiving carrier identification

Cross-substrate consciousness, temporal coexistence and closure | protocol 4 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 120: Cross-substrate consciousness, temporal coexistence and closure; original author status: Meta-adapter / closure operator. Native state: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

EVALUATION OBJECTIVE

Receiving carrier identification. Predict the destination, path and delay of consequential burdens before measuring post-state outcomes.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Any local adapter, species, institution or observer claiming closure from its own schema alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. The relevant receiving carriers are: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the....

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked system-of-systems closure test across multiple adapters. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Hidden compensation accounting

Cross-substrate consciousness, temporal coexistence and closure | protocol 5 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 120: Cross-substrate consciousness, temporal coexistence and closure; original author status: Meta-adapter / closure operator. Native state: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

EVALUATION OBJECTIVE

Hidden compensation accounting. Measure which agent, reserve, budget, ecosystem or later period silently carries the load.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Any local adapter, species, institution or observer claiming closure from its own schema alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. The relevant receiving carriers are: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the....

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked system-of-systems closure test across multiple adapters. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Native units and dimensional integrity

Cross-substrate consciousness, temporal coexistence and closure | protocol 6 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 120: Cross-substrate consciousness, temporal coexistence and closure; original author status: Meta-adapter / closure operator. Native state: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

EVALUATION OBJECTIVE

Native units and dimensional integrity. Define each carrier outcome in its own units; avoid offsetting unlike harms with unsupported scalar sums.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Any local adapter, species, institution or observer claiming closure from its own schema alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. The relevant receiving carriers are: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the....

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked system-of-systems closure test across multiple adapters. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Capacity specification

Cross-substrate consciousness, temporal coexistence and closure | protocol 7 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 120: Cross-substrate consciousness, temporal coexistence and closure; original author status: Meta-adapter / closure operator. Native state: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

EVALUATION OBJECTIVE

Capacity specification. Identify the binding receiver constraint and specify the smallest permissible constructive action.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Any local adapter, species, institution or observer claiming closure from its own schema alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. The relevant receiving carriers are: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the....

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked system-of-systems closure test across multiple adapters. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Intervention design

Cross-substrate consciousness, temporal coexistence and closure | protocol 8 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 120: Cross-substrate consciousness, temporal coexistence and closure; original author status: Meta-adapter / closure operator. Native state: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

EVALUATION OBJECTIVE

Intervention design. Choose randomized, natural, observational or mechanistic evidence compatible with safety and identifiability.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Any local adapter, species, institution or observer claiming closure from its own schema alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. The relevant receiving carriers are: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the....

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked system-of-systems closure test across multiple adapters. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Incumbent parity contest

Cross-substrate consciousness, temporal coexistence and closure | protocol 9 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 120: Cross-substrate consciousness, temporal coexistence and closure; original author status: Meta-adapter / closure operator. Native state: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

EVALUATION OBJECTIVE

Incumbent parity contest. Compare against strongest existing local, boundary-aware, causal and resource-constrained methods.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Any local adapter, species, institution or observer claiming closure from its own schema alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. The relevant receiving carriers are: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the....

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked system-of-systems closure test across multiple adapters. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Time-window and recurrence

Cross-substrate consciousness, temporal coexistence and closure | protocol 10 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 120: Cross-substrate consciousness, temporal coexistence and closure; original author status: Meta-adapter / closure operator. Native state: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

EVALUATION OBJECTIVE

Time-window and recurrence. Measure early, intermediate and delayed outcomes, including reserve depletion and changing bottlenecks.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Any local adapter, species, institution or observer claiming closure from its own schema alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. The relevant receiving carriers are: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the....

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked system-of-systems closure test across multiple adapters. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Ethical and legal safety

Cross-substrate consciousness, temporal coexistence and closure | protocol 11 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 120: Cross-substrate consciousness, temporal coexistence and closure; original author status: Meta-adapter / closure operator. Native state: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

EVALUATION OBJECTIVE

Ethical and legal safety. Determine affected rights, vulnerable groups, irreversible harms, consent, permissions and safety constraints.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Any local adapter, species, institution or observer claiming closure from its own schema alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. The relevant receiving carriers are: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the....

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked system-of-systems closure test across multiple adapters. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Sensitivity and negative controls

Cross-substrate consciousness, temporal coexistence and closure | protocol 12 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 120: Cross-substrate consciousness, temporal coexistence and closure; original author status: Meta-adapter / closure operator. Native state: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

EVALUATION OBJECTIVE

Sensitivity and negative controls. Vary unmeasured confounding, missingness, cost, thresholds and adverse spillover assumptions.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Any local adapter, species, institution or observer claiming closure from its own schema alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. The relevant receiving carriers are: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the....

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked system-of-systems closure test across multiple adapters. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Replication and provenance

Cross-substrate consciousness, temporal coexistence and closure | protocol 13 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 120: Cross-substrate consciousness, temporal coexistence and closure; original author status: Meta-adapter / closure operator. Native state: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

EVALUATION OBJECTIVE

Replication and provenance. Version the kernel, code, data, transformation decisions, negative findings and external replication.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Any local adapter, species, institution or observer claiming closure from its own schema alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. The relevant receiving carriers are: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the....

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked system-of-systems closure test across multiple adapters. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).

120: Cross-domain invariant contest

Cross-substrate consciousness, temporal coexistence and closure | protocol 14 of 14

RESEARCH PROTOCOL / NO NEW EMPIRICAL RESULT

ORIGINAL DOMAIN AND PRIOR RECORD

Adapter 120: Cross-substrate consciousness, temporal coexistence and closure; original author status: Meta-adapter / closure operator. Native state: The entire coupled consequential state across Adapters 001-119, including dependencies, time horizons and current consciousness-legibility.

EVALUATION OBJECTIVE

Cross-domain invariant contest. Keep K_R fixed while swapping M_d; test whether new measurements produce predictions beyond domain incumbents.

SCHEMA AND CANDIDATE BIAS

Currently described receiver representation: Any local adapter, species, institution or observer claiming closure from its own schema alone. Distinguish genuine measurement failure from domain states that do not exist.

CAUSAL PATHWAY TO DISCRIMINATE

Atlas candidate bottleneck: A newly recognized truth, capability or optimization becomes locally beneficial while dependent carriers lack the capacity to absorb its consequences through time. The relevant receiving carriers are: All materially coupled conscious carriers reached through direct, indirect and delayed dependency paths.

INTERVENTION WHOSE EFFECT MATTERS

Proposed constructive design: Coordinated capacity construction across relevant adapters, phased through time, while preserving openness to newly legible consciousness and dependencies. Show a feasible native action rather than restating a goal.

WHAT TO REPORT

Record exact sources, units, time, local and consequential objectives, strongest comparator, effect sizes, uncertainty and documented harms. Reference native goal: Increasing consciousness-legibility and viable coexistence across the coupled system, with no material carrier forced to become the....

RETEST AND SCOPE CONTROL

Candidate decisive method: Time-locked system-of-systems closure test across multiple adapters. Determine whether the fixed operator adds results beyond domain science, and retain negative findings.

CONSEQUENCE OF NON-PASS

Make the next constraint legible: missing causal evidence, insufficient capacity, unacceptable cost, physical impossibility, or external protected-carrier burden. Do not claim success without verified outcomes.

Source: Atlas Adapter 120, pp. 895-901 (Rufino 2026).